Universal Fault-Tolerant Computation on Three-Dimensional Layer Codes at Optimal Distance

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Three-dimensional layer codes combine local stabilizer checks with high storage density, but computation also requires local access to their logical operators. We give an explicit local realization of the Williamson–Baspin layer code for any CSS input whose checks have weight at least two, and prove that it encodes exactly the input logical qubits, that every pure-X logical class has a representative on a single lattice layer, at every height of the block, and that every pure-Z class has one in a single transverse plane, at every position. A running (prefix) parity along each check sheet gives these representatives. The slabs support logical Pauli measurements by nearest-neighbor three-dimensional grid gauging. A constant number of ancillas per grid edge preserves the full code distance, whereas a measurement layer of bounded thickness provably cannot: a thin interface contains a wall of few edges that a few faults can cross. Circuit-level simulations on explicit layer-code blocks confirm both statements; as the blocks grow, failures through the grid fall and failures through a one-row interface rise. We then observe these distance mechanisms on IQM and IBM superconducting processors, with three-qubit blocks and 6.9 million samples. Widening the interface from one edge to two lowers the error of a joint measurement 35-fold on IQM (from 2.4% to 0.070%, with no failure in 11,529 samples at three edges) and 9.7-fold on IBM, and two parallel ancillas on one edge lower it 4.6-fold on IQM. Repeated rounds protect the reported bit as a calibrated model predicts when every round uses fresh ancillas. When ancillas are measured and reset instead, errors that survive the reset, most likely leakage, leave the three-round error 68 times above the calibrated Pauli model, a failure mode outside local-stochastic noise. Lifting the input distance through the slabs shows that the realization has the optimal distance $\Theta(N^2)$ for good inputs. Combining the measurements with encoded resource transfer, distillation, and a local detector schedule gives a universal fault-tolerant architecture under local-stochastic Pauli noise. At fixed noise below an existential threshold, a depth-T circuit on W logical sites in a bounded-aspect box uses $O(Wf + f^{3/2})$ physical qubits and $O(Tf^{3/2}\Psi) \subseteq O(Tf^{5.41})$ depth, where $f = \Theta(\log(WT/\varepsilon))$ and $\Psi = O(f^{\log_2 15})$ is the distillation depth factor; a concrete estimate places the storage advantage at distances of order a thousand. We also derive a fragmentation correction for general regions and port-packing bounds saturated by balanced layer codes.

I. INTRODUCTION

Quantum error-correcting codes must protect stored information and make that information accessible to computation. These demands can pull in different directions: a code with favorable storage parameters may still require costly logical measurements or physical connections that are difficult to realize. Three-dimensional layer codes provide a natural setting in which to address this problem. The construction of Williamson and Baspin embeds a CSS code into a local network of surface-code sheets and defects [1]. For balanced good low-density parity-check inputs of length N, the resulting blocks have

$$n = \Theta(N^3), \qquad K = \Theta(N), \qquad d = \Theta(N^2), \quad (1)$$

with bounded-weight, geometrically local stabilizers. Here n is the physical block size, K its number of logical qubits, and d its distance.

The central result of this work is a geometric interface for processing that stored information. We give an explicit lattice realization of the layer code with complete check matrices, and show that every pure-X logical class has a representative on a single layer of the lattice and every pure-Z class one in a single transverse plane. The layer can be placed at any height, and the plane at any position, independently of N and of the chosen logical operator. This converts an algebraic choice of logical Pauli into a spatially localized measurement port.

We use this interface to construct universal fault-tolerant computation on the native three-dimensional nearest-neighbor lattice. The argument has three steps. A prefix-parity construction first produces the slab representatives. Relative expansion of a cubic grid then gives a local gauging measurement with distance $\Omega(d)$. Finally, repeated extraction, padded composition, and encoded resource transfer give a complete circuit under local-stochastic Pauli noise. Figure 1 summarizes the architecture.

Logical measurement by gauging, surgery, and homological constructions has become a general route to processing quantum LDPC codes [2–6]. Adapters, extractors, and universal architectures address complementary questions of connectivity, parallelism, and resource overhead [7–10]. Here the organizing constraint is three-dimensional Euclidean locality: logical access, resource delivery, and syndrome extraction use the same nearest-neighbor geometry. The layer-code slab theorem supplies the missing connection between the memory construction and these processing primitives.

Locality also fixes the achievable overhead. In three dimensions a local stabilizer code obeys $Kd = O(n)$ [11], so $\Theta(d)$ physical qubits per logical qubit is the best possible storage cost, and layer codes attain it. Surface-code

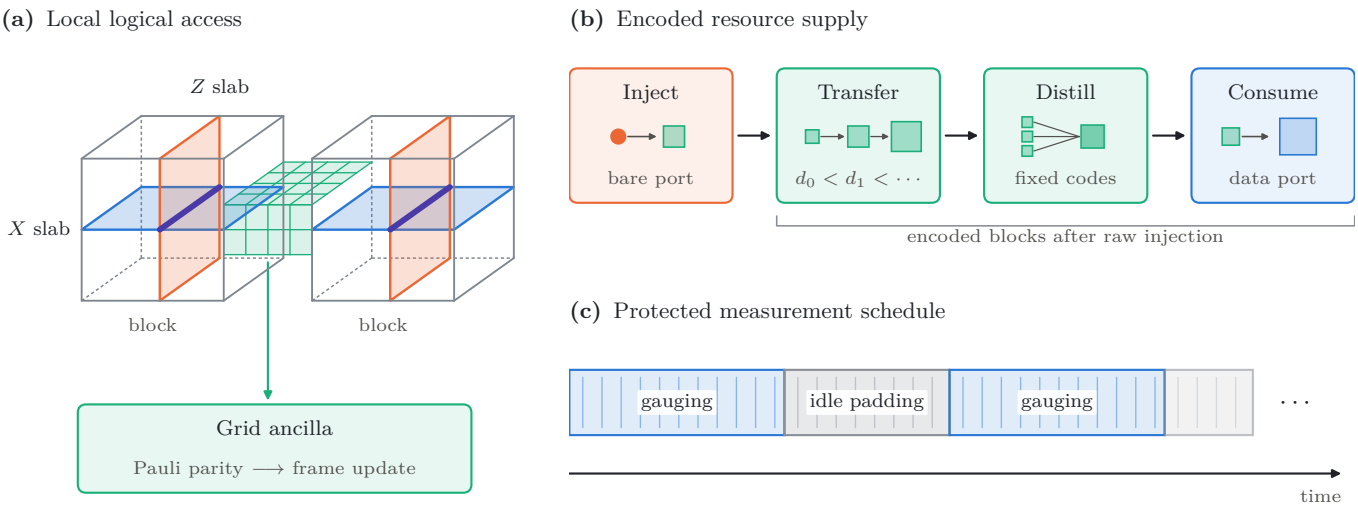


FIG. 1. Geometry-preserving processing. (a) A layer-code block exposes transverse slab ports for pure-X and pure-Z logical operators. A local grid ancilla measures the selected Pauli; neighboring blocks use the same interface for joint measurements. (b) A raw resource is injected into an encoded block, transferred through blocks of increasing distance, distilled, and consumed by the data path. (c) Repeated extraction and idle padding connect successive active measurements. All drawings are schematic; the encoded blocks and grid ancillas occupy three-dimensional regions.

patches processed by lattice surgery use $\Theta(d^2)$ qubits per logical qubit [12, 13], and three-dimensional surface and color codes trade larger volume for transversal gates [14]. Local simulations of LDPC codes [15], local fault-tolerance constructions [16], wire codes [17], and planar logical measurements [18] address related geometric questions, while constant-overhead schemes [19–21] and LDPC magic-state injection [22, 23] rely on nonlocal connectivity. The architecture below keeps the $\Theta(d)$-per-logical storage cost during computation, up to the cost of one encoded cell.

The distance mechanisms behind the gauging measurements already act in circuits small enough for present processors, so we also observe them on hardware (Sec. VII and Fig. 6). On IQM and IBM superconducting processors, with three-qubit blocks in place of layer-code blocks, a joint measurement through an interface of one edge fails 35 times more often than through an interface of two edges on IQM, and 9.7 times more often on IBM, as the cut obstruction predicts; two parallel ancillas on the single edge lower the error 4.6-fold on IQM; and repeated rounds protect the reported bit. The runs also expose a limit of the noise model. On IBM, ancilla errors that survive a reset, most likely leakage, repeat the same wrong outcome in every round and leave the three-round error 68 times above a calibrated Pauli model. Such correlated faults lie outside the local-stochastic model of our theorem, so an implementation of its schedule would need leakage reduction.

The main text develops the construction and the resource accounting, with proof sketches at each step, and reports a complete circuit-level simulation of one two-block logical measurement on explicit layer-code blocks and the hardware demonstrations of its distance mechanisms. Complete proofs, cell coordinates, detector templates, and numerical controls appear in the Supplemental Material [24]. We also show that the same slab mobility saturates a logical-port packing bound and restricts shallow code-preserving unitaries. These structural consequences are independent of the fault-tolerance proof.

II. LOGICAL OPERATORS IN CONSTANT-THICKNESS SLABS

A. The layer code as glued sheets

A layer code is assembled from surface-code sheets [1]. Each qubit q of the input CSS code becomes a *data sheet*, the plane $y = Y_q$; each input Z check becomes a Z-check sheet, a plane $x = \text{const}$; and each input X check becomes an X-check sheet, a plane $z = \text{const}$. Every sheet is an ordinary surface code on a rectangle of the cubic lattice, so it carries string operators. A string of X operators that runs from one side of a sheet to the other commutes with all checks of that sheet; a string that stops inside the sheet leaves a violated check, a *defect*, at its end. A Z-check sheet meets the data sheets of the qubits in its check along vertical lines, the *junctions*. Ordered along the check, these are its first, middle, and last junctions. The X-check sheets are attached in the same way, with X and Z exchanged.

A junction is a rule that says which strings may end on the line without leaving a defect. For the X strings used below only their presence matters. At a first or last junction the Z-check sheet ends on the data sheet, and a string may pass from one sheet to the other: the check-sheet string is present exactly when the data string is. At

a middle junction three strings meet the line: the data string, which crosses it, and the check-sheet strings on its two sides. They leave no defect exactly when an even number of the three is present. Writing 1 for a present string, and listing the check-sheet strings first (left, then right) and the data string last,

$$H_{\text{end}} = \begin{bmatrix} 1 & 1 \end{bmatrix}, \qquad \ker H_{\text{end}} = \{00, 11\},$$
$$H_{\text{mid}} = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}, \quad \ker H_{\text{mid}} = \{000, 110, 101, 011\}. \quad (2)$$

Williamson and Baspin state these rules as sets of condensed anyons, that is, as the combinations of string endpoints that disappear at the line; Eq. (2) is the same information written for X strings (Supplemental Sec. SI B). The word 110 is the important one. It lets a string on the check sheet pass a data sheet without any string on that data sheet. If a middle junction allowed only 000 and 111, every check-sheet string would drag a string onto each data sheet it passes, and a general logical operator could not be built; such a lattice in fact fails to encode the input qubits (Supplemental Sec. SII).

B. An explicit lattice for the junctions

We build every junction on the lattice (Supplemental Sec. SI A). Sheets own distinct qubits, and each junction rule adds one qubit of one sheet to one check of the other. At a middle junction the Z-check-sheet face beside the line gains the data sheet's edge on the line. That face then contains three qubits of interest: the check-sheet edge on the line, where the string from one side ends; the next check-sheet edge, where the string on the other side begins; and the added data edge. Its check is therefore exactly the middle rule of Eq. (2). The data stars on the line gain the check sheet's edges on the same side, which keeps them commuting with the modified faces. At a first or last junction the check sheet ends on the data line and shares its edges, which gives the two-bit rule.

One more repair is needed. An X-check sheet and a Z-check sheet cross along a line, and there the junction additions of the two sheets make a few stars and faces anticommute, once for each data qubit shared by the two input checks. A row of one-qubit additions along the crossing line, which we call a *wall*, removes these conflicts. It exists exactly when the shared number is even, which is the commutation of the two input checks. The slab operators below never use wall qubits, so the walls play no further role. Every check lies within unit distance of one lattice vertex and has weight at most six. Supplemental Fig. S1 draws one junction of each kind and a wall. The junctions have the condensed anyons of Ref. [1]; we do not claim that the lattice is identical to the drawings there, and all results below are proved for this explicit lattice, whose complete check matrices the source package generates.

C. Slab representatives

Theorem II.1 (Uniform slab representatives)**.** *For every input CSS code with checks of weight at least two, the explicit realization encodes exactly K logical qubits. Every pure-X logical class has a representative supported on the single layer of z-edges spanning $[z_0, z_0{+}1]$, for every height z_0 of the block, and every pure-Z class has a representative in the plane $x = x_0$, for every x_0. The heights and planes are chosen independently of the logical classes.*

The operator is built at one fixed height z_0, one input check at a time. Choose a pure-X logical $x \in \ker H_Z$ of the input code, and put an X string at height z_0 on the data sheet of every qubit with $x_q = 1$. Each such string crosses the junction of every Z-check sheet attached to its data sheet, and on its own would leave a defect there. Along one Z-check sheet the selected junctions must therefore be joined, in pairs, by strings on the check sheet at the same height; we call these check-sheet strings the *backbone*. Pairing is possible because x commutes with the input check, so the check contains an even number of selected qubits. Walk along the check from its first qubit and switch the backbone on or off at every selected junction. With ordered incidences $x_1, \ldots, x_k$, the backbone after the ith junction is present exactly when an odd number of selected qubits lies behind it, a running count that we call the *prefix parity*,

$$p_0 = 0, \qquad p_i = \sum_{h=1}^{i} x_h, \qquad p_k = 0, \qquad (3)$$

where $p_k = 0$ is the commutation $\sum_i x_i = 0$ over $\mathbb{F}_2$. At every middle junction $p_{i-1} + p_i + x_i = 0$, so none or two of the three strings are present, as Eq. (2) requires; at the ends $p_1 = x_1$ and $p_{k-1} = x_k$. No defect remains anywhere (Fig. 2). Every piece lies in the same layer of vertical edges, so the operator is one lattice layer thick, whatever the block size and whichever logical is chosen. Building it at another height gives another representative of the same class. In the explicit realization the check is exact and local: every stabilizer meets the operator in one of the forms $x_1 + p_1$, $p_{i-1} + p_i + x_i$, $p_{k-1} + x_k$ of a single incidence, or not at all.

The pure-Z operators are built in the same way in a vertical plane $x = x_0$, with backbones on the X-check sheets. An X slab operator and a Z slab operator share only one data edge for each input qubit selected by both, so they anticommute exactly when the two input logicals do. This gives K independent logical qubits. A count shows that there are no others. On its own, each data sheet would encode one qubit and each check sheet would remove one; the junctions and walls add neither qubits nor checks; and every redundancy among the lattice checks comes from a redundancy among the input checks. In formulas, each data sheet contributes $+1$ and each check sheet -1 to $n - \#X - \#Z$, and the relations among the lattice checks inject into those among the input checks.

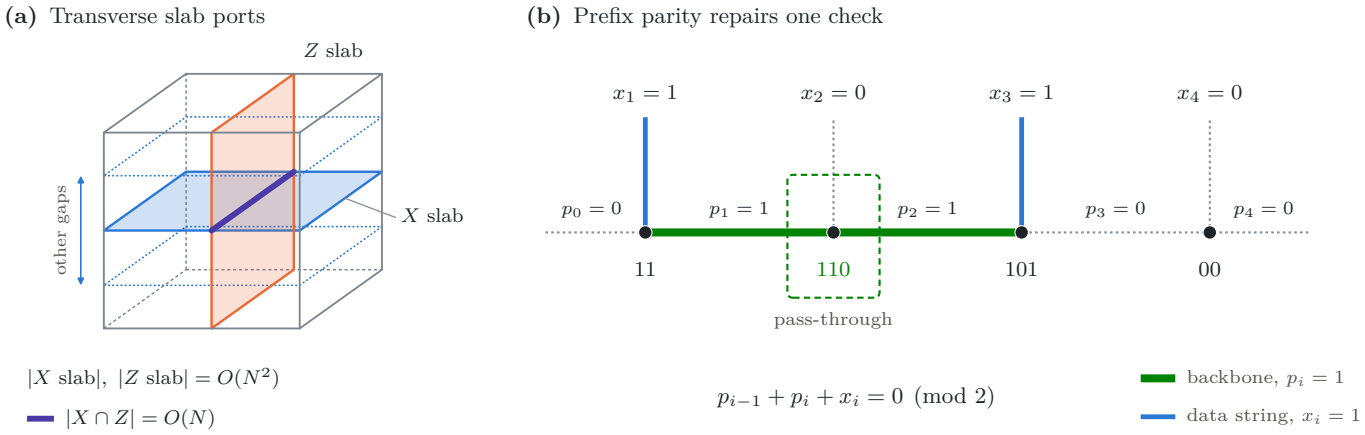


FIG. 2. Slab mobility and its local mechanism. (a) Complete X and Z ports occupy transverse single-layer slabs. Each slab contains $O(N^2)$ qubits, and their intersection contains $O(N)$ qubits. Translating to another height (or plane) gives another representative of the same logical class. (b) For the even incidence word $(1,0,1,0)$, the running parity is $(0,1,1,0,0)$. The highlighted middle cell uses 110: the backbone remains occupied although the data incidence is absent. Endpoint cells use the two-bit relation in Eq. (2).

Hence the realization encodes at most, and therefore exactly, K qubits, and the slab operators represent every pure-type class. A certificate on complete codes with up to 10,936 qubits, including the linear forms above, exhaustive negative controls, and the single-chart comparison, checks each step (Supplemental Sec. SII). The proofs are Supplemental Proposition SI.2 and Theorem SI.3.

Williamson and Baspin already obtain the layer-code logical operators by quasi-concatenating input logicals [1]. The new ingredients are an explicit realization in which this construction can be verified check by check, a single-layer thickness independent of the block size and of the chosen logical, and its uniformity over every height, which supplies the disjoint ports used below.

The same ports give the distance, by a simple idea: a logical operator must be visible in every plane that contains a complete port of the other type. Let O be a nontrivial X logical. It anticommutes with some Z logical, and that Z logical has a slab representative in each of the L_x+1 disjoint planes $x=x_0$, so O must place qubits in every one of these planes. To count them, cut the Z slabs of one plane into their pieces: a vertical data column for each input qubit, and the backbone segments. Record, for each piece, whether O meets it an odd number of times. Read a data piece as its input qubit, and a backbone piece as the qubits of its input X check that lie behind it, the first t of them for the tth segment. The records then spell out a nontrivial X logical of the input code, which has at least d_X^{in} entries. The first t qubits of a check may be exchanged for the remaining ones, because the whole check is a stabilizer of the input code, so a backbone piece stands for at most $\eta_X = \max\{1, \lfloor w_X/2 \rfloor\}$ qubits, where w_X is the largest input X-check weight. Each piece met an odd number of times holds at least one qubit of O, so each plane contains at least $\lceil d_X^{\mathrm{in}}/\eta_X \rceil$ qubits of O, and

$$d_X \geq (L_x+1)\left\lceil \frac{d_X^{\mathrm{in}}}{\eta_X} \right\rceil, \tag{4}$$

dually for Z. For balanced good inputs the realization therefore has $n=\Theta(N^3)$, $K=\Theta(N)$, and $d=\Theta(N^2)$, the parameters of Eq. (1) (Supplemental Theorem SI.6). For the $[[4,2,2]]$ and $[[6,4,2]]$ inputs the bound equals the exact distance in every computed instance.

For balanced inputs the block has $\Theta(N)$ heights and $\Theta(N)$ transverse positions, so there are

$$r_X = \Theta(N), \qquad r_Z = \Theta(N) \tag{5}$$

pairwise-disjoint slab regions within each orientation. Each region is a complete port for its logical polarization. This repeated access will also give the structural bounds in Sec. V.

III. LOCAL GAUGING AND UNIVERSAL LOGICAL OPERATIONS

A. Why a three-dimensional grid preserves distance

Gauging measures a logical operator by measuring many small checks whose product is that operator [2]. Let $\overline{P} = X(D)$ be the selected logical representative. Attach each data qubit of D to its own vertex of a connected grid $G=(V,E)$, call these vertices the *ports* U, and place an ancilla qubit on every edge. The new check at a vertex, its *vertex star*, is X on the edges at that vertex and on the attached data qubit, if any. Every edge belongs to two vertex stars, so the product of all vertex stars is $\overline{P}$, and the product of their measured outcomes is the logical outcome; yet each measurement involves only a

few neighboring qubits. The grid also carries Z checks around its unit squares, and every original Z check that meets D is extended along short grid paths joining its ports, so that it commutes with the stars. Preparing the edge qubits in $|0\rangle$, measuring the new checks, and finally reading the edges in Z returns the data to its original code with an updated Pauli frame. The X/Z-exchanged construction measures pure-Z logicals.

The danger is that the added qubits create new, shorter logical operators. The quantity that controls this is the expansion of the grid relative to its ports,

$$\beta(G,U) = \min_{S\subseteq V} \frac{|\delta S|}{\min\{|S\cap U|, |U\setminus S|\}}, \tag{6}$$

where δS is the set of edges leaving S and only positive denominators are included. We call δS a *wall* (not to be confused with the lattice walls of Sec. II): it separates the ports in S from the others, and β is the smallest number of wall edges paid per port separated. For a code with CSS distances d_X, d_Z, the deformed distances satisfy

$$\widetilde{d}_X \geq \min\{1, \beta(G,U)\} d_X, \qquad \widetilde{d}_Z \geq d_Z. \tag{7}$$

The reason is short. An X logical of the deformed code may put part of its weight on the ancilla edges, but those edges must form a wall δS; any other edge set anticommutes with some square Z check. Multiplying by the vertex stars in S removes the wall and puts X instead on the data qubits attached in S, or, using the complement of S, on those attached outside it. Either way the result is a logical of the original code, so it has weight at least d_X. The trade removes $|\delta S|$ edges and, on the better side, adds at most $m = \min\{|S\cap U|, |U\setminus S|\}$ data qubits, while $|\delta S| \geq \beta m$. Hence the deformed logical weighs at least $\min\{1,\beta\}$ times a logical of the original code. If $\beta \geq 1$, no logical saves weight by moving into the ancilla. The Z sector cannot use the ancilla at all: a Z logical reduces to an original Z logical times grid cycles.

The slab geometry makes β a constant, and this is where three dimensions matter. A slab representative lies in one plane and has $\Theta(N^2)$ qubits, hence $\Theta(N^2)$ ports. If the grid were only that plane, a straight line across it would wall off half of the ports with $\Theta(N)$ edges, so $\beta = O(1/N)$. If the grid is a box as thick as it is wide, a wall must be a surface. Either it separates roughly the same region in every layer of the box, and then it cuts through all the layers, or the separated region changes from layer to layer, and every change costs edges between layers. In both cases its area grows in proportion to the number of ports it separates (Supplemental Lemma SIII.8). If U occupies at most t parallel planes of a box with aspect ratio at most A, this gives

$$\beta(G,U) \geq \frac{1}{2At}. \tag{8}$$

Constant refinement assigns distinct local port labels to distinct grid vertices and increases t only by a constant. Unit-square cycles, bounded pairing paths, and bounded local density keep the check weight, diameter, and incidence constant. Thus one block or two blocks at fixed bounded cell separation admit local logical measurement using $O(n)$ ancillas and retaining distance at least $\beta_0 d$ for a fixed $\beta_0 > 0$. Section III B shows that $\beta_0 = 1$ is attainable. The proofs are in Supplemental Sec. SIII.

B. Keeping the distance throughout the computation

A logical measurement can lower the fault distance, the smallest number of faults that cause an undetected logical error, in three ways. The added qubits can carry a short logical operator; a few wrong measurement outcomes can flip the reported result; and a few faults can exploit the join between two consecutive operations. Figure 3 shows how the construction blocks each of them.

The first danger is real for thin interfaces. Consider a joint measurement $\overline{X}_a\overline{X}_b$ of two blocks. Once it has been measured, $\overline{X}_a\overline{X}_b$ is a stabilizer, so $\overline{X}_a$ and $\overline{X}_b$ act in the same way on the code. Take any wall δT with the ports of block a on one side and those of block b on the other. The product of the vertex stars in T is $\overline{X}_a$ times X on the wall, so $X(\delta T)$ alone is a logical operator equivalent to $\overline{X}_a$ [Fig. 3(a)], and

$$\widetilde{d}_X \leq \min\{|\delta T| :\ T\cap U = U_a\}. \tag{9}$$

For layer-code blocks of side $\Theta(N)$, the distance $d = \Theta(N^2)$ is comparable to the area of a block face. A measurement layer of bounded thickness h meets the plane between the blocks in a strip of $O(hN)$ edges, which is a wall, so it would reduce the distance of every joint measurement to $O(hN)$. Joint measurements cannot be avoided: they implement CNOT, phase teleportation, and resource transfer. Supplemental Table S6 confirms the loss on explicit codes: a single added sheet leaves distance at most L where the undeformed code has L^2. A complete circuit-level simulation on explicit layer-code blocks (Sec. VI and Fig. 5) finds the same effect in the fault distance of the measurement circuit, exactly at two block sizes and, with routed nearest-neighbor extraction, for three input codes, and in its failure rate at four sizes. Small hardware runs show it as well (Sec. VII).

Four ingredients keep the distance. First, the grid fills the participating blocks, so every wall between their ports is a surface with $\Omega(N^2)$ edges [Fig. 3(b)]; this is the constant relative expansion of Eq. (8), which the constant slab thickness makes possible. Second, carrying $m_0 = \lceil 2At \rceil$ parallel ancillas on every grid edge multiplies every wall by m_0, so $\beta \geq 1$ and Eq. (7) gives $\widetilde{d}_X \geq d_X$: each measurement keeps the full static distance at constant ancilla cost (Supplemental Corollary SIII.4). Third, the measurement is repeated in time [Fig. 3(c)]. A single wrong outcome of one vertex star flips the reported product, so the new checks are measured for $R = \Theta(d)$ rounds, and a wrong outcome that is not repeated is caught by

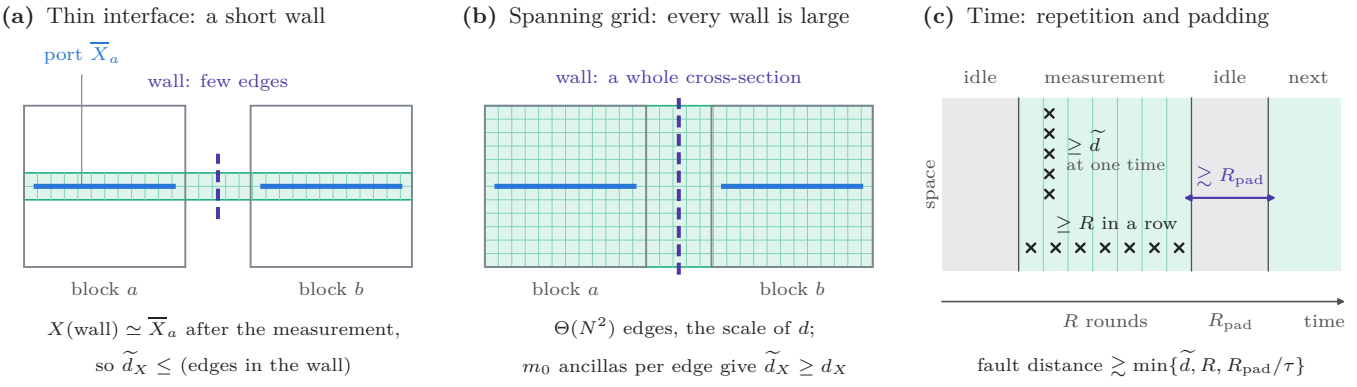


FIG. 3. Why the measurement keeps the fault distance, in a side view of two blocks (squares) with their X slab ports (blue) and the ancilla grid (green). (a) If the grid joins the blocks only through a thin bridge, X on a wall across the bridge (dashed) is equivalent to $\overline{X}_a$ once $\overline{X}_a\overline{X}_b$ is measured, so a few faults cause a logical error. (b) If the grid fills both blocks, every wall between their ports is a whole cross-section, with $\Theta(N^2)$ edges, the same scale as the code distance; m_0 ancillas per edge make every wall at least as heavy as the ports it separates, and the full distance is kept. (c) In time, an undetected error inside a measurement is either a logical at one time step, with at least $\widetilde{d}$ faults, or one outcome wrong in each of the R repeated rounds. Idle rounds between operations force any pattern that links two of them to stretch across the padding.

comparing consecutive rounds. To flip the result undetected, faults must corrupt the same outcome in every one of the R rounds. Williamson and Yoder proved the resulting fault-distance bound for gauging [2]. Because our operations start and end in open idle rounds rather than at perfect boundaries, we give a short direct proof (Supplemental Proposition SV.17). Split the faults by type. Faults of the measured type (X) leave, at the end, an error on the data and a set of flipped edge readouts; together these form one operator of the deformed code, which is a nontrivial logical whenever they cause an error. Every qubit of it carries at least one fault, so there are at least $\widetilde{d}_X$ faults. Faults of the other type either leave a logical error on the data, again with at least d_Z faults, or flip the reported outcome. In that case either every one of the R rounds contains a wrong outcome, or a round without one shows that the errors accumulated so far form a logical operator that anticommutes with the measured one, with at least d_Z faults. Hence the measurement has fault distance at least $\min\{\widetilde{d}_X, d_Z, R\}$, and exact minimization on small instances shows that this bound is attained. Fourth, successive measurements on a block are separated by $\Theta(d)$ idle rounds, so faults that straddle two operations also cost $\Omega(d)$ (Sec. IV).

Resource transfer is the one place where blocks of different distance meet. Joining a small block to a large one cannot expose the large block: the same wall argument, applied to everything except the small block's own logical qubits, shows that every other logical operator keeps the distance of the larger block (Supplemental Lemma SIII.6). The small courier blocks below therefore carry only one traveling resource each, and their errors are charged to distillation. Supplemental Table S9 lists the resulting bound for every operation: each logical qubit that is read later, other than a resource in transit, is protected at every step by fault distance $\Omega(d)$.

C. Gates and encoded resources

Pure-X/Z measurements suffice for the logical gate set. A CNOT uses an ancilla a in $|+\rangle$ and the measurements

$$Z_cZ_a\ (u), \qquad X_aX_t\ (v), \qquad Z_a\ (w), \tag{10}$$

followed by the Pauli-frame update $Z_c^vX_t^{u\oplus w}$. Outcome bit zero denotes eigenvalue $+1$.

For phase teleportation, prepare a in $R_\theta|+\rangle$, where $R_\theta = \mathrm{diag}(1, e^{i\theta})$, measure Z_dZ_a with bit m and X_d with bit n, and retain a. The correction

$$C_{mn} = R_{2\theta}^mZ_a^nX_a^m \tag{11}$$

produces $R_\theta|\psi\rangle$. For $\theta = \pi/2$ this consumes $|Y\rangle = S|+\rangle$; for $\theta = \pi/4$ it consumes $|A\rangle = T|+\rangle$, with a possible S correction using one additional $|Y\rangle$. Exchanging X and Z gives $S_X = HSH$, and $SS_XS = e^{i\pi/4}H$. These identities generate Clifford+T using the one- and two-block measurements above. Complete branch maps and resource counts appear in Supplemental Sec. SIV.

A physical resource enters at a designated bare port. The raw qubit is teleported into an encoded block by a joint ZZ measurement and an X readout. Its weight-one error class is the intentional resource degree of freedom; all spectator logical classes retain distance $\Omega(d)$ in the quotient by that class. Subsequent transfers use only encoded blocks, with distances growing by fixed factors. We call this sequence an encoded courier. Each courier block is dedicated to one traveling resource; its other logical qubits are prepared in fixed states and never read.

The resource supply uses fixed Steane $7\to1$ and punctured Reed–Muller $15\to1$ factories, implemented by the same logical primitives [25–27]. Resource errors are modeled as local-stochastic defective slots: a defective candidate may be in any state, entangled with other candidates or with the environment. Phase teleportation

with any such resource applies, relative to the ideal rotation, an operator in span$\{I, Z\}$ on the coordinate it feeds. A predetermined recovery therefore returns exactly the ideal output whenever at most one coordinate of a run is defective, so each run always produces an output, and the outputs are again a defective family with

$$q_{r+1} \le C_{\mathrm{dist}} q_r^2. \tag{12}$$

The constant counts pairs among the fifteen principal and fifteen correction slots of a run, $C_{\mathrm{dist}} = \binom{30}{2} = 435$. Disjoint runs use disjoint candidate sets; their errors need not be independent, and no twirl or diagonal error basis is assumed. The raw boundary, encoded transfers, and factories are included in the circuit analysis rather than supplied as ideal external resources.

IV. FAULT TOLERANCE AND RESOURCE SCALING

A. From local distance to a complete circuit

We consider local-stochastic Pauli circuit noise: any specified set F_0 of fault mechanisms at distinct locations occurs with probability at most $p^{|F_0|}$. A round of check measurements has constant depth and uses only nearest-neighbor gates on a finite ancilla schedule. Each circuit fault acts like at most $c_1 = O(1)$ elementary faults of the simpler phenomenological model, in which a fault is a data error or a wrong measurement outcome. We count faults in that model and divide by c_1 at the end.

The argument that the whole computation keeps fault distance $\Omega(d)$ has a simple shape. An undetected logical error needs a fault pattern that no detector notices, that is, no comparison of consecutive outcomes and no entry, exit, preparation, or readout check. Split such a pattern into connected pieces, where two faults are connected when they can trigger a common detector; at least one piece causes the error. A piece that lies inside one operation and the idle rounds around it needs at least $\min\{d, R\}$ faults, by Sec. III B. Successive operations on a block are separated by R_{pad} idle rounds. Two connected faults are at most $\tau = O(1)$ rounds apart, so a piece that reaches from one operation to the next must stretch across the padding and needs about R_{pad}/τ faults. Precisely, with δ_g^{rel} the fault distance of measurement window g and $\delta_{\mathrm{mem}}^{\mathrm{rel}}$ that of idle memory, both defined below, the composed phenomenological distance satisfies

$$f_{\mathrm{ph}} \ge \min\left\{ \min_g \delta_g^{\mathrm{rel}}, \delta_{\mathrm{mem}}^{\mathrm{rel}}, 1 + \left\lfloor \frac{R_{\mathrm{pad}} - \tau}{\tau} \right\rfloor \right\}, \tag{13}$$

with $\delta_g^{\mathrm{rel}} \ge \min\{\beta_0 d, R_g\}$ for a window of R_g rounds and $\delta_{\mathrm{mem}}^{\mathrm{rel}} \ge d$. Choosing R_g and R_{pad} of order d makes every term $\Omega(d)$, and the bounded spread c_1 converts this into a circuit-distance bound.

The technical work lies in making "inside one operation" precise without assuming that each operation starts from a perfect state, since earlier faults leave their traces. We cut the schedule in time at idle rounds and carry three pieces of data across each cut: the syndrome s left by earlier errors and the logical Pauli correction λ accumulated so far, which together describe the quantum state, and the outcomes ρ of the last measurement round, which the next detectors compare against. The records must be carried separately because a wrong outcome in the last round before a cut changes ρ but not the state. With these three variables, adjacent operations compose exactly; a direct simulation of an idle → gauging → idle → gauging → idle schedule on explicit layer-code blocks agrees with the composed map on every single phenomenological fault mechanism, and gluing without ρ fails. Each piece of a fault pattern is then judged by its effect *relative* to the same schedule without it: subtracting the fault-free evolution of the same incoming data removes every dependence on earlier errors. This includes a less obvious one: when the logical representatives at two cuts differ by stabilizers, the propagated frame acquires a term Cs_{in} linear in the incoming syndrome. A piece that lies between two cuts placed in idle padding, with one round free of it before the right cut, leaves no syndrome and no records at either cut, so its relative effect is its true effect and the window bounds above apply to it (Supplemental Lemma SV.14).

Resource errors need one more step, because the courier blocks have smaller distance. An error that a resource already carries is passed along with the other incoming data. A new error first created at encoded courier stage j needs a connected fault pattern of size $\Omega(d_j)$ when that stage's measurement and padding times are $\Theta(d_j)$; the raw injection stage is charged separately, with a constant. Distinct resources are kept far apart in spacetime, so one short connected pattern cannot corrupt two of them. The reason is again the padding: a chain of connected faults can leave the courier blocks of one resource only through its transfer into the factory block, and from there it must stay on that block for the R idle rounds that separate the transfer from every other operation on it, whatever cells it visits afterwards (Supplemental Lemma SV.25). A script builds a complete factory schedule by these rules, with cell reuse and adaptive branches, and checks them and the resulting separation on it. Summing over the stages gives a raw-to-encoded resource error parameter $q_0 = O(\sqrt{p})$ uniformly in the terminal distance, and Eq. (12) then suppresses the resource error quadratically.

Every detector has bounded support, so the graph that connects faults through common detectors has bounded degree, uniformly over the finitely many branches at adaptive steps. The committed decoder chooses a minimum-cardinality fault set consistent with the detector history through a fixed buffer beyond the commitment time. A wrong decision about a protected outcome requires a large connected set of fault locations, a fixed fraction of which are actual faults. Counting such connected sets in a bounded-degree graph therefore yields a positive threshold. This is an information-theoretic decoder; efficient classical decoding is not assumed. Supplemental Sec. SV

gives the local detector, relative-distance, resource, and cluster arguments in full.

B. Architecture theorem

The input family is a balanced good CSS LDPC family with boundedly dense available sizes: every sufficiently large target length has an available member within a fixed multiplicative factor. Fixed-degree Quantum Tanner families provide the existential input used here [28, 29]. Let $A \subset \mathbb{Z}^3$ be the finite allocated logical region, $W = |A|$, and T the logical circuit depth. Allocated working sites are included in W; physical gate ancillas and live factory resources are included in the bound below.

To state the geometric overhead, partition the logical lattice into coarse boxes of side $b(f) = \Theta(f^{1/6})$. Define $B_f(A)$ as the number of boxes that meet A but are not completely filled. This counts partial occupancy at the block-packing scale, including boundaries, holes, and fragmentation.

Theorem IV.1 (Universal fault-tolerant architecture)**.** *Use the explicit layer-code realization of Theorem II.1, a balanced good input family with boundedly dense sizes, and the local measurement, resource, and extraction schedules described above. There are constants $p_{\mathrm{th}}, c, f_{\min} > 0$ such that for $0 < p < p_{\mathrm{th}}$ and target failure $0 < \varepsilon < 1/2$, any finite nearest-neighbor three-dimensional logical circuit can be simulated with*

$$f = \max\left\{ f_{\min}, \left\lceil \frac{c \log(WT/\varepsilon)}{\log(p_{\mathrm{th}}/p)} \right\rceil \right\}, \tag{14}$$

using

$$Q_{\mathrm{phys}} = O\Big(Wf + (B_f(A) + 1) f^{3/2} \Big), \tag{15}$$

$$D_{\mathrm{phys}} = O(T f^{3/2} \Psi), \qquad \Psi = O(f^{\log_2 15}). \tag{16}$$

Persistent data remain in layer-code blocks, and physical operations use nearest-neighbor connectivity. For a connected bounded-aspect logical box,

$$Q_{\mathrm{phys}} = O(Wf + f^{3/2}). \tag{17}$$

The scaling follows directly from the code parameters. Choose $d = \Theta(f)$ and measurement, padding, and commitment buffers of order f. Equation (1) then gives

$$N = \Theta(\sqrt{f}), \qquad K = \Theta(\sqrt{f}), \qquad n = \Theta(f^{3/2}). \tag{18}$$

A full coarse logical box fits into one cell. Counting full and partial boxes requires $O(W/K + B_f(A) + 1)$ data/factory cells. Multiplication by n proves Eq. (15). Neighboring logical sites map to the same cell or adjacent cells, and a bounded coloring schedules inter-cell gates.

For a bounded-aspect logical box, $B_f(A) = O((W/\sqrt{f})^{2/3} + 1)$, so the partial-cell term is absorbed into Eq. (17). Figure 4 shows the crossover at $W \sim \sqrt{f}$. Thin or fragmented regions retain the general correction.

For time, a distilled resource takes $O(f\Psi)$ rounds, where ℓ distillation levels give $\Psi = O(15^\ell)$ and $2^\ell = O(f)$. One logical layer requires $O(K)$ resources per data block, and serial production gives $O(Kf\Psi) = O(f^{3/2}\Psi)$ rounds per layer, that is $O(f^{3/2+\log_2 15}) \subseteq O(f^{5.41})$. This serial schedule is chosen to keep the space bound; parallel factories would trade space for depth. A depth-first factory schedule stores $O(\ell)$ candidates at once, which fits within the reserved logical capacity. The complete location count is polynomial in f times $WT\Psi$; the exponential cluster bound and final-resource union bound are both at most $\varepsilon/2$ for the choice in Eq. (14). The detailed theorem and constant ledger appear in Supplemental Theorem SV.31.

The depth deserves as much attention as the space. Per logical layer the minimal-space schedule takes $O(Kf\Psi) = O(f^{3/2+\log_2 15})$ rounds, against $O(f)$ for lattice surgery. With P factories per cell the space becomes $O(Wf P + f^{3/2})$ and the depth $O(TKf\Psi/P)$, so qubit-time is unchanged: the serial schedule minimizes space, not space–time volume. A concrete estimate with stated constants, explicit integer allocations, and distances chosen from each schedule's complete runtime (Supplemental Sec. SXVI) makes the scales explicit. For $W = 100$ sites, $T = 10^6$ layers, $\varepsilon = 10^{-2}$, $p = 10^{-3}$, and injected resource error 10^{-2}, the always-output distillation covered by the theorem does not converge: Eq. (12) requires an injected error below $1/C_{\mathrm{dist}} \approx 2.3 \times 10^{-3}$, and even the ideal factory, whose error map starts at $105q_c^2$ with the per-coordinate error q_c including the S corrections, requires about 5×10^{-3}. With optimistic postselected distillation, which the theorem does not cover, the serial layer-code schedule needs about 10^9 qubits for two months, against 4×10^5 qubits for three days for a serial surface-code layout and 2×10^8 qubits for 23 s for a throughput-limited one. At these distances a layer-code block holds one to three logical qubits; with the same accounting on both sides it stores a logical more cheaply than a surface-code patch only for $d \gtrsim 1100$ (between about 600 and 2300 across the distance scenarios). The asymptotic advantage is therefore far from the practical regime until these constants, and the distillation protocol, improve.

The same architecture argument applies to any three-dimensional CSS family with the parameter scaling in Eq. (1), bounded local density and aspect ratio, boundedly dense sizes, and uniformly slab-supported pure logicals; the explicit realization supplies all of these. These are substantive geometric assumptions; Supplemental Corollary SV.32 states the transfer precisely.

V. PORT PACKING AND SHALLOW LOGICAL GATES

Slab mobility also constrains how logical information can be spatially accessible. For a region A, let $g_X(A)$ be the dimension of the pure-X logical space supported in A, and define $g_Z(B)$ analogously. The rank $r(A, B)$ of

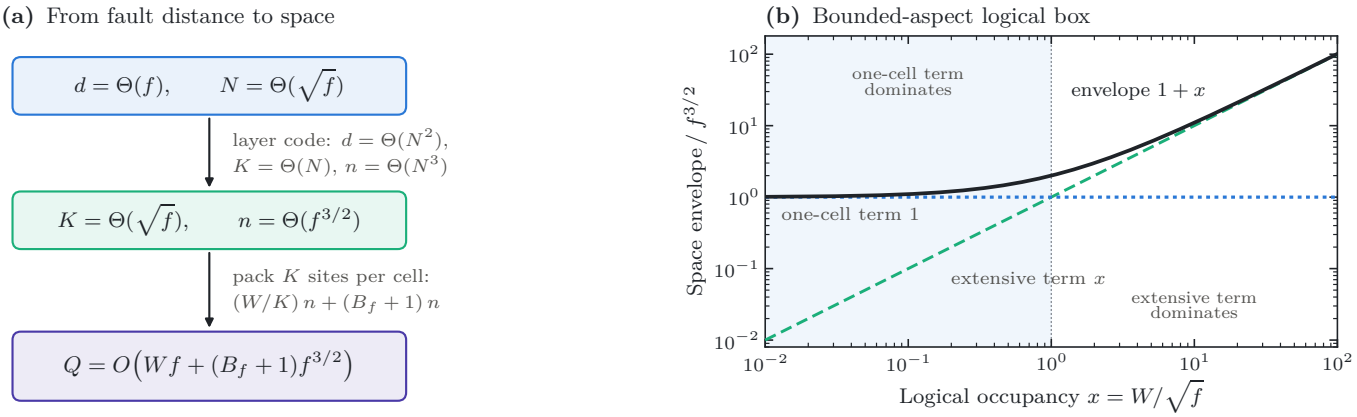


FIG. 4. Origin and crossover of the resource bound. (a) Target fault distance f fixes the cell parameters; packing $K = \Theta(\sqrt{f})$ logical sites per cell yields an extensive Wf term plus the cost of partially occupied cells. (b) For bounded-aspect boxes, the normalized scaling envelope is proportional to $1 + x$, where $x = W/\sqrt{f}$. The one-cell term dominates for $x \ll 1$ and the extensive term for $x \gg 1$. Constants are suppressed: this is an illustration of the asymptotic bound, not a numerical qubit estimate.

their commutation pairing obeys

$$\max\{0, g_X(A) + g_Z(B) - K\} \le r(A, B) \le |A \cap B|. \quad (19)$$

The upper bound follows because all cross commutation factors through the intersection; the lower bound follows from nondegeneracy of the full logical X/Z pairing.

If $A_1, \ldots, A_{r_X}$ are disjoint complete X ports and $B_1, \ldots, B_{r_Z}$ disjoint complete Z ports, summing Eq. (19) over all pairs gives the crossbar bound

$$r_X r_Z K \le n. \quad (20)$$

Every crossing carries rank K, and the crossings are pairwise disjoint. For the balanced layer-code slabs, $r_X, r_Z = \Theta(N)$, $K = \Theta(N)$, and $n = \Theta(N^3)$, so the bound is saturated up to constants. Each transverse intersection has size $\Theta(K)$, while each complete slab has size $\Theta(d)$. Thus the cubic physical size is optimal among CSS families required to provide $\Theta(K)$ disjoint complete ports in both polarizations. This is a bound on simultaneous spatial access, distinct from a storage lower bound.

There is a complementary constraint on unitary processing. Pair the ith X-slab representative and the ith Z-slab representative of any logical Pauli. Each qubit belongs to at most two resulting representatives, giving disjointness $\Delta = \Omega(N)$. The logical-weight bounds sharpen this to

$$\Delta = \Theta(N), \qquad d_\downarrow = \Theta(N^2), \qquad d_\uparrow = \Theta(N^2), \quad (21)$$

where $d_\downarrow$ and $d_\uparrow$ are the smallest and largest minimum weights among nontrivial logical Pauli classes. The Jochym-O'Connor–Kubica–Yoder criterion then forces every fixed-depth, bounded-arity, code-preserving unitary on a sufficiently large static block to be logical Clifford [30]. The result holds even without geometric locality of those gates. The architecture uses adaptive measurements, resource states, and changing codes, which lie outside that unitary setting. Complete packing and disjointness proofs are in Supplemental Secs. SI C and SI D.

VI. FINITE NUMERICAL EVIDENCE

The supplied simulations test finite interfaces in the construction. Three campaigns run complete circuits on explicit layer-code blocks (Fig. 5); the largest campaign, which shows a clean crossing, uses surface codes. That campaign follows a single encoded planar surface code of distance L, with $n = L^2 + (L-1)^2$ data qubits, through an algebraic base-code → gauged-strip → base-code circuit, without resetting the data between stages. The gauged strip is a path of L ancilla qubits attached to a weight-L logical representative, so the campaign tests the gauging interface and the detector bookkeeping of a code switch rather than the layer-code geometry. Its entry and exit satisfy the binary stabilizer transition identities. Noisy repeated extraction produces one detector history, decoded by minimum-weight perfect matching from exact graphlike detector error models.

The SI1000-like model uses two-qubit depolarizing strength p, single-qubit and ordinary idle strength $p/10$, measurement flips $5p$, reset noise $2p$, and data idle-wait noise $2p$. The archived campaign comprises 255 parameter points and 4.534×10^6 shots for sizes $L = 3, 5, 7, 9, 11$ and gauged durations $G = d, 2d, 3d$, where $d = L$ is the code distance. All 15 noiseless controls have zero failures.

For every duration, the failure rate decreases monotonically with L across all five sizes at every tested $p \le 0.004$ (Supplemental Fig. S14). Consecutive-size Wilson intervals are separated at every tested p from 0.0025 to 0.00375 for all three durations. At $p = 0.0035$, increasing L from 3 to 11 reduces the failure rate from 0.1111 to 0.0324 for $G = d$, and from 0.1326 to 0.0428 for $G = 3d$. At $p = 0.0015$ and $G = d$, the reduction is approximately 6.5×10^2.

The smallest-versus-largest crossing medians are 4.471, 4.431, and 4.328×10^{-3} for $G = d, 2d, 3d$, with conditional 95% posterior bands $[4.426, 4.522]$, $[4.392, 4.466]$,

and $[4.265, 4.393] \times 10^{-3}$. Tripling the gauged duration shifts the median by about 3.2%. These are finite-size diagnostics of a surface-code gauging model; they do not estimate the architecture theorem's asymptotic threshold, which concerns layer codes and a committed minimum-weight decoder.

A complete joint measurement on explicit layer-code blocks tests what the surface-code campaigns cannot: the reported bit, both logical sectors, spectator logicals, and an entangled input. Two $[[4, 2, 2]]$-input blocks ($n = 224$, two logical qubits each, distances 5 and 4) measure $\overline{X}_1^{(a)}\overline{X}_1^{(b)}$ by gauging on a planar grid attached to their single-layer slabs, with ancilla-based extraction of every check and uniform circuit noise. Each logical qubit starts maximally entangled with a noiseless reference, and eight observables certify the output Choi state. Without noise every detector and observable is deterministic, so the circuit implements the projective measurement with the correct outcome and leaves the spectators untouched. We then compute, exactly, the minimum number of faults that flips each observable undetectably. With a bridge of $w = 1, 2, 3$ grid rows between the blocks, the fault distance of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ is exactly w; for $w = 1$ a single fault suffices [Fig. 5(b)]. This is the cut obstruction of Sec. III B at circuit level. A grid spanning both blocks restores it to exactly 5, the blocks' X distance, while the fault distances of the reported bit, set by the number of rounds, and of the spectators do not depend on the interface (Supplemental Sec. SXIV).

A second campaign repeats the measurement on blocks of spacing $s = 2$ to 5 (block distances up to $d_X = 11$, up to 1400 qubits per block), with $2s + 1$ rounds per phase and 3.0×10^6 shots, decoded in each CSS sector separately (Supplemental Sec. SXIV A); on identical shots the full-model BP+LSD decoder of the first experiment (Supplemental Table S17) fails less often through the grid (half as often in the X experiment) and about as often through the bridge, so the contrast below is, if anything, understated. At $p = 10^{-3}$ the failure rate of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ through the one-row bridge rises from 3.9% at $s = 2$ to 8.8% at $s = 5$, because the undetectable bridge fault has more rounds in which to occur, while through the spanning grid it falls from 8.2×10^{-4} to about 10^{-4} and agrees with the same blocks left idle [Fig. 5(a)]. For the spanning grid each unit of spacing suppresses failures by a factor 4.8 in the X experiment and 2.6 in the Z experiment at $p = 10^{-3}$, falling to one near $p \approx 6 \times 10^{-3}$ [Fig. 5(c) and Supplemental Fig. S16]. At $s = 3$ the fault distance of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ through a bridge of $w = 1, 2, 3$ rows is still exactly w, although the blocks' X distance is now 7.

A third campaign replaces the colocated ancillas by the routed nearest-neighbor schedule of Supplemental Proposition SV.8, with SWAP-shuttled ancillas, colored conflict classes, and idle noise in every time step, and grows the input at $s = 2$: $[[4, 2, 2]]$, a six-qubit band code, and the Steane code, giving blocks of 224, 720 and 1792 qubits and circuits of up to 16,293 qubits (Supplemental Sec. SXIV B). Without noise all 28 measurement circuits implement the measurement exactly, and the 12 idle controls also give silent detectors and correct observables. Through the one-row bridge the fault distance of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ is exactly 1 for all three inputs, although the blocks' X distance grows from 5 to at least 18 [Fig. 5(b)]. On the $[[4, 2, 2]]$ blocks with $R = 5$ rounds per phase the spanning grid gives exactly 5, the same as the blocks left idle; through both interfaces the reported bit has exactly $5 = R$, and the spectator X frames have exactly $4 = d_Z$, as in idle memory. Where both are resolved, routing leaves the distances of the colocated circuit unchanged. On the band blocks the reported bit has an undetectable pattern of weight 6 at $R = 7$. The block itself explains this weight: the lightest Z-type operator that anticommutes with the measured logical has weight 6, so no schedule gives the bit more than $\min\{R, 6\}$; whether routing adds lighter patterns is open. At the tested noise strengths the sampled failure rates of these long routed circuits do not distinguish the interfaces, so we use only their exact fault distances.

Complementary memory, local-gauging, factory, and stage-composed tests are reported in Supplemental Sec. SXIII. The deterministic checks cover the complete-code slab certificate, refinement identities, gate maps, 10,008 single-location faults in the finite extraction catalog, and the thin-versus-thick interface comparison of Sec. III B. These checks support the individual interfaces and remain separate from the arbitrary-size proofs.

VII. HARDWARE DEMONSTRATIONS

Three mechanisms of Sec. III B are small enough to run on present superconducting processors: the cut obstruction, its removal by parallel ancillas, and the protection of the reported bit by repeated rounds. We ran them on the 54-qubit IQM Emerald processor through the Open Quantum platform [31] and on the 156-qubit IBM Heron processors `ibm_fez` and `ibm_kingston`, with 6.9 million analyzed samples. The top row of Fig. 6 shows what was run and the bottom row what was measured. The blocks are three-qubit repetition codes, not layer codes, so these runs test the mechanisms and not the architecture. Apart from the routed `ibm_fez` circuits, every circuit was placed without SWAP gates [Fig. 6(a) and Supplemental Fig. S18], and before each submission the exact circuit fault distance of the circuit as sent was computed. A script computes every number below from the archived raw outputs (Supplemental Sec. SXV).

a. Cut obstruction. Two blocks with Z checks Z_0Z_1, Z_1Z_2 and $\overline{X} = X_0X_1X_2$ start in $|000\rangle$ and $\overline{X}_a\overline{X}_b$ is measured by gauging on a graph whose bridge between the blocks has w edges, so that X on the bridge is a logical of weight w [Eq. (9)]; the deformed $\overline{Z}_a\overline{Z}_b$ must stay $+1$. The postselected error falls 35-fold, from 2.4% at $w = 1$ to 0.070% at $w = 2$, on IQM (pooled over three

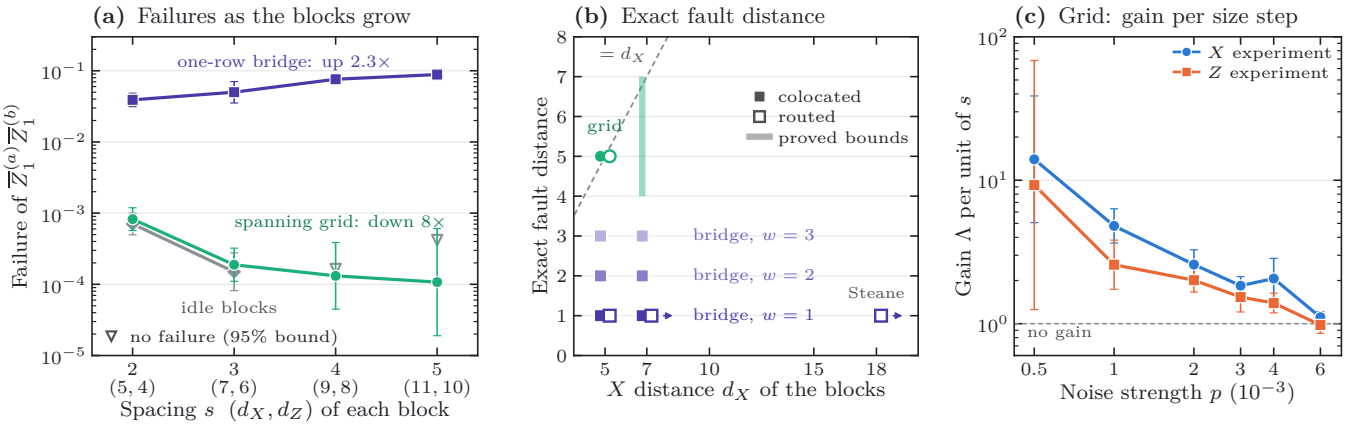


FIG. 5. Two-block logical measurement on explicit layer-code blocks: the cut obstruction at scale, in simulation (the hardware counterpart is Fig. 6). (a) Failure of the measured pair $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ at $p = 10^{-3}$ as the block spacing s grows, through a one-row bridge (violet) and a spanning grid (aqua), and for the same blocks left idle (grey), with Wilson 95% intervals (open triangles: Wilson upper ends of zero counts). (b) Exact circuit fault distance of the measured pair against the X distance of the blocks, computed by CP-SAT. Filled: colocated ancillas at $s = 2, 3$; open: routed nearest-neighbor extraction at $s = 2$ with the $[[4, 2, 2]]$, band and Steane inputs, whose blocks have X distance 5, at least 7 and at least 18 (arrows); bar: the proved lower bound and the weight of an undetectable pattern where the solver stopped. Through a bridge of w rows the distance stays at w while d_X grows; through the grid it equals d_X (dashed) where it is resolved. (c) Factor Λ by which each unit of spacing suppresses failures of any observable through the spanning grid, from a weighted fit over $s = 2$ to 5, with 95% intervals.

runs on two sets of qubits; $p \leq 9.5 \times 10^{-19}$ in each of the two runs with both widths), and from 5.8% to 0.59% and 0.14% at $w = 1, 2, 3$ on IBM, by factors of 9.7 and 4.3 [Fig. 6(a,d)]. When the cycle checks that the extra bridge edges create are ignored, the error returns to the width-one level (2.2 to 3.3% on IQM, 7.2 to 7.9% on IBM), so the gain comes from the larger wall and not from discarding more samples. On IQM, width three had no failure in 11,529 kept samples, but its difference from width two is not significant ($p = 0.071$, in a test fixed before the run).

b. Parallel ancillas. Two parallel ancillas on a single bridge edge ($m_0 = 2$) give 0.53% instead of 2.4%, a 4.6-fold gain ($p = 5.1 \times 10^{-10}$), and 2.3% when the new two-edge check is ignored [Fig. 6(b,e)]. The first CNOT order we tried had circuit fault distance one: a fault on the vertex ancilla between its two bridge CNOTs put X on both parallel edges, which the two-edge check cannot see. Separating the two CNOTs restores distance two. This is the constant-factor loss from circuit faults that Sec. IV allows for; a suitable CNOT order avoids it.

c. Repeated rounds and errors that survive a reset. For the reported bit we use phase-flip blocks ($d_Z = 3$) joined by one edge and measure the two vertex stars and the four block checks in R rounds [Fig. 6(c)]. With fresh ancillas in each round (IQM, which does not allow a measured qubit to be reused), the error falls 20-fold, from 6.2% at $R = 1$ to 0.31% at $R = 2$, close to the model's 6.9% and 0.34% ($p = 2.4 \times 10^{-188}$, fixed in advance). On IBM, where the ancillas are measured and reset between rounds, it falls from 4.6% to 1.1% and 0.59% for $R = 1, 2, 3$ (both steps $p < 10^{-300}$, fixed in advance), but far less than the calibrated model predicts (0.18% and 0.0086% at $R = 2, 3$); at $R = 3$ the error is 68 times the prediction [Fig. 6(f)]. The cause is an ancilla error that survives the reset: a check ancilla reads 1 in two consecutive rounds, with no data error to explain it, in 1.15% of samples, 23 times the rate that independent errors give, and a second reset removes only 3% of these events ($p = 8.8 \times 10^{-6}$) [Fig. 6(g)]. Such a fault, most likely leakage, corrupts the same outcome in every round. It lies outside the local-stochastic model of Sec. IV, which charges every round separately.

d. What the runs show. Together, the runs reproduce at small scale the three facts that the distance argument of Sec. III B combines: the smallest wall bounds the distance of a joint measurement (as Fig. 5 shows in simulation on layer-code blocks), parallel ancillas widen every wall that crosses their edge, and repeated rounds protect the reported bit. They also show which assumption of the theorem present hardware does not meet: faults that persist across rounds. They do not test layer codes, the grid expansion, the threshold, or the resource scaling, which need far larger devices.

VIII. DISCUSSION

The main architectural contribution is a local route from layer-code storage to universal processing. Prefix-parity repair exposes logical operators in single-layer slabs of an explicit realization; grid gauging turns those slabs

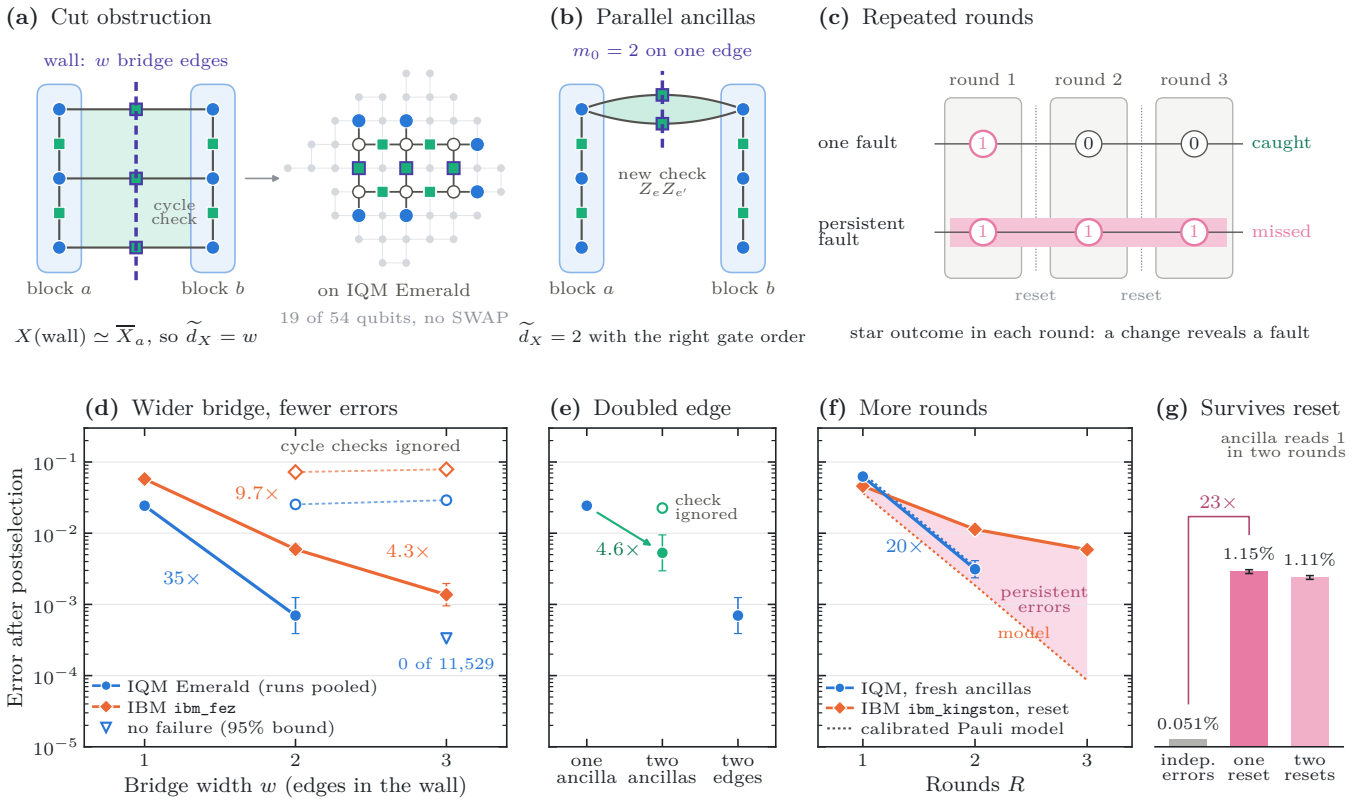


FIG. 6. Hardware demonstrations of the three distance mechanisms. Top row: what was run (schematics). (a) Two three-qubit blocks (blue) joined by a bridge of w edges, each with an edge ancilla (aqua squares); X on the wall of bridge edges (violet) is equivalent to $\overline{X}_a$, so the deformed X distance and the exact circuit fault distance are both w. Shaded: the cycle checks that the extra bridge edges create. Right: the circuit for $w = 3$ as placed on IQM Emerald, with the couplers it uses. (b) Two parallel ancillas on one bridge edge ($m_0 = 2$) add the two-edge check $Z_e Z_{e'}$. (c) A wrong star outcome in one round is revealed by the change in the next round; a fault that persists across rounds repeats the same wrong outcome and is missed. Bottom row: what was measured, as the error after postselection on every detector, with Wilson 95% intervals (open triangle: Wilson upper end of a zero count). (d) IQM runs pooled; open symbols: the same samples with the cycle checks ignored. (e) IQM, one bridge edge with one or two ancillas, and two bridge edges; open symbol: the two-edge check ignored. (f) Reported-bit error against the number of rounds R, with the calibrated Pauli models (dotted); IQM uses fresh ancillas in every round, `ibm_kingston` measures and resets them. Shaded: the excess over the model (run A). (g) Rate at which a check ancilla on `ibm_kingston` (run B) reads 1 in two consecutive rounds with no data error to explain it, against the rate that independent errors give, with one or two resets between rounds. Every run separately: Supplemental Fig. S17.

into protected measurements; and a composed detector schedule supports resource transfer and distillation in the same geometry. Because a layer-code distance is comparable to the area of a block face, every joint measurement must cross a constant fraction of that face; a thick grid does so and, with a constant ancilla multiplicity, keeps the full distance, whereas a thin measurement layer cannot. Simulations on layer-code blocks and small hardware runs show this contrast (Figs. 5 and 6). For bounded-aspect logical regions, the resulting space bound separates an extensive Wf cost from the minimum $f^{3/2}$ cost of one encoded cell.

The port-packing result gives a second interpretation of the layer-code parameters. Their cubic physical volume supports linearly many disjoint complete ports of each polarization, saturating the crossbar bound. Growing mobility also explains why adaptive measurement and resource injection are natural tools: shallow unitaries preserving a static large block are restricted to the Clifford group.

The next steps are a practical decoder for deformation histories, with changing checks, inherited frames and records, and joint measurements (the layer-code decoders of Ref. [32] are a natural starting point, although their memory guarantees do not yet cover code switching); routed circuit-level benchmarks on inputs that grow into the balanced regime, with a shorter routed schedule and a decoder close to minimum weight; bounds on how many logical measurements can share the slab interfaces, compared with parallel code surgery [3] under three-dimensional locality; and factories whose constants make the space–time estimate competitive, possibly in a

different code with a protected local transfer. Random layer codes built from random CSS inputs [32] are a natural candidate for reducing those constants; the explicit realization applies to any CSS input, but the resource accounting would need their parameter scaling. The present theorem uses minimum-weight committed decoding and local-stochastic Pauli noise. Extensions to coherent or leakage noise require an additional fault-path analysis; the IBM runs of Sec. VII show that ancilla errors that persist across rounds already limit the benefit of repeated rounds on present hardware, so leakage reduction would belong in any implementation of the extraction schedule. Deterministic selection of the input component codes is a separate constructive problem. The storage benchmark in Supplemental Sec. SVI makes the restricted regime for comparison with an extensive lower bound explicit, including the condition $W = \Omega(\sqrt{f})$ when the one-cell term is to be subleading.

DATA AVAILABILITY

The simulation datasets, finite algebraic certificates, verification scripts, and figure-generation code are included in the accompanying source and reproducibility package. The Supplemental Material describes the noise models, decoders, stopping rules, and scope of each dataset. The hardware circuits, job identifiers, raw outcomes, calibration snapshots, and analysis scripts are in the folder `hardware` of the package.

ACKNOWLEDGMENTS

The author thanks Dominic J. Williamson for helpful correspondence. Williamson pointed out that adding logical-measurement layers to a layer code can, in principle, reduce its fault distance, noted that this question had been left open in an earlier idea of performing universal computation on layer codes by adding such layers, and suggested that random layer codes [32] may be useful here. Section III B and the corresponding supplemental results (the cut obstruction, the edge-multiplicity corollary, the encoded-port quotient lemma, and the distance ledger) were added in response. He later suggested that the junction construction and the fault-distance argument be explained in plainer terms, and Secs. II, III, and IV A were rewritten for that purpose. The correspondence did not include a detailed review of the proofs, and any errors are the author's own. The IQM experiments used the Open Quantum platform [31]. We acknowledge the use of IBM Quantum services for this work. The views expressed are those of the author, and do not reflect the official policy or position of IBM or the IBM Quantum team.

ACKNOWLEDGMENT OF AI ASSISTANCE

Claude Code assisted with manuscript organization, language editing, figure preparation, and scripts from the supplied research materials and numerical data.

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Supplemental Material for "Universal Fault-Tolerant Computation on Three-Dimensional Layer Codes at Optimal Distance"

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This supplement contains the complete geometric, algebraic, and fault-tolerance arguments supporting the companion manuscript, followed by the detector and resource catalogs and the extended numerical evidence. It gives an explicit layer-code realization with complete check matrices and its slab and distance theorems (Secs. SI–SII), local grid gauging (Sec. SIII), and the fault-tolerance argument (Sec. SV), whose boundary interface carries the inherited syndrome, the last reported check records, and the logical frame across every cut, and whose resource layer allows defective candidates in arbitrary, possibly entangled states. It then reports a complete circuit-level two-block measurement on explicit layer-code blocks (Sec. SXIV) and a concrete space–time resource estimate with an explicit integer allocation (Sec. SXVI). The changes made in each revision, and the certificates they superseded, are collected in the archive of Sec. SXX. The supplied source package contains the finite certificates, simulation datasets, and figure code. Equations carrying A, R, S, J, or T tags retain their original technical labels; other proof equations use P labels. All section, theorem, figure, and table references in this document are internal unless stated otherwise.

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SI. SLAB CONSTRUCTION, EXPLICIT REALIZATION, AND STRUCTURAL CONSEQUENCES

a. Overview. The slab theorem is stated and proved for an explicit local realization of the Williamson–Baspin (WB) layer code [1] with complete check matrices, defined below. It carries the WB line-junction condensation data (Sec. SI B), encodes exactly the input logical qubits for every input (Proposition SI.2), has single-layer slab representatives at every height (Theorem SI.3), and has distance $\Theta(N^2)$ for balanced good inputs (Theorem SI.6). Figure S1 shows the junction and wall geometry, and Table S1 lists the rules. An earlier single-chart transcription of the WB cells, and why it does not encode the input logical qubits, is recorded in Sec. SXX.

b. The idea in plain terms. Every sheet is a surface code, so an X string that crosses a sheet from side to side commutes with its checks, and a string that stops inside a sheet leaves a violated check. A junction line decides which strings may end on it without leaving one. At a first or last junction of a Z check, a string passes between the data sheet and the check sheet. At a middle junction the data string and the two check-sheet strings on either side of the line must be present an even number of times; the case "both check-sheet strings, no data string" (110) lets a check-sheet string pass a data sheet that is not selected. A slab operator puts strings at one height on the selected data sheets and joins the selected junctions of every Z check in consecutive pairs by check-sheet strings, the backbone, which is possible because a Z check contains an even number of selected qubits. The explicit realization below enforces exactly these rules: at a middle junction, one face of the check sheet gains the data edge on the line, so that face checks the parity of the incoming backbone, the outgoing backbone, and the data string. The remaining rules keep the stars and faces commuting; the walls do so where an X-check sheet crosses a Z-check sheet, and they never touch a slab operator. The main text, Sec. II, gives the same picture with less notation.

A. The explicit two-sheet realization

Let $H_X \in \mathbb{F}_2^{m_X \times n_{\rm in}}$ and $H_Z \in \mathbb{F}_2^{m_Z \times n_{\rm in}}$ be the check matrices of an input CSS code with $H_X H_Z^T = 0$, every check of weight at least two, and $K = n_{\rm in} - \operatorname{rank} H_X - \operatorname{rank} H_Z$. Fix an integer spacing $s \geq 2$ and set

$$\begin{aligned} &Y_q = sq, \quad X_j = s(j+1), \quad Z_i = s(i+1), \\ &L_x = s(m_Z+1), \quad L_z = s(m_X+1), \end{aligned} \tag{P1}$$

with $q = 1, \dots, n_{\rm in}$, $j = 0, \dots, m_Z - 1$ and $i = 0, \dots, m_X - 1$. For a check c write $q_1^{(c)} < \cdots < q_{k_c}^{(c)}$ for its support in increasing order; $q_1^{(c)}$ is its *first*, $q_{k_c}^{(c)}$ its *last*, and the others its *middle* incidences.

a. Sheets. Every sheet is an edge surface code on a rectangle of the cubic lattice: one qubit per unit edge of the sheet, an X check (*star*) on each vertex acting on the sheet edges at that vertex, and a Z check (*face*) on each unit square acting on its boundary edges.

- Data sheet D_q ($q = 1, \dots, n_{\rm in}$): the plane $y = Y_q$ with $0 \leq x \leq L_x$, $0 \leq z \leq L_z$. The rows $z = 0$ and $z = L_z$ carry no x-edges and no stars.
- Z-check sheet C_j: the plane $x = X_j$ with $Y_{q_1} \leq y \leq Y_{q_k}$ ($q_t = q_t^{(Z_j)}$) and $0 \leq z \leq L_z$. The rows $z = 0, L_z$ carry no y-edges and no stars.
- X-check sheet A_i: the plane $z = Z_i$ with $Y_{q_1} \leq y \leq Y_{q_k}$ ($q_t = q_t^{(X_i)}$) and $0 \leq x \leq L_x$; every vertex carries a star.

Distinct sheets own distinct qubits, also on coincident lattice edges, with one exception: the first and last z-edge columns of C_j *are* the z-lines $(X_j, Y_{q_1}, \cdot)$ and $(X_j, Y_{q_k}, \cdot)$ of D_{q_1} and D_{q_k}, and the stars of C_j on those columns are merged into the corresponding data stars. A sheet pair without an input incidence therefore crosses transparently. At most two qubits share a lattice edge.

b. Line junctions. Write $d_\mu(v)$, $c_\mu(v)$, $a_\mu(v)$ for the μ-directed edge $v \to v + \hat{\mu}$ of D, C, A respectively. The input incidences act by adding one qubit to existing checks (Table S1):

(Z) for $q = q_t^{(Z_j)}$ and every vertex $u = (X_j, Y_q, z)$ of the z-line: the D_q star at u gains $c_y(u)$ if $t < k$ and $c_y(u - \hat{y})$ if $t = k$; if q is a middle incidence, the C_j face on $[Y_q, Y_q + 1] \times [z, z + 1]$ gains $d_z(u)$. (For the first and last incidence the adjacent C_j face already contains d_z through the shared column.)

(X) for $q \in \operatorname{supp} X_i$ and every vertex $u = (x, Y_q, Z_i)$ of the x-line: the A_i star at u gains $d_z(u)$, and the D_q face on $[x, x + 1] \times [Z_i, Z_i + 1]$ gains $a_x(u)$.

c. Walls on crossing lines. A Z-check sheet C_j and an X-check sheet A_i cross along the y-line $\ell_{ij} = \{(X_j, y, Z_i)\}$ for y in the overlap $[\ell, h]$ of their y-ranges. On this line we allow two further one-qubit additions, with binary coefficients σ_y ($\ell \leq y \leq h$) and τ_y ($\ell \leq y < h$): the A_i star at (X_j, y, Z_i) gains $\sigma_y\, c_z(X_j, y, Z_i)$, and the C_j face on $[y, y + 1] \times [Z_i, Z_i + 1]$ gains $\tau_y\, a_y(X_j, y, Z_i)$. The coefficients are the unique solution of the commutation conditions of Lemma SI.1.

Lemma SI.1 (Commuting checks and point defects)**.** *With the junction rules, every star commutes with every face except possibly an A_i star on a crossing line ℓ_{ij} and a C_j face adjacent to it. For each crossing line the commutation conditions for these pairs are linear in (σ, τ) and form a path $\sigma_\ell, \tau_\ell, \sigma_{\ell+1}, \dots, \tau_{h-1}, \sigma_h$ whose two ends are fixed. The system has a solution, which is then unique, if and only if $|\operatorname{supp} X_i \cap \operatorname{supp} Z_j|$ is even. Hence, for every CSS input, all checks commute.*

TABLE S1. Junction and wall rules of the explicit realization. Each rule adds one qubit of another sheet to one existing check. The last column lists the condensed anyons of the junction (Sec. SI B); e denotes a star violation and m a face violation, and subscripts $1, 2$ denote the two sides of a sheet cut by the line.

Junction	Modified checks	Boundary relation of the slab operator	Condensed anyons
Z first (q_1)	C_j column $= D_{q_1}$ z-line; D star $+ c_y(u)$	$x_{q_1} + p_1 = 0$	$e_C e_{D_1}$, $e_{D_1} e_{D_2}$, $m_C m_{D_1} m_{D_2}$
Z middle (q_t)	D star $+ c_y(u)$; C face on $[Y_q, Y_q{+}1]$ $+ d_z(u)$	$p_{t-1} + p_t + x_{q_t} = 0$	$e_{D_1} e_{D_2}$, $e_{C_1} e_{C_2} e_D$, $m_{C_1} m_{C_2}$, $m_{C_2} m_{D_1} m_{D_2}$
Z last (q_k)	C_j column $= D_{q_k}$ z-line; D star $+ c_y(u - \hat{y})$	$p_{k-1} + x_{q_k} = 0$	as first
X any ($q \in X_i$)	A star $+ d_z(u)$; D face above the line $+ a_x(u)$	$r_{t-1} + r_t + z_{q_t} = 0$ ($r_0 = r_k = 0$)	$X \leftrightarrow Z$ dual of the Z rows
Wall on ℓ_{ij}	A star $+ \sigma_y c_z$; C face above the line $+ \tau_y a_y$	none (slab operators avoid these qubits)	switches at each $q \in X_i \cap Z_j$

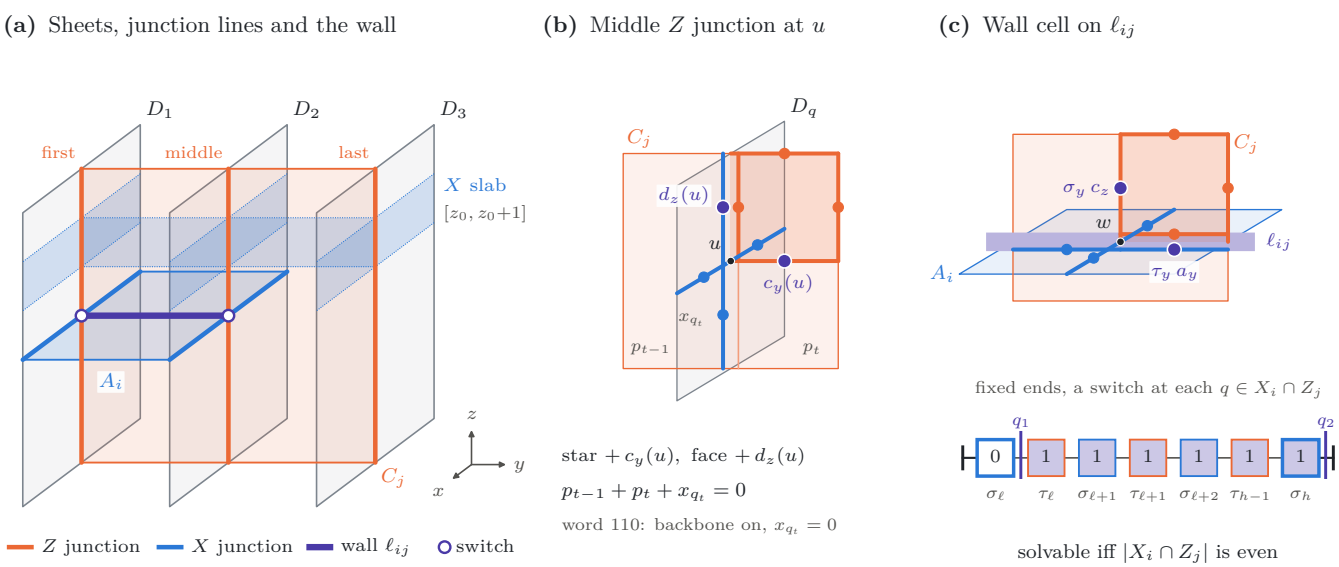


FIG. S1. Junction and wall geometry of the explicit realization. (a) A miniature instance with three data sheets D_1, D_2, D_3 (planes $y = Y_q$), one Z-check sheet C_j (plane $x = X_j$) with support $\{q_1, q_2, q_3\}$, and one X-check sheet A_i (plane $z = Z_i$) with support $\{q_1, q_2\}$. The Z junctions are the z-lines where C_j meets the data sheets (first, middle, last), the X junctions are the x-lines where A_i meets them, and the wall is the crossing line ℓ_{ij} of C_j and A_i; its coefficients switch at the two data qubits of $X_i \cap Z_j$. A single-layer X slab is shaded. (b) A middle Z junction: the D_q star at u gains the C_j edge $c_y(u)$ and the C_j face beside the line gains the data edge $d_z(u)$, which enforces $p_{t-1} + p_t + x_{q_t} = 0$ and lets the backbone pass an unselected data sheet (word 110). (c) A wall cell on ℓ_{ij}: the A_i star gains $\sigma_y c_z$ and the C_j face gains $\tau_y a_y$; the coefficients form a path with fixed ends, solvable exactly when $|X_i \cap Z_j|$ is even. The rules are listed in Table S1.

Proof. Unmodified checks of one sheet commute as in the surface code, and checks of different sheets share no qubit unless a rule added one. It remains to inspect the finitely many pairs touched by a rule. (i) A D star on a z-line and a C face containing d_z: their common qubits are d_z and the C edge c_y added to the star (or, at a first/last column, the shared z-edge and the merged C edge), two in each case. (ii) An A star on an x-line and a D face above that line: common qubits $d_z(u)$ and one a_x edge. (iii) The added qubits c_y, a_x lie in no other check of the opposite type, so no further pair is affected. (iv) An A_i star on ℓ_{ij} contains $d_z(u)$ when u lies on a data line of X_i, and a C_j face contains d_z of the same line when that data qubit is a middle incidence of Z_j (face on the $+y$ side) or its first/last incidence (adjacent face through the shared column). Hence the base overlap of such a pair is 1 exactly once for each common qubit $q \in X_i \cap Z_j$, and 0 otherwise. The wall additions change the overlap of the A star at y' and the C face on $[y, y+1]$, $y' \in \{y, y+1\}$, by $\sigma_{y'} + \tau_y$. If the C face on $[\ell - 1, \ell]$ exists it fixes σ_ℓ; otherwise ℓ is the first column of C_j, which is shared with a data line, and commutation with the two D faces adjacent to that line forces $\sigma_\ell = 0$; the right end is fixed in the same way. Summing the path equations telescopes to $\sigma_\ell + \sigma_h$ plus the sum of all base overlaps, which is $|X_i \cap Z_j|$ modulo two. A path with both ends fixed is solvable exactly when this sum vanishes, and then uniquely. □

Proposition SI.2 (The realization encodes exactly K qubits)**.** *For every input CSS code with check weights at least two and every $s \geq 2$, the explicit realization is a CSS code with $k = K$.*

Proof. Euler count. A data sheet has $L_x(L_z-1)+(L_x+1)L_z$ qubits, $(L_x+1)(L_z-1)$ stars and L_xL_z faces, so it contributes $+1$ to $n-\#X-\#Z$. A C sheet of width w has $w(L_z-1)$ y-edges, $(w-1)L_z$ own z-edges, $(w-1)(L_z-1)$ stars and wL_z faces, and an A sheet of width w has $L_x(w+1)+(L_x+1)w$ edges, $(L_x+1)(w+1)$ stars and L_xw faces; each contributes -1. Junction and wall rules add no qubits or checks. Hence $n-\#X-\#Z = n_{\rm in}-m_X-m_Z$.

X relations. Let a be a set of stars whose product is the identity. Edges of A_i lie only in the two A_i stars at their ends, so a is constant, α_i, on the stars of A_i. On the data column $x=0$ of D_q, the edge $d_z(0,Y_q,z)$ lies in the D_q stars at its ends (when present) and, for $z=Z_i$ with $q\in X_i$, in one A_i star; the rows $z=0,L_z$ carry no stars. Summing along the column gives $\sum_{i\ni q}\alpha_i=0$, i.e. $\alpha^T H_X=0$. Conversely α determines a: the D stars are determined column by column from $z=0$ and are constant along rows (a D x-edge lies only in the two stars at its ends), and the C stars are determined column by column from $z=0$, where $c_z(y,0)$ lies in a single star. Thus the space of X relations injects into the left kernel of H_X.

Z relations. Let b be a set of faces with trivial product. The edges of C_j lie only in C_j faces, two each, so b is constant, γ_j, on C_j. A D x-edge lies only in the faces above and below it, so b is constant, $\beta_q(x)$, on each column of faces of D_q. A data z-edge at column x lies in the D faces at $x-1$ and x and, for $x=X_j$ with $q\in Z_j$, in one C_j face; the edges at $x=0$ and $x=L_x$ lie in one face. Hence $\beta_q(0)=\beta_q(L_x-1)=0$ and $\beta_q(x-1)+\beta_q(x)=\sum_{j:X_j=x,\,q\in Z_j}\gamma_j$, which sum to $\gamma^T H_Z=0$. The A faces are determined by γ from the column $x=0$, where each a_y lies in one face. So the Z relations inject into the left kernel of H_Z.

Therefore, writing $H_X^{\rm lc}, H_Z^{\rm lc}$ for the check matrices of the explicit realization, $k=n-\operatorname{rank}H_X^{\rm lc}-\operatorname{rank}H_Z^{\rm lc}=(n-\#X-\#Z)+\dim(\text{relations})\le n_{\rm in}-\operatorname{rank}H_X-\operatorname{rank}H_Z=K$. Theorem SI.3 below exhibits K independent logical pairs, so $k=K$. □

d. Prefix-parity slab operators. For $x\in\ker H_Z$ and a height $0\le z_0<L_z$ let

$$\widetilde{X}(x;z_0)=\sum_q x_q\,E_q^D(z_0)+\sum_j\sum_{t=1}^{k_j-1}p_t^{(j)}\,E_{j,t}^C(z_0),$$
$$p_t^{(j)}=\sum_{h=1}^{t}x_{q_h^{(Z_j)}},\qquad\text{(P2)}$$

where $E_q^D(z_0)$ is the set of D_q z-edges spanning $[z_0,z_0+1]$ (all x), and $E_{j,t}^C(z_0)$ is the set of C_j-owned z-edges spanning $[z_0,z_0+1]$ with $Y_{q_t}<y\le Y_{q_{t+1}}$, excluding a shared last column. Commutation of x with Z_j is

$$\sum_{t=1}^{k_j}x_{q_t}=0\pmod 2,\qquad\text{(P3)}$$

so $p_{k_j}^{(j)}=0$ and $p_{t-1}+p_t+x_{q_t}=0$ at every incidence ($p_0=0$), i.e., with $H_{\rm end}=[1\ 1]$ and $H_{\rm mid}=[1\ 1\ 1]$ (Eq. (P10)),

$$(p_{t-1},p_t,x_{q_t})\in\ker H_{\rm mid},$$
$$(p_1,x_{q_1}),\,(p_{k-1},x_{q_k})\in\ker H_{\rm end}.\qquad\text{(P4)}$$

Dually, for $z\in\ker H_X$ and a column $0\le x_0\le L_x$ let $\widetilde{Z}(z;x_0)$ be Z on the D_q z-edges of column x_0 (all z) for $z_q=1$, and on the A_i y-edges of column x_0 with $Y_{q_t}\le y<Y_{q_{t+1}}$ for $r_t^{(i)}=\sum_{h\le t}z_{q_h^{(X_i)}}=1$.

Theorem SI.3 (Single-layer slab representatives)**.** *In the explicit realization, for every height z_0 and every column x_0:*

1. (gluing forms) *for every face h, the overlap parity $\langle h,\widetilde{X}(x;z_0)\rangle$ is either 0 or exactly one of the linear forms $x_{q_1}+p_1$, $p_{t-1}+p_t+x_{q_t}$, $p_{k-1}+x_{q_k}$ of a single incidence of a single check; each such form occurs. The dual statement holds for stars and $\widetilde{Z}(z;x_0)$;*
2. *$\widetilde{X}(x;z_0)$ commutes with every stabilizer for all $x\in\ker H_Z$, and $\widetilde{Z}(z;x_0)$ for all $z\in\ker H_X$;*
3. *$\langle\widetilde{X}(x;z_0),\widetilde{Z}(z;x_0)\rangle=x\cdot z$;*
4. *every pure-X logical class has a representative supported on the single layer of z-edges spanning $[z_0,z_0+1]$, and every pure-Z logical class has one in the plane $x=x_0$.*

Proof. (1) A face meets the edges of $\widetilde{X}$ only if it is a D face or a C face in the row $[z_0,z_0+1]$; A faces contain only A edges. A D face there contains two z-edges of the same data string, and the qubit a_x added above an x-line is not in $\widetilde{X}$; its form is 0. A C face on $[y,y+1]$ contains the C_j columns y and $y+1$. Inside a segment both carry p_t; the wall qubit a_y is not in $\widetilde{X}$. At a middle incidence q_t the $+y$ face contains the line edge $c_z(Y_{q_t})$, which belongs to the segment ending there and carries p_{t-1}, the next column (p_t) and the added d_z (x_{q_t}). At the first (last) incidence the face adjacent to the shared column contains that data edge (x_{q_1}, resp. x_{q_k}) and one segment column. Each incidence therefore contributes exactly one face per height. The dual argument uses the D stars of column x_0 (two consecutive data edges, and c_y not in $\widetilde{Z}$) and the A stars of column x_0, whose added edge $d_z(u)$ carries z_q at an incidence row; the wall qubit c_z is not in $\widetilde{Z}$. (2) follows from (1) and Eq. (P4). (3) The C edges of $\widetilde{X}$ and the A edges of $\widetilde{Z}$ are disjoint, and the data strings meet in the single edge $d_z(x_0,Y_q,z_0)$ for each q with $x_q=z_q=1$. (4) Applied to a symplectic basis of the input code, (2)–(3) give K independent logical pairs. By Proposition SI.2 they exhaust the logical space, so $x\mapsto[\widetilde{X}(x;z_0)]$ maps $\ker H_Z$ onto the pure-X logical classes; an x in the row space of H_X pairs trivially with every $\widetilde{Z}$ and is therefore mapped to a stabilizer. The same holds for $\widetilde{Z}$. □

The support of $\widetilde{X}(x;z_0)$ is a single layer of z-edges, so the slab thickness is $t_0 = 1$, and $\widetilde{Z}(z;x_0)$ lies in one plane. No admissibility condition on z_0 or x_0 is needed: the construction works on check layers as well as between them, because the wall rules add only qubits that the slab operators do not use, the Z-junction rules are the same at every height and the X-junction rules the same at every column, and the remaining junction rules (X junctions for $\widetilde{X}$, Z junctions for $\widetilde{Z}$) add only qubits that the slab operator does not use.

Corollary SI.4 (Uniform-gap slab mobility)**.** *For a balanced input family with $m_X, m_Z = \Theta(N)$, there are*

$$r_X = L_z = \Theta(N), \qquad r_Z = L_x + 1 = \Theta(N) \tag{P5}$$

pairwise disjoint complete X ports (one per height) and complete Z ports (one per column).

Lemma SI.5 (Locality of the explicit realization)**.** *Every check of the explicit realization is supported within the unit L^∞ ball of one lattice vertex, has weight at most six, and every qubit lies in at most three checks of each type; at most two qubits share a lattice edge. These bounds are independent of the input code, of s, and of the block size.*

Proof. Stars have at most four in-sheet edges, plus at most one junction qubit and, for A stars, one wall qubit; faces have at most four edges plus at most one junction qubit and one wall qubit. All added qubits are edges at the same vertex, or edges of the same face position, as the check they join. A data line edge lies in two data faces and at most one C face; an a_x edge lies in two A faces and at most one D face; the remaining edge types are counted in the same way. Two sheets meet along a line, so at most two qubits share an edge. □

Theorem SI.6 (Distance of the explicit realization)**.** *Let $d_X^{\mathrm{in}}, d_Z^{\mathrm{in}}$ be the distances of the input code, w_X, w_Z its largest X- and Z-check weights, and*

$$\eta_X = \max\{1, \lfloor w_X/2 \rfloor\}, \qquad \eta_Z = \max\{1, \lfloor w_Z/2 \rfloor\}. \tag{P6}$$

The explicit realization has

$$d_X \geq (L_x + 1)\left\lceil \frac{d_X^{\mathrm{in}}}{\eta_X} \right\rceil, \qquad d_Z \geq L_z \left\lceil \frac{d_Z^{\mathrm{in}}}{\eta_Z} \right\rceil, \tag{P7}$$

and, from the slab representatives of a minimum-weight input logical, $d_X \leq (L_x+1)d_X^{\mathrm{in}} + s\, m_Z n_{\mathrm{in}}$ and $d_Z \leq L_z d_Z^{\mathrm{in}} + s\, m_X n_{\mathrm{in}}$. For a balanced good input family with bounded check weight, $d_X, d_Z = \Theta(N^2)$.

Proof. Let O be a nontrivial X-type logical. By Theorem SI.3 the $\widetilde{X}(x;z_0)$ represent every X class, so O is equivalent to some $\widetilde{X}(x;z_0)$ with $x \in \ker H_Z$ outside the row space of H_X, and $\langle O, \widetilde{Z}(z;x_0)\rangle = x \cdot z$ for every $z \in \ker H_X$ and every column x_0. Fix x_0. The support of $\widetilde{Z}(z;x_0)$ is a union of disjoint pieces lying in the plane $x = x_0$: the data column of D_q when $z_q = 1$ and the column-x_0 segment t of A_i when $r_t^{(i)} = \sum_{h \leq t} z_{q_h} = 1$. Let a_q and $b_{i,t}$ be the parities of O on these pieces. Then

$$\langle O, \widetilde{Z}(z;x_0)\rangle = \sum_q z_q a_q + \sum_{i,t} r_t^{(i)} b_{i,t} = z \cdot v(x_0),$$
$$v(x_0) = \sum_q a_q e_q + \sum_{i,t} b_{i,t}\, \pi_{i,t},$$

where $\pi_{i,t}$ is the indicator of the *prefix* $\{q_1^{(X_i)}, \ldots, q_t^{(X_i)}\}$ of check X_i. Since $z \cdot v(x_0) = z \cdot x$ for all $z \in \ker H_X$, the vector $v(x_0)$ lies in $x + \operatorname{row} H_X$. For a check of weight k, the prefix $\pi_{i,t}$ and the complementary suffix $\pi_{i,t} + X_i$ differ by the row X_i of H_X, and the shorter of the two has at most $\min\{t, k-t\} \leq \lfloor k/2 \rfloor$ entries. Replace every occupied prefix by its shorter side:

$$v'(x_0) = \sum_q a_q e_q + \sum_{i,t} b_{i,t}\, \pi'_{i,t}, \qquad \pi'_{i,t} \in \{\pi_{i,t}, \pi_{i,t} + X_i\}.$$

This adds a sum of rows of H_X, so $v'(x_0) \in x + \operatorname{row} H_X$ is still a nontrivial X logical of the input code and $|v'(x_0)| \geq d_X^{\mathrm{in}}$. Each nonzero a_q contributes one entry of $v'(x_0)$ and each nonzero $b_{i,t}$ at most η_X entries, so the number of pieces of odd parity is an integer of at least d_X^{in}/η_X, hence at least $\lceil d_X^{\mathrm{in}}/\eta_X \rceil$, and each such piece contains a qubit of O. The planes $x = x_0$, $0 \leq x_0 \leq L_x$, are disjoint; summing the integer bound over them gives $|O| \geq (L_x+1)\lceil d_X^{\mathrm{in}}/\eta_X \rceil$. The Z bound is the same argument with the layers of z-edges at the heights $0 \leq z_0 < L_z$, the X slabs, and the prefixes of the Z checks. The upper bounds count the data strings of a minimum-weight input logical and at most $s\, n_{\mathrm{in}}$ backbone edges per check. For balanced good inputs, $L_x, L_z = \Theta(N)$ and $d^{\mathrm{in}} = \Theta(N)$. □

The argument lifts the input distance through the complete ports of Theorem SI.3: each of the $\Theta(N)$ disjoint planes of one orientation must carry a whole input logical's worth of the operator. Two features give the constant. A backbone piece represents a prefix of a check, which may be exchanged for the complementary suffix modulo that check, so a piece costs at most half a check weight; and the count of occupied pieces is rounded up in each plane before summing over planes. The script `distance_lower_bound_check.py` checks each step on every input of Table S5 up to $n_{\mathrm{in}} = 15$. Every basis slab is exactly the union of the pieces used above, lies in its own plane or layer, and obeys the upper bound (1010 slabs). For random representatives (slab operators times random stabilizers) at every column and height (3858 cases), both $v(x_0)$ and its shortest-side replacement $v'(x_0)$ land in the right coset, $|v'(x_0)|$ is at most η times the number of occupied pieces, and that number is at least $\lceil d^{\mathrm{in}}/\eta \rceil$. The script also compares Eq. (P7) with the exact distances of Table S2. For the inputs $[[4,2,2]]$ and $[[6,4,2]]$ ($d^{\mathrm{in}} = 2$, $\eta = 2$ and 3) the bound is $d_X \geq L_x + 1$, $d_Z \geq L_z$, and it equals the exact distance in every instance. At $s = 5$, where the CP-SAT search stopped at the brackets 10–11 and 9–10, the theorem closes them: $d_X = 11$ and $d_Z = 10$.

TABLE S2. Code-capacity distances of the explicit realization (`explicit_layer_code_distance.py`). The upper bound is an explicit undetectable logical found by Stim's search, the lower bound is proved with CP-SAT, and the last two columns give the bound of Theorem SI.6. At $s = 5$ the CP-SAT bounds are 10 and 9, and the theorem supplies the matching lower bounds (†).

Input	s	n	$L_x = L_z$	d_X	d_Z	Eq. (P7)	
$[[4, 2, 2]]$	2	224	4	5	4	5	4
$[[4, 2, 2]]$	3	504	6	7	6	7	6
$[[4, 2, 2]]$	4	896	8	9	8	9	8
$[[4, 2, 2]]$	5	1400	10	$11^\dagger$	$10^\dagger$	11	10
$[[6, 4, 2]]$	2	352	4	5	4	5	4
$[[6, 4, 2]]$	3	792	6	7	6	7	6
$[[6, 4, 2]]$	4	1408	8	9	8	9	8

For an input of distance three the exact comparison is still open. For the Steane $[[7, 1, 3]]$ input at $s = 2$ ($n = 1792$, $\eta = 2$) the bound is $d_X \geq 18$ and $d_Z \geq 16$; two hours of CP-SAT per sector bracket d_X between 14 and 20 and d_Z between 14 and 19 (`run_distances.py` in `scaling_campaign/campaign/`, `numerical/scaling_campaign/code_distance.csv`). The search found no logical operator lighter than the bound.

Lemma SI.7 (Symplectic correspondence)**.** *For the representatives of Theorem SI.3,* $\langle \widetilde{X}(a; z_0), \widetilde{Z}(b; x_0) \rangle = a \cdot b \pmod 2$ *for all heights and columns.*

Proof. This is item (3) of Theorem SI.3. □

e. Why the slab theorem is the bottleneck. The middle-junction relation $p_{t-1} + p_t + x_{q_t} = 0$ is the nontrivial local move: it lets backbone parity cross an unselected data sheet (110) without creating a companion string. A realization that enforces only the all-equal relation at middle junctions cannot represent a general logical in a slab, and indeed does not encode the input logical qubits (Sec. SII).

B. Relation to the published junction cells

a. Condensation data. WB specify each line junction by condensed anyon generators. For the first $\hat{z}$ junction, with m-occupations (p, x_L, x_R) of the check sheet and the two sides of the data sheet, the condensed e generators $e_C e_{D_1}$ and $e_{D_1} e_{D_2}$ give, by trivial braiding,

$$G_{\text{end}} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \tag{P8}$$

and for a through-going data string ($x_L = x_R$) the relation $H_{\text{end}} = [1\ 1]$. For a middle junction with occupations (p_L, p_R, x_L, x_R) the generators $e_{C_1} e_{C_2} e_{D_1}$ and $e_{D_1} e_{D_2}$ give

$$G_{\text{mid}} = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}, \tag{P9}$$

and $H_{\text{mid}} = [1\ 1\ 1]$. Thus

$$\begin{aligned} H_{\text{end}} &= \begin{bmatrix} 1 & 1 \end{bmatrix}, & \ker H_{\text{end}} &= \{00, 11\}, \\ H_{\text{mid}} &= \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}, & \ker H_{\text{mid}} &= \{000, 110, 101, 011\}. \end{aligned} \tag{P10}$$

The X-check junctions are the $X \leftrightarrow Z$ duals. The condensed anyons of the explicit junctions (Table S1) are exactly these data: at a middle $\hat{z}$ junction, Z on the two data edges across the line passes through the star $(e_{D_1} e_{D_2})$; Z on the two C edges across the line leaves one violation of the data star, which carries c_y $(e_{C_1} e_{C_2} e_D)$; X on the C line edge flips the two C faces beside it ($m_{C_1} m_{C_2}$, the 110 pass-through); and X on the data line edge flips the two data faces and the $+y$ C face ($m_{C_2} m_{D_1} m_{D_2}$). The first and last junctions, where C_j shares the data line, condense all e pairs and $m_C m_{D_1} m_{D_2}$. Each set generates a Lagrangian subgroup of the four (three) incident half-sheets.

b. The single-chart transcription (negative control). A transcription of WB Sec. VI B items 3–8 into one unit-cubic chart (Table S3) lets the check sheet and the data sheet share the line edges and makes every junction star the union of the two sheet stars. Glued, these cells form the plain union of the sheets. At a middle junction the plain union condenses every e pair and only the fourfold m product, so a slab operator must satisfy $p_{t-1} = p_t = x_{q_t}$: the all-equal relation $\{000, 111\}$, not $\ker H_{\text{mid}}$. The complete single-chart codes do not encode K qubits (Table S5), which makes the transcription a useful negative control (an earlier version of this work used it, with a certificate that did not detect the problem; Sec. SXX). The transcription's "novel" S^Z entries (P^+_{yz} at middle $\hat{z}$ cells, P^+_{xz} at $\hat{x}$ cells) are exactly the faces that, in the explicit realization, carry the other sheet's line edge. This suggests that the published cells use two sheets with distinct line qubits, as the explicit realization does. We do not claim literal identity with the published drawings; the theorem above is stated for the explicit realization, whose complete check matrices are generated by `layer_code_explicit.py`.

c. Microscopic inputs. Table S4 lists every local input of Theorem SI.3 and its status.

Lemma SI.8 (Interface inputs for the slab theorem)**.** *Theorem SI.3 uses only the sheet definitions, the junction and wall rules of Table S1, and Proposition SI.2; it uses no property of the published lattice drawings and no finite certificate.*

Proof. Every step of the proof of Theorem SI.3 is a statement about the checks meeting one layer of z-edges (or one column), and these checks are listed in Table S1. □

d. Port separation and constant refinement. The gauging construction of Sec. SIII maps distinct port qubits to distinct grid vertices. At most two qubits share a lattice edge (Lemma SI.5), so a grid of twice the lattice resolution separates them, changing the grid constants

TABLE S3. Single-chart unit-cubic transcription of the novel WB Sec. VI B line-junction checks, kept as the negative control of Sec. SII; the explicit realization of Sec. SI A replaces it. A dash means that no additional check of that type is listed.

Source item	Cell	Novel S^X support	Novel S^Z support
3	first $\hat{z}$	$\{q_{x-}, q_{x+}, q_{y+}, q_{z-}, q_{z+}\}$	—
4	middle $\hat{z}$	$\{q_{x-}, q_{x+}, q_{y-}, q_{y+}, q_{z-}, q_{z+}\}$	P^+_{yz}
5	last $\hat{z}$	$\{q_{x-}, q_{x+}, q_{y-}, q_{z-}, q_{z+}\}$	—
6	first $\hat{x}$	$\{q_{x-}, q_{x+}, q_{y+}, q_{z-}, q_{z+}\}$	P^+_{xz}
7	middle $\hat{x}$	$\{q_{x-}, q_{x+}, q_{y-}, q_{y+}, q_{z-}, q_{z+}\}$	P^+_{xz}
8	last $\hat{x}$	$\{q_{x-}, q_{x+}, q_{y-}, q_{z-}, q_{z+}\}$	P^+_{xz}

TABLE S4. Local inputs to Theorem SI.3 and their status.

Input	Status	Where it is used
Sheet surface codes and boundaries	Defined here (Sec. SI A)	Euler count, relation injectivity, slab strings
Z/X line-junction rules	Defined here (Table S1); condensation data equal to WB's (Sec. SI B)	Gluing forms, $k = K$
Wall rules at crossing lines	Derived here (Lemma SI.1); solvable iff input commutes	Commutation of all checks
H_{end}, H_{mid}	Derived from WB condensation generators (Eqs. (P8)–(P10)); realized by one face per incidence and height (Theorem SI.3(1))	Prefix recurrence
Prefix-parity assembly	Proved here (Eqs. (P2)–(P4))	Every pure logical class
$k = K$	Proved here (Proposition SI.2)	Completeness of the slab representatives
$d = \Theta(N^2)$	Proved here (Theorem SI.6); exact small-instance values in Table S2	Architecture theorem only
Complete finite certificate	`slab_gluing_certificate.py` (Sec. SII)	Independent check of all of the above on finite inputs

and the port thickness t by $O(1)$; the two-block simulation of Sec. SXIV uses exactly this map. The slab theorem itself needs no refinement. The stabilizer-equivalent subdivisions below remain available for enlarging ordinary sheet regions.

Lemma SI.9 (Constant subdivision refinement is stabilizer equivalent)**.** *Fix a refinement factor c. Let $\mathcal{S}$ be the stabilizer group before refinement and let $\mathcal{A}_c$ be the product-state stabilizers of the finitely many new $|+\rangle$ and $|0\rangle$ edge qubits. The subdivision refinement admits a geometrically local Clifford circuit U_c of depth and light-cone size depending only on c and the finite local catalog such that*

$$\mathcal{S}_{\mathrm{ref}} = U_c(\mathcal{S} \otimes \mathcal{A}_c)U_c^\dagger. \tag{R1}$$

Consequently the encoded logical dimension is exactly unchanged. If s_c bounds the Pauli light-cone size of both U_c and $U_c^\dagger$, then

$$\frac{d}{s_c} \le d_{\mathrm{ref}} \le s_c d. \tag{R2}$$

Proof. For an edge split, add a in $|+\rangle$ and apply $U_E = \mathrm{CNOT}_{a\to q}$, so that $X_a \mapsto X_a X_q$ and $Z_q \mapsto Z_a Z_q$: the first image is the new degree-two star, and every face containing q acquires $Z_a Z_q$. For a face split along paths P_1, P_2, add a in $|0\rangle$ and apply $\prod_{q\in P_1} \mathrm{CNOT}_{q\to a}$, so that $Z_a \mapsto Z_a Z(P_1)$ and the two refined faces are generated; in the X sector each endpoint star of P_1 acquires X_a. Bounded-degree moves are colored into $O_c(1)$ layers. Local Clifford equivalence with initialized ancillas preserves K, and light cones give (R2). □

C. Symplectic overlap of transverse logical ports

The fixed slabs expose not only local representatives but also a quantitative constraint on how two logical polarizations can overlap. For an $[[n, K]]$ CSS code, let

$$\mathcal{L}_X = \ker H_Z / \operatorname{row} H_X, \tag{P11}$$
$$\mathcal{L}_Z = \ker H_X / \operatorname{row} H_Z, \tag{P12}$$

so $\dim \mathcal{L}_X = \dim \mathcal{L}_Z = K$ and the physical dot product induces a perfect bilinear commutation pairing $\langle\cdot,\cdot\rangle : \mathcal{L}_X \times \mathcal{L}_Z \to \mathbb{F}_2$. For a physical region R, define $\mathcal{X}(R) \subseteq \mathcal{L}_X$ and $\mathcal{Z}(R) \subseteq \mathcal{L}_Z$ to be the logical classes admitting, respectively, pure-X and pure-Z representatives supported in R, and set

$$g_X(R) = \dim \mathcal{X}(R), \qquad g_Z(R) = \dim \mathcal{Z}(R). \tag{P13}$$

For two regions A, B, write

$$r(A, B) = \operatorname{rank}\Big(\langle\cdot,\cdot\rangle\big|_{\mathcal{X}(A)\times\mathcal{Z}(B)}\Big). \tag{P14}$$

Theorem SI.10 (Overlap-corrected CSS cleaning inequality)**.** *For every pair of physical regions A, B in an $[[n, K]]$ CSS stabilizer code,*

$$\max\{0, g_X(A) + g_Z(B) - K\} \le r(A, B) \le |A \cap B|, \tag{P15}$$

and therefore

$$g_X(A) + g_Z(B) \le K + |A \cap B|. \tag{P16}$$

Proof. Put $C = A \cap B$, $a = g_X(A)$, and $b = g_Z(B)$. Choose bases of the two supported logical spaces and pure representatives $x_i \subseteq A$ and $z_j \subseteq B$. Their logical commutation matrix has entries $M_{ij} = x_i \cdot z_j$. Outside C, at least one factor vanishes, so if $X_C \in \mathbb{F}_2^{a\times|C|}$ and $Z_C \in \mathbb{F}_2^{b\times|C|}$ collect the restrictions, then $M = X_C Z_C^T$ and hence $r(A,B) = \operatorname{rank} M \le |C|$.

For the lower bound, put $U = \mathcal{X}(A)$ and $V = \mathcal{Z}(B)$. The kernel of the restricted pairing map $U \to V^*$ is $U \cap V^\perp$. Because the full X/Z logical pairing is nondegenerate, $\dim V^\perp = K - b$, and therefore

$$r(A,B) = a - \dim(U \cap V^\perp) \ge a - (K - b) = a + b - K.$$

Combining this with nonnegativity gives Eq. (P15), and the upper and lower bounds together give Eq. (P16). □

For $B = A^c$, the overlap penalty vanishes. The usual stabilizer cleaning lemma sharpens the resulting inequality to the familiar complementary-region identity in the appropriate logical sectors [2]. Equation (P16) is useful here because it applies to arbitrary overlapping regions.

Theorem SI.11 (Port-packing inequality)**.** *Let $A_1, \dots, A_r$ and B be physical regions of an $[[n, K]]$ CSS code. Suppose each physical qubit belongs to at most c of the regions A_i. Then*

$$\sum_{i=1}^{r} g_X(A_i) \le r\big(K - g_Z(B)\big) + c|B|. \tag{P17}$$

If B is a complete pure-Z port, $g_Z(B) = K$, this reduces to

$$\sum_{i=1}^{r} g_X(A_i) \le c|B|. \tag{P18}$$

For pairwise-disjoint A_i, $c = 1$. If in addition every A_i is a complete pure-X port, then

$$rK \le |B|. \tag{P19}$$

The $X \leftrightarrow Z$ statement holds by duality.

Proof. Apply Theorem SI.10 to each pair (A_i, B):

$$g_X(A_i) \le K - g_Z(B) + |A_i \cap B|.$$

Summing and counting the intersections qubit by qubit gives

$$\sum_i |A_i \cap B| = \sum_{q\in B} \#\{i : q \in A_i\} \le c|B|,$$

which proves Eq. (P17) and its special cases. □

The complete-B case also has a direct proof that does not rely on choosing mutually compatible representatives. Put $U_i = \mathcal{X}(A_i)$ and $C_i = A_i \cap B$. Choose a basis of U_i, choose one pure-X representative supported in A_i for each basis vector, and extend linearly. Restricting this representative map to C_i gives

$$\rho_i : U_i \longrightarrow \mathbb{F}_2^{C_i}.$$

If $\rho_i(u) = 0$, the chosen representative of u is disjoint from B. Because $g_Z(B) = K$, every pure-Z logical class has a representative in B, so u commutes with the entire Z-logical space. Nondegeneracy of the CSS logical pairing forces $u = 0$. Thus ρ_i is injective and $g_X(A_i) \le |C_i|$. Summing gives Eq. (P18) directly.

Theorem SI.12 (Symplectic crossbar bound)**.** *Let $A_1, \dots, A_r$ and $B_1, \dots, B_s$ be physical regions of an $[[n, K]]$ CSS code. Suppose each physical qubit belongs to at most c_X of the regions A_i and at most c_Z of the regions B_j. Put $a_i = g_X(A_i)$ and $b_j = g_Z(B_j)$. Then*

$$\sum_{i=1}^{r}\sum_{j=1}^{s} \max\{0, a_i + b_j - K\} \le \sum_{i=1}^{r}\sum_{j=1}^{s} r(A_i, B_j) \le c_X c_Z n. \tag{P20}$$

In particular, if every A_i is a complete pure-X port and every B_j is a complete pure-Z port, then

$$rsK \le c_X c_Z n. \tag{P21}$$

For pairwise-disjoint families, $c_X = c_Z = 1$ and hence

$$rsK \le n. \tag{P22}$$

Proof. Theorem SI.10 gives, for every pair (i, j),

$$\max\{0, a_i + b_j - K\} \le r(A_i, B_j) \le |A_i \cap B_j|.$$

For a physical qubit q, set $\mu_X(q) = \#\{i : q \in A_i\}$ and $\mu_Z(q) = \#\{j : q \in B_j\}$. Summing the intersection sizes and then counting qubitwise gives

$$\sum_{i,j} |A_i \cap B_j| = \sum_q \mu_X(q)\mu_Z(q) \le c_X c_Z n,$$

which proves Eq. (P20). The complete-port and disjoint-family statements follow immediately. □

Definition SI.13 (CSS symplectic-overlap number)**.** Define

$$\chi_{XZ}(Q) = \min_{\substack{A,B:\\ g_X(A)=K,\ g_Z(B)=K}} |A \cap B|. \tag{P23}$$

Thus χ_{XZ} is the smallest physical intersection through which a complete pure-X/Z logical polarization can realize its full commutation pairing. Theorem SI.10 gives

$$\chi_{XZ}(Q) \ge K. \tag{P24}$$

Corollary SI.14 (Layer-code crossbar saturation and transverse rank)**.** *For the balanced layer-code family (explicit realization), choose $r_X = \Theta(N)$ pairwise-disjoint*

complete X slabs $A_1, \dots, A_{r_X}$ and $r_Z = \Theta(N)$ pairwise-disjoint complete Z slabs $B_1, \dots, B_{r_Z}$. Then, for every i, j,

$$r_X K \le |B_j| = \Theta(N^2) = \Theta(K^2), \tag{P25}$$

$$r(A_i, B_j) = K, \qquad |A_i \cap B_j| = \Theta(K), \tag{P26}$$

and the full two-polarization family obeys

$$r_X r_Z K \le n = \Theta(N^3) = \Theta(K^3), \qquad r_X r_Z K = \Theta(n). \tag{P27}$$

Thus the symplectic crossbar bound is saturated up to constants. The $r_X r_Z = \Theta(K^2)$ crossings are pairwise disjoint, each carries the full logical commutation rank K, and together they occupy $\Theta(n)$ physical qubits. Moreover,

$$\chi_{XZ}(Q_N) = \Theta(K) = \Theta(N) = \Theta(n/d). \tag{P28}$$

Proof. Corollary SI.4 supplies both pairwise-disjoint slab families. Fix j. Equation (P19) gives $r_X K \le |B_j|$. A constant-thickness slab across a bounded-aspect three-dimensional block has $|B_j| = O(N^2)$ by bounded local density, while $r_X = \Theta(N)$ and $K = \Theta(N)$, so the bounds match.

For one crossing $C_{ij} = A_i \cap B_j$, Theorem SI.10 gives $r(A_i, B_j) \ge K$, while the pairing rank cannot exceed K; hence $r(A_i, B_j) = K$ and $|C_{ij}| \ge K$. Geometrically C_{ij} is a constant-cross-section tube of length $O(N)$, so $|C_{ij}| = O(N) = O(K)$ and therefore $|C_{ij}| = \Theta(K)$. Theorem SI.12 gives $r_X r_Z K \le n$; both sides scale as $\Theta(N^3)$, proving Eq. (P27). If $(i, j) \ne (i', j')$, then either $A_i \cap A_{i'} = \varnothing$ or $B_j \cap B_{j'} = \varnothing$, so the corresponding crossings are disjoint. Summing their $\Theta(K)$ sizes gives $\Theta(N^3) = \Theta(n)$. Finally the same single-pair argument gives $\chi_{XZ} = \Theta(K)$, and $n/d = \Theta(N)$. □

a. Rate–addressability consequence. For pairwise-disjoint complete port families, write $R = K/n$ for the code rate. Equation (P22) is equivalently

$$R r_X r_Z \le 1. \tag{P29}$$

Thus if $r_X, r_Z = \Omega(K)$ then $n = \Omega(K^3)$. The balanced layer-code family has $r_X, r_Z = \Theta(K)$ and $n = \Theta(K^3)$, so its physical-qubit scaling is asymptotically optimal among CSS families required to provide $\Theta(K)$ pairwise-disjoint complete ports in each logical polarization. Conversely, a constant-rate CSS family must have $r_X r_Z = O(1)$ for such disjoint complete ports.

b. Mobility–interface tradeoff. The same counting argument is dimension-independent. If a D-dimensional CSS family of linear size L has $r(L)$ pairwise-disjoint complete X ports and a complete Z region of size $O(L^{D-1})$, then

$$K(L) = O\left(\frac{L^{D-1}}{r(L)}\right). \tag{P30}$$

In particular, $r(L) = \Omega(L)$ implies $K = O(L^{D-2})$. For $D = 3$ the layer-code family has $r = \Theta(N)$ and $K = \Theta(N)$, so it saturates this scaling. This is a mobility/interface constraint, not a stronger replacement for the usual distance-based BPT tradeoff.

Figure S2 summarizes the pairwise overlap-cleaning mechanism and the layer-code scaling it enforces; Theorem SI.12 upgrades the same rank counting to the full two-polarization family.

Section SXVII records extensions that are not needed by the architecture theorem: a sharp factor-of-two analog for arbitrary stabilizer codes, a symplectic rederivation of the Bravyi–Poulin–Terhal local-code tradeoff, the relation between χ_{XZ} and the price of a code, bipolar mobility–distance bounds and their exponent consequences, and the tile-code comparison.

D. Repeated slabs, disjointness, and a shallow-unitary obstruction

We next use the stronger uniform-gap statement in a different way. The same bipolar mobility that saturates the crossbar bound also forces many low-overlap representatives of every logical Pauli. The result concerns a *static* layer-code block and ordinary unitary circuits on its physical qubits; it does not constrain adaptive measurements, injected resources, ancillas, or code deformation used later in the architecture. Figure S3 summarizes the geometric mechanism: many disjoint same-orientation slab ports give a growing disjointness invariant, which in turn feeds directly into the Jochym-O'Connor–Kubica–Yoder shallow-circuit theorem.

For the single-qubit partition, let $\mathcal{L}$ be the nontrivial logical Pauli cosets. Following Jochym-O'Connor, Kubica, and Yoder [3], define

$$d_\downarrow = \min_{G \in \mathcal{L}} d(G), \qquad d_\uparrow = \max_{G \in \mathcal{L}} d(G), \tag{P31}$$

where $d(G)$ is the minimum weight in the logical coset G. A set of representatives of one G is c-disjoint if each physical qubit lies in the support of at most c representatives, and

$$\Delta_c(G) = \frac{1}{c} \max\{|\mathcal{A}| : \mathcal{A} \subset G \text{ is } c\text{-disjoint}\},$$
$$\Delta = \max_{c \ge 1} \min_{G \in \mathcal{L}} \Delta_c(G). \tag{P32}$$

Proposition SI.15 (Bipolar mobility lower-bounds disjointness)**.** *Let an $[[n, K]]$ CSS code have r_X pairwise-disjoint complete pure-X ports and r_Z pairwise-disjoint complete pure-Z ports. For the single-qubit partition,*

$$\Delta(Q) \ge \frac{1}{2} \min\{r_X, r_Z\}. \tag{P33}$$

Proof. Put $r = \min\{r_X, r_Z\}$ and fix a nontrivial logical Pauli coset $G = [X(u)Z(v)]$. For $1 \le i \le r$, choose a pure-X representative $X_i(u)$ in the ith X port and a pure-Z representative $Z_i(v)$ in the ith Z port, omitting a factor when its polarization is trivial, and set

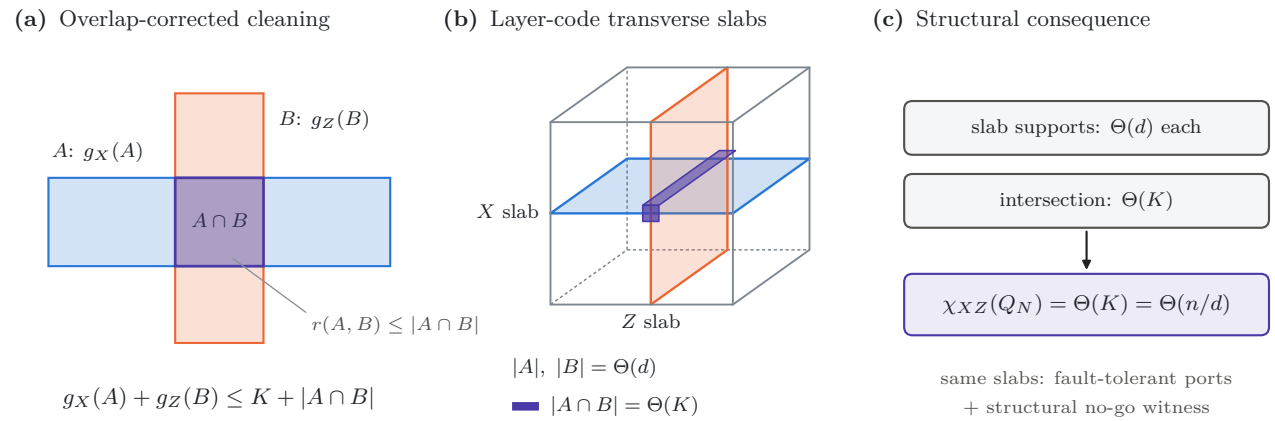


FIG. S2. Symplectic overlap of transverse logical ports. **(a)** General CSS overlap-cleaning picture: if region A supports $g_X(A)$ pure-X logical classes and region B supports $g_Z(B)$ pure-Z logical classes, then the commutation they realize is constrained by the physical overlap, giving $g_X(A) + g_Z(B) \le K + |A \cap B|$. In particular, any complete pure-X/pure-Z polarization pair obeys $\chi_{XZ}(Q) \ge K$. **(b)** In a layer code, the fixed X and Z slabs are both of size $\Theta(d)$ and intersect in a constant-cross-section tube of size $\Theta(K)$. **(c)** The native slab pair is therefore simultaneously efficient in support size and overlap size: it attains $\chi_{XZ}(Q_N) = \Theta(K) = \Theta(n/d)$ while also serving as the local logical port structure used by the fault-tolerance architecture.

$P_i = X_i(u)Z_i(v) \in G$. Each physical qubit occurs in at most one of the X_i and at most one of the Z_i, hence in at most two of the P_i. The P_i are distinct: in the mixed case $P_i = P_j$ would force the pure-X operator X_iX_j to equal the pure-Z operator Z_iZ_j up to phase, and therefore both to be identity, contradicting disjoint nonempty supports for $i \neq j$; the pure cases are immediate. Thus every nontrivial logical coset has a 2-disjoint set of size r, proving Eq. (P33). □

Theorem SI.16 (Layer-code disjointness and logical-weight symmetry)**.** *For the balanced good layer-code family considered in the architecture theorem (explicit realization, Theorem SI.6), with the single-qubit partition,*

$$\Delta(Q_N) = \Theta(N), \qquad d_\downarrow = \Theta(N^2), \qquad d_\uparrow = \Theta(N^2). \tag{P34}$$

In particular $d_\uparrow/d_\downarrow = \Theta(1)$.

Proof. Corollary SI.4 gives $r_X, r_Z = \Theta(N)$ mutually disjoint complete X- and Z-slab regions. Proposition SI.15 therefore gives $\Delta \ge \Omega(N)$.

For the reverse bound, Lemma 2 of Ref. [3] proves $\Delta \le n_{\text{phys}}/d_\uparrow$ for the single-qubit partition. Every logical Pauli class has a representative obtained by multiplying one fixed X-slab representative and one fixed transverse Z-slab representative; bounded density and balanced block dimensions give weight $O(N^2)$. Hence $d_\uparrow = O(N^2)$. Since $d_\downarrow$ is the ordinary code distance and the family has $d_\downarrow = \Theta(N^2)$, necessarily $d_\uparrow \ge d_\downarrow = \Omega(N^2)$, so $d_\uparrow = \Theta(N^2)$. With $n_{\text{phys}} = \Theta(N^3)$, the published upper bound gives $\Delta = O(N)$, completing the proof. □

Corollary SI.17 (Fixed-depth bounded-arity native unitaries are Clifford)**.** *Fix constants $q, h < \infty$. There is $N_0(q,h)$ such that, for every $N \ge N_0(q,h)$, any code-preserving logical unitary implemented on one static layer-code block by a q-local physical circuit of depth h lies in the logical Clifford group. The physical gates need not be geometrically local. In particular, sufficiently large blocks admit no within-block transversal logical T gate, and more generally no bounded-arity constant-depth within-block non-Clifford logical unitary.*

Proof. Theorem 9 of Jochym-O'Connor, Kubica, and Yoder states that a q-local depth-h circuit can implement only a level-M Clifford-hierarchy logical unitary whenever

$$d_\uparrow q^{(2^M-1)h} < d_\downarrow \Delta^{M-1} \tag{P35}$$

[3]. Set $M = 2$. Theorem SI.16 gives

$$\frac{d_\uparrow q^{3h}}{d_\downarrow \Delta} = O(N^{-1}) \longrightarrow 0. \tag{P36}$$

Thus Eq. (P35) holds for all sufficiently large N, and the induced logical unitary lies in the second level, the Clifford group. Transversal unitaries are the special case $q = 1, h = 1$. By contrast, the Bravyi–König geometric-locality theorem permits third-level logical gates in three dimensions in general [4]; the stronger restriction here comes from the growing disjointness and does not assume geometric locality of the circuit. □

More generally, Proposition SI.15 and the same Jochym-O'Connor–Kubica–Yoder criterion show that if $m = \min\{r_X, r_Z\}$ and $d_\uparrow/d_\downarrow \le \rho$, then the sufficient condition

$$m > 2\rho q^{3h} \tag{P37}$$

forces every q-local depth-h code-preserving logical unitary into the Clifford group. Thus growing two-polarization mobility simultaneously supplies many measurement ports and strengthens the obstruction to static shallow non-Clifford unitary gates.

Corollary SI.4 supplies the disjoint slab families used by this corollary.

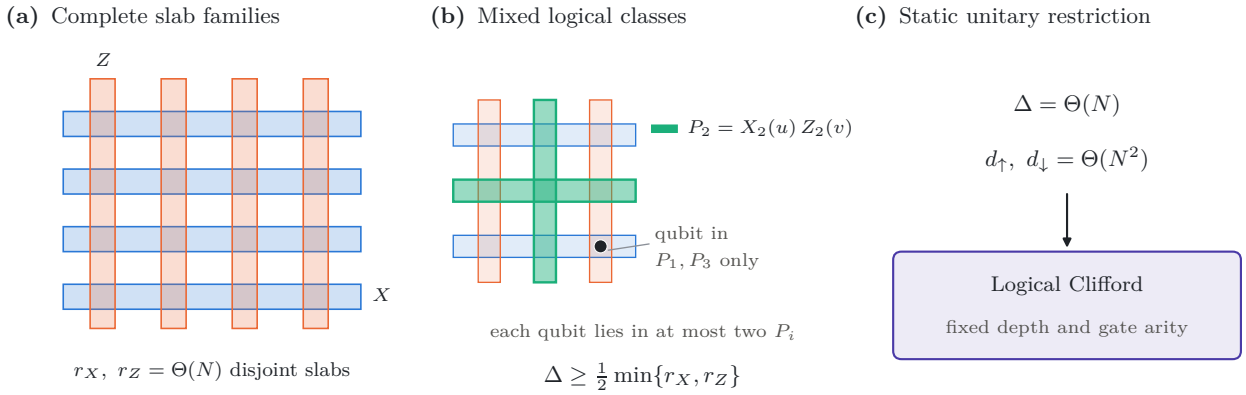


FIG. S3. From uniform slab mobility to a Clifford-only shallow-unitary obstruction. **(a)** Every height (for X) and every transverse position (for Z) supports every pure logical class, so balanced families provide $\Theta(N)$ mutually disjoint X-slab regions and $\Theta(N)$ mutually disjoint Z-slab regions. **(b)** For a logical Pauli coset $G = [X(u)Z(v)]$, taking the ith pure-X and pure-Z representatives gives $P_i = X_i(u)Z_i(v)$. Because any qubit can lie in at most one same-orientation slab of each type, the family $\{P_i\}$ is 2-disjoint. **(c)** Consequently $\Delta(Q_N) = \Theta(N)$ and $d_\uparrow, d_\downarrow = \Theta(N^2)$, so the Jochym-O'Connor–Kubica–Yoder criterion implies that any fixed-depth bounded-arity code-preserving unitary on a sufficiently large static layer-code block is logical Clifford.

SII. SLAB CERTIFICATE ON COMPLETE CHECK MATRICES

Theorem SI.3 and Proposition SI.2 are proved for arbitrary inputs. The script `slab_gluing_certificate.py` checks the same statements independently on complete finite codes, with negative controls. Every computation is exact GF(2) linear algebra on the complete check matrices produced by `layer_code_explicit.py`; no check is omitted and no boundary is treated as open. (Its part A re-examines an archived cell certificate of an earlier version; that material is in Sec. SXX.)

a. Part B: complete codes. For seven input codes (Table S5), including a code with $K = 7$, the surface code, and two random hypergraph-product codes, and for spacings $s = 2, 3$ where feasible, the script builds the complete explicit realization and verifies: all checks commute; the wall system of every crossing line is solved; $n - \#X - \#Z$ equals $n_{\text{in}} - m_X - m_Z$ and the relation spaces have the input dimensions (the two steps of Proposition SI.2); and $k = K$. The same inputs built in the single-chart transcription give valid CSS codes (a plain union of sheets is a cell complex), but with $k \neq K$ in every case.

b. Part C: gluing forms. The prefix operator is written as a sum of binary variables, one per data string (x_q) and one per backbone segment $(p_t^{(j)})$, each a fixed edge set. For every stabilizer the script computes the linear form in these variables given by its overlap parity and compares the set of nonzero forms with the first/middle/last relations of Eq. (P4). In the explicit realization the forms are exactly these relations, one face per incidence, at every height z_0 and, dually, at every column x_0 (Theorem SI.3(1)). In the single-chart transcription every middle incidence produces the two forms $x_{q_t} + p_{t-1}$ and $x_{q_t} + p_t$, i.e. the all-equal relation.

c. Part D: exhaustive words, negative controls, and logical checks. For a single check of weight $2 \leq k \leq 6$ every incidence word and every backbone assignment (2^{2k-1} assignments) is tried. In the explicit realization the zero-syndrome assignments are exactly the 2^{k-1} even words with their prefix backbones; every other assignment, including odd words and every single-segment flip, violates at least one stabilizer at the offending incidence. In the single-chart transcription only the constant words survive. For every input code, every basis vector of $\ker H_Z$ ($\ker H_X$), and every height (column), the slab operator commutes with every stabilizer; stabilizer inputs map to stabilizers; the pairing with the dual slabs equals the input pairing; and its rank is K.

d. Part E: point defects. Walls close exactly when the input checks commute. For five pairs of one X and one Z check with overlaps $2, 1, 3, 4, 3$ (built without the CSS requirement), the wall system is inconsistent for every odd overlap and solvable for every even one, as Lemma SI.1 states.

e. Rank criterion. For any finite CSS code the following criterion decides whether a region S supports every pure logical class; the certificate does not need it, because it constructs the representatives, but it remains a convenient independent test. With $\rho_X = \operatorname{rank} H_X$, $\rho_Z = \operatorname{rank} H_Z$ and $\overline{S}$ the complement, every pure-X class has a representative in S if and only if

$$|S| - \operatorname{rank}(H_Z|_S) - \rho_X + \operatorname{rank}(H_X|_{\overline{S}}) = k, \qquad \text{(P38)}$$

and dually

$$|S| - \operatorname{rank}(H_X|_S) - \rho_Z + \operatorname{rank}(H_Z|_{\overline{S}}) = k. \qquad \text{(P39)}$$

The left side of Eq. (P38) is the dimension of the image of $\ker H_Z \cap \mathbb{F}_2^S$ in $\ker H_Z / \operatorname{row} H_X$.

TABLE S5. Complete-code certificate (`slab_cert_B_complete_codes.csv`, `slab_cert_C_forms.csv`). "Forms" counts heights (columns) at which the nonzero gluing forms are exactly the prefix relations; "ops" counts slab operators, over all basis inputs and positions, that commute with every stabilizer; "rank" is the rank of the X/Z slab pairing. The last two columns are the single-chart transcription of the same input.

Input	s	n	K	k	anticomm.	X forms	Z forms	X ops	Z ops	rank	k (1-chart)	X ops (1-chart)
$[[4,2,2]]$	2	224	2	2	0	4/4	5/5	12/12	15/15	2	4	0/12
$[[4,2,2]]$	3	504	2	2	0	6/6	7/7	18/18	21/21	2	4	0/18
$[[6,4,2]]$	2	352	4	4	0	4/4	5/5	20/20	25/25	4	6	0/20
$[[6,4,2]]$	3	792	4	4	0	6/6	7/7	30/30	35/35	4	6	0/30
Steane $[[7,1,3]]$	2	1792	1	1	0	8/8	9/9	32/32	36/36	1	10	0/32
Steane $[[7,1,3]]$	3	4032	1	1	0	12/12	13/13	48/48	52/52	1	10	0/48
Hamming $[[15,7,3]]$	2	6600	7	7	0	10/10	11/11	110/110	121/121	7	29	0/110
surface $[[13,1,3]]$	2	10314	1	1	0	14/14	15/15	98/98	105/105	1	15	0/98
random HGP 1	2	10936	2	2	0	14/14	15/15	98/98	120/120	2	20	0/98
random HGP 2	2	10512	2	2	0	14/14	15/15	98/98	120/120	2	18	0/98

SIII. GRID EXPANSION AND LOCAL GAUGING

a. The idea in plain terms. Gauging measures a logical $\overline{P} = X(D)$ through many local vertex checks on an ancilla grid whose product is $\overline{P}$. The added qubits may create short logical operators, and the whole section controls one quantity that rules them out. An X logical of the deformed code can only place its ancilla part on a *wall*, the set δT of edges leaving some vertex set T. Multiplying by the vertex checks in T trades that wall for X on the data qubits attached in T, which turns the operator into a logical of the original code. This trade never helps a short operator if every wall has at least as many edges as the ports it separates, which is the relative expansion $\beta \geq 1$ (Proposition SIII.1). Conversely, a small wall between the ports of two blocks is itself a short logical (Proposition SIII.2): a thin interface fails. The slab theorem puts the ports in one plane, and in a three-dimensional box as thick as it is wide every wall is a surface whose area grows with the number of ports it separates, so β is a constant (Lemmas SIII.7 and SIII.8); a constant number of ancillas per edge then gives $\beta \geq 1$ (Corollary SIII.4).

b. Notation for ports and relative expansion. Let C be a CSS stabilizer code, let $\overline{P} = X(D)$ be a nontrivial X-type logical representative with data support D, and let $G = (W, E)$ be a connected graph. A port map is an injection $f : D \hookrightarrow W$, and we write $U = f(D)$. Vertices in $W \setminus U$ are called dummy vertices. For $T \subseteq W$, let δT be the set of edges with exactly one endpoint in T. For $U \subseteq W$ define

$$\beta(G,U) = \min_{\substack{T \subseteq W \\ \min\{|T\cap U|, |U\setminus T|\}>0}} \frac{|\delta T|}{\min\{|T\cap U|, |U\setminus T|\}}. \tag{P40}$$

We use the convention $\beta(G,U) = +\infty$ when $|U| \leq 1$.

c. Gauged generators. For every vertex u, let

$$A_u = X_{f^{-1}(u)} \prod_{e \ni u} X_e, \qquad u \in U, \tag{P41}$$

$$A_u = \prod_{e \ni u} X_e, \qquad u \notin U. \tag{P42}$$

Use $B_c = \prod_{e\in c} Z_e$ for a cycle-space generating set, retain the X checks of C, and replace a Z check s by $\widetilde{s} = s\prod_{e\in\mu_s} Z_e$, where $\partial\mu_s = f(\mathrm{supp}(s) \cap D)$. This boundary is even because s commutes with the measured logical. Connectedness of G supplies such μ_s. These are the usual gauging generators of Ref. [5].

d. Deformed-code distance. The following formulation separates data qubits from graph vertices and also records the two degenerate cases used below.

Proposition SIII.1 (Distance under gauging)**.** *Let C have CSS distances d_X, d_Z. Gauge the representative $\overline{P} = X(D)$ using a connected graph $G = (W, E)$ and an injective port map $f : D \hookrightarrow W$. Put $U = f(D)$ and $\beta = \beta(G, U)$. The deformed code is CSS and satisfies*

$$d_X(\widetilde{C}) \geq \min\{1, \beta\} d_X, \qquad d_Z(\widetilde{C}) \geq d_Z. \tag{P43}$$

Dummy vertices require no separate hypothesis: β in (P43) *is computed on the final graph, including all dummy vertices.*

If $D = \varnothing$, no logical is gauged. The graph ancilla generated by all vertex stars and by a basis of the cycle space is a stabilizer state, so the resulting code is C tensored with that state and has the same logical distances as C.

If $C = C_1 \otimes C_2$ and the measured Pauli is a product of one logical from each factor, the same conclusion holds with d_X and d_Z replaced by the minima of the corresponding distances of the two blocks. The statement with X and Z exchanged also holds.

With the Cheeger constant $h(G)$ in place of β, this is Lemma 2 (space fault-distance) in the Methods of the published version of Ref. [5]; the relative form below follows the same cut argument.

Proof. All displayed generators of the gauged code are of pure X or pure Z type, so the code is CSS. Let $L' = X(A)X(F)$ be an X-type logical of the deformed code, where A is a set of data qubits and $F \subseteq E$. Commutation with every cycle generator says that F is orthogonal to the cycle space. Because G is connected, the orthogonal complement of its cycle space is its cut space; hence $F = \delta T$ for some $T \subseteq W$. Multiplication by $\prod_{u\in T} A_u$

removes the edge support and gives

$$L'' = X\big(A \triangle f^{-1}(T \cap U)\big). \tag{P44}$$

This operator commutes with every check of C. If L' is nontrivial, neither L'' nor $L''\overline{P}$ is an X stabilizer of C; otherwise L' would be generated by the deformed stabilizers. Consequently both have weight at least d_X. Choose between T and $W \setminus T$ so that the chosen port side has cardinality $m = \min\{|T \cap U|, |U \setminus T|\}$. The corresponding one of L'' and $L''\overline{P}$ has support contained in the symmetric difference of A with a set of m data qubits. Put $b_0 = \min\{1, \beta\}$. For every cut, including a cut with $m = 0$, $|\delta T| \geq b_0 m$. Therefore

$$\begin{aligned} |L'| &= |A| + |\delta T| && \text{(P45)}\\ &\geq |A| + b_0 m && \text{(P46)}\\ &\geq \min\{1, \beta\}(|A| + m) && \text{(P47)}\\ &\geq \min\{1, \beta\} d_X. && \text{(P48)} \end{aligned}$$

Now let $L' = Z(B)Z(F)$ be a Z-type logical. The unchanged X checks imply that $Z(B)$ lies in the normalizer of C. Commutation with every vertex operator gives

$$\partial F = f(B \cap D), \tag{P49}$$

and hence $|B \cap D| = |\partial F| = 0 \pmod 2$. Thus $Z(B)$ commutes with $\overline{P}$. If $Z(B)$ is a stabilizer of C, multiplying L' by the corresponding deformed Z checks leaves a pure edge cycle, which is a product of the chosen cycle generators; in that case L' is trivial. Otherwise $|L'| \geq |B| \geq d_Z$.

For $D = \varnothing$, the graph ancilla has $|W| - 1$ independent vertex stars and $|E| - |W| + 1$ independent cycle generators, for a total of $|E|$ independent stabilizers on $|E|$ edge qubits. It therefore encodes no qubit. The product-code claim follows from the same proof because a nontrivial logical on $C_1 \otimes C_2$ has weight at least the minimum of the two component distances. □

e. Why added measurement layers can lose distance. Proposition SIII.1 is a lower bound. The converse below shows that the relative expansion is the quantity that must be controlled, and it makes precise the concern that attaching an additional measurement layer to a code can lower its fault distance. When the ports of two independent logical factors are separated by a small cut of the ancilla graph, the deformed code acquires a correspondingly short logical operator.

Proposition SIII.2 (Cut obstruction)**.** *In the setting of Proposition SIII.1, suppose that $D = D_1 \sqcup D_2$, that $X(D_1)$ commutes with every check of C, and that $X(D_1) \notin \langle S_X(C), \overline{P}\rangle$. If $T \subseteq W$ satisfies $T \cap U = f(D_1)$, then $X(\delta T)$ is a nontrivial X-type logical operator of the deformed code and represents the same class as $X(D_1)$. Consequently*

$$d_X(\widetilde{C}) \leq \min\big\{|\delta T| :\ T \subseteq W,\ T \cap U = f(D_1)\big\}, \tag{P50}$$

the size of a minimum edge cut of G separating $f(D_1)$ from $f(D_2)$.

Proof. Because $T \cap U = f(D_1)$, the product of the vertex operators in T is $\prod_{u\in T} A_u = X(D_1)X(\delta T)$. The operator $X(D_1)$ has no edge support and commutes with every check of C. It therefore commutes with every cycle generator B_c and with every deformed check $\widetilde{s}$, whose data part is s. Thus $X(D_1)$ lies in the deformed normalizer, and so does $X(\delta T) = X(D_1)\prod_{u\in T} A_u$, which represents the same class. Suppose this class were trivial. Then $X(D_1) = \sigma \prod_{u\in T'} A_u$ for some $\sigma \in S_X(C)$ and $T' \subseteq W$. The edge support of the right side is $\delta T'$, which must vanish, so connectedness gives $T' \in \{\varnothing, W\}$. Since $\prod_{u\in W} A_u = \overline{P}$, this places $X(D_1)$ in $\langle S_X(C), \overline{P}\rangle$, contrary to assumption. □

Corollary SIII.3 (Thin interfaces for joint measurements)**.** *Let $C = C_a \otimes C_b$ and gauge $\overline{P} = \overline{X}_a\overline{X}_b$, where $\overline{X}_a = X(D_a)$ and $\overline{X}_b = X(D_b)$ are nontrivial logical operators of the two blocks. Then $d_X(\widetilde{C})$ is at most the minimum number of ancilla edges whose removal separates $f(D_a)$ from $f(D_b)$ in G, however large the distances of C_a and C_b are.*

Proof. Apply Proposition SIII.2 with $D_1 = D_a$. The operator $X(D_a)$ commutes with every check of $C_a \otimes C_b$. It is not in $S_X(C)$, because its restriction to block a is the nontrivial $\overline{X}_a$. It is not in the coset $S_X(C)\overline{P}$, because every element of that coset restricts to $\overline{X}_b$ times a stabilizer on block b, whereas $X(D_a)$ acts trivially there. □

f. Geometric consequence for layer codes. For a geometrically local ancilla of bounded density, every path joining the two port sets crosses any surface separating the two blocks. If the ancilla joins two adjacent blocks only through a layer of thickness h and width w, Corollary SIII.3 therefore gives $d_X(\widetilde{C}) = O(hw)$. For the balanced layer codes, a block has linear size $\Theta(N)$ and $d = \Theta(N^2)$ is of the same order as the area of one block face. This is generic for distance-optimal three-dimensional codes, since $d = O(L^2)$ for any local stabilizer code of linear size L in three dimensions [6]. A measurement layer of bounded thickness thus loses a factor $\Theta(N)$ of distance in every joint measurement, and a joint measurement that keeps distance $\Theta(d)$ must cross a constant fraction of the full $\Theta(N^2)$ interface between the blocks. Joint measurements are unavoidable in the gate set of Sec. SIV (CNOT and phase teleportation) and in resource transfer (Sec. SV). The grid box of Theorem SIII.11 spans both blocks, so its separating cuts have size $\Omega(N^2)$; this is the content of Corollary SIII.9. Corollary SIII.3 does not exclude thin layers for measuring a single logical inside one block, but the construction does not rely on that case.

g. Explicit check. Table S6 applies Propositions SIII.1 and SIII.2 to two periodic three-dimensional toric codes of linear size L. Their X logical is a membrane of weight L^2 supported in one plane, a small stand-in for a slab representative. The joint logical $\overline{X}_a\overline{X}_b$ is gauged with a $2L \times L \times H$ cubic-grid ancilla whose plane number $\lfloor H/2 \rfloor$ (counting the H planes from

TABLE S6. Deformed X distance for the joint measurement $\overline{X}_a\overline{X}_b$ of two periodic three-dimensional toric codes, bracketed between the relative-expansion lower bound and the explicit cut logical (or $\overline{X}_a$). Here H is the ancilla thickness and m the number of ancilla qubits per grid edge. Source: `measurement_interface_check.csv`.

L	d_X	$H=1$	$H=2$	$H=L$	$H=L$, $m=2$
3	9	$[2,3]$	$[3,6]$	$[5,9]$	9
4	16	$[2,4]$	$[4,8]$	$[8,16]$	16
5	25	$[3,5]$	$[5,10]$	$[13,25]$	25
6	36	$[3,6]$	$[6,12]$	$[18,36]$	36

zero) contains both membranes. For each instance the script `measurement_interface_check.py` constructs the deformed code, checks $H_X H_Z^{\mathsf{T}} = 0$, and verifies over GF(2) that $X(\delta T)$ is a nontrivial deformed logical equivalent to $\overline{X}_a$, while $\overline{X}_a\overline{X}_b$ is a deformed stabilizer. The lower end of each bracket is Proposition SIII.1 with $\beta \geq m\min\{1, H/(2L)\}$ (Lemmas SIII.7 and SIII.8 with one port plane). A single added sheet ($H=1$) leaves a deformed distance between $L/2$ and L, whereas the thick box keeps at least $L^2/2$. With two ancillas per grid edge (Corollary SIII.4) the thick box keeps exactly L^2. For $L=3$ an exact CP-SAT computation closes every bracket at its upper end: the deformed distances are 3, 6 and 9 for $H=1,2,3$ (Table S8).

h. Full distance at constant ancilla multiplicity. Proposition SIII.1 loses the factor $\min\{1,\beta\}$ only when $\beta<1$. Because the grid relative expansion is a positive constant, a constant number of parallel ancillas per edge removes this loss.

Corollary SIII.4 (Edge multiplicity restores the full distance)**.** *Let $G^{(m)}$ be obtained from the connected graph G by replacing every edge with m parallel edges, each carrying its own ancilla qubit. Then $\beta(G^{(m)},U) = m\,\beta(G,U)$. Gauge $\overline{P}$ on $G^{(m)}$ with vertex operators acting on all parallel copies, pairing paths on one designated copy of each edge, and cycle generators consisting of the chosen cycle generators of G on the designated copies together with the weight-two bundle cycles $Z_eZ_{e'}$ of parallel edges. If $m \geq 1/\beta(G,U)$, then*

$$d_X(\widetilde{C}) \geq d_X, \qquad d_Z(\widetilde{C}) \geq d_Z. \tag{P51}$$

The same holds in the $X \leftrightarrow Z$ dual convention, and the factor $\min\{1,\beta\}$ in Lemmas SIII.5 and SIII.6 is likewise replaced by $\min\{1,m\beta\}$.

Proof. Every cut of $G^{(m)}$ consists of all m copies of the edges of the corresponding cut of G, so $|\delta^{(m)}T| = m|\delta T|$ for all $T \subseteq W$, and the relative expansion scales by m. The displayed generators span the cycle space of $G^{(m)}$: replacing every copy of an edge by its designated copy using bundle cycles leaves an even-degree edge set of G. The proof of Proposition SIII.1 uses only connectedness and the fact that the cut space is the orthogonal complement of the cycle space. Both hold for multigraphs, since an edge set orthogonal to every bundle cycle contains all or none of the copies of each edge. Hence $d_X(\widetilde{C}) \geq \min\{1,m\beta\}d_X = d_X$. The Z bound is unchanged, and the two quotient lemmas use the same cut estimate. □

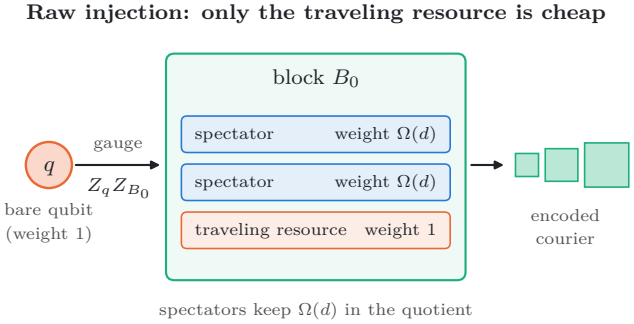


FIG. S4. Bare-port intuition. Raw injection creates a weight-one traveling-resource class, while all spectator logicals remain $\Omega(d)$ in the quotient (Lemma SIII.5). Only encoded courier stages appear downstream.

i. Raw-injection intuition. Before stating the bare-port quotient lemma, record the intended picture (Fig. S4): appending an unchecked physical resource qubit creates a weight-one logical class, but that class is precisely the *intentional traveling resource*. Every spectator logical of the data block remains at distance $\Omega(d)$ after gauging, by the quotient statement below; subsequent encoded-courier stages never reintroduce a bare port.

j. A bare injection port. The distance-one product code obtained by appending an unchecked qubit cannot be inserted directly into Proposition SIII.1 to obtain a block-distance bound. The required statement is a relative distance assertion.

Lemma SIII.5 (Bare-port quotient distance)**.** *Let C_b be a CSS block with X and Z distances at least d, append an unchecked qubit q, and gauge $M = \overline{Z}Z_q$ using the $X \leftrightarrow Z$ dual convention on a connected graph with relative expansion β. In the deformed logical Pauli group, every class outside the subgroup generated by $[Z_q]$ has minimum weight at least $\min\{1,\beta\}d$. The exceptional class has the weight-one representative Z_q, and the new stabilizer relation $M=1$ identifies its action with that of $\overline{Z}$ on the block. The analogous statement holds with X and Z exchanged.*

Proof. In this dual convention the Z-type logicals admit the cut-space reduction. Multiplying by the Gauss operators removes the edge support and produces a Z normalizer on $C_b \otimes q$, with the two complementary reductions differing by M. For a class outside $\langle[Z_q]\rangle$, the block part of each reduction is outside $\langle S_Z(C_b), \overline{Z}\rangle$: otherwise the reduced operator, and hence the original class, is generated by M, Z_q, and stabilizers. Each block part consequently has weight at least d. Choosing the smaller

port side and using the preceding finite $b_0 = \min\{1, \beta\}$ estimate proves the $b_0 d$ bound for these Z classes.

For an X-type logical, the cycle-space reduction restricts its data part to $X(B)X_q^b$, where $X(B)$ is a block normalizer. Commutation with the product of the dual Gauss operators gives $b = \langle X(B), \overline{Z}\rangle$. If $X(B)$ is a block stabilizer, then $b = 0$ and the remaining edge cycle is trivial modulo the deformed checks. Thus every nontrivial X logical has a nontrivial block restriction and weight at least d.

The code is CSS, so any Pauli normalizer factors as $L = L_X L_Z$, with both pure components separately in its normalizer. If $[L] \notin \langle [Z_q]\rangle$, either $[L_X]$ is nontrivial or $[L_Z] \notin \langle [Z_q]\rangle$. The corresponding component has weight at least $b_0 d$, and $\mathrm{wt}(L) = |\operatorname{supp}(L_X) \cup \operatorname{supp}(L_Z)|$ is at least that component's weight. This proves the mixed-Pauli assertion too.

Finally Z_q commutes with every deformed generator. It is nontrivial: choose a block X logical anticommuting with $\overline{Z}$, append X_q, and extend it over edges using connectedness to solve the even boundary constraint. The resulting deformed normalizer anticommutes with Z_q. The stabilizer M gives $[Z_q] = [\overline{Z}]$. □

k. Encoded ports of unequal distance. Resource transfer joins a small courier block to a larger block (Sec. SV). Proposition SIII.1 then gives only the smaller distance. The quotient statement below shows that the smaller block exposes only its own logical classes; every other logical class keeps the distance of the larger block. Lemma SIII.5 is the special case in which C_1 is a single unchecked qubit.

Lemma SIII.6 (Encoded-port quotient distance)**.** *Let C_1 and C_2 be CSS codes, where C_2 has X and Z distances at least d_2 and C_1 is arbitrary. Let $\overline{Z}_1 = Z(D_1)$ and $\overline{Z}_2 = Z(D_2)$ be nontrivial logical representatives, and gauge $M = \overline{Z}_1\overline{Z}_2$ on $C_1 \otimes C_2$ in the $X \leftrightarrow Z$ dual convention, using a connected graph with relative expansion β with respect to $U = f(D_1 \sqcup D_2)$. Let $\mathcal{Q}_1$ be the group of deformed logical classes obtained from logical operators of C_1 that commute with $\overline{Z}_1$; an X-type operator $X(B_1)$ is completed by edge operators $X(F)$ with $\partial F = f(B_1 \cap D_1)$. Then $[\overline{Z}_2] = [\overline{Z}_1] \in \mathcal{Q}_1$, and every deformed logical class outside $\mathcal{Q}_1$ has minimum weight at least $\min\{1, \beta\} d_2$.*

Proof. In the dual convention the vertex operators are $A_u = Z_{f^{-1}(u)} \prod_{e \ni u} Z_e$ (without the data factor at dummy vertices), the cycle generators are X-type, the Z checks of $C_1 \otimes C_2$ are retained, and each X check s is replaced by $\widetilde{s} = s \prod_{e \in \mu_s} X_e$. The relation $\prod_u A_u = M$ gives $[\overline{Z}_2] = [\overline{Z}_1]$.

X-type classes. Let $L' = X(B)X(F)$ be a deformed logical operator and write $X(B) = X(B_1)X(B_2)$ by block. Commutation with the vertex operators gives $\partial F = f(B \cap D)$, so $X(B)$ commutes with M, and commutation with the retained Z checks shows that $X(B)$ commutes with every check of $C_1 \otimes C_2$. Two deformed operators with the same data part differ by an edge cycle, and data parts that differ by an X stabilizer σ of $C_1 \otimes C_2$ are related by the deformed check $\widetilde{\sigma}$ up to a cycle. Hence $[L']$ depends only on the logical class of $X(B)$. If $X(B_2)$ is a stabilizer of C_2, then $X(B_1)$ commutes with $\overline{Z}_1$ and $[L'] \in \mathcal{Q}_1$. Otherwise $X(B_2)$ is a nontrivial logical operator of C_2, and $|L'| \geq |B_2| \geq d_2$.

Z-type classes. Let $L' = Z(A)Z(F)$. Commutation with the X-type cycle generators makes F orthogonal to the cycle space, so $F = \delta T$ for some $T \subseteq W$. Multiplying by $\prod_{u \in T} A_u$ or by $\prod_{u \in W \setminus T} A_u$ gives two data reductions, L'' and $L''M$. Both are Z-type operators commuting with every check of $C_1 \otimes C_2$, and both represent $[L']$. If the C_2 part of L'' lies in $\langle S_Z(C_2), \overline{Z}_2\rangle$, then L'' is a Z-type operator on C_1 times a stabilizer times a power of $\overline{Z}_2$. A Z-type operator on C_1 commutes with $\overline{Z}_1$ and needs no edge completion, so $[L'] \in \mathcal{Q}_1$. Otherwise the C_2 parts of L'' and of $L''M$ are both nontrivial logical operators of C_2, so both reductions have weight at least d_2. The reduction associated with the smaller port side is supported in the symmetric difference of A with a set of $m = \min\{|T \cap U|, |U \setminus T|\}$ data qubits. As in Proposition SIII.1, with $b_0 = \min\{1, \beta\}$,

$$|L'| = |A| + |\delta T| \geq |A| + b_0 m \geq b_0(|A| + m) \geq b_0 d_2.$$

Mixed classes. The deformed code is CSS, so every normalizer element factors as $L = L_X L_Z$ with both factors in the normalizer. The group $\mathcal{Q}_1$ is generated by pure-type classes, because the X part of a logical operator of C_1 commuting with $\overline{Z}_1$ also commutes with $\overline{Z}_1$. A class outside $\mathcal{Q}_1$ therefore has a pure component outside $\mathcal{Q}_1$, and the bound for that component applies to the whole operator. □

l. Cheeger bound for a rectangular grid.

Lemma SIII.7. *Let $G_0 = [a] \,\square\, [b]$ be the rectangular grid with $a \leq b$. For every $S \subseteq V(G_0)$ with $|S| \leq ab/2$,*

$$|\delta S| \geq \frac{|S|}{b}. \tag{P52}$$

In particular, $h(G_0) \geq 1/b$.

Proof. Call a row or column full, empty, or mixed according to its intersection with S. If a full and an empty row both exist, every column contains a boundary edge, so $|\delta S| \geq b \geq |S|/b$. If no row is full, every row meeting S contains a boundary edge, and at least $|S|/b$ rows meet S. In the remaining case some row is full and no row is empty. Every column meets S, while at most $|S|/a \leq b/2$ columns are full. Hence at least $b/2$ columns are mixed. Since $b/2 \geq |S|/a \geq |S|/b$, the result follows. □

m. Thickening relative to several layers.

Lemma SIII.8 (Thickening)**.** *Let $G = G_0 \square P_\ell$, where G_0 is connected and P_ℓ is a path on ℓ vertices. If $U \subseteq V(G)$ is contained in the union of at most t layers, then*

$$\beta(G, U) \geq \frac{\min\{1, \ell h(G_0)\}}{2t}. \tag{P53}$$

If U is contained in a single layer, the factor 2 can be omitted.

Proof. Fix a set $T \subseteq V(G)$ occurring in (P40). Let $T_j \subseteq V(G_0)$ be its intersection with layer j. Put $J = \{j : U \text{ meets layer } j\}$, so $|J| \le t$, choose $j_0 \in J$, and define

$$m = \max_j |T_j \triangle T_{j_0}|, \qquad \mu = \min\{|T_{j_0}|, |V(G_0) \setminus T_{j_0}|\}. \tag{P54}$$

For each vertex of $T_j \triangle T_{j_0}$, the vertical column joining layers j and j_0 contains an edge of δT. These columns are distinct, so the vertical part of the boundary has size at least m. For every j,

$$\min\{|T_j|, |V(G_0) \setminus T_j|\} \ge \max\{0, \mu - m\}. \tag{P55}$$

Summing the Cheeger bound over all layers shows that the horizontal part has size at least $\ell h(G_0) \max\{0, \mu - m\}$. The horizontal and vertical boundary edges are disjoint. With $c = \min\{1, \ell h(G_0)\}$,

$$|\delta T| \ge m + c \max\{0, \mu - m\} \ge c \max\{\mu, m\}. \tag{P56}$$

On the other hand,

$$|T \cap U| \le |J|\big(|T_{j_0}| + m\big), \tag{P57}$$
$$|U \setminus T| \le |J|\big(|V(G_0) \setminus T_{j_0}| + m\big). \tag{P58}$$

Therefore

$$\min\{|T \cap U|, |U \setminus T|\} \le |J|(\mu + m) \le 2t \max\{\mu, m\}. \tag{P59}$$

Combining this with (P56) proves (P53). If $t = 1$, choose j_0 to be the unique occupied layer. Then the denominator is at most μ, so the factor 2 is unnecessary. □

n. Cubic boxes.

Corollary SIII.9. *Let $B = [a_1] \times [a_2] \times [a_3]$ be a cubic-lattice box, let $a_{\max} = \max_i a_i$ and $a_{\min} = \min_i a_i$, and let Γ_B be its grid graph. If U is contained in at most t coordinate planes perpendicular to one axis, then*

$$\beta(\Gamma_B, U) \ge \frac{a_{\min}}{2t a_{\max}}. \tag{P60}$$

Proof. Take G_0 to be the rectangular cross-section parallel to the occupied planes and let ℓ be the side perpendicular to them. By Lemma SIII.7, $h(G_0) \ge 1/a_{\max}$, while $\ell \ge a_{\min}$. Since $a_{\min}/a_{\max} \le 1$, Lemma SIII.8 gives the claim. □

o. Injective placement of ports.

Lemma SIII.10 (Reserved local port labels)**.** *Assume bounded qubit and check density per coarse lattice cell. Give each possible port in a coarse cell its complete local label: the layer species, the subsite in that cell, and the local check-incidence label. Let $\Delta_p = O(1)$ be the maximum number of such labels in a cell. There is a constant refinement factor λ_{ref} and an injective map from all data ports to refined-grid vertices such that every port is moved by at most a constant number r_1 of coarse lattice units.*

Proof. Choose λ_{ref} so that each refined coarse cell contains at least Δ_p reserved microvertices. Fix an injection from the finite set of local labels to those microvertices, and use the same lookup rule in every coarse cell. Ports with different local labels in one cell have different images, while ports in different coarse cells lie in disjoint refined blocks. The map is therefore injective. Its displacement is bounded by the diameter of one refined coarse cell, which is a constant independent of the code size. No requirement that the image remain in the same coordinate plane is needed; bounded normal displacement only enlarges the number of occupied planes by a constant. □

p. Local pairing paths and congestion. For each bounded-weight check s, pair the even number of ports in $f(\mathrm{supp}_Z(s) \cap D)$ by a fixed lookup rule for its local check geometry. Join each pair by a shortest refined-grid path inside a fixed-radius neighborhood of $\mathrm{supp}(s)$. There are only finitely many local check geometries, so the path length is bounded by a uniform constant ℓ_μ. Moreover, if an edge e lies on such a path, then the corresponding check lies in a fixed-radius ball about e. Bounded check density therefore implies that at most a constant c_μ of these paths use any one edge. This supplies the congestion conclusion without assuming that all ports of a check lie in a unique plane.

q. Local gauging theorem.

Theorem SIII.11. *Assume that the measured pure-type logical representatives are supported in at most $t_0 = O(1)$ coarse coordinate planes per block, with a common normal axis for a given Pauli type across the participating blocks. Consider one block, two equal-size blocks at a fixed bounded cell separation, or a courier pair consisting of a block and a smaller block placed against one of its faces. Use the refined cubic-grid box spanning the participating blocks, with $m \ge 1$ parallel ancilla qubits on every grid edge, the port map of Lemma SIII.10, unit-square and bundle cycle generators, and the local pairing paths described above. Suppose all such boxes have aspect ratio at most $A = O(1)$. Then there are constants $\beta_1 > 0$ and $w_0 < \infty$, independent of block size, such that with $\beta_0 = \min\{1, m\beta_1\}$:*

1. *every deformed stabilizer has weight at most w_0 and bounded diameter, every qubit belongs to at most w_0 stabilizers, and the ancilla contains $O(mN^3) = O(n)$ qubits;*
2. *without a bare injection qubit, the deformed distance is at least $\beta_0 d$, where d is the least distance of the blocks involved. In particular, the multiplicity $m_0 := \lceil 1/\beta_1 \rceil$ gives $\beta_0 = 1$, so the deformed code keeps the full distance d;*
3. *with one bare injection qubit, the deformed code has distance one, but every logical class outside the injected-qubit subgroup has weight at least $\beta_0 d$;*
4. *for a courier pair with larger-block distance d_{large}, every logical class outside the group $\mathcal{Q}_1$ of*

Lemma SIII.6 generated by the smaller block has weight at least $\beta_0 d_{\mathrm{large}}$;

5. *the protocol implements the projective measurement of the selected logical Pauli.*

Proof. A refined-grid vertex has degree at most six, so a vertex Gauss operator has weight at most $1+6m$. A unit-square cycle operator has weight four and a bundle cycle weight two. Let $w_C = O(1)$ be the maximum check weight of the participating blocks. A modified data check has weight at most $w_C + \lfloor w_C/2 \rfloor \ell_\mu = O(1)$, because its at most w_C ports are joined in pairs by paths of length at most ℓ_μ. Each edge belongs to two vertex stars, at most four unit squares, at most $m-1$ bundle cycles, and at most $c_\mu = O(1)$ modified data checks. The analogous incidence bounds for vertices and data qubits follow from bounded check density. All generators have bounded geometric diameter. Unit squares generate the cycle space of a rectangular grid box: plaquettes may be successively peeled from any cycle because the box is contractible. The box has $\Theta(mN^3)$ edge qubits, and the layer code has $n = \Theta(N^3)$ qubits. For a courier pair, the box of the larger block extended across the adjacent face has aspect ratio at most twice that of the larger block's box.

Let r_1 be the maximum port displacement in coarse units. After constant refinement, the image of a representative from at most two blocks is contained in at most

$$t_p \le 2t_0\lambda_{\mathrm{ref}}(2r_1+3) = O(1) \tag{P61}$$

refined coordinate planes. The harmless additive constant covers cell boundary conventions. Since the box aspect ratio is at most A, Corollary SIII.9 yields, for the grid with single edges,

$$\beta(G,U) \ge \frac{1}{2At_p} =: \beta_1 > 0, \tag{P62}$$

and Corollary SIII.4 gives $\beta(G^{(m)},U) \ge m\beta_1$. Proposition SIII.1 therefore gives ordinary deformed distance at least

$$\beta_0 d, \qquad \beta_0 = \min\{1, m\beta_1\}. \tag{P63}$$

For two blocks, use the product-code clause of that proposition; for a courier pair, Lemma SIII.6 gives item 4. For a bare injection qubit, Lemma SIII.5 gives the claimed relative distance and identifies the exceptional injected logical subgroup (every other class has weight at least $\beta_0 d$). In the ideal protocol, initialize edge qubits in $|0\rangle$, measure the deformed checks, read the product of the Gauss outcomes, measure the edge qubits in Z, and update the data Pauli frame from those edge outcomes. Finally, the standard gauging-measurement argument of Ref. [5] applies because the graph is connected and the port map is injective: the product of all Gauss operators is the measured Pauli, initialization and final edge readout remove the ancilla, and the recorded Gauss parity is its projective measurement outcome. □

Clifford+T from pure-X/Z measurements

Gate	Measurements / resources	Correction
S	phase teleportation $+\,\|Y\rangle$	$Z^{m\oplus n}X^m$
T	phase teleportation $+\,\|A\rangle$	$S^m Z^n X^m$
H	$S \to S_X \to S$ $(3\,\|Y\rangle)$	Pauli frame
CNOT	$Z_c Z_a,\ X_a X_t,\ Z_a$	$Z_c^v X_t^{u\oplus w}$
CZ	$H_t \cdot \mathrm{CNOT} \cdot H_t$	$\le 6\,\|Y\rangle$

Only pure-X or pure-Z logical measurements are required.

FIG. S5. Measurement-based compilation of S, T, H, CNOT, and CZ from pure-X/Z logical measurements, $|Y\rangle/|A\rangle$ resources, and Pauli-frame updates. The displayed circuits are ideal identities; their fault-tolerant realization uses the geometric and spacetime conditions developed in Secs. SIII–SV.

SIV. PURE-X/Z COMPILATION OF CLIFFORD+T

The following formulas make the ideal logical gate algebra explicit. Fault-tolerant realization additionally depends on the geometric and spacetime conditions developed in Secs. SIII and SV. An outcome bit $m=0$ denotes eigenvalue $+1$, and $m=1$ denotes eigenvalue -1. Products of correction operators act from right to left. All formulas use pure-X or pure-Z measurements and Pauli frame updates. Output wires and candidate resource costs are specified below.

a. Phase teleportation. Let $R_\theta = \mathrm{diag}(1, e^{i\theta})$ and prepare an auxiliary logical qubit a in $R_\theta|+\rangle$. Measure $Z_d Z_a$ with bit m, then X_d with bit n, discard d, and retain a. Apply

$$C_{mn} = R_{2\theta}^m Z_a^n X_a^m. \tag{P64}$$

The output is $R_\theta|\psi\rangle$ up to a global phase. For $\theta = \pi/2$ this implements S using one $|Y\rangle$ with correction $Z^{m\oplus n}X^m$. For $\theta = \pi/4$ it implements T using one $|A\rangle$, with correction $S^m Z^n X^m$. The possible S correction uses at most one further $|Y\rangle$ and another phase teleportation. Preallocate that candidate resource when a branch-independent resource count is needed.

Proof. Write $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$. The unnormalized retained state after the two measurements is

$$m=0: \quad \tfrac{1}{2}\big(\alpha|0\rangle + (-1)^n e^{i\theta}\beta|1\rangle\big), \tag{P65}$$

$$m=1: \quad \tfrac{1}{2}\big((-1)^n\beta|0\rangle + e^{i\theta}\alpha|1\rangle\big). \tag{P66}$$

Applying Eq. (P64) gives respectively $R_\theta|\psi\rangle/2$ and $e^{i\theta}R_\theta|\psi\rangle/2$. Each branch has probability $1/4$, independent of the input, so the same identities hold with an arbitrary reference entangled with the input. □

b. An X-axis phase and a Hadamard. Define $S_X = HSH$. Prepare a in $Z|Y\rangle$, measure X_dX_a with bit m, then Z_d with bit n. Retain a and apply $X^{m\oplus n}Z^m$. This implements S_X up to a global phase: it is the X/Z-exchanged phase-teleportation identity, since $H|Y\rangle = e^{i\pi/4}Z|Y\rangle$. Therefore the sequence

$$S \longrightarrow S_X \longrightarrow S \tag{P67}$$

implements H up to a global phase, because

$$SS_XS = e^{i\pi/4}H. \tag{P68}$$

This uses three $|Y\rangle$ states and requires no physical Hadamard as an input primitive.

c. CNOT. For control c, target t, and ancilla a initialized to $|+\rangle$, measure in this order

$$Z_cZ_a\ (u), \qquad X_aX_t\ (v), \qquad Z_a\ (w). \tag{P69}$$

Discard a and retain (c,t). The correction is

$$Z_c^vX_t^{u\oplus w}. \tag{P70}$$

For a direct verification, write $\Pi_m(P) = (I+(-1)^mP)/2$. The unnormalized branch map before correction is

$$K_{uvw} = \langle w|_a\,\Pi_v(X_aX_t)\Pi_u(Z_cZ_a)\,|+\rangle_a. \tag{P71}$$

Expanding the two projectors gives

$$Z_c^vX_t^{u\oplus w}K_{uvw} = \frac{e^{i\phi_{uvw}}}{2\sqrt{2}}\,\mathrm{CNOT}_{c\to t} \tag{P72}$$

for a branch-dependent global phase $e^{i\phi_{uvw}}$. This establishes the channel identity, with probability $1/8$ for each of the eight branches. No resource state is required.

d. CZ and universality. Apply H_t, then $\mathrm{CNOT}_{c\to t}$, then H_t. This implements CZ with at most six $|Y\rangle$ states using the preceding constructions. Together with preparation of $|0\rangle$ and $|+\rangle$, the verified CNOT, H, S and T formulas generate Clifford+T. All parity measurements involve at most two retained or ancillary logical qubits and can be scheduled as the one-block or two-block measurements covered by the architecture. Each teleported output must be assigned to its new logical wire and its Pauli frame carried forward.

e. Phase teleportation with a defective resource. The same computation with an arbitrary resource shows what a bad candidate can do. Let the auxiliary qubit a and a system E be in any joint state $|\Phi\rangle_{aE} = |0\rangle_a|\phi_0\rangle + |1\rangle_a|\phi_1\rangle$ (a mixed resource, or one entangled with other resources, is covered by taking E to contain a purifying system). In branch (m,n) the retained state is, before correction,

$$\tfrac{1}{\sqrt{2}}\sum_{b\in\{0,1\}}(-1)^{nb}\,\psi_b\,|b\oplus m\rangle_a\otimes|\phi_{b\oplus m}\rangle_E, \tag{P73}$$

where $|\psi\rangle = \psi_0|0\rangle + \psi_1|1\rangle$. After X_a^m this is $\sum_b \psi_b|b\rangle_a \otimes |\epsilon_b\rangle_E$ for vectors $|\epsilon_b\rangle$ independent of ψ, that is $(I\otimes|\epsilon_+\rangle + Z\otimes|\epsilon_-\rangle)|\psi\rangle$ with $|\epsilon_\pm\rangle = (|\epsilon_0\rangle \pm |\epsilon_1\rangle)/2$. The remaining corrections $R_{2\theta}^mZ^n$ are diagonal. Hence, relative to the ideal output $R_\theta|\psi\rangle$, every branch applies an operator in $\mathrm{span}\{I,Z\}\otimes\mathcal{B}(\mathcal{H}_E)$ on the retained wire, whatever the resource state and whatever it is entangled with. The same holds when the S correction of a T teleportation is itself teleported with a defective $|Y\rangle$ slot, because a product of diagonal operators is diagonal. Section SV uses this in place of a resource-basis twirl (see Lemma SV.3).

f. Algebraic verification. The script `reproducibility_checks.py` evaluates all four S branches, four T branches, four S_X branches, and eight CNOT branches. It checks proportionality to the target unitary, branch probabilities, and the Hadamard composition to tolerance 10^{-12}. The script `resource_defect_check.py` checks Eq. (P73) for random resource states entangled with an environment. The symbolic identities above establish the channel formulas, and the finite matrix checks provide reproducible cross-checks of every branch.

SV. FAULT TOLERANCE AND UNIVERSAL ARCHITECTURE THEOREM

a. The argument in plain terms. This section is long because it handles every operation of the schedule, but its shape is short, and the main text (Secs. III B and IV A there) states it in words. An undetected logical error needs a fault pattern that no detector notices. Split such a pattern into pieces connected through common detectors; at least one piece causes the error. (i) A piece that lies inside one operation and its surrounding idle rounds needs at least $\min\{\beta_0 d, R\}$ faults: either the errors it leaves form a logical of the code at one time, every qubit of which carries a fault, or a wrong outcome is repeated through all R rounds (Lemma SV.18 and the direct slice argument of Proposition SV.17). (ii) A piece that reaches from one operation to the next must stay on one block for the whole padding between them, and two connected faults are at most $\tau = O(1)$ rounds apart, so it needs about R_{pad}/τ faults (Lemma SV.19 and Theorem SV.20); the same lemma keeps the neighborhoods of distinct resources R/τ apart (Lemma SV.25). (iii) To apply (i) without assuming perfect states at the start of an operation, every time cut carries the syndrome, the logical frame, and the last measurement records, and each piece is judged by its effect relative to the fault-free evolution of that incoming data (Lemmas SV.13 and SV.14). (iv) Small courier blocks can expose only the resource they carry, and errors created there are charged to the stage that creates them and then to distillation (Lemmas SV.22 and SV.24). (v) A minimum-weight decoder fails only on a large connected set of fault locations, a fixed fraction of which are faults, and counting such sets gives the threshold. Figure S7 shows how the lemmas depend on each other.

Detector convention. Physical detector vertices are only the bounded-support rows of D^{loc}. The full left-kernel space $\mathcal{R}^{\mathrm{full}}$ is used to check boundary algebra; its additional, possibly macroscopic relations are never inserted into the decoder graph or used as physical detectors.

b. Conventions for faults and adaptive branches. A circuit location carries a finite set of nonidentity Pauli labels. In the local-stochastic Pauli model considered here, each faulty location is the ideal operation followed by one selected Pauli on its support. Thus a realized fault set contains at most one mechanism above each circuit location. If A is a set of mechanisms supported on distinct locations, local stochastic noise means

$$\Pr[F \supseteq A] \le p^{|A|}; \tag{A1}$$

if A contains two labels above one location, the event on the left is empty. We write M for the total number of mechanism vertices and reserve F for the realized random set.

We formulate the decoder using a fixed causal location skeleton and a *local branch-uniform detector catalog*. For a realized branch h write D_h^{loc} for the physical detector rows selected from that catalog, O_h for protected effects, and Γ_h for the resulting fault graph. Let Γ_{un} be the graph generated by the union of the same local catalog over the finite branch alternatives. Only D^{loc} defines Γ_h, Γ_{un}, connected components, cluster counting, the temporal edge range τ, or any separation length below.

This physical detector catalog is distinct from the algebraic closure of stabilizer relations at a code-deformation boundary. For a primitive g with boundary check matrices H_g^- and H_g^+ we write

$$\mathcal{R}_g^{\mathrm{full}} := \{(u,v) : uH_g^- = vH_g^+\}. \tag{A2}$$

The space $\mathcal{R}_g^{\mathrm{full}}$ is used only to audit the boundary-state and gluing algebra. Its elements are not, merely by virtue of belonging to this left kernel, detector vertices of the fault graph. In particular a macroscopic stabilizer dependency is never inserted into D^{loc}. At a controlled location the classical history selects one operation from a fixed finite list of bounded-support Clifford templates, and every branch selects a subcollection of the fixed local detector catalog. Thus the union graph is fixed and has bounded degree without conditioning the local-stochastic noise law on a prior global branch history.

c. Heterogeneous and biased Pauli rates. The single parameter p in (A1) is only a domination parameter. Suppose each mechanism a has a possibly location- and Pauli-dependent parameter p_a and, for every set A of mechanisms on distinct circuit locations, the physical distribution obeys

$$\Pr[F \supseteq A] \le \prod_{a\in A} p_a, \qquad \sup_a p_a \le p.$$

Then (A1) follows immediately, so every threshold and resource statement below continues to hold with this p. Thus arbitrary fixed Pauli bias, unequal rates for gates, preparations, measurements and resets, and local correlations already covered by this joint domination inequality require no change to the proof. Preparation and readout faults are explicit mechanisms in templates T3–T5 of Sec. SIX. This observation does not extend the theorem to coherent CPTP faults or leakage, which require additional arguments.

Lemma SV.1 (Bounded fault-graph degree)**.** *There are constants $a_{\mathrm{det}}, b_{\mathrm{det}}, a_{\mathrm{loc}} < \infty$, depending only on the fixed syndrome-extraction circuits, such that every detector contains at most a_{det} circuit mechanisms, every mechanism flips at most b_{det} detectors, and every circuit location has at most a_{loc} Pauli labels. Consequently*

$$\deg(\Gamma_{\mathrm{un}}) \le b_{\mathrm{det}}(a_{\mathrm{det}} - 1) + a_{\mathrm{loc}} - 1 =: z = O(1). \tag{A3}$$

Proof. Each row of D^{loc} has bounded spacetime support by Proposition SV.9 below, so only locations in a bounded neighborhood can flip it. Conversely a Pauli at one location propagates through only a bounded number of gates before the next round boundary and hence flips only boundedly many detectors. A neighbor of a mechanism either has another label at the same location or shares one of its at most b_{det} detectors, proving (A3). The same bounds hold after taking the union over the finite list of branch operations. □

d. Resource conventions. We use outcome bit 0 for eigenvalue $+1$ and bit 1 for eigenvalue -1. Every compilation identity of Sec. SIV is understood as an equality of Kraus maps, up to global phase, and specifies the retained output wire and its Pauli frame. Resource errors are described by defective subsystems rather than by a fixed error basis (Definition SV.2 below).

e. Uniform resource-state assumption. The resource layer uses only fixed-size logical factory codes and the same local layer-code primitives as the data path. The Steane $7\to1$ and punctured Reed–Muller $15\to1$ circuits below do not grow with the target distance; their coordinates are implemented by protected logical operations on layer-code blocks, and distillation is iterated by moving candidate states through encoded courier stages of increasing block distance. No perfect external magic-state source or nonlocal factory connectivity is assumed. The sole distance-one exception is the raw injection boundary, which is isolated by the quotient-distance and earliest-stage charging lemmas before later encoded transfers are counted.

f. Explicit one-output factory primitives. We use the standard code-based $7 \to 1$ $|Y\rangle$ factory on the Steane $[[7,1,3]]$ code and the $15 \to 1$ $|A\rangle$ factory on the punctured Reed–Muller $[[15,1,3]]$ code [7–9]. At the logical level both are fixed in the same encode–transversal-phase–recover–decode form

$$\mathcal{F}_Q = \mathrm{Dec}_Q \circ \mathcal{R}_Q \circ \mathrm{Syn}_Q \circ U_Q^{\mathrm{tr}} \circ \mathrm{Enc}_Q. \tag{A4a}$$

For the Steane factory, Enc_Q prepares encoded $|+\rangle$ and U_Q^{tr} is the fixed transversal $S/S^\dagger$ orientation that maps it

to encoded $|Y\rangle$; its seven physical phase gates are implemented by the phase-teleportation identity of Sec. SIV, one preassigned $|Y\rangle$ candidate per coordinate. For the Reed–Muller factory, U_Q^{tr} is the conventional transversal $T/T^\dagger$ layer of the punctured Reed–Muller code, again implemented coordinatewise by the fifteen preassigned $|A\rangle$ candidates; the fixed T versus $T^\dagger$ convention is absorbed into the known Clifford/frame convention of the factory. Syn_Q extracts the CSS syndrome, $\mathcal{R}_Q$ applies a predetermined minimum-weight Pauli recovery, and Dec_Q returns the single retained logical output. Thus the factory always produces an output rather than postselecting; in the zero-error branch Eq. (A4a) is exactly the standard ideal $|Y\rangle$ or $|A\rangle$ factory map. Tables S11–S13 give the coordinate matrices and the candidate-to-syndrome map used by $\mathcal{R}_Q$; `factory_map_audit.py` checks the ideal transversal phase orientation and writes `factory_map_audit.csv`.

Definition SV.2 (Local-stochastic defective resource family)**.** Let $\mathcal{S}$ be a finite set of resource slots, each an encoded logical qubit that should hold an ideal resource $|R_i\rangle \in \{|A\rangle, |Y\rangle\}$ in the Pauli frame of the realized branch. The family is *local-stochastic defective with parameter q* if its joint state, with everything else in the computation collected into a system E, has the form

$$\rho = \sum_{B\subseteq\mathcal{S}} \pi(B) \left(\bigotimes_{i\notin B} |R_i\rangle\langle R_i|\right) \otimes \sigma_{B,E}, \tag{A4}$$

where π is a probability distribution with $\sum_{B\supseteq A} \pi(B) \le q^{|A|}$ for every $A \subseteq \mathcal{S}$, and each $\sigma_{B,E}$ is an arbitrary state of the defective slots B together with E.

A defective slot may therefore hold any state, including a state entangled with other defective slots, with data, or with the environment; no error basis is assumed. A good slot is exactly ideal and in tensor product with everything else.

Lemma SV.3 (Defective-resource phase teleportation)**.** *In the phase-teleportation gadget of Sec. SIV with target rotation R_θ, replace the resource by an arbitrary state of the auxiliary qubit and a system E. Then in every measurement branch the retained output equals $(R_\theta \otimes I_E)K$ applied to the input, with $K \in \mathrm{span}\{I, Z\} \otimes \mathcal{B}(\mathcal{H}_E)$ independent of the input state. The same holds for $\theta = \pi/4$ when the conditional S correction is teleported with an arbitrary $|Y\rangle$-slot state.*

Proof. This is Eq. (P73) and the remark following it: after the X^m correction the retained state is $\sum_b \psi_b |b\rangle \otimes |\epsilon_b\rangle_E$ with $|\epsilon_b\rangle$ independent of ψ, and the remaining corrections, and the S correction whether ideal or teleported with a defective slot, are diagonal in the computational basis of the retained wire. □

Lemma SV.4 (Single-defect recovery)**.** *Let Q be the $[[7, 1, 3]]$ Steane code or the punctured $[[15, 1, 3]]$ Reed–Muller code used by the factory map (A4a), with the coordinatewise convention of Tables S11 and S12: the seven Steane candidates and fifteen Reed–Muller candidates are assigned to coordinates i, and a conditional sign-correction slot is assigned to the same coordinate as the injection it corrects. Call coordinate i defective if its principal candidate or its correction slot is defective. Assume the compiled nonresource operations of the run act as their ideal logical channels. If at most one coordinate is defective, the decoded output is exactly the ideal resource and is in tensor product with all other systems, whatever the states of the defective slots and whatever they are entangled with. Hence a defective output requires defective slots at two distinct coordinates, and in particular at least two defective slots.*

Proof. Condition on the branch of all teleportation outcomes. By Lemma SV.3, good coordinates receive their ideal rotation and the defective coordinate i (if any) receives it followed by an operator $I \otimes K_I + Z_i \otimes K_Z$ with K_I, K_Z acting on the defective slots and E only. The state after the transversal layer is therefore $|\overline{\psi}\rangle \otimes |\kappa_I\rangle + Z_i|\overline{\psi}\rangle \otimes |\kappa_Z\rangle$, where $|\overline{\psi}\rangle$ is the ideal encoded output. The X-check syndrome of Z_i is the nonzero column v_i of Tables S11 and S12, and of I it is zero, so the syndrome measurement selects one of the two terms and the predetermined recovery $\mathcal{R}_Q$ maps it to $|\overline{\psi}\rangle$. The result is $|\overline{\psi}\rangle \otimes |\kappa'\rangle$ in every branch, so the decoded output is the ideal resource and carries no correlation with E. Distinct columns are needed only for two different coordinates; equivalently, if Z_i and Z_j had the same syndrome then Z_iZ_j would be a Z-type normalizer element of weight two, hence a stabilizer since $d_Z(Q) \ge 3$. □

All possible conditional correction resources are assigned in the fixed schedule before a run, whether or not they will be used; each is produced just before its coordinate is processed, and an unused one is discarded. The fixed assignment, not live storage, is what the lemmas need. Each principal or correction slot consumes a distinct candidate, and different distillation runs use disjoint candidate sets, so no candidate is reused across coordinates or runs. If c_Y is the maximum number of conditional correction slots of a fifteen-to-one run ($c_Y = 15$ for one S correction per coordinate), set

$$C_A = \tbinom{15+c_Y}{2}, \quad C_Y = \tbinom{7}{2}, \quad C_{\mathrm{dist}} = \max(C_A, C_Y). \tag{A6}$$

Lemma SV.5 (Always-output distillation)**.** *Suppose the candidate slots at level r form a local-stochastic defective family with parameter q_r (Definition SV.2), distinct runs use disjoint slot sets, and the compiled nonresource operations act as their ideal logical channels. Then the outputs of the runs form a local-stochastic defective family with parameter*

$$q_{r+1} \le C_{\mathrm{dist}}\, q_r^2. \tag{A8}$$

Proof. Condition on the defective set B of the input family. By Lemma SV.4, every run whose slots meet B in at most one coordinate outputs its ideal resource in tensor product

with everything else, whatever $\sigma_{B,E}$ is. Let B' be the set of runs whose slots meet B in at least two slots. Averaging over B with fixed B' gives a state of the form (A4) for the outputs, with defective set B'. If $B' \supseteq A'$, then B contains a pair of slots of each run in A'; these pairs are disjoint, so a union bound over the at most C_A or C_Y pairs per run gives $\sum_{B' \supseteq A'} \pi'(B') \le \prod_{u \in A'} C_u\, q_r^2 \le (C_{\mathrm{dist}} q_r^2)^{|A'|}$. This proves (A8) without postselection, without independence of resource errors, and without assuming any error basis for a defective slot. □

No resource-basis twirl is used, and Lemmas SV.4 and SV.5 do not need one (Sec. SXX records the twirl of an earlier version and why it fails). The script `resource_defect_check.py` simulates both factories with one defective slot in an arbitrary state entangled with an environment. The decoded output equals the ideal output in every branch of the Steane factory, and in sampled branches of the Reed–Muller factory with every branch at the defective coordinate enumerated (good coordinates contribute branch-independent ideal operations). Defects at two coordinates fail.

g. Injected resources form a defective family. Under the local-stochastic Pauli model, conditional on the realized fault set every raw resource is $P_\iota|R\rangle$ for a Pauli P_ι (a fault after the physical preparation or rotation is a Pauli, and a Pauli before the rotation either acts trivially on $|+\rangle$ or becomes Z), every later operation is a Clifford measurement circuit, and all frame updates are Pauli. In the realized frame each delivered resource is therefore $P_\iota|R\rangle$ with P_ι the residual logical Pauli on the traveling slot, and the joint state is a mixture over fault sets of product states. Let B be the set of sites with $P_\iota \neq I$; a site lies in B exactly when E_ι occurs. Theorem SV.7, Eq. (A13), gives $\Pr[B \supseteq A,\, G] \le q_{\mathrm{inj}}^{|A|}$, and conditioning on G multiplies this by at most two when $\Pr[G^c] \le 1/2$, so $\Pr[B \supseteq A \mid G] \le (2q_{\mathrm{inj}})^{|A|}$. Grouping the mixture by B puts it in the form (A4) with parameter $q_0 \le 2q_{\mathrm{inj}}$. Lemma SV.5 is then applied level by level on G, which is contained in the event G_{res} on which every noninjection compiled operation acts as its ideal logical channel (paragraph "Residual components and the event G_{res}" below Theorem SV.7).

h. A location-level committed-decoding bound. At commitment time t_j, the decoder uses all available branch detectors through $t_j + b$ and selects a minimum-cardinality mechanism set with the observed syndrome. Every observable committed at t_j is a parity of outcomes available by t_j. The mechanisms selected up to t_j are committed as the decoder's correction; the *residual* is the symmetric difference of the realized faults and the committed correction.

Lemma SV.6 (Committed witness)**.** *If a committed observable is wrong, Γ_{un} contains a connected set S with $|S \cap F| \ge |S|/2$ such that either S has empty full syndrome and the corresponding observable effect, or*

$$|S| \ge 1 + \left\lfloor \frac{b - \tau}{\tau} \right\rfloor, \tag{A10}$$

where adjacent mechanisms differ in time by at most τ.

Proof. Symmetric-difference the realized prefix with the minimum-cardinality correction and decompose it using adjacency through available detectors and shared locations. Each component has zero available syndrome. Minimality gives at least half realized faults in every component, since exchanging the realized and corrected mechanisms of a component with fewer realized faults would give a smaller correction with the same available syndrome; in particular this holds for every component whose removal changes the committed value. Such a component either has no unavailable boundary detector and hence zero full syndrome, or connects the causal support of the observable at time at most t_j to a mechanism within τ of the boundary $t_j + b$. A path in the union graph gives (A10). □

For an injection site ι let E_ι be the event that the resource delivered at ι carries a nontrivial residual logical Pauli in the realized frame, that is, that some connected component of the residual has a nonzero effect on the traveling resource of ι; we call this a wrong injection commitment.

Theorem SV.7 (Adaptive cluster bound with weighted injection certificates)**.** *Suppose every noninjection observable has protected mechanism fault distance at least f, the buffer obeys $b \ge \tau(f+1)$, and distinct injection neighborhoods have union-graph distance at least ρ. Partition each injection neighborhood as $N_\iota = \bigcup_a N_{\iota,a}$. Let G be the event that Γ_{un} contains no connected quarter-dense set (a set S with $|S \cap F| \ge |S|/4$;* half-dense *means $|S \cap F| \ge |S|/2$) of size at least $f_\star := \min(f, \rho)$. Assume that, on G, every wrong injection commitment ι has a connected half-dense witness anchored in some $N_{\iota,a}$, of size at least s_a, and witnesses for distinct wrong sites do not touch. If $p \le p_\star := (4ez)^{-4}$, then*

$$\Pr[G^c] \le 4M(p/p_\star)^{f_\star/4}, \tag{A12}$$

$$\Pr\left[\bigcap_{\iota \in I} E_\iota \cap G\right] \le q_{\mathrm{inj}}^{|I|}, \tag{A13}$$

$$q_{\mathrm{inj}} \le \sum_a |N_{\iota,a}| \sum_{s \ge s_a} (ez)^{s-1} 2^s p^{s/2}. \tag{A14}$$

The supremum of the right side of (A14) over ι is understood.

Proof. There are at most $M(ez)^{s-1}$ connected size-s sets with a chosen anchor. For a fixed set, (A1) and a union bound over subsets show that quarter density costs at most $2^s p^{s/4}$. All connected-set enumeration is performed in the single fixed union graph Γ_{un} before reference to

any realized adaptive history; at no point is (A1) replaced by, or inferred for, a distribution conditioned on a branch. Summing the geometric tail gives (A12). On G, Lemma SV.6 and the distance hypotheses exclude all wrong noninjection commitments. A wrong injection witness of size at least f would also have been excluded; a smaller zero-syndrome witness is therefore assigned to its unique injection neighborhood. If two assigned witnesses touched, their union would be connected, meet two neighborhoods, have size at least ρ, and be quarter-dense, again contradicting G.

For fixed pairwise nontouching witnesses, choose at least half their vertices as realized mechanisms. These vertices lie at distinct physical locations, including across witnesses, so (A1) applies and the sum factorizes. Anchoring in stage a, counting connected sets, and summing over sizes gives one factor (A14) per site, proving (A13). No independence of injection events is used. □

Conditioning on G multiplies q_{inj} by at most two whenever $\Pr[G^c] \le 1/2$.

i. Residual components and the event G_{res}. Let G_{res} be the event that every noninjection compiled operation, including memory between production and consumption of resources, acts as its ideal logical channel: every committed outcome of a data or factory operation is correct, and the residual has no component with a nonzero effect on a data or factory logical other than a traveling resource slot. (Raw and courier teleportation outcomes act only on the frame of the traveling resource; a wrong one is part of the event E_ι.) The effect of a component is evaluated at idle cuts placed around it, the right one after a round free of it; by Lemma SV.14 the component leaves zero syndrome and zero records there, so its effect does not depend on the choice of logical representatives. With the parameter choice $\alpha \min\{\beta_0 d, R\} \ge f$ of (A19), used in Theorem SV.31, $G \subseteq G_{\mathrm{res}}$. Indeed, decompose the residual as in the proof of Lemma SV.6: by minimality every component is half-dense, and a component either reaches the boundary of the available detectors, and then $|S| \ge 1 + \lfloor (b-\tau)/\tau \rfloor \ge f+1$ by (A10) and $b \ge \tau(f+1)$, or has zero full syndrome. A component of the second kind with a nonzero effect on a committed data or factory outcome, or on a data or factory logical away from a resource boundary, has size at least f by item 1 of Theorem SV.26. At a resource boundary only the terminal transfer touches a data or factory block (the raw stage and the courier stages $j < J$ involve dedicated courier blocks only), and there an effect on a logical of the factory block other than the designated slot has size at least $\alpha \min\{\beta_0 d, R\} \ge f$ by item 2 of Theorem SV.26 and (A19). In all cases the component is a connected quarter-dense set of size at least $f_\star$, which G excludes. Under Pauli noise, correct outcomes and residual components without logical effect mean that the logical action is the ideal one. For the events E_ι the same decomposition, the exclusion of the (A10) alternative, and items 3–5 of Theorem SV.26 (stage sizes, anchoring in $N_{\iota,j}$, and nontouching) give the half-dense witness hypothesis of Theorem SV.7.

j. Canonical nearest-neighbor extraction schedule. We now fix the physical schedule rather than leave it as a certificate. For any CSS check set appearing in an idle, gauged, factory, birth, death, or readout window, use local ancillas to measure all Z checks and then all X checks. Theorem SIII.11 gives a uniform bound on check weight, qubit/check incidence, and geometric diameter.

Proposition SV.8 (Nearest-neighbor extraction from local templates)**.** *There is a constant-depth nearest-neighbor syndrome-extraction schedule for every check set in the canonical catalog. A circuit Pauli fault in one such round propagates to at most $c_1 = O(1)$ phenomenological data-Pauli or check-outcome mechanisms, uniformly in block size.*

Proof. Let R_* be a uniform radius containing the support of every check in the fixed refined catalog. Because the refinement is fixed and every support has bounded size, reserve within the same ball one routing ancilla site and, for each possible support label, a contact microvertex adjacent to that data qubit. The number of such reserved sites is $O(1)$ per local geometry. Choose the lexicographically first spanning tree of the reserved routing graph, rooted at the ancilla site, and move the ancilla label by nearest-neighbor SWAPs along a depth-first traversal. On the first visit to the contact microvertex assigned to a support qubit, apply the required nearest-neighbor data–ancilla CNOT. After the last interaction, retrace the traversal before readout, restoring all moved labels to their canonical sites.

If V_* is the maximum number of routing sites in such a ball and w_0 the maximum check weight, one extraction word contains at most $2(V_* - 1)$ shuttle SWAPs, w_0 data–ancilla CNOTs, and $O(1)$ preparation/readout locations. Hence its depth and fault light cone are bounded by a constant $L_* = O(V_* + w_0)$ independent of block size. Two extraction words conflict only when their radius-R_* balls overlap. Bounded local density therefore gives a bounded-degree conflict graph, so a constant coloring executes all checks in $O(1)$ sequential color classes. A circuit Pauli fault crosses at most $O(L_*)$ subsequent local gates before the round boundary and consequently produces at most $c_1 = O(1)$ phenomenological mechanisms. This canonical construction fixes a nearest-neighbor route for every bounded local support. □

k. Finite single-fault audit. The ancillary script regression-tests the circuit-to-phenomenological reduction on both displayed T8 CSS master families by inserting every nonidentity Pauli at every preparation, CNOT, SWAP, and measurement location and propagating it through the remaining Clifford word. The bundled audit covers check weights 1–8 and reversible shuttle stress words of lengths 0–2, totaling 10,008 single-location faults; its largest phenomenological image has weight 10. The companion `t8_master_routing_catalog.csv` records all 48 audited words, including the operation word, return-to-canonical flag, location count, and per-word maximum

fault image. Proposition SV.8 supplies the complete routing construction analytically; these finite words provide executable regression tests of its local fault bookkeeping.

Adaptive corrections are compiled into preallocated slots. The physical slot at a spacetime cell is one member of a finite local catalog: idle extraction, one- or two-block gauging, logical X/Z readout, encoded $|0\rangle/|+\rangle$ birth, encoded discard, raw injection, or encoded courier transfer. A fixed target logical circuit fixes the global locations of these slots; classical outcomes only select among finite local alternatives.

Proposition SV.9 (Uniform local detector generation)**.** *For every canonical primitive g and every branch, there is a physical matrix D_g^{loc} and constants $r_{\mathrm{det}}, a_{\mathrm{det}}, b_{\mathrm{det}} = O(1)$, independent of block size, such that: (i) each detector row is contained in a radius-r_{det} spacetime ball; (ii) each row touches at most a_{det} phenomenological mechanisms; and (iii) each mechanism touches at most b_{det} detector rows. The catalog is generated by finitely many local cell/template types. It does not contain an arbitrary basis of $\mathcal{R}_g^{\mathrm{full}}$.*

Proof. Section SIX displays the catalog as master templates T0–T8 rather than only naming template classes. T0 is the consecutive-round row $[1\ 1 \mid \chi_C]$ for a persistent check; T1 is a bounded same-slice meta-check; T2 is the explicit bounded merge/split row; T3 and T4 are the gauging entry and exit rows involving s_j, $\widetilde{s}_j$, B_p, and edge preparation/readout bits; T5 gives one-sided birth/readout endpoints; and T6 states that injection/courier/factory slots introduce no new row species. The banded matrix T7 displays an entire modified-check timeline, and T8 gives the nearest-neighbor ancilla/SWAP/CNOT circuit words generated by the canonical spanning-tree shuttle construction. These are the same local detector generators used in the spacetime-code construction of Ref. [5], supplemented by bounded same-slice local relations when present.

There are only finitely many labeled local geometries after the fixed refinement. Hence the displayed templates have uniform maxima for row diameter and row mechanism count, defining r_{det} and a_{det}. Conversely a mechanism can affect only displayed circuit words whose supports meet a fixed-radius neighborhood of that mechanism. Bounded qubit/check density and bounded template depth therefore give a uniform column-incidence bound b_{det}. Raw injection, encoded courier, and factory slots are finite compositions of these same rows. The constants depend on this finite physical catalog, not on the dimension of the block-dependent full relation space. □

Propositions SV.8 and SV.9 justify Lemma SV.1; in particular the fault graph is the graph of D^{loc}, not of $\mathcal{R}^{\mathrm{full}}$.

l. Boundary-state linearization and algebraic closure. Fix one realized causal branch. At every cut between adjacent catalog primitives choose an independent stabilizer generating set and a complete symplectic logical Pauli basis for every retained encoded block. For a Pauli deviation P at that cut let $s(P)$ be its stabilizer-syndrome vector and $\lambda(P)$ the vector of its commutators with the chosen logical representatives, taken in the Pauli frame of the realized branch (outcome-dependent byproducts, such as the gauging byproduct fixed by the edge readouts, are applied to the frame). The pair (s, λ) determines the Pauli class modulo the boundary stabilizer group: the kernel of $P \mapsto (s(P), \lambda(P))$ is exactly that stabilizer group. We use deviations from the ideal branch, so all relations are homogeneous over $\mathbb{F}_2$.

m. Quantum class and classical record. The pair (s, λ) describes the quantum state at a cut; it does not describe the measurement record. For one persistent check C, let f_t be the data deviation before round t and μ_t the flip of its record in round t. The physical syndrome, the reported record (as a deviation from the ideal record) and the detector are

$$s_t = s_{t-1} + \chi_C(f_t), \qquad r_t = s_t + \mu_t, \qquad d_t = r_{t-1} + r_t, \tag{A15-rec}$$

so $d_t = \mu_{t-1} + \mu_t + \chi_C(f_t)$ is the T0 row of the catalog. A row that compares the last record before a cut with the first record after it (a T0 row of a persistent check, or a T2–T4 transition row) therefore reads the record flip μ of the last round before the cut. That flip is classical data: it is not a function of the Pauli class, and it cannot be absorbed into s. At every cut we therefore also carry the vector ρ of reported-record deviations of the last round before the cut, one entry for each check measured in that round; in a fault-free round $\rho = \Pi s$, where Π maps s to the syndromes of the checks measured in that round, and in general $\rho = \Pi s + \mu_\partial$ with μ_∂ the record flips of that round. Every catalog row compares at most two consecutive rounds, so records of earlier rounds never enter a row that straddles a cut. The complete boundary variable at a cut is (s, λ, ρ).

The frame coordinate depends on the chosen representatives whenever $s(P) \neq 0$. With stabilizer X_1X_2, the operators X_1 and X_2 represent the same logical class, but Z_1 anticommutes with the first and commutes with the second; changing representatives changes $\lambda(Z_1)$ by the syndrome bit. In general, if an output representative pulled back through the ideal primitive equals the input representative times the stabilizer product $\prod_i S_i^{c_i}$, the frame acquires $\sum_i c_i s_i$.

For a primitive g, let e_g be its phenomenological fault vector, δ_g the values of its detector rows (the rows whose later record lies in g, including the rows that straddle its input cut), and o_g its protected committed outcomes. Propagation through the fixed Clifford measurement schedule gives

$$\delta_g = D_g^{\mathrm{loc}} e_g + A_g^{\mathrm{loc}} s_g^{\mathrm{in}} + P_g^{\mathrm{loc}} \rho_g^{\mathrm{in}}, \tag{A15a}$$

$$\lambda_g^{\mathrm{out}} = L_g \lambda_g^{\mathrm{in}} + C_g s_g^{\mathrm{in}} + E_g e_g, \tag{A15b}$$

$$o_g = Q_g \lambda_g^{\mathrm{in}} + G_g s_g^{\mathrm{in}} + F_g e_g, \tag{A15c}$$

together with the output boundary variables

$$s_g^{\mathrm{out}} = \Sigma_g(s_g^{\mathrm{in}}, \lambda_g^{\mathrm{in}}) + J_g e_g, \qquad \rho_g^{\mathrm{out}} = \Pi_g s_g^{\mathrm{out}} + M_g e_g, \tag{A15a$'$}$$

where Π_g expresses the syndrome of each check of the last round through the generating set and M_g picks out the record flips of that round. A fault pattern is detector-trivial when every δ_g vanishes. In a fault-free inherited state $\rho^{\mathrm{in}} = \Pi s^{\mathrm{in}}$ and the input rows of (A15a) vanish, so an inherited syndrome is not a detection event; an inherited record error, $\rho^{\mathrm{in}} \neq \Pi s^{\mathrm{in}}$, is. The committed outcomes of the catalog are parities of records measured inside the primitive, so o_g does not depend on ρ_g^{in}.

The matrices C_g and G_g are fixed by the ideal branch and the representatives chosen at the two cuts. Linearity does not force them to vanish. For a unitary primitive one may choose the output representatives as the images of the input ones, which gives $C_g = 0$; a code deformation generally has to multiply the pushed-forward representatives by input stabilizers to make them commute with the new checks, so we retain C_g and G_g throughout. Separately, embed the two boundary check groups in their common qubit set, adjoining known product-state stabilizers at births and final readout observables at deaths. The full transition-relation space is the algebraic closure $\mathcal{R}_g^{\mathrm{full}}$ of Eq. (A2). Section SIX lists both objects explicitly: the local template catalog is used for decoding, whereas the full left kernel is used only as an algebraic audit of boundary relations.

Proposition SV.10 (Local generators suffice for boundary restriction)**.** *For every canonical transition, the rows of D_g^{loc} together with the complete boundary variables (s, λ, ρ) contain every constraint required to restrict and glue the physical detector/effect system. Every local merge/split or endpoint parity used by the scheduled-circuit proof is a row of this local catalog. An additional element of $\mathcal{R}_g^{\mathrm{full}}$ is a derived algebraic relation among ideal boundary checks; it is not added as a fault-graph detector vertex.*

Proof. Unchanged checks are connected across a transition by same-check local comparisons. Where the check set changes, the transition is a tiling by the finite local merge/split, grid-plaquette, birth, readout, or courier endpoint templates listed in Sec. SIX. These are precisely the local parities used by the cleaning and rectangle arguments. The variables (s, λ) record the residual Pauli class at the cut, and ρ records the classical data that the rows straddling the cut read, so no perfect-boundary condition or macroscopic parity row is needed to continue propagation into the next primitive. Summing local identities may of course produce further elements of $\mathcal{R}_g^{\mathrm{full}}$, and the full left kernel enumerates all such algebraic relations, including possible macroscopic dependencies. Such derived relations supply no additional physical measurement record and therefore remain in the algebraic audit space rather than in D_g^{loc} or the fault graph. □

Lemma SV.11 (Canonical local detector/effect gluing)**.** *For the canonical schedule, the global* physical *detector/effect system on a realized causal branch is the fiber product of the primitive systems (A15a)–(A15c) and (A15a$'$), obtained by identifying the complete (s, λ, ρ) variables on every retained wire. This exact gluing uses D^{loc} for detector rows and does not require promoting the full relation space to physical detectors.*

Proof. A Clifford circuit with Pauli faults is linear on Pauli classes. At a cut, (s, λ) is a complete coordinate for that class modulo stabilizers, so the future quantum evolution, and hence every future record and logical effect, depends on the past faults only through these variables. A future detector row can in addition read records measured before the cut; every catalog row compares at most two consecutive rounds, so the only such records are those of the last round before the cut, which are ρ. Conversely each physical detector row belongs to one bounded slot template of Proposition SV.9; all cross-slot dependence is therefore mediated by the retained boundary variables. Identifying them and eliminating internal ones reproduces exactly the physical circuit equations. Proposition SV.10 ensures that every local merge/split parity needed by the schedule is present. The full space $\mathcal{R}_g^{\mathrm{full}}$ may be used to check the algebraic consistency of this boundary description, but its extra derived rows are not required for, and are not inserted into, the fiber product of physical detector rows. □

n. Worked gluing example (two adjacent idle primitives). The following miniature illustrates the boundary variables; it does *not* replace the general proof of Proposition SV.10 or Lemma SV.11. Take a single persistent check C and two consecutive idle primitives g_1, g_2 of one round each, glued on the mid-cut, with fault bits $e_t = (f_t, \mu_t)^\top$ (data Pauli anticommuting with C and with the logical partner, and record flip). The boundary variables are the syndromes $s_{\mathrm{in}}, s_{\mathrm{mid}}, s_{\mathrm{out}}$, the last-round records $\rho_{\mathrm{in}}, \rho_{\mathrm{mid}}, \rho_{\mathrm{out}}$, and the frames $\lambda_{\mathrm{in}}, \lambda_{\mathrm{mid}}, \lambda_{\mathrm{out}}$. By (A15-rec) the two primitives are

$$\begin{aligned} g_1:\ & \delta_1 = \rho_{\mathrm{in}} + s_{\mathrm{in}} + f_1 + \mu_1, \quad s_{\mathrm{mid}} = s_{\mathrm{in}} + f_1, \\ & \rho_{\mathrm{mid}} = s_{\mathrm{mid}} + \mu_1, \quad \lambda_{\mathrm{mid}} = \lambda_{\mathrm{in}} + f_1, \\ g_2:\ & \delta_2 = \rho_{\mathrm{mid}} + s_{\mathrm{mid}} + f_2 + \mu_2, \quad s_{\mathrm{out}} = s_{\mathrm{mid}} + f_2, \\ & \rho_{\mathrm{out}} = s_{\mathrm{out}} + \mu_2, \quad \lambda_{\mathrm{out}} = \lambda_{\mathrm{mid}} + f_2. \end{aligned} \tag{A15-ex1}$$

Eliminating $(s_{\mathrm{mid}}, \rho_{\mathrm{mid}}, \lambda_{\mathrm{mid}})$ gives

$$\delta_1 = \rho_{\mathrm{in}} + s_{\mathrm{in}} + f_1 + \mu_1, \qquad \delta_2 = \mu_1 + f_2 + \mu_2, \tag{A15-ex2}$$

$$\begin{aligned} & s_{\mathrm{out}} = s_{\mathrm{in}} + f_1 + f_2, \qquad \rho_{\mathrm{out}} = s_{\mathrm{out}} + \mu_2, \\ & \lambda_{\mathrm{out}} = \lambda_{\mathrm{in}} + f_1 + f_2, \end{aligned} \tag{A15-ex3}$$

which are the detector/effect equations of the two-round idle circuit on C written directly. The syndrome is a quantum variable and changes only with f; the records carry μ. For $f_1 = f_2 = 0$, $\mu_1 = \mu_2 = 1$ and $s_{\mathrm{in}} = \rho_{\mathrm{in}} = 0$,

the syndrome is zero at every cut, $\rho_{\mathrm{mid}} = \rho_{\mathrm{out}} = 1$, and a clean following round gives the detector history $(1, 0, 1)$. Gluing with (s, λ) alone, that is, setting $\rho_{\mathrm{mid}} = s_{\mathrm{mid}}$, would miss the μ_1 term of δ_2. The ancillary script `glue_example_check.py` verifies the fiber-product identity on all 2^7 assignments of $(f_1, \mu_1, f_2, \mu_2, s_{\mathrm{in}}, \rho_{\mathrm{in}}, \lambda_{\mathrm{in}})$, the history $(1, 0, 1)$, and that (s, λ)-only gluing disagrees exactly when $\mu_1 = 1$.

o. Direct versus composed map on explicit blocks. The script `boundary_composition_check.py` tests Lemma SV.11 on the paper's own objects. Two explicit $[[4, 2, 2]]$-input blocks ($s = 2$, 448 data qubits) with the spanning grid of Sec. SXIV (184 edge qubits) run the schedule idle $\to$ gauging $\to$ idle $\to$ gauging $\to$ idle, each gauging primitive consisting of an undeformed entry round, edge preparation, two deformed rounds measuring $\overline{X}_1^{(a)}\overline{X}_1^{(b)}$, edge readout, and an undeformed exit round. The detectors are the catalog rows T0, T3 and T4 (6462 rows, all deterministic in the noiseless circuit). The *direct* map is a Stim frame simulation of the complete schedule in the phenomenological model (each check measured as a multi-qubit Pauli product, with data and edge Paulis, record flips and edge-readout flips as the fault mechanisms); the *composed* map simulates each primitive in isolation from (s, λ, ρ) alone, with the logical representatives changed at every cut so that the terms $C_g s$ are exercised. For every one of the 20,000 single phenomenological fault mechanisms, and for every unit inherited input (s_0, λ_0, ρ_0) at the first cut (896 inputs), the two maps agree on all detectors, both reported measurement bits, and the final boundary variables; by linearity the maps are equal, including inherited syndromes and record errors at the cuts. Gluing with (s, λ) only disagrees on exactly the 1776 record flips in the last round before the four internal cuts, and the pattern of record flips on both sides of a cut gives the history $(1, 0, 1)$ in both maps.

Lemma SV.12 (Componentwise local-detector triviality)**.** *Let the phenomenological fault graph join mechanisms when they share a row of D^{loc} or the same physical location. If a global fault vector has zero local-detector syndrome, every connected component of its support separately has zero local-detector syndrome.*

Proof. All mechanisms appearing in one row of D^{loc} are pairwise adjacent. Hence the nonzero mechanisms contributing to one detector row lie in a single connected component. The global parity of that row is zero, so its parity in that component is zero. This holds for every local detector row. □

Lemma SV.13 (Rectangle restriction for the canonical schedule)**.** *Let C be a connected component of a globally D^{loc}-trivial fault pattern and suppose its temporal span intersects at most one active primitive. Choose cuts in the surrounding idle pads containing every mechanism of C. After all other components are set to zero, there exist inherited boundary variables (s_L, λ_L, ρ_L) and (s_R, λ_R, ρ_R) for which the restricted subchain is a valid relative rectangle solution with exactly the protected relative effect contributed by C. No condition $s_L = s_R = 0$ or $\rho_L = \rho_R = 0$ is imposed.*

Proof. Lemma SV.12 makes the mechanism vector of C locally detector-trivial by itself. Apply Lemma SV.11 with every other mechanism set to zero. The resulting global solution supplies definite boundary variables at each cut; retaining only the variables in the chosen subchain preserves all local primitive and gluing equations. Equations (A15a)–(A15c) are linear, so subtracting the fault-free propagation of the same left boundary variables, including the terms $A_g s_L$, $P_g \rho_L$, $C_g s_L$ and $G_g s_L$, leaves exactly the contribution of C. Thus a nonzero inherited syndrome, or an inherited record error, can cross either artificial cut without being reinterpreted as a new fault. When the cuts lie in idle pads, contain every mechanism of C, and the right cut follows a round that contains no mechanism of C, Lemma SV.14 shows in addition that $s_L = \rho_L = 0$ and $s_R = \rho_R = 0$. □

Lemma SV.14 (Contained components leave no cut syndrome or record)**.** *Let C be a detector-trivial connected component and let two cuts lie in idle rounds of every block that C touches, with every mechanism of C strictly between them and at least one complete round containing no mechanism of C immediately before the right cut. With all other components set to zero, the boundary variables propagated by C have $s = 0$ and $\rho = 0$ at both cuts. Consequently the global protected effect of C equals its relative effect (A15-rel) in the enclosed subchain, and the terms $A_g s$, $P_g \rho$, $C_g s$ and $G_g s$ never contribute to a contained component.*

Proof. At the left cut no mechanism has occurred, so $(s_L, \lambda_L, \rho_L) = 0$; a block born inside the subchain has no left variables and receives the fixed product-state input. At the right cut take any check of the boundary generating set of a retained block, and let t^* be the last round before the cut, which contains no mechanism of C. The check's record in round t^* is linked to the left cut, or to the birth of its block, by a chain of local detector rows. Within idle and deformed windows the chain consists of consecutive T0 rows; at a deformation it uses the T3a entry row and the T4a exit row. At a merge or split, an output check whose value is fixed by the input checks is chained back through its T2 row to the chains of those inputs; for the remaining, genuinely new output checks, the first records of a basis fix the realized frame, so their chains start there with value zero. At a birth the chain ends on the T5a row of a check with deterministic initial value, which plays the role of $\rho_L = 0$; for a check whose first record after birth is random, that record fixes the realized frame in the same way. Every row of the chain is satisfied because C is detector-trivial. Summing the chain telescopes by (A15-rec): each interior record occurs in two consecutive rows and cancels. The terms that occur only once are $\rho_L = 0$, the record r_{t^*}, the preparation bits of a T5a

birth row, and, for a check s_j deformed in a gauging window, the edge terms of its T3a and T4a rows, whose sum over $e \in \mu_j$ is the deviation of the reported edge readouts on μ_j. Add the T3b, T0 and T4b rows of the flux checks of a 2-chain of unit squares and parallel-copy cycles whose boundary is μ_j plus the tree path joining the ports of s_j (these checks generate the cycle space of the grid). Each flux chain contributes the reported-readout deviation around its cycle, so this replaces the sum over μ_j by the reported-readout deviation along the tree path, which is the commutator of the gauging byproduct with s_j. The telescoped sum is therefore the record of the check in round t^* in the realized frame, and it vanishes. No record flip of C occurs in round t^*, so $s_{t^*} = r_{t^*} = 0$, and no data mechanism of C occurs after round t^*, so $s_R = s_{t^*} = 0$ and $\rho_R = r_{t^*} = 0$. A block discarded inside the subchain has no right variables; its protected readout outcome is part of the relative effect. The global effect of C is the fault-free propagation of $(0, \lambda_R, 0)$ through the rest of the circuit, which is determined by λ_R alone; and λ_R, together with the recorded outcomes inside the subchain, is the relative effect of C because the incoming variables are zero. The exhaustive check in `frame_propagation_check.py` confirms $s = 0$ at the output cut on all 2^{16} patterns of a two-check, four-round idle window when a fault-free round follows the cut, and `boundary_composition_check.py` confirms the record-aware gluing on explicit blocks. □

p. Relative local rectangle distance. For a primitive or macro-rectangle with incoming variables (s, λ, ρ), define its *relative* detector values and protected effects by subtracting the fault-free propagation of the same incoming variables:

$$\Delta\delta = \delta + As^{\mathrm{in}} + P\rho^{\mathrm{in}}, \qquad \Delta\lambda = \lambda^{\mathrm{out}} + L\lambda^{\mathrm{in}} + Cs^{\mathrm{in}},$$
$$\Delta o = o + Q\lambda^{\mathrm{in}} + Gs^{\mathrm{in}}. \tag{A15-rel}$$

Because the circuit is linear on Pauli deviations, $(\Delta\delta, \Delta\lambda, \Delta o) = (De, Ee, Fe)$ depends only on the internal fault vector, not on the inherited boundary variables. Without the Cs^{in} and Gs^{in} terms this independence fails: on an explicit layer-code block whose two cuts use slab representatives at different heights, the reduced law $\lambda^{\mathrm{out}} = L\lambda^{\mathrm{in}}$ fails for 986 of 1979 random inherited deviations with nonzero syndrome, while (A15b) holds for all 2000 and (A15-rel) returns the internal effect in every case (`frame_propagation_check.py`). Without the $P\rho^{\mathrm{in}}$ term an inherited record error would be charged to the rectangle as a detection event of its own faults.

q. Relative spacetime cleaning. Williamson–Yoder Theorem 4 (fault tolerance), with the detailed spacetime-code statement in Theorem 5 (spacetime fault-distance) of their Supplementary Information and its Lemmas 4–7 (spacetime code, local spacetime stabilizers, time fault-distance, decoupling of space and time faults), in the published version of Ref. [5], lower-bounds the gauging fault distance by separating detector-trivial histories into spacelike and timelike representatives. We use that cleaning argument through the explicit WY1–WY6 interface below, adapted to inherited boundary classes. Table S7 gives the one-to-one correspondence with the present local detector schedule.

Proposition SV.15 (Spacetime-cleaning interface for the canonical schedule)**.** *Consider one ordinary gauging macro-rectangle, including its undeformed entry round, R_g repeated deformed-code rounds, and undeformed exit round. Let $\mathcal{C}_0$ and $\mathcal{C}_g$ be the undeformed and deformed check sets. The spacetime-cleaning theorem of Ref. [5] applies to the relative problem used here because the following hypotheses hold.*

(WY1) *The detector set contains consecutive-round comparison rows for every check repeated within $\mathcal{C}_0$ or $\mathcal{C}_g$, the local undeformed–deformed entry rows, and the local deformed–undeformed exit rows. Bounded same-slice relations, when present, are included as additional local detectors.*

(WY2) *For every physical qubit and every adjacent pair of time slices, the usual local spacetime stabilizer exists: equal Pauli faults on the two slices, together with measurement faults on precisely the intervening checks that anticommute with that Pauli. At entry and exit the corresponding initialization/readout versions exist. These moves are detector-trivial and have zero protected relative effect.*

(WY3) *The deformed check set is repeated for exactly R_g complete rounds, so a genuinely timelike measurement fault must connect the two deformation boundaries and has phenomenological weight at least R_g.*

(WY4) *Every nontrivial spacelike protected class of the deformed code has weight at least $d_g^{\mathrm{space}} \geq \beta_0 d$.*

(WY5) *The incoming boundary variables (s, λ, ρ) are held fixed. Hence the relevant quantity is the relative detector pattern and output effect, not an absolute syndrome condition at an artificial perfect cut.*

(WY6) *Phenomenological weight counts one for each data-Pauli, initialization/readout, or check-record mechanism. Cleaning preserves the parity of occupied space/initialization faults on every fixed spatial timeline. Consequently, if cleaning leaves w occupied spacelike timelines, the original fault vector has weight at least w.*

Under these hypotheses every detector-trivial fault vector with nonzero relative protected effect is equivalent, by multiplication with local spacetime stabilizers, to a product of a spacelike protected fault and a purely timelike fault. Therefore its weight is at least $\min\{d_g^{\mathrm{space}}, R_g\}$.

Proof. We verify the six cleaning conditions directly in the present detector and boundary notation. For WY1, template T0 of Sec. SIX is the repeated-check detector of the gauging spacetime code; T3 and T4 are respectively the initialization/deformation and deformation/readout

detector rows, including the B_p and $\widetilde{s}_j$ endpoint parities. Template T1 adds only bounded space-only local relations. This is exactly the modification allowed when a code has local meta-checks: adding valid detector rows cannot create a new detector-trivial fault.

For WY2, the fixed bounded-depth extraction circuit gives the standard adjacent-time spacetime stabilizers locally. A Pauli P inserted on a qubit at time t and the same P at $t+1$, together with flips of the intervening check records that anticommute with P, changes neither any physical detector nor the propagated encoded Pauli class. At the deformation boundaries an edge-initialization fault is paired with the corresponding first-slice Pauli, and a last-slice Pauli is paired with the corresponding edge readout fault. These are precisely the local generators used in the spacetime-stabilizer construction of Ref. [5]. Because they are actual identity moves of the recorded Clifford schedule, they remain trivial for the additional T1 local-dependency detectors as well.

WY3 is part of the canonical macro-rectangle definition. Once the spacelike Pauli support has been cleaned to one time slice, any residual measurement-only logical is a temporal string with endpoints on the start and end of the deformation window. Consecutive-check detectors force such a string through all R_g repeated rounds, so it contains at least R_g measurement/init/readout mechanisms. WY4 is exactly the static distance statement of Theorem SIII.11; the corresponding idle-memory value is d.

For WY5, Proposition SV.10 and Lemma SV.13 allow arbitrary inherited variables (s, λ, ρ) at the left cut and compare the faulty rectangle with the fault-free propagation of the same variables, including the syndrome and record terms of Eq. (A15-rel). Subtracting the two solutions therefore produces the zero-input relative problem required by cleaning, without imposing $s = 0$ at a real boundary.

Finally, the weight step in WY6 is the point that prevents a hidden weight-increasing projection. The adjacent-time generators let us sweep every paired data-Pauli occupation forward or backward along its spatial timeline. At the first and last slices, the entry/exit versions terminate exactly the occupations paired with the corresponding preparation/readout variables. After this elimination, choose one reference slice and sweep every remaining non-endpoint data occupation to that slice. What remains separates into (i) a single-slice Pauli representative carrying the spacelike protected class and (ii) record/init/readout variables only. Consecutive-round detector equations force any nontrivial part of (ii) to form a temporal chain from one deformation boundary to the other; otherwise its first or last endpoint would flip a local T0/T3/T4 detector. Hence the temporal part has weight at least R_g.

The same sweep preserves mod-2 occupancy on each fixed spatial timeline. Every occupied timeline in the single-slice representative therefore contains at least one mechanism of the original fault vector, including the endpoint convention. If the spacelike part has nonzero protected effect it has at least d_g^{space} occupied timelines; if it is protected-trivial, all nonzero relative effect resides in the temporal part and costs at least R_g. If both parts are present, either lower bound already applies to the original vector. Thus the desired relative distance is at least $\min\{d_g^{\mathrm{space}}, R_g\}$. This is the Williamson–Yoder cleaning argument written directly in the present boundary and detector notation; the citation supplies its original spacetime-code formulation rather than an additional unverified hypothesis. □

Corollary SV.16 (Canonical rectangle meets the Williamson–Yoder interface)**.** *The canonical gauging macro-rectangle of Sec. SV (undeformed entry, R_g repeated deformed rounds, and undeformed exit) satisfies WY1 to WY6 of Proposition SV.15. Therefore every detector-trivial fault with nonzero relative protected effect has weight at least* $\min\{d_g^{\mathrm{space}}, R_g\}$.

Proof. Table S7 gives the direct correspondence. WY1 and WY2 are implemented by the local detector/extraction catalog of Propositions SV.8 and SV.9; WY3 is the repeated-round definition; WY4 is Theorem SIII.11; WY5 is the relative (s, λ, ρ) reduction of Proposition SV.10 and Lemma SV.13; and WY6 is the timeline-parity sweep proved in Proposition SV.15. Proposition SV.15 then gives the stated distance bound. □

r. A direct slice proof. Proposition SV.15 reaches the rectangle bound through the cleaning argument of Ref. [5], and its weight step (WY6) is the delicate point: cleaning must not make the fault vector heavier at any boundary. The proposition below proves the same bound, in a slightly sharper form, without cleaning. Its only weight comparison is the elementary fact that a qubit in the support of an accumulated error carries at least one fault. It is the result used in the rest of this section; Proposition SV.15 records the correspondence with Ref. [5].

We use the phenomenological model of a rectangle with rounds $1, \ldots, T$: $a \geq 1$ undeformed rounds, deformed rounds $t_{\mathrm{in}}, \ldots, t_{\mathrm{out}}$ with $R = t_{\mathrm{out}} - t_{\mathrm{in}} + 1$, and $b \geq 1$ undeformed rounds. A data fault is a Pauli on a qubit in the gap before round g, for $1 \leq g \leq T + 1$; an edge qubit exists in the gaps t_{in} to $t_{\mathrm{out}} + 1$, and an X fault in its first (last) gap is its preparation (readout) fault. A record fault flips one check record in one round. Write E_t for the product of all data faults in the gaps up to t, the error present when round t is measured, and $\sigma_c(E) \in \mathbb{F}_2$ for the commutator of E with a check c. Relative to the fault-free run with the same random outcomes, the record of c in round t is $\Delta r_{c,t} = \sigma_c(E_t) + m_{c,t}$, with $m_{c,t}$ its record fault. The relative problem has zero input at the left cut, so $\Delta r_{c,1} = 0$ for every check measured in round 1, and the right cut has zero syndrome and zero records.

Proposition SV.17 (Slice bound for a gauging rectangle)**.** *Consider the relative problem of one rectangle that gauges an X-type logical $\overline{P} = X(D)$ of a CSS code C (one*

TABLE S7. Correspondence between the Williamson–Yoder spacetime-cleaning ingredients and the canonical schedule used here. The final column records the two extensions needed for the present circuit: bounded local dependency rows and inherited boundary classes.

Label	Williamson–Yoder ingredient	Verification in this manuscript	Modification in the present schedule
WY1	Local consecutive-round and deformation-boundary detectors (spacetime-code generators)	Proposition SV.15, WY1; Proposition SV.9; templates T0, T3, T4 and banded matrix T7 of Sec. SIX	We additionally allow bounded same-slice T1 dependency rows. They only shrink the detector-trivial set and are generated locally.
WY2	Local spacetime stabilizers used for cleaning	Proposition SV.15, WY2; the bounded-depth circuit of Proposition SV.8 realizes the adjacent-time identity moves	Entry/exit versions include edge preparation/readout. Complete boundary variables (s, λ, ρ) are retained rather than imposing a perfect artificial cut.
WY3	Timelike fault-distance lower bound	Proposition SV.15, WY3; R_g repeated deformed rounds and open temporal endpoints	The macro-rectangle contains explicit undeformed entry/exit rounds; the protected temporal string must still cross all R_g deformed rounds.
WY4	Spacelike distance of the deformed code	Proposition SV.15, WY4; Theorem SIII.11, giving $d_g^{\mathrm{space}} \geq \beta_0 d$ (or the corresponding quotient statement at the raw port)	Grid ancillas are placed by the local port map; no expander-style nonlocal auxiliary connectivity is assumed.
WY5	Zero-input / relative problem (perfect initial-final boundary convention)	Proposition SV.15, WY5; Proposition SV.10; Lemma SV.13	Cleaning is applied to the *relative* problem obtained by subtracting fault-free propagation of the same inherited (s, λ, ρ) variables.
WY6	Weight / timeline-parity preservation under cleaning	Proposition SV.15, WY6; not needed after Proposition SV.17	Timeline-parity preservation is stated explicitly: every occupied cleaned spatial timeline contains at least one original phenomenological mechanism. The direct slice argument of Proposition SV.17 proves the bound without any sweep.

block, or $C = C_a \otimes C_b$ for two blocks), with zero input at the left cut and zero syndrome and zero records at the right cut. Let $\widetilde{d}_X$ be the X distance of the deformed code and d_Z the Z distance of C. Every detector-trivial phenomenological fault vector e with nonzero relative effect satisfies

$$|e| \geq \min\{\widetilde{d}_X, d_Z, R\}. \tag{A15s}$$

More precisely, the X-type part of e (the X components of its data and edge faults and its Z-check record faults) has a nonzero effect only if it contains at least $\widetilde{d}_X$ faults, and the Z-type part only if it contains at least $\min\{d_Z, R\}$. An idle rectangle ($R = 0$) obeys $|e| \geq \min\{d_X, d_Z\} = d$. The statement with X and Z exchanged holds for the gauging of Z-type logicals. The bound is attained.

Proof. Sectors. Every detector row and every effect is either X-type or Z-type: X-type faults enter only the rows of Z-type checks, the edge readouts and the X frame, and Z-type faults only the rows of X-type checks, the reported outcome and the Z frame. A Y fault counts once and contributes to both parts. Hence each part of a detector-trivial e is detector-trivial, the effect of e is the sum of the effects of its parts, and each part has at most $|e|$ faults. It suffices to bound each part.

X-type part. Every Z-type check has a deterministic record when it is first measured: the Z checks of C in round 1 (zero input), and the deformed checks $\widetilde{s} = s\, Z(\mu_s)$ and the unit cycles in round t_{in}, where the entry rows compare them with round $t_{\mathrm{in}} - 1$ and with the $|0\rangle$ edges. Chaining the consecutive and entry rows from the left cut, and the consecutive rows back from the right cut, where the records vanish, gives $\Delta r_{c,t} = 0$ for every Z-type check in every round. Let $z_e \in \mathbb{F}_2$ be the parity of the X faults on edge e during its life, which is its relative readout. The exit rows, $\Delta r_{s,t_{\mathrm{out}}+1} = \Delta r^{\sim}_{s,t_{\mathrm{out}}} + \sum_{e\in\mu_s} z_e$ and $\Delta r_{c,t_{\mathrm{out}}} = \sum_{e\in c} z_e$ for a unit cycle c, then give $\sum_{e\in\mu_s} z_e = 0$ for every Z check s and $\sum_{e\in c} z_e = 0$ for every cycle. So z is orthogonal to the cycle space, and $z = \delta T$ for a vertex set T, determined up to complement. Let E_{d} be the X error on the data at the right cut; the right-cut condition gives $\sigma_s(E_{\mathrm{d}}) = 0$ for every Z check s of C. The frame update that the protocol computes from the readouts changes by $X(f^{-1}(T \cap U))$, and the complementary choice of T changes this by $\overline{P}$, which acts trivially after the measurement. So the effect of the part is the class of $L = E_{\mathrm{d}}\, X(f^{-1}(T \cap U))$ modulo $\langle S_X(C), \overline{P}\rangle$.

Now consider the single-time operator $O = E_{\mathrm{d}}\, X(\delta T)$ on data and edges. It commutes with every deformed check $\widetilde{s} = s\, Z(\mu_s)$, since $\sigma_s(E_{\mathrm{d}}) + |\mu_s \cap \delta T| = 0$, and with every cycle, since $|c \cap \delta T|$ is even. So O lies in the X normalizer of the deformed code, and $O \prod_{u\in T} A_u = L$. If O were a deformed stabilizer, $O = \sigma \prod_{u\in W} A_u$ with $\sigma \in S_X(C)$, comparing edge supports would give $\delta W = \delta T$, hence $W \in \{T, V \setminus T\}$ and $L \in \langle S_X(C), \overline{P}\rangle$; the converse is immediate. Hence a nonzero effect means that O is a nontrivial deformed logical and $|O| \geq \widetilde{d}_X$. Every data qubit in the support of E_{d} and every edge with $z_e = 1$ carries at least one X-type fault, and these faults are distinct. The part therefore has at least $|E_{\mathrm{d}}| + |\delta T| = |O| \geq \widetilde{d}_X$ faults. Record faults are not needed in this count.

Z-type part. The X checks of C are measured in every round, with zero input at the left cut, so their consecutive rows give $\sigma_a(E_t) = m_{a,t}$ for every X check a and every round. The vertex stars A_u have random first records and are compared only between consecutive deformed rounds, so $\Delta r_{u,t} = b_u$ for every deformed round t, and the reported outcome is flipped by $o = \sum_u b_u$. Let E_{d} be the Z error on the data at the right cut; by the right-cut condition it commutes with every X check of C, and the effect of the part is the pair $(o, [E_{\mathrm{d}}])$. If $[E_{\mathrm{d}}] \neq 0$, then E_{d} is a nontrivial Z logical of C, so $|E_{\mathrm{d}}| \geq d_Z$, and every qubit of its support carries a Z fault. Otherwise $o = 1$. If every deformed round contains a record fault of some X-type check, the part has at least R faults. If not, let t^* be a deformed round without one. Then $\sigma_a(E_{t^*}) = 0$ for every X check a of C and $\sigma_{A_u}(E_{t^*}) = b_u$ for every vertex u. Write $E_{t^*} = Z(B)Z(F)$ with B on the data and F on the edges. Every edge lies in two stars and every port in one, so $\sum_u \sigma_{A_u}(E_{t^*}) = |B \cap D| \pmod 2$, which equals $o = 1$. Thus $Z(B)$ commutes with every X check of C and anticommutes with $\overline{P}$: it is a nontrivial Z logical of C, $|B| \geq d_Z$, and every qubit of B carries a Z fault in a gap before round t^*. In all cases the part has at least $\min\{d_Z, R\}$ faults.

Idle rectangle and attainment. With $R = 0$ there are no edges and no outcome, and the two end-slice arguments give $|E_{\mathrm{d}}| \geq d_X$ and $|E_{\mathrm{d}}| \geq d_Z$. The bound is attained by a minimum-weight deformed logical inserted in one deformed round, by a minimum-weight Z logical of C inserted before the window, and by one vertex-star record flipped in all R rounds. □

Since $\widetilde{d}_X \geq \beta_0 d$ (Theorem SIII.11) and $d_Z \geq d$, Eq. (A15s) gives $\min\{\beta_0 d, R\}$. For two blocks, d_Z is the smaller block distance, and the cut obstruction is visible in the X-type part: a thin interface makes $\widetilde{d}_X$ small, while the Z-type part is unaffected. The script `rectangle_slice_check.py` builds the relative problem from the phenomenological model alone, with the frame update computed from the edge readouts along a spanning tree as in the protocol, and finds by exact CP-SAT minimization that each part attains its bound exactly (Table S8).

The proof uses only the detector rows, the static normalizer structure, and the count of faults on the support of one operator. The same argument therefore gives the quotient statements used below: if effects are recorded modulo a subgroup $\mathcal{Q}$ in which the reported outcome acts only on a designated resource frame, a nonzero effect modulo $\mathcal{Q}$ forces one of the two end-slice operators (O, or the Z error E_{d}) to represent a class outside $\mathcal{Q}$. For O the static quotient bounds of Lemmas SIII.5 and SIII.6 then apply, and for E_{d}, a logical of the undeformed blocks acting outside $\mathcal{Q}$, the distance of the block it acts on.

Lemma SV.18 (Protected relative rectangle distance)**.** *For an ordinary one- or two-block gauging rectangle with R_g repeated rounds, any detector-trivial phenomenological fault vector with nonzero protected relative effect that leaves zero syndrome and zero records at the right cut (as a contained component does, Lemma SV.14) has weight at least*

$$\delta_g^{\mathrm{rel}} \geq \min\{\beta_0 d, R_g\}. \tag{A15d}$$

An idle-memory rectangle of any length obeys $\delta_{\mathrm{mem}}^{\mathrm{rel}} \geq d$, and the one-sided encoded birth and protected logical-readout rectangles obey the corresponding bound for the logical eigenvalue or readout they protect.

TABLE S8. Exact minimum number of faults with a nonzero effect in the relative problem of one gauging rectangle, for each CSS part, computed by CP-SAT from the phenomenological model (`rectangle_slice_check.py`, `rectangle_slice_check.csv`). The codes are periodic three-dimensional toric codes with $L = 3$ ($d_X = 9$, $d_Z = 3$), measured on the membrane $\overline{X}$ (two blocks: $\overline{X}_a\overline{X}_b$) with a grid of thickness H and m ancillas per edge; "pads" gives the undeformed rounds before and after the R deformed rounds, and $R = 0$ is idle memory. $\widetilde{d}_X$ is the exact static X distance of the deformed code (for $R = 0$ the bound of the X part is d_X). Every entry is optimal and equals the bound of Proposition SV.17.

Blocks	H	R	pads	$\widetilde{d}_X$	X part	d_Z	$\min\{d_Z, R\}$	Z part
2	1	0	1,1	—	9	3	3	3
2	1	1	1,1	3	3	3	1	1
2	1	2	1,1	3	3	3	2	2
2	1	4	1,1	3	3	3	3	3
2	2	2	1,1	6	6	3	2	2
2	2	4	1,1	6	6	3	3	3
2	3	2	1,1	9	9	3	2	2
2	3	4	1,1	9	9	3	3	3
2	3	3	2,2	9	9	3	3	3
2	3, $m = 2$	1	1,1	9	9	3	1	1
2	3, $m = 2$	2	1,1	9	9	3	2	2
2	3, $m = 2$	4	1,1	9	9	3	3	3
1	1	2	1,1	9	9	3	2	2
1	3	4	1,1	9	9	3	3	3

Proof. Subtract the fault-free propagation of the same incoming boundary variables (s, λ, ρ). The difference has zero relative input and the same internal fault vector, and by hypothesis zero syndrome and zero records at the right cut, so Proposition SV.17 applies and gives $\min\{\widetilde{d}_X, d_Z, R_g\} \geq \min\{\beta_0 d, R_g\}$. Idle memory of any length is the case $R = 0$, with bound d; the idle rounds on either side of a window are part of the rectangle in that proposition. The zero-syndrome condition at the right cut is needed: without it a single data Pauli in the last round would change the frame of an open cut. For an encoded birth in the X basis the X checks have deterministic first records, so the Z-type argument applies with the chain starting at the birth, while X-type errors act trivially on the prepared eigenstates; for a birth in the Z basis exchange X and Z. For a logical readout in the X basis

the transversal measurement returns every X check and the logical; the T5 readout rows force the Z error just before the readout to commute with the X checks, and it flips the reported logical only if it is a nontrivial Z logical, of weight at least d. Proposition SV.15 gives the same bounds by cleaning. □

s. Schedule invariants and paths between primitives. Every block occupies a fixed cell from its birth to its discard. A primitive P involves the cells of its participating blocks, $\beta(P)$, and its window $W(P)$ is the set of rounds from the last undeformed round before its entry to the first undeformed round after its exit. The *region* of P is the set of mechanisms on the cells of $\beta(P)$, including its grid ancillas, at rounds in $W(P)$. Adjacency in Γ_{un} means sharing a row of D^{loc} in some branch, or sharing a location. The canonical schedule satisfies three invariants.

(I1) *Cell locality.* Every physical qubit, including every extraction ancilla and routing site, belongs to one cell. Every row of D^{loc} either involves mechanisms of a single block only, or is a row of one primitive P and involves only mechanisms in the region of P. No row involves both a block discarded from a cell and a block later born in it.

(I2) *Padding.* If distinct primitives $P \neq P'$ involve the same block B, every round of $W(P)$ is at least $R_{\mathrm{pad}}(B)$ rounds from every round of $W(P')$, where $R_{\mathrm{pad}}(B) = R$ for a data or factory block and R_j for a courier block B_j. In particular a block is in at most one window at a time.

(I3) *Dedication.* A primitive that involves a dedicated courier block or the bare qubit of an injection site involves no block or bare qubit of another site, and each terminal transfer serves exactly one site.

(I1) holds because the extraction words of Proposition SV.8 use routing sites within a fixed radius R_* of their check: giving every cell a margin of width R_*, a constant overhead, keeps every idle word inside its block's cell and every word of a primitive inside the cells it involves, where its grid box lies. Births and discards use the one-sided T5 rows of a single block, which start from product states or end in transversal readouts. (I2) is the padding rule of the schedule, applied to every block including the factory block before and after each terminal transfer; in a parallel layout, primitives that share a block are placed in different time slots by a bounded-degree coloring, which changes the depth by a constant factor. The hypothesis $R_{\mathrm{pad}}(B) > \tau$ of the lemma below holds because the parameters are chosen with $R > \tau$ and $d_0 > \tau$ (Eq. (A19) and the courier estimate). (I3) is the courier convention. The branch alternatives at an adaptive slot share the slot's cells and window, so (I1) holds for the union graph.

t. An executable schedule certificate. The invariants are rules for building the schedule, so they can be checked on a schedule built by them. The script `schedule_verifier.py` builds one complete serial schedule on a fixed layout of 13 cells: sixteen Steane $7 \to 1$ runs and one Reed–Muller $15 \to 1$ run with an adaptive S slot after each T teleportation, feeding one data T gate with its own adaptive S slot. Each of the 127 raw candidates has its own bare qubit and courier chain ($d_0 = 2$, $d_1 = 3$, $R = 5$); two courier lanes per factory alternate, so courier cells are reused 246 times and courier chains run while the factory blocks work (488 pairs of primitives on disjoint blocks overlap in time). Births and discards are primitives with their own windows. The script exports the schedule (1292 primitives over 4496 rounds, with blocks, cells, sites, slots, windows and branch flags) and, from that export alone, generates the detector rows of templates T0, T3, T4 and T5 with one vertex per owner and round. It then checks the lifecycle (one live block per cell, no primitive on a dead owner), the geometry (disjoint cells; every routing ball inside its cell, or inside the cells of its primitive, with margin 2), the factory slots (one waiting resource per slot, every consumption reading the slot its resource was delivered to), (I1) for six branch assignments and for the union graph, (I2) with $\tau = 1$, and (I3); and, in the union graph, the conclusions of Lemma SV.19 for every primitive and of Lemma SV.25 for every pair of sites. All checks pass; the closest two site neighborhoods are 16 edges apart, against the required $R/\tau = 5$. Because the rows of any branch assignment are a subset of the union rows, and distances in a subgraph are not shorter, the union check covers all 2^{16} assignments. Nine negative controls, each breaking one rule (padding on the factory block, a shared courier block, early cell reuse, a two-block primitive across non-adjacent cells, a routing radius larger than the margin, a row joining two blocks, courier distance equal to τ, an adaptive branch reaching another block, a consumption from the wrong slot), are each detected. The rows are coarse, one vertex per owner and round, so the script certifies the schedule rules and this schedule, not the lattice-level rows or the arbitrary-size theorems; the earlier routing sketch `factory_3d_routing.py` is schematic, its coordinates overlap, and it is not used as a physical allocation.

Lemma SV.19 (Paths between primitives)**.** *Assume (I1) and (I2), with $R_{\mathrm{pad}}(B) > \tau$. (a) An edge of Γ_{un} with an endpoint outside every primitive region joins two mechanisms of one block. (b) No edge joins the regions of two distinct primitives. (c) Let γ be a path that starts in the region of P and reaches the region of some other primitive; let v' be its first vertex in the region of a primitive $Q \neq P$, and v its last vertex in the region of P before v'. Then v and v' lie on one block B involved in both P and Q, and are joined by a subpath on B with at least $R_{\mathrm{pad}}(B)/\tau$ edges.*

Proof. By (I2) a mechanism of a block lies in the region of at most one primitive, and grid mechanisms belong to their primitive. (a) A mechanism outside every region occurs in no primitive row, so its rows and its location involve one block. (b) A primitive row puts both mecha-

nisms in the region of that primitive. A single-block row joins mechanisms at most τ rounds apart on one block B; if they lay in the regions of $P \neq P'$, then B would be in both, and (I2) would separate them by $R_{\mathrm{pad}}(B) > \tau$ rounds. (c) By the choice of v and v', every vertex strictly between them lies outside every region, and by (b) there is at least one such vertex. By (a) the edges from v to v' stay on one block B, so v and v' are mechanisms of B in the regions of P and Q, and B is involved in both. Their rounds lie in $W(P)$ and $W(Q)$, at least $R_{\mathrm{pad}}(B)$ apart by (I2), and each edge changes the round by at most τ. □

The lemma excludes short paths through any number of intermediate cells, shared or adjacent: between two primitives a path must stay on one block for the whole padding of that block.

Theorem SV.20 (Padded protected composition)**.** *Synchronize interacting active windows into single catalog primitives and separate successive primitives sharing a retained block by R_{pad} complete idle rounds. Let τ bound the time separation of adjacent vertices in the phenomenological fault graph. Then the protected phenomenological fault distance of the composed data/factory schedule is at least*

$$f_{\mathrm{ph}} \geq \min\left\{ \min_g \delta_g^{\mathrm{rel}}, \delta_{\mathrm{mem}}^{\mathrm{rel}}, 1 + \left\lfloor \frac{R_{\mathrm{pad}} - \tau}{\tau} \right\rfloor \right\}. \quad \text{(A15e)}$$

Proof. Let a detector-trivial fault pattern have nonzero protected effect and choose a connected component C with nonzero contribution to that effect. If $|C|$ is below the third term of (A15e), then $|C|-1 \leq \lfloor (R_{\mathrm{pad}}-\tau)/\tau \rfloor - 1$, and a path of at most $|C|-1$ fault-graph edges bounds its temporal span by $\tau(|C|-1) \leq R_{\mathrm{pad}} - 2\tau$. It therefore cannot fill an idle pad, and by Lemma SV.19(c) it meets the region of at most one active primitive, also when two primitives on disjoint blocks run at the same time. If it meets an active primitive, each pad flanking that primitive keeps at least $2\tau \geq 2$ rounds free of C on its far side. If it meets none, it lies in idle rounds of every block it touches: when a round free of C follows it before the next active primitive, use the idle-memory rectangle with cuts just before C and after that round; otherwise C ends immediately before an active primitive g, keeps at least $2\tau \geq 2$ rounds free of it on its left, and the rectangle of g with cuts in its flanking pads contains it. Place the cuts in these pads, before the first mechanism of C and after a round free of C. Lemma SV.13 then turns C into a valid relative idle or active rectangle solution, and Lemma SV.14 shows that its cut syndromes and records vanish, so its relative effect equals its nonzero global contribution. Lemma SV.18 forces $|C|$ to be at least the corresponding local relative distance. If the component is not short, the third term already lower-bounds it. This proof uses inherited boundary variables rather than perfect artificial cuts. □

u. Explicit resource courier. Let B_j be a resource block of distance d_j. Choose fixed constants $1 < \lambda_- \leq \lambda_+ < \infty$ and an encoded-stage subsequence satisfying

$$\lambda_- d_j \leq d_{j+1} \leq \lambda_+ d_j. \quad \text{(A15e}')$$

Initialize the traveling slot of every courier block B_j ($j \geq 0$), and the designated slot t of the factory block F below, to $|+\rangle$. The fixed-factor geometric choice (A15e′) is compatible with the bounded-density code-family hypothesis used below: if $d(N)$ lies between fixed positive multiples of N^2, request the next available size above μN_j for a fixed μ chosen large enough; bounded density gives a fixed upper overshoot and hence constants $\lambda_- > 1$ and $\lambda_+ < \infty$. The raw stage injects a physical resource qubit q into B_0 by measuring $Z_q Z_{B_0}$ with the bare-port gauging construction, then measuring X_q and discarding q. Encoded courier stage $j \geq 0$ transfers the state from B_j to B_{j+1} by measuring $Z_{B_j} Z_{B_{j+1}}$, then measuring X_{B_j} and discarding B_j. The one-bit teleportation correction is kept in the Pauli frame. Encoded stage j uses $R_j = d_j$ deformed rounds and is followed by R_j idle rounds. Courier blocks are dedicated: every logical qubit of B_j other than the traveling slot is prepared by encoded birth in a fixed state, is never read, and is discarded together with B_j. The terminal transfer moves the resource from B_J into a designated slot t of a factory block F of the full distance d, using the courier-pair gauging of Theorem SIII.11 with R deformed rounds; F keeps its standard padding of R idle rounds before and after the transfer. The distance-one bare port occurs only in the raw stage; all later transfers are between encoded blocks.

Lemma SV.21 (Raw quotient spacetime distance)**.** *Let $\mathcal{R}$ be the Pauli subgroup generated by the retained traveling resource logical at the output of the raw injection macro-rectangle. Modulo $\mathcal{R}$, every detector-trivial phenomenological fault vector with nonzero relative protected effect has weight at least*

$$\delta'^{\mathcal{R}}_{\mathrm{raw}} \geq \min\{\beta_0 d_0, R_0\}. \quad \text{(A15f)}$$

Thus below this weight the only possible logical effect of the raw boundary is on the traveling resource; every spectator logical and protected committed outcome is unchanged.

Proof. Apply the slice argument of Proposition SV.17 in the $X \leftrightarrow Z$ dual convention, recording effects modulo $\mathcal{R}$. The reported outcome of $Z_q \overline{Z}$ and the record of the final physical X_q measurement enter only the one-bit teleportation correction of the traveling resource, so their flips lie in $\mathcal{R}$. A nonzero effect modulo $\mathcal{R}$ therefore forces one of the two end-slice operators to represent a class outside $\mathcal{R}$. For the end-slice operator of the gauged type, a normalizer element of the deformed code, this class lies outside the exceptional subgroup of Lemma SIII.5, so the operator, and hence the part of the fault vector that produces it, has weight at least $\beta_0 d_0$. For the error of the other type left on B_0 at the right cut (q has been measured and discarded), a class outside $\mathcal{R}$ is a nontrivial logical of B_0, of weight at least $d_0 \geq \beta_0 d_0$. In both cases every qubit of the support carries a fault, as in Proposition SV.17. Supplemental

Eq. (S6) then transports every remaining exceptional input/output Pauli into the designated traveling-resource subgroup. The bound $\min\{\beta_0 d_0, R_0\}$ stated in (A15f) is weaker than what this argument gives and is kept for uniformity. □

Lemma SV.22 (Encoded courier relative distance)**.** *Let $r_j \in \mathbb{F}_2^2$ be the incoming traveling-resource Pauli frame at encoded courier stage j, and let T_j be the fault-free one-bit teleportation map on that frame (the identity in the convention of Supplemental Eq. (S6)). Define the new frame change caused in this stage by $\Delta r_j = r_{j+1} + T_j r_j$. There is a constant $c_{\mathrm{cour}} > 0$ such that a detector-trivial phenomenological fault component producing $\Delta r_j \neq 0$ has weight at least*

$$\delta^{\mathrm{rel}}_{\mathrm{cour},j} \geq c_{\mathrm{cour}} \min\{\beta_0 d_j, R_j\}. \tag{A15g}$$

The incoming resource Pauli is a boundary condition and is not charged as a fault. Effects on the other logical qubits of full-distance blocks are controlled separately by Lemma SV.23.

Proof. The parity measurement $Z_{B_j} Z_{B_{j+1}}$ is a protected two-block gauging rectangle whose product-code distance is at least d_j; the following logical X_{B_j} readout is a protected single-block rectangle. Separate the two by, and follow them with, R_j idle rounds. Theorem SV.20 applied to this macro-rectangle gives a constant-factor lower bound $\Omega(\min\{\beta_0 d_j, R_j\})$ for any relative change in the protected logical/output data. The one-bit teleportation identity transports r_j to $T_j r_j$ in the absence of new faults, so a nonzero Δr_j is exactly such a relative change. This proves (A15g) after absorbing the fixed padding and catalog constants into c_{cour}. □

Lemma SV.23 (Spectator protection at courier boundaries)**.** *At an encoded stage $j < J$ both participating blocks are dedicated courier blocks, so the only logical effect of the stage that is ever read is the traveling-resource frame change Δr_j of Lemma SV.22. At the terminal transfer from B_J into slot t of a full-distance block F, let $\mathcal{R}_{\mathrm{term}}$ be the group generated by all logical operators of B_J and by the slot logicals $\overline{X}_t, \overline{Z}_t$. There is a constant $\alpha' > 0$ such that every detector-trivial phenomenological fault vector whose protected effect is nonzero modulo $\mathcal{R}_{\mathrm{term}}$ has weight at least $\alpha' \min\{\beta_0 d, R\}$.*

Proof. The first statement restates the courier convention. The logical qubits of B_j other than the traveling slot are prepared in fixed states, never read, and discarded. A logical Pauli on one of them commutes with the logical operators of the traveling slot, and hence with the stage's parity measurement and readout.

For the terminal transfer, take a connected detector-trivial component C with nonzero effect modulo $\mathcal{R}_{\mathrm{term}}$. If $|C| \geq 1 + \lfloor (R-\tau)/\tau \rfloor$ there is nothing to prove. Otherwise the temporal span of C is below R, so by Lemma SV.19 it meets no active primitive on F other than the terminal transfer. Place cuts in the idle pads of F before and after the transfer, and a cut on B_J at the start of the terminal macro-rectangle. The part of C outside the enclosed subchain lies on B_J and on earlier dedicated courier blocks, because these blocks interact with F only through the terminal transfer and with no other block at all, by (I1) and (I3). By Lemma SV.11, restricting the global solution to the enclosed subchain gives a valid relative solution whose inherited boundary variables on B_J are produced by that outside part. Their record component enters only the detector rows (the term $P_g \rho$ of (A15a)), which the relative problem subtracts. In the fault-free terminal transfer every inherited class of B_J, including its syndrome component, is mapped into $\mathcal{R}_{\mathrm{term}}$: the syndrome terms $C_g s$ and $G_g s$ of an inherited class supported on B_J act only on frames and recorded outcomes of B_J and of the traveling slot, and the one-bit teleportation identity (S6) carries the traveling slot of B_J to slot t, and all other logical qubits of B_J are discarded. Hence the restricted fault vector, a subset of C, has the same nonzero relative effect modulo $\mathcal{R}_{\mathrm{term}}$.

Apply the slice argument of Proposition SV.17 to the restricted vector, recording effects modulo $\mathcal{R}_{\mathrm{term}}$ as in Lemma SV.21. The recorded outcomes of $\overline{Z}_J \overline{Z}_t$ and of the final $\overline{X}_J$ readout enter only the Pauli frame of slot t, so their flips, including every chain of record faults through the window, are trivial modulo $\mathcal{R}_{\mathrm{term}}$. A nonzero effect modulo $\mathcal{R}_{\mathrm{term}}$ therefore forces an end-slice operator outside $\mathcal{R}_{\mathrm{term}}$. The one of the gauged type is a deformed logical class outside $\mathcal{R}_{\mathrm{term}}$, hence outside the group $\mathcal{Q}_1$ of Lemma SIII.6 with $C_1 = B_J$ and $C_2 = F$, and item 4 of Theorem SIII.11 gives it weight at least $\beta_0 d$. The one of the other type is a logical of F (the block B_J is discarded) acting outside slot t, of weight at least d. Every qubit of either operator carries a fault of the restricted vector. Taking the smaller of this bound and the padding term gives α'. □

Lemma SV.24 (Earliest-stage charging)**.** *For a connected detector-trivial component C that contributes a resource Pauli at a consumed-resource boundary, define*

$$j(C) = \min\{j : \Delta r_j(C) \neq 0\}$$

when such an encoded stage exists. If the resource Pauli is already created at the raw boundary and every encoded $\Delta r_j(C)$ vanishes, charge C only to the raw stage. Otherwise, for $j(C) \geq 0$,

$$|C| \geq \gamma \min\{\beta_0 d_{j(C)}, R_{j(C)}\} \tag{A15h}$$

for a constant $\gamma > 0$. No inherited resource Pauli is charged again at a later fault-free transfer.

Proof. By definition of $j(C)$, the incoming resource frame at stage $j(C)$ may contain the propagated contribution of C from earlier stages, but all previous encoded stage changes of that contribution vanish. Thus the first new resource-frame change is precisely the relative quantity controlled by Lemma SV.22. If the component fits inside that courier macro-rectangle, (A15g) applies directly. If

it also meets a distinct active courier window, its connected temporal span crosses an adjacent idle separation. For $j(C) \geq 1$, the fault-graph range bound costs $\Omega(R_{j(C)-1}) = \Omega(R_{j(C)})$, because $R_j = d_j$ and the fixed-factor bound gives $R_{j(C)-1} \geq R_{j(C)}/\lambda_+$. For $j(C) = 0$, the first-stage padding of R_0 rounds gives $\Omega(R_0)$. Taking the smaller fixed constant in the two cases gives γ. A component whose only new resource effect is at the raw boundary is instead assigned there and receives only the constant raw charge; later zero-Δr_j propagation never creates a second charge. □

Lemma SV.25 (Resource-boundary separation)**.** *For an injection site ι let A_ι be the set of mechanisms on its bare qubit and on its dedicated courier blocks $B_0^\iota, \ldots, B_J^\iota$ during their lifetimes, together with the regions of the raw injection, of the courier transfers, and of the terminal transfer T_ι of ι (the last on B_J^ι, on the factory block F at rounds of $W(T_\iota)$, and on its grid). Every stage set $N_{\iota,j}$ and $N_{\iota,\mathrm{term}}$ lies in A_ι. Under (I1)–(I3), for distinct sites $\iota \neq \iota'$ every path in Γ_{un} from A_ι to $A_{\iota'}$ has at least R/τ edges, so*

$$\rho \geq \kappa R, \qquad \kappa = 1/\tau. \tag{A15i}$$

In particular, a connected witness meeting the neighborhoods of two sites has more than R/τ mechanisms.

Proof. Let γ be such a path, let w be its first vertex in $A_{\iota'}$, and let v be its last vertex in A_ι before w. First, every neighbor in Γ_{un} of a mechanism on the bare qubit or on a dedicated block of ι, or on the grid of one of its primitives, lies in A_ι. By (I1) such a neighbor shares a single-block row with it, and then lies on the same dedicated block, or shares a row of a primitive that involves that block or grid; by (I3) this primitive is one of the raw injection, courier transfers or terminal transfer of ι, whose regions lie in A_ι. The neighbor of v on γ lies outside A_ι, so v is none of these mechanisms; it is a mechanism of the factory block F in the region of T_ι. In the same way w is a mechanism of a full-distance block in the region of $T_{\iota'}$, and $T_\iota \neq T_{\iota'}$ by (I3).

The subpath from v to w therefore starts in the region of T_ι and reaches the region of another primitive. Its vertices after v lie outside A_ι and hence outside the region of T_ι, so in Lemma SV.19(c), applied with $P = T_\iota$, the last vertex in the region of T_ι is v itself. The lemma gives a block B containing v and a subpath on B with at least $R_{\mathrm{pad}}(B)/\tau$ edges. The block containing v is F, a factory block, so $R_{\mathrm{pad}}(B) = R$. □

The proof uses no property of the layout beyond (I1)–(I3). A path that leaves the courier chain of ι can do so only through T_ι, and it must then stay on F for the whole padding that separates T_ι from every other primitive on F, whatever cells it visits afterwards. Reuse of a physical cell by a later courier block creates no path, by the last clause of (I1), and the small padding R_j of the courier blocks never enters the bound, because the separating block is always a factory block. The terminal stage set $N_{\iota,\mathrm{term}}$ is taken to be the region of T_ι, which has $O(d^{3/2}R)$ mechanisms as Theorem SV.26 requires; the idle pads of F around T_ι are protected memory of F, covered by item 1 of that theorem.

v. How the composition lemmas fit together. The next theorem is a direct assembly of six protections established above. Lemma SV.18 supplies the local relative distance of each active rectangle. Theorem SV.20 prevents a short detector-trivial component from canceling protected effects across gadget boundaries. Lemma SV.21 isolates the unique distance-one raw-resource exception by quotienting out the traveling-resource subgroup. Lemmas SV.22 and SV.24 ensure that inherited resource errors are transported as boundary data while the first stage that creates a new encoded resource error pays that stage's distance, and Lemma SV.23 shows that the small courier blocks never lower the distance of any logical qubit of a full-distance block. Lemma SV.25 keeps distinct resource boundaries at fault-graph distance at least κR, proportional to the padding scale. With the bounded circuit-to-phenomenological spread of the extraction schedule, these statements make the global circuit-distance bound below a composition theorem rather than a new case analysis.

Theorem SV.26 (Complete scheduled-circuit distance and courier locality)**.** *For the canonical extraction schedule and courier above there are constants $\alpha, C_0, \gamma, \kappa > 0$, independent of block size and circuit length, such that the complete adaptive scheduled circuit has the following properties.*

1. *Any detector-trivial circuit fault set that flips a protected gauging outcome or logical readout, or induces a nontrivial residual logical Pauli on a data or factory subsystem away from a resource boundary, has size at least*

$$\alpha \min\{\beta_0 d, R\}. \tag{A16}$$

2. *Below this bound, every detector-trivial effect at a raw-injection or courier boundary acts only on the designated traveling resource logical or on never-read logical qubits of dedicated courier blocks. Every other logical qubit, including every logical qubit of a data or factory block other than the designated slot, has trivial effect.*

3. *If a resource error is first created by faults at encoded courier stage $j \geq 0$, then any connected detector-trivial witness that creates it has size at least*

$$s_j \geq \max\{1, \gamma \min(\beta_0 d_j, R_j)\}. \tag{A17}$$

   The physical raw-injection stage has a separate constant bound $s_{\mathrm{raw}} \geq 1$.

4. *For the committed minimum-weight decoder, every wrong resource commitment on the complement of the large-cluster event has a connected half-dense witness S_ι anchored in a stage set $N_{\iota,j}$ with*

$$|N_{\iota,j}| \leq C_0 d_j^{3/2} R_j, \qquad |S_\iota| \geq s_j. \tag{A17a}$$

At the terminal stage, $|N_{\iota,\mathrm{term}}| \le C_0 d^{3/2} R$, which is $O(f^{5/2})$ for the choice $d, R = \Theta(f)$ made in the proof of Theorem SV.31.

5. *The neighborhoods of distinct resource sites are at union-fault-graph distance at least κR. Hence, after excluding a connected quarter-dense set meeting two such boundaries, witnesses assigned to distinct resource sites do not touch.*

Consequently the complete universal schedule, including gadget boundaries, raw resource injection, and encoded courier stages, has protected circuit fault distance $\Omega(\min\{\beta_0 d, R\})$ together with the weighted resource-locality bounds used below.

Proof. First replace every circuit fault by at most c_1 phenomenological mechanisms having the same complete detector and logical-effect vector. This follows from the bounded depth and bounded fan-in/fan-out of the canonical extraction circuit, and division by c_1 only changes constants.

For data and factory operations, Lemma SV.18 and Theorem SV.20 give a phenomenological distance $\Omega(\min\{\beta_0 d, R\})$ across the complete schedule, including cancellation opportunities at shared gadget boundaries. The proof uses the actual inherited boundary Pauli class supplied by Lemma SV.13, so nonzero inherited syndrome, record, and frame data are carried through the composition explicitly.

At a raw injection boundary, Lemma SV.21 excludes every sub-distance logical effect outside the traveling-resource subgroup. At the terminal courier transfer, Lemma SV.23 excludes every effect outside $\mathcal{R}_{\mathrm{term}}$ below $\alpha' \min\{\beta_0 d, R\}$, and earlier courier stages expose only never-read logical qubits of dedicated blocks. At encoded courier stages, Lemma SV.22 controls every new resource-frame change and Lemma SV.24 assigns each connected witness to the first stage at which that change is created. This proves the first three clauses after the circuit-level constant is absorbed into α and γ.

A stage-j courier neighborhood contains a constant number of layer-code blocks of volume $\Theta(d_j^{3/2})$ for R_j rounds and bounded-degree ancilla circuitry, so its number of mechanism vertices is at most $C_0 d_j^{3/2} R_j$. Lemma SV.6 supplies the half-dense connected minimum-weight-decoder witness. The same argument at terminal distance gives $C_0 d^{3/2} R = O(f^{5/2})$. Finally, Lemma SV.25 derives the linking length $\rho \ge \kappa R$, with $\kappa = 1/\tau$, from the schedule invariants (I1)–(I3) alone, for every path through any number of intermediate cells. The local catalog is finite and branch-uniform, so all constants are uniform over the realized causal branch. This proves the theorem. □

Corollary SV.27 (Circuit-distance parameters)**.** *For the schedule of Theorem SV.26, the padded composed circuit has injection locality at scale*

$$f_{\mathrm{circ}} = \lceil \alpha \min(\beta_0 d, R) \rceil \tag{A18}$$

and linking length $\rho = \Omega(R)$. A courier witness first created at encoded stage j has size at least $s_j = \lceil \gamma \min(\beta_0 d_j, R_j) \rceil$, while the physical raw stage has $s_{\mathrm{raw}} \ge 1$.

If the phenomenological-to-circuit hook replacement uses at most c_1 mechanisms per circuit fault, the constant α above already contains the factor $1/c_1$. Equivalently, a target circuit distance f is obtained by choosing

$$\beta_0 d \ge f/\alpha, \qquad R = R_{\mathrm{pad}} \ge \max\{f/\alpha, \tau + 1\}. \tag{A19}$$

The condition $R > \tau$ is the hypothesis $R_{\mathrm{pad}}(B) > \tau$ of Lemma SV.19 for data and factory blocks; it costs nothing asymptotically, since $\tau = O(1)$.

w. Distance throughout the computation. Table S9 collects, for every slot of the canonical catalog, the logical qubits it protects and the fault-distance bound that applies to them. Every logical qubit that is read later, other than a resource in transit, is protected at every step by circuit fault distance at least $\alpha \min\{\beta_0 d, R\} \ge f$. With the edge multiplicity m_0 of Theorem SIII.11, $\beta_0 = 1$, so gauging costs no fraction of the static distance. The only objects of smaller distance are the raw port and the dedicated courier blocks. Their errors can affect only the traveling resource; Lemma SV.24 charges them to the stage that creates them, and distillation suppresses them. Thin measurement layers, which would lower the distance of joint measurements, are excluded by Corollary SIII.3.

x. Courier estimate. Let $M_j \le C_0 d_j^{3/2} R_j$, $R_j = d_j$, and use the fixed-factor encoded sequence $\lambda_- d_j \le d_{j+1} \le \lambda_+ d_j$ with $1 < \lambda_- \le \lambda_+ < \infty$. Let $\gamma > 0$ satisfy $s_j \ge \gamma d_j$ for $j \ge 0$. Choose the fixed encoded starting distance d_0 large enough that $\gamma d_0 \ge 1$ and $d_0 > \tau$. Since $R_j = d_j \ge d_0$, every courier block then has $R_{\mathrm{pad}}(B_j) > \tau$, as Lemma SV.19 requires. The raw stage is kept separate: its anchor has constant volume $M_{\mathrm{raw}} = O(d_0^{5/2}) = O(1)$ and $s_{\mathrm{raw}} \ge 1$. For the terminal stage, $M_{\mathrm{term}} = O(f^{5/2})$ and $s_{\mathrm{term}} \ge \gamma d_J$. With $a = 2ez$ and absolute constants $C_{\mathrm{raw}}, C_2 > 0$,

$$\begin{aligned} q_{\mathrm{inj}} &\le C_{\mathrm{raw}}(a\sqrt{p}) + C_2 \sum_{j=0}^{J} d_j^{5/2} (a\sqrt{p})^{\gamma d_j} \\ &\quad + C_2 f^{5/2} (a\sqrt{p})^{\gamma d_J}. \end{aligned} \tag{A20}$$

Set $p_0 := \min\{p_\star/2,\, 1/(2a^2)\}$ and define

$$\eta = \inf_{0<p\le p_0} \frac{\log(1/(a\sqrt{p}))}{\log(1/p)} > 0. \tag{A21}$$

If

$$d_J \ge d_{\mathrm{base}} + C_{\mathrm{chain}} \frac{\log f}{\log(1/p)}, \qquad C_{\mathrm{chain}} > \frac{5}{2\gamma\eta}, \tag{A22}$$

then the terminal term is $O(p)$ after increasing d_{base} by an absolute constant. The raw term is $O(\sqrt{p})$ and, because $\gamma d_0 \ge 1$, the geometrically growing encoded-stage sum

TABLE S9. Distance ledger for the canonical schedule. Phenomenological bounds are listed; the circuit-level bounds follow on dividing by the spread c_1 of Proposition SV.8. Here d is the common distance of data and factory blocks, d_j the distance of courier block B_j, R the number of repeated and padding rounds, and $\beta_0 = 1$ for the edge multiplicity m_0.

Slot	Blocks	Protected logical qubits	Bound	Source
Idle extraction	data, factory	all	d	Lemma SV.18
One- or two-block gauging (gates, factories, consumption)	data, factory	all, and the measured outcome	$\min\{\beta_0 d, R\}$	Thm. SIII.11, Lemma SV.18
Encoded birth and readout	data, factory	all	$\min\{d, R\}$	Lemma SV.18
Raw injection	bare qubit, dedicated B_0	traveling resource only	$s_{\mathrm{raw}} \geq 1$ (charged)	Lemma SV.21
Courier stage $j < J$	dedicated B_j, B_{j+1}	traveling resource only	$\gamma \min\{\beta_0 d_j, R_j\}$ (charged)	Lemmas SV.22, SV.24
Terminal transfer	B_J and factory block F	resource: $\gamma\beta_0 d_J$ (charged); every other logical of F	$\alpha' \min\{\beta_0 d, R\}$	Lemma SV.23
Successive primitives on one block	any	all	$1 + \lfloor (R - \tau)/\tau \rfloor$	Thm. SV.20

is also $O(\sqrt{p})$. Hence $q_{\mathrm{inj}} \leq C_{\mathrm{inj}}\sqrt{p}$ uniformly in f for $p \leq p_0$.

The distillation constant in (A8) is C_{dist} of (A6), and the injected family has parameter $q_0 \leq 2q_{\mathrm{inj}} \leq C_{\mathrm{inj}}\sqrt{p}$ after enlarging C_{inj} by the factor two of the conditioning on G. One explicit safe threshold choice is

$$p_{\mathrm{th}} := \min\left\{p_0,\ (4C_{\mathrm{dist}}C_{\mathrm{inj}})^{-2}\right\}. \tag{A23}$$

y. Main architecture theorem and resource scaling. For this section, a *balanced good CSS family* means a CSS LDPC family indexed by its input-qubit length N with $k = \Theta(N)$, $d_X, d_Z = \Theta(N)$, uniformly bounded stabilizer weights and qubit/check degrees, and chosen bounded-weight X- and Z-generator lists containing $n_X, n_Z = \Theta(N)$ rows. Independence of every listed row is not required: the layer-code construction itself depends on the chosen check presentation [1]. The balance condition is exactly what is needed in the layer-code parameter map $[[\Theta(Nn_Xn_Z), k, \Omega(d\min\{n_X, n_Z\})]]$ (for bounded input check weight), giving $n = \Theta(N^3)$, $K = \Theta(N)$, and $d = \Theta(N^2)$ here. A family has *boundedly dense sizes* when there is c_{len} such that every sufficiently large target x has an available member with $N \in [x, c_{\mathrm{len}}x]$.

Lemma SV.28 (A bounded-density Quantum Tanner input family)**.** *A fixed-degree Quantum Tanner code family of Leverrier and Zémor [10], of the type used as a good CSS LDPC input for the optimal layer-code construction of Ref. [1], can be chosen to satisfy all of the preceding hypotheses. More explicitly, for each fixed sufficiently large degree $\Delta = q + 1$ with q an odd prime power, Dinur* et al. *[11] supply (and Leverrier–Zémor recall) an infinite family of left–right Cayley complexes over $G_i = \mathrm{PSL}_2(q^i)$. By Leverrier–Zémor Theorem 1, if finite component codes C_A, C_B satisfy the stated distance/robustness hypotheses, then the Tanner construction on that infinite complex family yields an asymptotically good CSS LDPC sequence. By their Theorem 2, for all large enough Δ a randomly chosen pair (C_A, C_B) succeeds with positive probability. The underlying left–right Cayley complexes are themselves explicit (LPS/Ramanujan-type Cayley graphs over $\mathrm{PSL}_2(q^i)$ as in Dinur* et al. *[11]). The random-method guarantee is the unconditional input used here; no deterministic derandomization is assumed. With one successful fixed pair in hand, if N_i denotes the input-code length then the available lengths are boundedly dense; one may take*

$$c_{\mathrm{len}} = q(q^2 + 1). \tag{A24}$$

Proof. The Quantum Tanner construction places qubits on the squares of a fixed-degree left–right Cayley complex. For each fixed large enough $\Delta = q + 1$, the underlying infinite family of left–right Cayley complexes over $G_i = \mathrm{PSL}_2(q^i)$ is that of Dinur *et al.* [11], as recalled by Leverrier and Zémor [10]. Leverrier–Zémor Theorem 1 states that if the finite component codes C_A, C_B meet the distance/robustness hypotheses, then the Tanner codes on that complex family are asymptotically good. Their Theorem 2 supplies, for all large enough Δ, a positive-probability existence proof (random method) that some fixed pair (C_A, C_B) works; we select one such pair existentially and reuse it for every i, without claiming a deterministic construction of the pair. Once q and that fixed finite component pair are chosen, both generator-set sizes are constant, so the number of squares, and hence the input-code length, has the form $N_i = c_{\mathrm{sq}}|G_i|$ for a constant $c_{\mathrm{sq}} > 0$ independent of i. The selected pair yields $k, d_X, d_Z = \Theta(N_i)$, while local degrees and component codes are fixed; hence stabilizer weights and qubit/check degrees are constant. In the standard local presentation a constant positive number of X- or Z-type constraint rows is attached per vertex of the corresponding bipartition. Since the number of vertices is $\Theta(|G_i|) = \Theta(N_i)$, the chosen generator lists satisfy $n_X, n_Z = \Theta(N_i)$.

Because $Q = q^i$ is odd, $|\mathrm{PSL}_2(Q)| = Q(Q^2-1)/2$. The factor $1/2$ and the constant c_{sq} cancel in successive length ratios, and an elementary calculation on this paper's part gives

$$\frac{N_{i+1}}{N_i} = q\,\frac{q^{2i+2}-1}{q^{2i}-1} = q\left(q^2 + \frac{q^2-1}{q^{2i}-1}\right) \le q(q^2+1). \tag{A24a}$$

Thus the next available length above any sufficiently large target is at most a factor $q(q^2+1)$ larger. With the existentially selected fixed component pair applied uniformly along the infinite sequence, this proves the bounded-density input-family existence needed for the resource theorem. □

z. Coarse fragmentation parameter. Let $A \subset \mathbb{Z}^3$ be the finite set of allocated logical sites and $W = |A|$. For an integer $b \ge 1$, partition the logical lattice into axis-aligned $b \times b \times b$ coarse boxes. Let $Q_b(A)$ be the number of coarse boxes that meet A, and let $B_b(A)$ be the number of those boxes that are not completely filled by sites of A. Thus $B_b(A)$ measures boundary and fragmentation at scale b, rather than microscopic surface area.

Lemma SV.29 (Architecture packing with an explicit fragmentation penalty)**.** *Let a layer-code cell encode K logical qubits. Choose an integer $b = \Theta(K^{1/3})$ with $b^3 \le K$. Any finite nearest-neighbor logical region $A \subset \mathbb{Z}^3$ can be assigned to*

$$O\left(\frac{W}{K} + B_b(A) + 1\right) \tag{A24b}$$

data cells. With all cells oriented using the same X- and Z-slab axes, every nearest-neighbor logical interaction is either intra-cell or between coarse-neighbor cells. Boundary interactions admit an $O(1)$ color schedule, and a constant-factor enlargement of the coarse cell lattice provides neighboring factory/courier positions while preserving the count in (A24b).

Proof. Assign one layer-code cell to every occupied coarse box. A full occupied box contains exactly $b^3 = \Theta(K)$ allocated sites and fits in one cell because $b^3 \le K$. If $F_b(A) := Q_b(A) - B_b(A)$ is the number of full boxes, then $F_b(A)b^3 \le W$, and hence

$$\begin{aligned} Q_b(A) &= F_b(A) + B_b(A) \\ &\le \frac{W}{b^3} + B_b(A) \\ &= O\left(\frac{W}{K} + B_b(A)\right). \end{aligned}$$

The additive one in (A24b) covers the one-cell convention at very small W.

Two nearest-neighbor logical sites lie either in the same coarse box or in boxes sharing a coarse face. Contracting occupied boxes therefore gives a subgraph of the bounded-degree cubic coarse lattice. Its inter-cell edges have an $O(1)$ edge coloring, so crossing gates can be serialized by only a constant factor and every interacting pair remains at fixed bounded cell separation, as required by Theorem SIII.11. A common physical orientation aligns the slab axes. Finally, replace each coarse lattice site by a fixed constant-size supercell containing one data position and a constant number of factory/courier reserve positions. This constant refinement preserves bounded separation and multiplies the cell count by only a constant. □

Corollary SV.30 (Bounded-aspect specialization)**.** *If A is a connected bounded-aspect three-dimensional grid box, then at the packing scale above*

$$B_b(A) = O\Big((W/K)^{2/3} + 1\Big),$$

and the number of data/factory cells is $O(W/K + 1)$.

Proof. Only coarse boxes intersecting the $O(1)$ outer faces can be partial. For a bounded-aspect box the number of such boxes is at most a constant times its coarse surface area, $O((W/b^3)^{2/3} + 1)$. Since $b^3 = \Theta(K)$ this is the stated bound. When $W \ge K$, $(W/K)^{2/3} \le W/K$; when $W < K$ the additive one dominates. □

aa. Constant and ceiling ledger. The constants β_0 (static gauging), c_1 (circuit-to-phenomenological spread), α (composed circuit distance), γ (earliest-stage courier charge), C_0 (stage volume), κ (resource-boundary linking), C_{dist} (distillation pairs), $p_\star$ (cluster counting), and the chosen p_{th} depend only on the fixed input family constants, finite local cell/template catalog, and fixed scheduling conventions. They are independent of W, T, N, f and of the realized adaptive branch. Ceilings change the displayed distance and padding inequalities only by additive $O(1)$ terms, which are absorbed by $f_{\min}$. The physical raw stage remains separate with $s_{\mathrm{raw}} \ge 1$ and is never assigned encoded distance.

bb. Informal corollary (connected bounded-aspect regions). For a connected bounded-aspect logical region A, Corollary SV.30 reduces the cell count to $O(W/K+1)$. Below threshold at target failure ε, with $f = \Theta(\log(WT/\varepsilon))$ at fixed $p < p_{\mathrm{th}}$, universal nearest-neighbor computation therefore uses roughly

$$O\big(Wf + f^{3/2}\big)$$

physical qubits, together with the polylogarithmic depth factor Ψ of (A27). The fully general fragmentation-aware resource theorem is stated next.

cc. Dependency map. Figure S7 separates the theorem chain from cited inputs and finite supporting checks.

Theorem SV.31 (Universal fault-tolerant resource theorem)**.** *Use the fixed-degree Quantum Tanner family of Lemma SV.28, or more generally any balanced good CSS family with boundedly dense sizes, and apply the explicit layer-code realization of Sec. SI A to it, using the slab representatives of Theorem SI.3 and the distance bound*

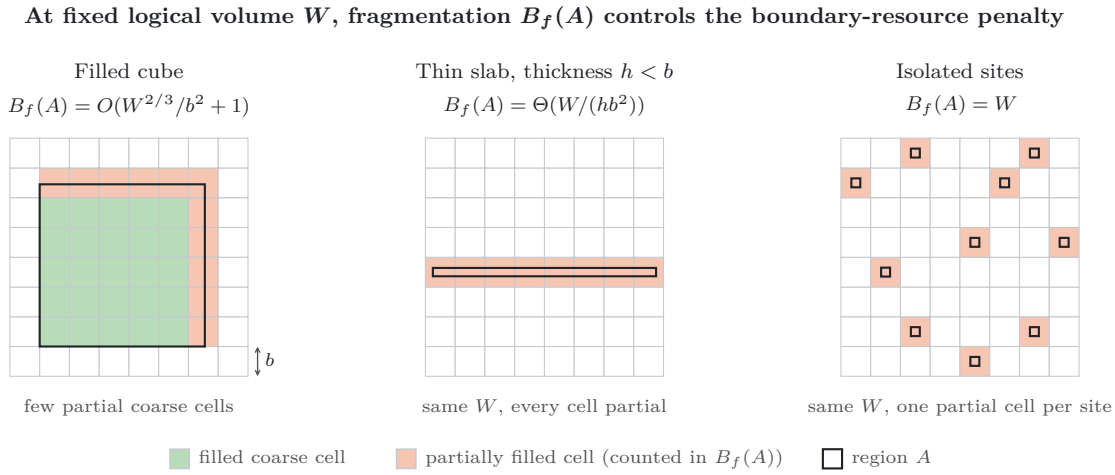


FIG. S6. Qualitative role of the coarse fragmentation term $B_f(A)$ at fixed logical volume W, with coarse boxes of side $b = \Theta(f^{1/6})$. A filled bounded-aspect cube has $B_f(A) = O(W^{2/3}/b^2 + 1)$, so the partial-cell term $f^{3/2}B_f$ is subleading once $W \gtrsim \sqrt{f}$. A thin slab of thickness $h < b$ has $B_f(A) = \Theta(W/(hb^2))$ and a partial-cell cost $\Theta(Wf^{7/6}/h)$, which exceeds the extensive Wf term. Isolated sites at mutual spacing at least b give $B_f(A) = W$ and cost $\Theta(Wf^{3/2})$. These terms enter the space bound $O\big(Wf + (B_f(A) + 1)f^{3/2}\big)$ of Theorem SV.31/Eq. (A26).

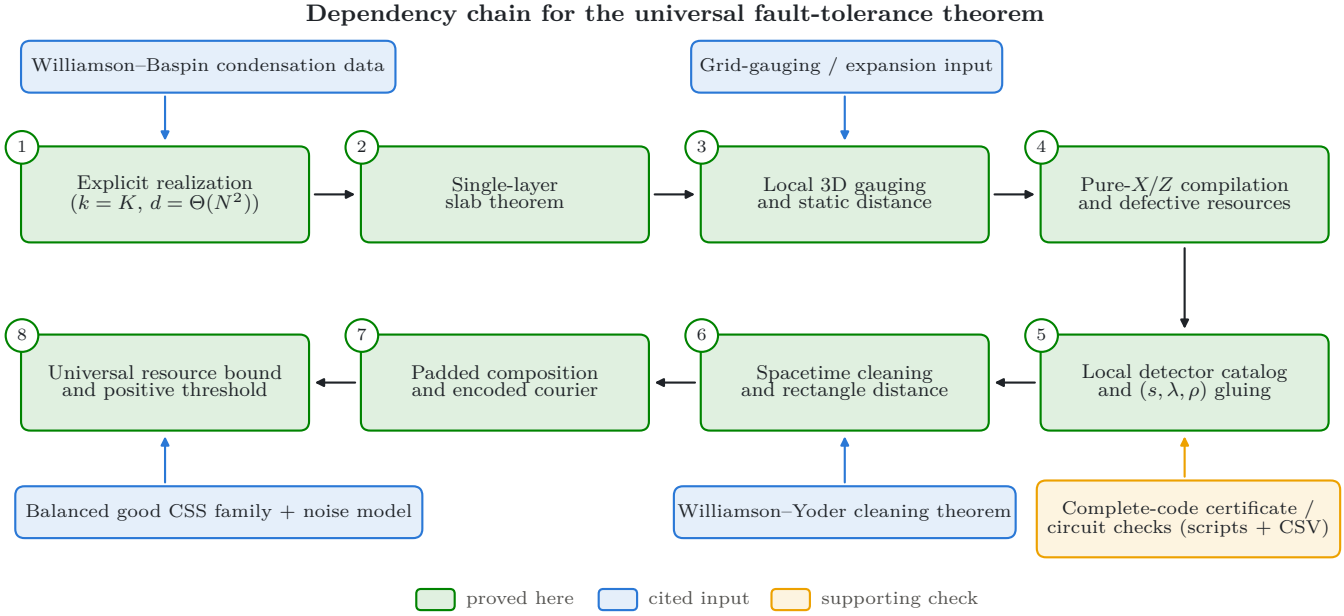


FIG. S7. Dependency map for the universal fault-tolerance theorem. Green boxes form the proof chain, blue boxes are cited inputs, and amber boxes are finite supporting checks. The supporting checks and numerical crossings are not hypotheses of Theorem SV.31.

of Theorem SI.6. Use the explicit resource-coordinate convention of Lemma SV.4 and Tables S11 and S12, together with the canonical extraction schedule and encoded courier of Theorem SV.26. Under the local-stochastic Pauli noise model (A1), there are constants $p_{\mathrm{th}}, c, f_{\min} > 0$ such that, for $0 < p < p_{\mathrm{th}}$, a nearest-neighbor three-dimensional logical circuit whose finite allocated region is $A \subset \mathbb{Z}^3$, with $W = |A|$, depth T, and target error $0 < \varepsilon < 1/2$ can be simulated with

$$f = \max\left\{f_{\min}, \left\lceil \frac{c\log(WT/\varepsilon)}{\log(p_{\mathrm{th}}/p)} \right\rceil\right\}. \tag{A25}$$

Let $K = \Theta(\sqrt{f})$ be the number of logicals in the selected layer-code cell, choose $b = b(f) = \Theta(K^{1/3}) = \Theta(f^{1/6})$ as in Lemma SV.29, and abbreviate $B_f(A) := B_{b(f)}(A)$.

The simulation uses

$$O\Big(Wf + (B_f(A)+1)f^{3/2}\Big) \tag{A26}$$

physical qubits and depth

$$O(Tf^{3/2}\Psi), \qquad \Psi = O(15^\ell) = O\big(f^{\log_2 15}\big), \tag{A27}$$

where the number ℓ of distillation levels obeys $2^\ell = O(\min\{f, \log(WT/\varepsilon)\})$. Thus the depth is $O(Tf^{3/2+\log_2 15}) \subseteq O(Tf^{5.41})$ for the minimal-space factory schedule. Every persistent data logical is stored in a layer-code block. Reserved gate ancillas and live factory resources are included in the physical-space count.

Proof. Fix the edge multiplicity m_0 of Theorem SIII.11, so that $\beta_0 = 1$; any other fixed multiplicity changes only constants. Choose the input size so that $\beta_0 d \geq f/\alpha$. By Theorem SV.26 the resource boundaries have linking $\rho \geq \kappa R$ for fixed $\kappa > 0$. Define

$$C_R := \max\{1, 1/\alpha, 1/\kappa\}, \qquad R = R_{\mathrm{pad}} = \lceil C_R f \rceil.$$

Then $\alpha \min\{\beta_0 d, R\} \geq f$ and $\kappa R \geq f$ exactly; finite endpoint offsets are absorbed in the already fixed $f_{\min}$. Lemma SV.28 supplies bounded density for the Quantum Tanner input family (or use the assumed bound for a general family), giving $N = \Theta(\sqrt{f})$, $K = \Theta(\sqrt{f})$, and $n = \Theta(f^{3/2})$. Lemma SV.29 gives $O(W/K + B_f(A) + 1)$ data/factory cells and makes the fixed-separation two-cell requirement explicit. Multiplying by the cell volume $n = \Theta(f^{3/2})$ gives (A26), since $(W/K)f^{3/2} = \Theta(Wf)$. Each factory's live courier uses only a constant number of additional blocks at any one time; these occupy the factory/courier reserve positions of the supercells of Lemma SV.29 and are already included in the cell count.

Apply Theorem SV.7, (A20), and the defective-resource lemmas. Conditional on the large-cluster complement, the injected resources form a local-stochastic defective family (Definition SV.2) with parameter $q_0 \leq C_{\mathrm{inj}}\sqrt{p}$ after enlarging the fixed constant if necessary, and (A8) makes the final resource error at most $\varepsilon/(2c_{\mathrm{res}}WT)$ after the stated number of levels, where $c_{\mathrm{res}} = O(1)$ bounds the number of final resources consumed per logical site and layer. The complete fixed skeleton, including factory and courier locations, has $M \leq C_M WT\Psi f^{a_M}$ mechanisms for fixed C_M, a_M. Put $X = WT/\varepsilon$, $L = \log(p_{\mathrm{th}}/p)$, and $\delta = \log(p_\star/p_{\mathrm{th}}) \geq \log 2$. Then

$$(p/p_\star)^{f/4} = \exp[-f(L+\delta)/4]. \tag{A28}$$

If the second term in (A25) determines f, the factor $e^{-fL/4}$ is at most $X^{-c/4}$. If instead $f = f_{\min}$, then $c\log X/L \leq f_{\min}$, so again $e^{-fL/4} \leq X^{-c/4}$. The remaining $f^{a_M}e^{-\delta f/4}$ is uniformly bounded for $f \geq f_{\min}$, and $\Psi = (\log X)^{O(1)}$. Choosing c larger than the fixed polynomial exponents therefore makes (A12) at most $\varepsilon/2$, uniformly for every $0 < p < p_{\mathrm{th}}$. A union bound over consumed final resources completes the correctness proof.

The recurrence may be stopped at the least ℓ satisfying $2^\ell \log(1/(C_{\mathrm{dist}}q_0)) \geq c' \log X$ for a fixed c'. Because $q_0 \leq C_{\mathrm{inj}}\sqrt{p}$ and (A23) gives $C_{\mathrm{dist}}C_{\mathrm{inj}} \leq 1/(4\sqrt{p_{\mathrm{th}}})$, we have $C_{\mathrm{dist}}q_0 \leq \frac{1}{4}\sqrt{p/p_{\mathrm{th}}}$ and hence $\log(1/(C_{\mathrm{dist}}q_0)) \geq \log 4 + L/2$. Minimality of ℓ therefore gives $2^\ell \leq 2c' \log X/(\log 4 + L/2)$. This is $O(\log X)$, and since (A25) gives $\log X \leq fL/c$ it is also at most $4c'f/c = O(f)$, including the regime $f = f_{\min}$. Consequently $\ell \leq \log_2 f + O(1)$ and $\Psi = O(15^\ell) = O((2^\ell)^{\log_2 15}) = O(f^{\log_2 15})$, with $\log_2 15 \approx 3.907$. A depth-first factory schedule stores only $O(\ell)$ completed candidate resources at once (with a constant factor for the branching protocol), which fits among $K = \Theta(\sqrt{f})$ factory logicals after increasing $f_{\min}$. One distilled resource uses $O(\Psi)$ serialized gadget slots, each of $O(f)$ rounds, hence costs $O(f\Psi)$ rounds. A logical layer requires $O(K)$ resources per data block; producing the batch costs $O(Kf\Psi) = O(f^{3/2}\Psi)$ rounds, proving (A27). This serial, depth-first schedule is chosen to keep the space bound (A26); running several factories in parallel per data block would shorten the depth at the cost of additional factory cells. □

For a connected bounded-aspect allocated box, Corollary SV.30 gives $B_f(A) = O((W/\sqrt{f})^{2/3}+1)$. If $W \geq \sqrt{f}$, its contribution $f^{3/2}B_f(A) = O(W^{2/3}f^{7/6} + f^{3/2})$ is $O(Wf + f^{3/2})$; if $W < \sqrt{f}$, one-cell/additive terms dominate. Thus the previous compact space bound $O(Wf + f^{3/2})$ is recovered in the bounded-aspect regime, while (A26) quantifies the extra cost of thin, holed, or fragmented regions.

Corollary SV.32 (Transfer to uniformly slab-supported 3D CSS families)**.** *The layer-code origin of the slabs is not used after Theorem SI.3. Let $\{\mathcal{C}_N\}$ instead be any three-dimensional local CSS family satisfying all of the following uniformly: (i) an embedding in a fixed finite-type three-dimensional lattice with bounded qubit/check density, check weight, check degree, and stabilizer diameter; (ii) physical block aspect ratio $O(1)$; (iii) parameters $n = \Theta(N^3)$, $K = \Theta(N)$, and $d_X, d_Z = \Theta(N^2)$; (iv) boundedly dense available values of N; and (v) fixed coordinate axes such that every pure X logical class and every pure Z logical class has a representative supported in at most $t_0 = O(1)$ parallel coordinate planes of the appropriate orientation. Then the local gauging, detector, spacetime-composition, courier, and threshold arguments of Secs. SIII–SV apply verbatim, and Theorem SV.31 holds for this family with the same asymptotic bounds, including the fragmentation term in (A26).*

Proof. The layer-code defect network is used in the architecture proof only to establish the uniform slab representatives and the parameter scaling. Under assumptions (i)–(v), those inputs are supplied directly. Corollary SIII.9 and Lemma SIII.10 give constant relative expansion after the same constant refinement; Theorem SIII.11 then gives constant-factor deformed distance and bounded local incidence. The finite-radius extraction and detector-template arguments of Proposition SV.8 and Proposition SV.9 use

only those uniform locality bounds. Boundary gluing, relative cleaning, padded composition, and the courier lemmas are code-family agnostic once the static space-distance and local-template hypotheses hold. Finally $n = \Theta(N^3)$, $K = \Theta(N)$, $d = \Theta(N^2)$ and boundedly dense sizes reproduce $N = \Theta(\sqrt{f})$ and the resource accounting in Theorem SV.31. □

For very small p, scaling statements should use $\max\{1, \log(WT/\varepsilon)/\log(1/p)\}$; otherwise a displayed qubit count would incorrectly tend to zero. Finally, the extensive lower bound $\Omega(Wf)$ matches (A26) only when $W = \Omega(\sqrt{f})$, so that the additive one-block term does not dominate. The occupancy condition $k = O(\sqrt{f})$ alone does not establish constant-factor optimality in the small-W regime.

SVI. STORAGE BENCHMARK

A. Statement and scope

Related volume limits on quantum storage appear in Ref. [12]. For comparison only, consider fully occupied unknown-data blocks stored through independent depolarizing noise with $0 < p \le 3/4$, target infidelity $0 < \varepsilon \le 1/2$, and maximally entangled logical inputs. Put $q = p(1-p)/3$. The reference-system Pauli-twirl argument below gives

$$F_e \le \prod_{b=1}^{B} \left(1 - q^{\lceil d_b/2 \rceil}\right), \tag{P74}$$

and therefore, whenever $F_e \ge 1 - \varepsilon$,

$$\sum_b q^{\lceil d_b/2 \rceil} \le -\log(1-\varepsilon) \le 2\varepsilon. \tag{P75}$$

For equal $[[n, k, d]]$ blocks with $B = W/k$, this implies $d \ge 2\log(W/(2k\varepsilon))/\log(1/q) - 2$. A valid nonnegative distance for the Bravyi–Poulin–Terhal (BPT) inequality $kd^{2/(D-1)} \le c_D n$ in D dimensions is

$$d_* = \max\left\{1,\ 2\left\lceil \frac{\log(1/(2\varepsilon))}{\log(1/q)} \right\rceil - 1\right\}, \tag{P76}$$

so a valid nonuniform volume benchmark is

$$\sum_b n_b \ge \frac{W}{c_D}\, d_*^{2/(D-1)}. \tag{P77}$$

This comparison keeps k inside $\log(W/(k\varepsilon))$ and does not yield a uniform $\log(W/\varepsilon)$ bound over arbitrary occupancies. Matching to the upper space bound is claimed only in the stated comparable regime, including $W = \Omega(\sqrt{f})$; reserved gate/factory ancillas are not counted as stored unknown data in the same W.

The proof of Eqs. (P74)–(P77) follows.

Proof. Choose a minimum-weight nonidentity logical Pauli L_b in block b, split its support into sets of sizes $a_b = \lceil d_b/2 \rceil$ and $d_b - a_b$, and let E_1, E_2 be the restrictions of L_b to those sets. For each Pauli F outside this support, the independent depolarizing channel assigns probabilities

$$P_1(F) = (p/3)^{a_b}(1-p)^{d_b - a_b}\Pr(F), \tag{P78}$$

$$P_2(F) = (p/3)^{d_b - a_b}(1-p)^{a_b}\Pr(F). \tag{P79}$$

Set $m_F = \min(P_1(F), P_2(F))$ and $M_b = \sum_F m_F$. Since $p/3 \le 1 - p$,

$$M_b = (p/3)^{a_b}(1-p)^{d_b - a_b} \ge q^{a_b}. \tag{P80}$$

Pairing these two errors decomposes the block channel into a component of weight $2M_b$ of the form $\mathcal{N}_b \circ \mathcal{D}_{L_b}$ and a remaining channel, where

$$\mathcal{D}_{L_b}(\rho) = \tfrac{1}{2}(\rho + L_b \rho L_b^\dagger) \tag{P81}$$

and $\mathcal{N}_b$ is a mixture of the errors $E_1 F$. On half of a maximally entangled logical state, $\mathcal{D}_{L_b}$ can equivalently act by L_b^T on the reference. It then commutes with every subsequent operation on the physical system.

Expand the product of block channels and let S be the subset of blocks selecting the twirled component. The $|S|$ independent reference Paulis map the maximally entangled target projector to $2^{|S|}$ orthogonal rank-one projectors with equal weights. The pinched target therefore has operator norm $2^{-|S|}$. The conditional output fidelity, even after an arbitrary joint recovery, is at most that norm. Averaging gives

$$F_e \le \mathbb{E}_S 2^{-|S|} = \prod_b (1 - M_b) \le \prod_b (1 - q^{a_b}). \tag{P82}$$

Taking logarithms and using $-\log(1-x) \ge x$ proves Eq. (P75). Inverting $q^{\lceil d/2 \rceil} \le u$ requires $\lceil d/2 \rceil \ge \lceil \log(1/u)/\log(1/q) \rceil$, hence $d \ge 2\lceil \log(1/u)/\log(1/q) \rceil - 1$. This yields Eq. (P76); the BPT tradeoff then gives the volume bound. □

a. Median-distance consequence. Put $a = \log(B/(4\varepsilon))/\log(1/q)$. Fewer than half the blocks satisfy

$$d_b < 2\lceil a \rceil - 1. \tag{P83}$$

Indeed, this inequality implies $\lceil d_b/2 \rceil < a$, so $q^{\lceil d_b/2 \rceil} > 4\varepsilon/B$; summing proves the count. The ceiling is essential. For example, one unencoded qubit with $B = n = k = d = 1$ and $\varepsilon = p = 0.01$ has $q = 0.0033$ and $2a \simeq 1.12670$; the integer cutoff in Eq. (P83) is instead 1.

b. Scope of the benchmark. The comparison keeps k inside $\log(W/(k\varepsilon))$. It does not imply a lower bound with $\log(W/\varepsilon)$ uniformly over arbitrary occupancies. For the bounded-aspect logical regions in which the architecture upper bound reduces to $O(Wf + f^{3/2})$, the $W < \sqrt{f}$ regime is dominated by one layer-code block; Eq. (P77) supplies only a bound of order Wf in the comparable

logarithmic regime. Thus constant-factor matching to this lower bound is restricted to $W = \Omega(\sqrt{f})$ as well as the stated occupancy, depth, geometry, and noise restrictions; irregular regions additionally incur the fragmentation term in (A26).

The storage benchmark applies to fully occupied unknown-data blocks. Reserved logical qubits used for known gate ancillas or factory states are not counted as stored unknown data in the same W; computational-space comparisons must therefore account for working logical capacity separately.

SVII. REUSABLE BOUNDARY-STATE COMPOSITION THEOREM

The complete-circuit proof uses the schedule-specific restriction lemma, Lemma SV.13. We record its abstract connected-component form here for reuse; this section introduces no additional hypothesis into Theorem SV.31.

Proposition SVII.1 (Canonical rectangle restriction)**.** *For every realized causal branch of the canonical finite slot catalog, the global detector/effect matrix satisfies the rectangle restriction property: if a connected detector-trivial component meets at most one active primitive, then its restriction to a surrounding padded subchain is a valid relative rectangle solution with the inherited boundary syndrome, last-round records, and logical frame left arbitrary, and with exactly the same protected relative effect.*

Proof. This is Lemma SV.13. The required physical rows are the uniformly local rows of D^{loc} from Proposition SV.9; Proposition SV.10 and Lemma SV.11 show that complete boundary variables suffice to glue and restrict them. The full space $\mathcal{R}_g^{\mathrm{full}}$ is only the algebraic closure used to audit boundary relations, so no nonlocal left-kernel basis is inserted into the fault graph and no perfect-cut assumption is introduced. □

Theorem SVII.2 (Abstract padded connected-component composition)**.** *Consider any linear phenomenological detector model with bounded temporal fault-graph range τ. Suppose successive active rectangles sharing support are separated by R_{pad} idle rounds, every connected component meeting at most one active rectangle has a valid relative restriction with its protected effect preserved, protected active rectangles have relative fault distances δ_g^{rel}, and idle rectangles have relative fault distance $\delta_{\mathrm{mem}}^{\mathrm{rel}}$. Then*

$$f_{\mathrm{ph}} \geq \min\left\{ \min_g \delta_g^{\mathrm{rel}},\ \delta_{\mathrm{mem}}^{\mathrm{rel}},\ 1 + \left\lfloor \frac{R_{\mathrm{pad}} - \tau}{\tau} \right\rfloor \right\}. \quad \text{(P84)}$$

If the detector/effect vector of every circuit mechanism is the sum of the vectors of at most c_1 phenomenological mechanisms, then

$$f_{\mathrm{circ}} \geq \lceil f_{\mathrm{ph}}/c_1 \rceil\,. \quad \text{(P85)}$$

Proof. Take a detector-trivial pattern with nonzero protected effect. By Lemma SV.12, one of its connected components C is itself detector trivial and has nonzero protected contribution. If C is smaller than the third term, its temporal span is below R_{pad} and it meets at most one active rectangle. The assumed relative restriction then places it in one active or idle rectangle without changing its protected relative effect, so its size is at least the corresponding local relative distance. Otherwise its size is already at least the padding term. The circuit-level statement follows by replacing each circuit mechanism by its bounded phenomenological image; connectivity of that image is not required. □

For the canonical schedule, Proposition SVII.1 and Lemma SV.18 verify the hypotheses of Theorem SVII.2, and Eq. (P84) is exactly the protected composition input used in Theorem SV.26. The raw resource boundary lies outside the ordinary protected class and is handled by Lemma SV.21; encoded courier changes are handled by Lemmas SV.22 and SV.24.

SVIII. REPRODUCIBILITY PACKAGE

The submission includes `reproducibility_checks.py`, `glue_example_check.py`, and `operational_anchor.py`. Running

```
python reproducibility_checks.py
```

checks the slab-cell boundary matrices derived below from the published condensation generators, including the 110 pass-through and their $X \leftrightarrow Z$ duals. It also tests all even prefix-parity words of lengths 2 to 8, longer deterministic stress words, and the GF(2) support-rank formula used by the slab criterion. For the logical and detector layers it checks the seven- and fifteen-qubit single-Z syndrome maps; the S, T, S_X, and CNOT Kraus identities; T0–T7 detector matrices; boundary-state and gluing identities; one-bit teleportation and Pauli transport; and $SS_XS = e^{i\pi/4}H$ up to global phase. Finally, it propagates single-location Paulis through the T8 master words for support weights 1–8 and shuttle lengths 0–2, transcribes the WB Sec. VI B items 3–8 junction supports in unit-cubic edge coordinates, checks the two surface-code subdivision identities, and checks the Steane and punctured Reed–Muller transversal phase orientations.

The scripts write `factory_map_audit.csv`, `t8_single_fault_audit.csv`, `t8_master_routing_catalog.csv`, `slab_cell_audit.csv`, `wb_junction_support_catalog.csv`, and `refinement_move_audit.csv`. The matrix comparisons use tolerance 10^{-12}. The package includes a pinned `requirements.txt` and `environment-known-good.json`; the latter records the Python and package versions used for the archived numerical campaign.

The slab-cell calculation derives $H_{\mathrm{end}} = [1\ 1]$ and $H_{\mathrm{mid}} = [1\ 1\ 1]$ (kernels $\{00, 11\}$ and $\{000, 110, 101, 011\}$, including the 110 pass-through) from the published condensation generators.

a. Construction, distance, composition, and resource scripts. `layer_code_explicit.py` generates the complete check matrices of the explicit realization (and of the single-chart transcription, as a negative control) for any CSS input. `slab_gluing_certificate.py` runs the certificate of Sec. SII and writes `slab_cert_A_archived_all_words.csv`, `slab_cert_A_archived_full_syndrome.csv`, `slab_cert_B_complete_codes.csv`, `slab_cert_C_forms.csv`, `slab_cert_D_exhaustive_words.csv`, and `slab_cert_E_wall_controls.csv`; two instances are exported with all check and qubit labels (`explicit_layer_code_422_s2.npz`, `explicit_layer_code_steane_s2.npz`). `explicit_layer_code_distance.py` computes the distances of Table S2 with the CP-SAT solver wrapped in `min_weight_logical.py`. `distance_lower_bound_check.py` checks the plane-restriction and shortest-side steps of Theorem SI.6 and compares its bound with these distances (`distance_lower_bound_check.csv`, `distance_bound_vs_exact.csv`). `frame_propagation_check.py` checks the syndrome terms of (A15b) and the contained-component lemma; `glue_example_check.py` checks the record-aware worked example; and `boundary_composition_check.py` compares the direct and composed detector/effect maps on explicit blocks (`boundary_composition_check.csv`). `resource_defect_check.py` checks the defective-resource lemmas, the single-defect exactness of both factories, and the malignant-pattern counts (`resource_defect_check.csv`). `two_block_measurement.py` builds and analyzes the circuits of Sec. SXIV, with recorded integer seeds (`tighten_two_block.py` reruns selected solves with a longer time limit); `resource_estimate.py` produces the estimate of Sec. SXVI; and `make_junction_figure.py` draws Fig. S1 from a generated instance. These scripts need NumPy, SciPy, Stim, ldpc (GF(2) rank and BP+LSD decoding), OR-Tools, and Matplotlib, pinned in `requirements-revision.txt` and installed with `pip install -r requirements-revision.txt`. The exact commands and the time limits of the CP-SAT runs are listed in `REVISION_RUNS.md`.

The archived scripts `wb_microscopic_bridge_audit.py`, `wb_exact_cell_patch_audit.py`, and `wb_layer_code_instance.py` are retained for provenance; Sec. SXX F explains why their certificates are superseded.

b. Operational-anchor files. Running `python operational_anchor.py` regenerates: (i) the incidence values $r_{\mathrm{inc}}^{(\infty)} = 1$, $r_{\mathrm{inc}}^{(G)} = 2$, and maximum support weight 6 from `wb_junction_support_catalog.csv`; (ii) `finite_instance_slab_certificate.csv` for a stacked 4×4 toric CSS code ($n = 96$, $k = 6$), with minimum passing thickness $t_{\min} = 3$; (iii) `phenomenological_mw_demo.csv` for distance-3 and 5 toric codes; and (iv) `operational_constants.csv`. The complete script sequence is listed in `run_verification.sh`.

SIX. CANONICAL SCHEDULE, LOCAL DETECTORS, AND BOUNDARY-STATE GLUING

For each primitive g, the physical decoder uses a uniformly local detector catalog D_g^{loc}. The complete stabilizer transition space $\mathcal{R}_g^{\mathrm{full}}$ is a separate algebraic closure for boundary-state consistency. Its elements are not automatically physical detector rows and are not inserted into the fault graph merely because they belong to a left kernel.

A. Binary Pauli boundary coordinates

Represent an n-qubit Pauli, modulo phase, by a binary symplectic vector $p = (x \mid z) \in \mathbb{F}_2^{2n}$. Let J denote the standard symplectic form. At every cut choose an independent stabilizer generator matrix $H \in \mathbb{F}_2^{r \times 2n}$ and a complete symplectic logical basis. The boundary syndrome is

$$s = HJp^T, \tag{S1}$$

and the logical-frame vector λ is the vector of commutators of p with the chosen logical partners. The map $p \mapsto (s, \lambda)$ has kernel exactly the stabilizer row space. Thus (s, λ) is a complete coordinate for the Pauli class modulo stabilizers; an inherited error at a fault-free cut need not have $s = 0$. The classical boundary datum is the vector ρ of reported-record deviations of the last round before the cut, $\rho = \Pi s + \mu_\partial$, where Π gives the syndrome of each measured check through the generators and μ_∂ are the record flips of that round (Eq. (A15-rec)). A cut carries (s, λ, ρ).

For each primitive g, let e_g denote its phenomenological fault bits and δ_g the values of its detector rows, including the rows that straddle its input cut. The physical local detector/effect system has the form

$$\delta_g = D_g^{\mathrm{loc}} e_g + A_g^{\mathrm{loc}} s_g^{\mathrm{in}} + P_g^{\mathrm{loc}} \rho_g^{\mathrm{in}}, \tag{S2}$$

$$\lambda_g^{\mathrm{out}} = L_g \lambda_g^{\mathrm{in}} + C_g s_g^{\mathrm{in}} + E_g e_g, \tag{S3}$$

$$o_g = Q_g \lambda_g^{\mathrm{in}} + G_g s_g^{\mathrm{in}} + F_g e_g, \tag{S4}$$

with the output variables of (A15a′). Here o_g contains any committed protected effect of the slot, and a pattern is detector-trivial when every δ_g vanishes. The syndrome terms C_g, G_g record how the representatives chosen at the two cuts differ by stabilizers (see the discussion of (A15b)); they vanish in the idle example below, where both cuts use the same representatives.

B. Uniformly local physical detector catalog

The local rows are generated from the following finite set of *master templates*. A block-size-dependent row is obtained only by translating one of these templates to one of the finitely many labeled local geometries of the refined lattice. Consequently there are constants $r_{\mathrm{det}}, a_{\mathrm{det}}, b_{\mathrm{det}} = O(1)$ such that every row is contained in a radius-r_{det} spacetime ball, every row touches at most a_{det} phenomenological mechanisms, and every mechanism touches at most b_{det} rows. These bounds arise from the finite template types, not from the dimension of a block-size-dependent relation space.

Write $m_t(C) \in \mathbb{F}_2$ for a flip of the reported eigenvalue of a measured Pauli check C at the indicated half-integer measurement time. If E is the net phenomenological Pauli mechanism in a bounded interval, write $\chi_C(E) = 1$ when E anticommutes with C after propagation to the relevant slice. For an edge ancilla, ι_e denotes a wrong Z_e eigenvalue at $|0\rangle$ preparation and ν_e a flipped final Z_e readout. All sums below are over $\mathbb{F}_2$. The $X \leftrightarrow Z$ dual templates are understood wherever the opposite CSS type is extracted.

a. T0: consecutive measurement of one persistent check. For any check C measured in two consecutive rounds,

$$d_C^t = m_{t-1/2}(C) + m_{t+1/2}(C) + \chi_C(E_t) = 0, \quad \text{(T0a)}$$

$$D_C^t = [\,1\;1 \mid \chi_C\,]. \quad \text{(T0b)}$$

This is the only temporal template in an idle window and is also used for A_v, B_p, and $\widetilde{s}_j$ during repeated gauged rounds.

b. T1: bounded same-slice dependency. If a bounded local set of simultaneously measured checks obeys $C_1 \cdots C_r = I$ (including the known sign convention), then

$$d_{\mathrm{dep}}^t = \sum_{i=1}^{r} m_t(C_i) = 0, \qquad D_{\mathrm{dep}}^t = [\,1\;1\;\cdots\;1\,]. \quad \text{(T1)}$$

The number r is uniformly bounded. These are the optional space-only meta-check rows mentioned in the spacetime-code construction; adding them can only further restrict the detector-trivial set.

c. T2: bounded merge/split relation. Suppose a local transition has input checks $A_1, \ldots, A_r$ and output checks $B_1, \ldots, B_s$ whose Pauli products agree on the common boundary, $\prod_i A_i = \prod_j B_j =: C$. The physical transition row is

$$d_{A\to B} = \sum_i m_-(A_i) + \sum_j m_+(B_j) + \chi_C(E_\partial) = 0, \quad \text{(T2a)}$$

$$D_{A\to B} = [\,\mathbf{1}_r\;\mathbf{1}_s \mid \chi_C\,]. \quad \text{(T2b)}$$

Only relations supported inside one bounded transition cell are instantiated. T0 is the special case $r = s = 1$ with no change of check type.

d. T3: explicit gauging-entry rows. Let the modified check be $\widetilde{s}_j = s_j \prod_{e\in\mu_j} Z_e$, where the pairing path μ_j has uniformly bounded length, and let p be a unit-square cycle. At the first deformed round the two endpoint rows are

$$d^{\mathrm{in}}_{\widetilde{s}_j} = m_-(s_j) + m_+(\widetilde{s}_j) + \sum_{e\in\mu_j} \iota_e + \chi_{\widetilde{s}_j}(E_{\mathrm{in}}) = 0, \quad \text{(T3a)}$$

$$d^{\mathrm{in}}_{B_p} = m_+(B_p) + \sum_{e\in p} \iota_e + \chi_{B_p}(E_{\mathrm{in}}) = 0. \quad \text{(T3b)}$$

There is no additional macroscopic entry row for the product of all A_v outcomes: that product is the committed logical-measurement effect. The A_v records use T0 once a second deformed round exists; their open temporal endpoint is exactly the endpoint used in the timelike-distance argument.

e. T4: explicit gauging-exit rows. At the return to the undeformed code,

$$d^{\mathrm{out}}_{\widetilde{s}_j} = m_-(\widetilde{s}_j) + m_+(s_j) + \sum_{e\in\mu_j} \nu_e + \chi_{s_j}(E_{\mathrm{out}}) = 0, \quad \text{(T4a)}$$

$$d^{\mathrm{out}}_{B_p} = m_-(B_p) + \sum_{e\in p} \nu_e + \chi_{B_p}(E_{\mathrm{out}}) = 0. \quad \text{(T4b)}$$

Together T3, T0 on the repeated deformed checks, and T4 are the local detector generators listed for gauging measurement in the Williamson–Yoder gauging construction [5]; T1 is added when the code has bounded local relations.

f. T5: product-state birth and terminal readout. For a bounded-support Z check $C = \prod_{q\in S} Z_q$ following $|0\rangle$ preparation, and for the dual terminal Z readout of the same support,

$$d_C^{\mathrm{birth}} = m_+(C) + \sum_{q\in S} \iota_q + \chi_C(E_{\mathrm{birth}}) = 0, \quad \text{(T5a)}$$

$$d_C^{\mathrm{death}} = m_-(C) + \sum_{q\in S} \nu_q + \chi_C(E_{\mathrm{death}}) = 0. \quad \text{(T5b)}$$

The X-basis versions are identical after exchanging X and Z. Encoded $|0\rangle/|+\rangle$ birth, protected logical readout, and encoded discard are tiled by T0–T2 and these endpoint rows.

g. T6: raw injection, courier, and factory slots. These introduce no new detector-row species. The protected $Z_q Z_{B_0}$ or $Z_{B_j} Z_{B_{j+1}}$ measurement is a one- or two-block instance of T0–T4, while the final X_q or X_{B_j} readout is an instance of T5. The two teleportation outcome bits and the product of Gauss outcomes are protected effects stored in o_g or the Pauli frame, not extra detector vertices. Factory slots are finite compositions of the same templates.

h. Displayed measurement/preparation submatrix. For one modified check with $\mu_j = \{e_1, \ldots, e_\ell\}$ and R

deformed rounds, order the measurement columns as $(m_0(s_j), m_1(\widetilde{s}_j), \dots, m_R(\widetilde{s}_j), m_{R+1}(s_j))$. Let

$$B_R = \begin{bmatrix} 1 & 1 & 0 & \cdots & 0 \\ 0 & 1 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & 1 & 1 \end{bmatrix} \in \mathbb{F}_2^{(R+1)\times(R+2)} \tag{T7a}$$

be the path-incidence matrix in time, and let $\mathbf{u}_1, \mathbf{u}_{R+1}$ denote the first and last standard basis columns of $\mathbb{F}_2^{R+1}$. Suppressing only the appended local Pauli-commutator columns, the exact preparation/measurement submatrix is

$$D^{(0)}_{\widetilde{s}_j} = \left[\, B_R \;\middle|\; \mathbf{u}_1 \mathbf{1}_\ell^T \;\middle|\; \mathbf{u}_{R+1} \mathbf{1}_\ell^T \,\right], \tag{T7b}$$

with column groups $(m_0, \dots, m_{R+1} \mid \iota_{e_1}, \dots, \iota_{e_\ell} \mid \nu_{e_1}, \dots, \nu_{e_\ell})$. Thus the first row is T3a, the last row is T4a, and all interior rows are T0. The B_p block is the same construction with the two undeformed-s_j measurement columns deleted; the A_v block is the interior bidiagonal temporal part because A_v has open deformation endpoints. Appending the χ columns leaves row weight and column incidence uniformly bounded because all supports and transition circuits are local.

i. Nearest-neighbor extraction circuit words. For each labeled support C choose the canonical radius-R_* interaction ball from the detailed extraction proposition, with its reserved routing ancilla and one reserved contact microvertex adjacent to each possible support label. On this refined nearest-neighbor routing graph, root the lexicographically first spanning tree at the ancilla site and take its depth-first traversal. The route segments $R_{C,i}$ below are the consecutive pieces of this fixed walk between first visits to the contact microvertices assigned to $q_1, \dots, q_w$. For $C = Z_{q_1} \cdots Z_{q_w}$ the measured circuit word is

$$\begin{aligned} &|0\rangle_a;\, R_{C,1};\, \mathrm{CNOT}_{q_1 \to a};\, R_{C,2};\, \cdots \\ &\quad \mathrm{CNOT}_{q_w \to a};\, R_{C,w+1};\, M_Z(a), \end{aligned} \tag{T8a}$$

and for $C = X_{q_1} \cdots X_{q_w}$ it is

$$\begin{aligned} &|+\rangle_a;\, R_{C,1};\, \mathrm{CNOT}_{a \to q_1};\, R_{C,2};\, \cdots \\ &\quad \mathrm{CNOT}_{a \to q_w};\, R_{C,w+1};\, M_X(a). \end{aligned} \tag{T8b}$$

The traversal is completed by retracing the tree, so every moved data/ancilla label returns to its canonical site. If V_* is the largest number of interaction sites in a radius-R_* ball, the shuttle uses at most $2(V_* - 1)$ nearest-neighbor SWAPs and at most $w_{\max}$ data–ancilla CNOTs. Thus the T8 word length is uniformly bounded by the explicit canonical routing construction. Bounded local density gives a bounded-degree conflict graph for these words, hence an $O(1)$ coloring, constant extraction depth, and bounded fault spread. Equations T0–T8 form the explicit local detector/extraction catalog used by the proof.

j. Catalog bounds. Let $w_{\max}$ be the largest extracted-check weight, $\ell_{\max}$ the largest pairing-path length, $r_{\max}$ the largest number of check records participating in one bounded local dependency/merge/split relation, and q_{ball} the largest number of phenomenological data-Pauli mechanisms whose propagated support can meet one catalog neighborhood during one slot. All four are constants fixed by the finite local geometry. The displayed rows then touch at most

$$\begin{aligned} a_{\mathrm{det}} \le \max\{&2 + q_{\mathrm{ball}},\, r_{\max},\, r_{\max} + q_{\mathrm{ball}}, \\ &2 + \ell_{\max} + q_{\mathrm{ball}},\, 5 + q_{\mathrm{ball}},\, 1 + w_{\max} + q_{\mathrm{ball}}\}. \end{aligned} \tag{T9}$$

Here the 5 is the worst unit-square endpoint row (one check record plus four edge preparation/readout bits); harmless zero terms are simply omitted for a particular template. Conversely, a record column occurs in at most two T0 rows plus $O(1)$ same-slice/transition rows, an edge preparation or readout column occurs only in the bounded set of endpoint rows incident on that edge, and a data-Pauli column can meet only catalog neighborhoods intersecting one constant-radius spacetime ball. If n_{nb} is the maximum number of such neighborhoods and c_{row} the maximum number of their rows containing one mechanism, then $b_{\mathrm{det}} \le 2 + n_{\mathrm{nb}} c_{\mathrm{row}} = O(1)$. Thus both row weight and column incidence are bounded by inspectable catalog constants, not by the dimension of the full relation space.

C. T8 single-fault circuit check

The circuit-level reduction in the preceding analysis is analytic, and the source package also contains an exhaustive finite regression check of the displayed T8 master words. A circuit location is faulted by each nonidentity Pauli on its support (three possibilities after a one-qubit preparation or before measurement, and fifteen after a CNOT or SWAP). The fault is propagated through the remaining Clifford word. The resulting row records

$$\begin{aligned} &v_{\mathrm{ph}} = (e_{\mathrm{bdry}}, m, \chi_C(e_{\mathrm{bdry}}), d_{T0}), \\ &d_{T0} = m + \chi_C(e_{\mathrm{bdry}}) \pmod 2. \end{aligned} \tag{T10}$$

The calculation instantiates both Z- and X-check T8 words for check weights $1 \le w \le 8$ and reversible nearest-neighbor shuttle words of route lengths $0, 1, 2$. The file `t8_master_routing_catalog.csv` records all 48 master words, including the operation word, return-to-canonical flag, location count, and maximum single-fault phenomenological image weight. Enumerating 10,008 single-location Pauli faults gives maximum image weight 10 on this regression family. The detailed spanning-tree construction supplies the route for any bounded local support.

TABLE S10. Allowed local adjacencies. At every retained wire the complete (s, λ, ρ) output is identified with the next input.

Left slot	Right slot
idle	idle / gauging entry / readout
gauging exit	idle / protected readout
birth	idle / gauging entry
raw injection	idle / encoded courier
encoded courier	idle / encoded courier / factory
factory	idle / gauging / readout
readout	discard or retained-wire idle

D. The full transition-relation space is an algebraic closure

Embed both sides of a transition in the union of qubits present at that boundary. Fresh ancillas are supplied with their known product-state stabilizers and departing qubits with their final readout observables. Let H^- and H^+ be the resulting binary symplectic generator matrices. The complete algebraic space of fault-independent stabilizer relations is

$$\mathcal{R}_g^{\text{full}} = \{(u, v) : uH^- = vH^+\}. \tag{S5}$$

Equivalently, it is the left kernel of the vertically stacked matrix $[H^-; H^+]$. Equation (S5) characterizes the *algebraic closure* of valid transition relations. We do not take an arbitrary basis of this space as the physical detector catalog. In particular, a macroscopic dependency in $\mathcal{R}_g^{\text{full}}$ is a derived algebraic relation and is not a fault-graph detector vertex.

Every local row listed above is an element of the appropriate transition relation space after its boundary coordinates are included, so Eq. (S5) is a useful consistency check. Conversely, the detailed restriction proof needs only the listed local rows together with the complete boundary variables (s, λ, ρ). The proof does not assert rowspan $D_g^{\text{loc}} = \mathcal{R}_g^{\text{full}}$ for an entire block. Additional full-space relations remain valid algebraic identities, but the restriction/gluing proof does not require choosing them as primitive rows or expressing them as physical detector vertices.

E. Canonical gluing table

Adjacent slots are glued only by identifying the complete syndrome, last-round records, and logical frame of every retained block. There is no identification for a discarded block, and a newly born block instead receives the fixed product-state input boundary specified above. The allowed local adjacencies are listed below; all others are compositions of these rows or are disjoint simultaneous slots.

The global *physical* detector/effect system is obtained by taking the block direct sum of Eqs. (S2)–(S4), identifying the complete (s, λ, ρ) variables on retained wires, and eliminating internal boundary variables. Because (s, λ) is a complete Pauli-class coordinate, future records and protected effects depend on the past faults only through it; because every catalog row compares at most two consecutive rounds, future detector rows read past records only through ρ. This fiber product is therefore exact for the local physical detector system, as `boundary_composition_check.py` confirms on explicit blocks. Equation (S5) separately checks the algebraic consistency of the boundary semantics; it is not a prescription to add all left-kernel rows to the graph.

a. Worked two-idle expansion of the detailed gluing example. In the notation of (S2)–(S4) and (A15a′), the two idle primitives of the worked example (one persistent check C, one round each, fault bits $e_t = (f_t, \mu_t)^\top$) have

$$\begin{aligned} D_t^{\text{loc}} &= [\,1\;1\,], \quad A_t^{\text{loc}} = P_t^{\text{loc}} = [1], \\ \delta_t &= \rho_t^{\text{in}} + s_t^{\text{in}} + f_t + \mu_t, \\ s_t^{\text{out}} &= s_t^{\text{in}} + f_t, \qquad \rho_t^{\text{out}} = s_t^{\text{out}} + \mu_t, \\ L_t &= [1], \quad C_t = 0, \quad E_t = [\,1\;0\,], \\ \lambda_t^{\text{out}} &= \lambda_t^{\text{in}} + f_t. \end{aligned}$$

Identifying (s, ρ, λ) at the mid-cut and eliminating it gives Eqs. (A15-ex2)–(A15-ex3), identical to the two-round idle circuit written as a single primitive. The script `glue_example_check.py` checks this on all 2^7 assignments of $(f_1, \mu_1, f_2, \mu_2, s_{\text{in}}, \rho_{\text{in}}, \lambda_{\text{in}})$, and reproduces the defect of the earlier form of the example, whose rows $f_t + \mu_t + s_t^{\text{in}} + s_t^{\text{out}} = 0$ accept a record error as a syndrome change. This is an illustration of local-catalog sufficiency with complete boundary variables; it does not replace the general gluing proof.

For restriction, set all fault components except one connected component C to zero and retain the induced boundary variables at cuts surrounding C. The restricted subchain still satisfies the same local physical equations. The comparison is with fault-free propagation of the same inherited boundary variables, so s_L, ρ_L, s_R and ρ_R may be nonzero. This is the precise sense in which local detector generation is sufficient for the detailed relative restriction theorem.

F. Courier identity and relative resource frame

For both the raw and encoded courier, the ideal logical algebra is one-bit teleportation. If m is the outcome bit of $Z_q Z_b$ and n is the outcome bit of X_q, with b initialized to $|+\rangle$, then

$$X_b^m Z_b^n \, \langle n_X|_q \, \Pi_m(Z_q Z_b) \big(|\psi\rangle_q |+\rangle_b\big) = e^{i\phi_{mn}} \frac{|\psi\rangle_b}{2}. \tag{S6}$$

The same identity holds for logical Paulis on encoded blocks. Therefore an incoming resource Pauli is a boundary condition transported to the retained block. The

stage-relative error variable is the difference between the actual outgoing frame and this fault-free transported frame; this is the Δr_j used by the encoded-courier and earliest-stage lemmas.

G. Scripts and outputs

The scripts `reproducibility_checks.py` and `glue_example_check.py` derive the slab-cell parity matrices, write `slab_cell_audit.csv`, enumerate the T8 single-location fault images in `t8_single_fault_audit.csv`, and write the 48 routing words to `t8_master_routing_catalog.csv`. The fault CSV records the circuit location, inserted Pauli, boundary Pauli image, measurement-record flip, commutator with the extracted check, and local T0 detector signature.

SX. SINGLE-ERROR RESOURCE-COORDINATE MAPS

The detailed factory map is

$$\mathcal{F}_Q = \mathrm{Dec}_Q \circ \mathcal{R}_Q \circ \mathrm{Syn}_Q \circ U_Q^{\mathrm{tr}} \circ \mathrm{Enc}_Q. \tag{S6a}$$

For the Steane $7 \to 1$ factory, $Q = [[7, 1, 3]]$, Enc_Q prepares encoded $|+\rangle$, and U_Q^{tr} is the fixed transversal $S/S^\dagger$ orientation whose ideal output is encoded $|Y\rangle$. The seven physical phase rotations are supplied by seven preassigned $|Y\rangle$ candidates through the phase-teleportation identity. For the punctured Reed–Muller $15 \to 1$ factory, $Q = [[15, 1, 3]]$ and U_Q^{tr} is the standard transversal $T/T^\dagger$ orientation whose ideal output is encoded $|A\rangle$, implemented coordinatewise with fifteen preassigned $|A\rangle$ candidates. The syndrome map Syn_Q is read from the X-check columns below, $\mathcal{R}_Q$ maps each weight-one syndrome to the corresponding single-Z correction, and Dec_Q retains the one logical output. This is the always-output version of the standard code-based factories: the syndrome is corrected rather than used for postselection. Script `factory_map_audit.py` enumerates the two CSS codeword cosets and checks directly that $(S^\dagger)^{\otimes 7}$ gives logical S on the Steane code while $(T^\dagger)^{\otimes 15}$ gives logical T on the punctured Reed–Muller code; it also rechecks uniqueness of all single-Z syndrome columns.

Use the following circuit-to-coordinate convention for the detailed single-error resource-coordinate lemma; coordinate permutations give equivalent descriptions. For the Steane block, label the seven coordinates by the nonzero vectors $v \in \mathbb{F}_2^3$; for the punctured Reed–Muller block, label the fifteen coordinates by the nonzero vectors $v \in \mathbb{F}_2^4$. The X-check syndrome of a single Z_i is the column v_i. Since all listed columns are nonzero and distinct, every single-coordinate Z error has a distinct nonzero syndrome.

A defective slot assigned to coordinate i, in any state and possibly entangled with other systems, acts after the pure-X/Z teleportation identity and Pauli-frame update as an operator in $\mathrm{span}\{I, Z_i\}$ on the code block, tensored with an operator on the other systems (Lemma SV.3). When a conditional sign-correction resource is preallocated, its slot is attached to the same coordinate i as the principal injection it corrects, so a defect of either slot, or of both, stays in this span. This is the explicit map used by the single-defect recovery argument (Lemma SV.4).

TABLE S11. Steane $[[7, 1, 3]]$ coordinate convention. The syndrome column is also the coordinate label v_i; a defective slot at coordinate i acts within $\mathrm{span}\{I, Z_i\}$.

i	$v_i \in \mathbb{F}_2^3$ (X-check syndrome of Z_i)
1	001
2	010
3	011
4	100
5	101
6	110
7	111

TABLE S12. Punctured Reed–Muller $[[15, 1, 3]]$ coordinate convention. The fifteen transversal rotation labels are the nonzero linear forms $x \mapsto v_i \cdot x$ on $\mathbb{F}_2^4$; v_i is also the single-Z_i syndrome column, and a defective slot at coordinate i acts within $\mathrm{span}\{I, Z_i\}$.

i	$v_i \in \mathbb{F}_2^4$ (X-check syndrome of Z_i)
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
10	1010
11	1011
12	1100
13	1101
14	1110
15	1111

a. What the tables establish. For either code, no two weight-one Z errors share an X-check syndrome, and every weight-one syndrome is nonzero. Thus any predetermined decoder that maps each weight-one syndrome to its weight-one representative returns the ideal encoded state when at most one coordinate is defective, whatever the defective state is. A defective output consequently requires defective slots at two distinct coordinates, which is the finite input needed for the quadratic local-stochastic recursion in the detailed fault-tolerance section. With

TABLE S13. Factory map fixed in the preceding analysis. The phase orientation (S versus $S^\dagger$, or T versus $T^\dagger$) is fixed once so that the ideal decoded output is the displayed resource; different conventional orientations differ only by known Clifford/frame updates.

Output	Code	Candidate layer	Syndrome used for recovery	Output rule
$\|Y\rangle$	Steane $[[7,1,3]]$	7 coordinatewise phase injections	3-bit X-check syndrome	correct, then decode
$\|A\rangle$	punctured RM $[[15,1,3]]$	15 coordinatewise $T/T^\dagger$ injections	4-bit X-check syndrome	correct, then decode

the recovery used here every pair of Z errors is malignant (105 pairs for the Reed–Muller factory, 21 for the Steane factory), so the output error of the ideal always-output Reed–Muller factory is $105q_c^2 + O(q_c^3)$, with q_c the per-coordinate error, not the $35q_c^3$ of postselected distillation; `resource_defect_check.py` enumerates both.

b. Role in the complete proof. These tables close the circuit-to-coordinate bookkeeping for the standard transversal candidate assignment. Together with the explicit factory map (S6a), courier identity, and finite slot catalog, they provide the resource-interface data used by the scheduled-circuit theorem (Theorem SV.26).

SXI. INCIDENCE AND REFINEMENT VERIFICATION

Table S3 lists the six junction supports, and Lemma SI.9 gives the two local-Clifford subdivision identities. The files `wb_junction_support_catalog.csv` and `refinement_move_audit.csv` record the corresponding coordinates and GF(2) row-space checks. These coordinates specify a combinatorial transcription; the asymptotic arguments require only a uniform finite cell radius.

SXII. OPERATIONAL OUTPUTS

The following table collects the numerical values referenced in the preceding analysis.

a. Incidence radius. Every endpoint listed in `wb_junction_support_catalog.csv` has L^∞ distance at most 1 from the junction vertex $(0,0,0)$ on the unit cubic lattice, while the cubic-lattice vertex-graph (L^1) distance is at most 2 (e.g. endpoint $(0,1,1)$), and every support weight is at most 6. The resulting constants are $r_{\mathrm{inc}}^{(\infty)} = 1$ (max L^∞ endpoint radius) and $r_{\mathrm{inc}}^{(G)} = 2$ (max vertex-graph distance), together with max support weight 6. The slab theorem uses only the resulting block-size-independent locality constant.

b. Finite-instance slab ledger. The stacked code is the direct sum of three independent periodic 4×4 toric codes. Columns of H_X and H_Z are partitioned into three contiguous layer blocks of 32 qubits. For $S =$ the first t blocks, Eqs. (P38) and (P39) give left-hand sides $2, 4, 6$ for $t = 1, 2, 3$ respectively, equaling $k = 6$ only at $t = 3$. Both CSS sectors agree. The CSV columns are `code_id`, `n`, `k`, `t`, `S_size`, `lhs_X`, `lhs_Z`, pass/fail flags, and a scope note.

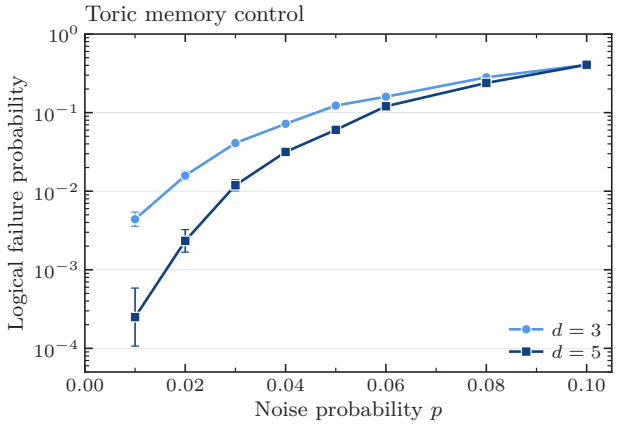


FIG. S8. Code-capacity MWPM sanity check on toric codes at distances 3 and 5. The plot shows the logical error rate under the finite code-capacity decoder test described in the surrounding text.

c. Code-capacity MWPM demo (toric). Independent bit-flip and phase-flip code-capacity noise of probability p is applied to each data qubit, with one perfect syndrome measurement; X and Z sectors are decoded separately by minimum-weight perfect matching on the toric check graph (PyMatching). A shot is a logical failure if either residual anticommutes with a nontrivial logical of the opposite type. Wilson score intervals are reported. The CSV records the decoding convention used for this finite sanity check; its legacy label "phenomenological" predates this description, and the noise is code-capacity as stated here.

d. Gauging-rectangle sanity check (MWPM). Script `gauging_rectangle_mwpm.py` attaches a lattice-surgery-style path-strip gauging graph to a rotated surface-code logical Z at $d \in \{3, 5, 7\}$, decodes the deformed CSS code with PyMatching under code-capacity X/Z data noise, and writes `gauging_rectangle_mwpm.csv`. Observed distance-like suppression: d=5 below d=3 for $p \leq 0.06$. This is a deformed surface-code/path-strip building-block sanity check under code-capacity X/Z data noise.

e. Analytic illustrations moved from the preceding analysis. The following plots visualize two closed-form

TABLE S14. Selected operational values used by the companion manuscript.

Anchor	Regenerated value	Scope
WB junction support / radii	max support weight 6; $r_{\mathrm{inc}}^{(\infty)} = 1$, $r_{\mathrm{inc}}^{(G)} = 2$	Six WB Sec. VI B line-junction cells; junction-incidence transcription
T8 extraction	48 master words; 10,008 single-location Pauli faults; max phenomenological image weight 10	T8 master-word family
Factory phase map	Steane $(S^\dagger)^{\otimes 7} \mapsto S_L$; RM15 $(T^\dagger)^{\otimes 15} \mapsto T_L$	codeword-phase and syndrome check
Slab-rank demo	stacked toric CSS instance, $n = 96$, $k = 6$, minimum passing $t = 3$	rank-criterion example
Explicit realization	seven inputs, n up to 10,936; $k = K$; slab operators at every height/column	complete-code certificate (Table S5)
Decoder sanity checks	toric $d = 3, 5$; gauging strip $d = 3, 5, 7$	MWPM sanity checks under simplified noise
Single-chart BP–OSD memory sweep (superseded)	$L = 3, 4, 5, 6$; finite crossing near $p \approx 0.10$	Corrected-CSS code-capacity memory data; not architecture schedule p_{th}
Numerical campaign scope	finite memory, primitive, stage-composed, and continuous algebraic code-switch diagnostics	Finite-size checks only; no architecture threshold is inferred

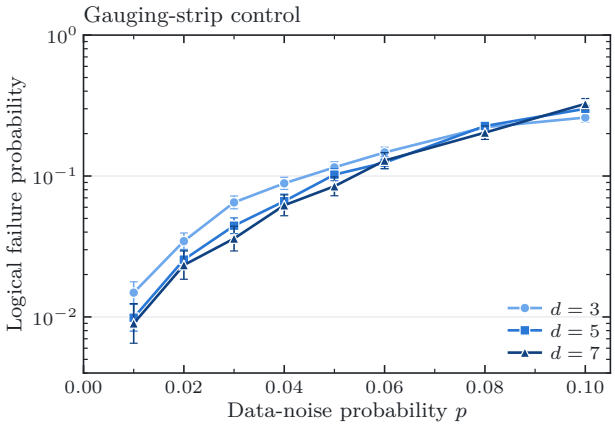


FIG. S9. MWPM sanity check for the gauging-rectangle building block at $d \in \{3, 5, 7\}$. The curves illustrate the expected distance-like suppression for the finite deformed surface-code/path-strip test instance.

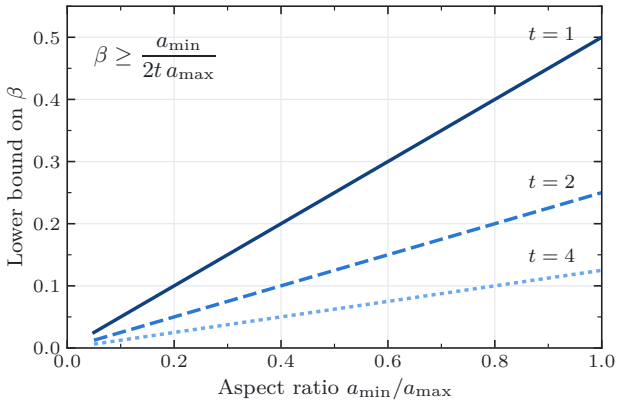


FIG. S10. Analytic relative-expansion bound used in the local gauging argument.

bounds used in the proofs; they contain no fitted or simulated data and are kept here so the companion manuscript can prioritize the slab and architecture schematics.

f. Verification workflow. The complete finite verification sequence is recorded in `run_verification.sh`; the cell-specific sequence is in `run_bridge_verification.sh`. The revised package also records the deterministic checks rerun during preparation of these documents.

SXIII. EXTENDED NUMERICAL RESULTS

The numerical campaign combines algebraic checks with Monte Carlo tests of memory decoding, a local gauging primitive, a stage-composed schedule, and a continuous algebraic base→gauged→base code switch with persistent encoded data. The scripts cover the source transcription, local incidence, detector identities, refinement moves, resource maps, and fault propagation. These calculations characterize the implemented finite-size models; they do not assign a numerical value to the asymptotic threshold in the detailed universal resource theorem.

The circuit-level campaigns that show a clean crossing all use surface codes. The gauging primitive uses the rotated surface code of distance d. The stage-composed and continuous campaigns use the unrotated planar surface code of distance L, built as the hypergraph product of two length-L repetition codes, with $n = L^2 + (L-1)^2$ data qubits, and gauge a weight-L logical representative with a strip of L ancilla qubits. They therefore test the gauging interface and the detector bookkeeping of a code switch, not the layer-code geometry. The complete two-block measurement of Sec. SXIV runs on explicit layer-code

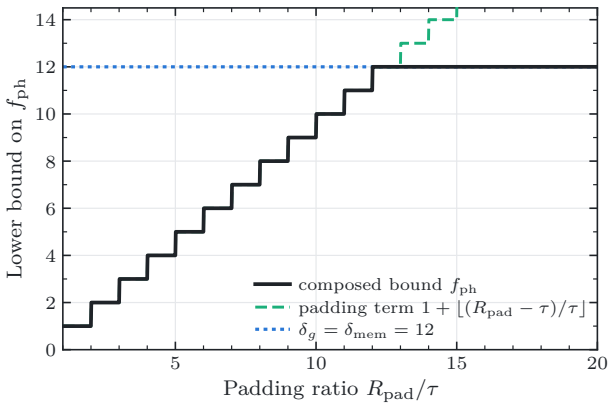


FIG. S11. Representative behavior of the padding term in the composed-distance bound.

blocks; the earlier truncated layer-code family, built from the single-chart cells, is retained only as the negative control of Sec. SXIII F.

A. Campaign setup

The continuous code-switch circuit removes the main weakness of the earlier stage-composed model: the data block is never reset between the undeformed and deformed stages. Its transition rules are algebraic rather than a gate-by-gate rendering of every Williamson–Baspin defect cell, so the three-dimensional realization still enters through the slab proof and junction-incidence construction.

The SI1000-like baseline assigns two-qubit depolarizing strength p, single-qubit and ordinary idle strength $p/10$, measurement flips $5p$, reset noise $2p$, and data idle-wait noise $2p$. Stim detector error models are decoded by PyMatching MWPM. The heavy and continuous headline runs use exact graphlike DEMs.

B. Deterministic checks

The source-cell script derives $H_{\rm end}$ and $H_{\rm mid}$ from the published condensation generators, including the 110 middle-cell pass-through and the $X \leftrightarrow Z$ duals. The slab theorem itself is checked on complete explicit codes (Sec. SII).

The prefix-parity recurrence is checked on all even words through length eight and on longer deterministic stress words. The rank criterion is evaluated on concrete CSS instances; the stacked-toric control has minimum passing slab thickness $t = 3$. The two surface-code subdivision identities used for constant refinement are checked at the stabilizer level.

For the detector schedule, the scripts check T0–T7 parity matrices, boundary-state coordinates, transition relations, and fiber-product gluing. The T8 routing calculation covers both CSS orientations, check weights 1–8, and reversible shuttle words. It enumerates 10,008 single-location nonidentity Pauli faults and finds maximum phenomenological image weight ten on this regression family. The bounded-radius construction in the preceding analysis supplies the general route.

The resource calculations separately check one-bit teleportation, Pauli transport, $SS_XS = e^{i\pi/4}H$ up to phase, and the Steane $7 \to 1$ and punctured Reed–Muller $15 \to 1$ phase orientations. The corrected factory circuit has zero logical failure in the noiseless checks. A legacy factory diagnostic that failed at $p = 0$ was traced to an encoding/convention mismatch and is not used.

The core scripts are `reproducibility_checks.py`, `slab_interface_audit.py`, `glue_example_check.py`, and `factory_map_audit.py`. CSV ledgers retain the slab cells, junction supports, refinement moves, T8 words and fault images, factory maps, and operational constants.

C. Memory, gauging, and stage-composed controls

a. Memory. The truncated single-chart incidence family (superseded; Sec. SXX F) $L = 3, 4, 5, 6$ was decoded under independent code-capacity X/Z noise with BP–OSD and the corrected CSS logical pairing. Increasing-size suppression is separated at several low-p points and the curves cross near $p \simeq 0.10$. A separate growing HGP memory family gives the same qualitative control behavior.

b. Gauging primitive. The local SI1000-like gauging primitive, on the rotated surface code, was extended to $d = 3, 5, 7, 9, 11$. All five sizes show monotone suppression at $p = 1.5, 2.5, 3.5 \times 10^{-3}$; the smallest-versus-largest crossing is $p_\times \simeq 4.12 \times 10^{-3}$.

c. Stage-composed schedule. Before the continuous transition circuit was built, a stage-composed protocol decoded the idle, noisy gauging, and final stages separately. The campaign used five sizes through $L = 11$, exact graphlike DEMs, and 3.955×10^6 Monte Carlo shots over 180 parameter points. For $R = d$ the five-size ordering is Wilson-separated at $p = 0.0015, 0.0025, 0.0035$, with crossing 4.64×10^{-3}. For $R = 2d$ the ordering remains separated through $p = 0.004$, with crossing 4.43×10^{-3}. In both schedules the fitted slope of $\log p_L$ changes sign across the crossing.

D. Continuous algebraic base→gauged→base code-switch circuits

The largest numerical experiment keeps the same encoded data block, a planar surface code of distance L, through the entire algebraic code switch. At entry, each old X stabilizer is represented by a modified gauged generator together with strip operators whose product reproduces the undeformed check. The deformed ZZ

TABLE S15. Summary of the checks and numerical campaigns.

Item	Test	Result
Explicit layer-code realization	complete check matrices, seven inputs; gluing forms, exhaustive words, walls	$k = K$; slab operators at every height (Table S5)
Archived cell certificate	all boundary words, full syndromes	non-discriminating; archived (Sec. SXX)
Two-block layer-code measurement	complete circuits, Choi-state input, exact fault distance	Table S17
T8 extraction	single-location Pauli propagation	10,008 faults; max image wt. 10
Resource maps	Steane and punctured RM15 algebraic maps	passed
Single-chart memory control (superseded)	BP–OSD, $L = 3, 4, 5, 6$	crossing near 0.10
Gauging primitive (rotated surface code)	exact-DEM SI1000-like MWPM, $d = 3, 5, 7, 9, 11$	crossing $\simeq 4.12 \times 10^{-3}$
Stage-composed (planar surface code)	exact-DEM, $L = 3, 5, 7, 9, 11$	4.64×10^{-3} ($R = d$), 4.43×10^{-3} ($R = 2d$)
Continuous code switch (planar surface code)	base→gauged→base, $L = 3, 5, 7, 9, 11$; 4.534×10^6 shots	Jeffreys crossing medians $4.47, 4.43, 4.33 \times 10^{-3}$ for $G = d, 2d, 3d$
Thin vs. thick interface	deformed-code construction on two 3D toric codes, $L = 3$–6	cut obstruction confirmed (Table S6)
Truncated single-chart SI1000-like (superseded)	incidence-based control	no clean four-size crossing

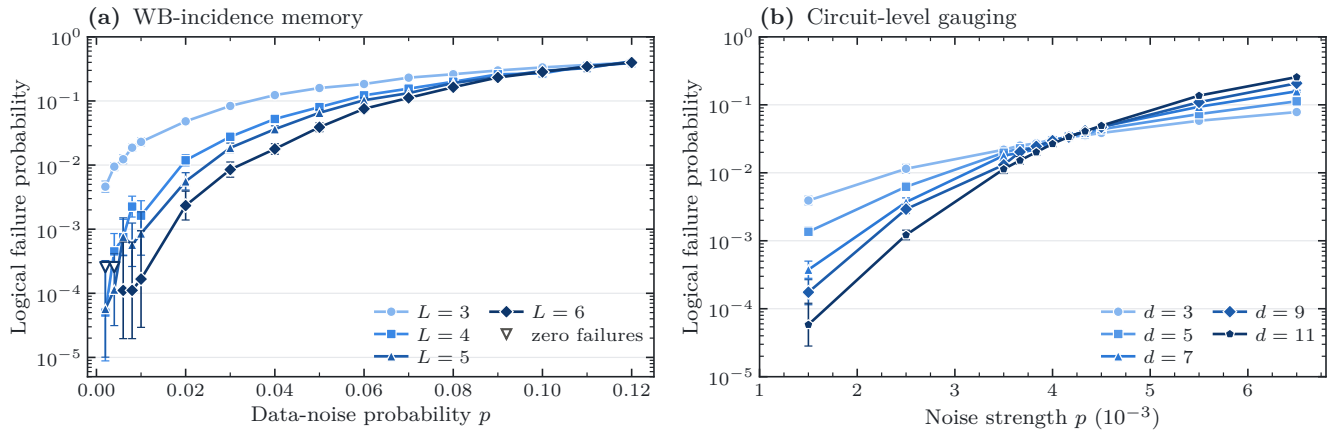


FIG. S12. Finite building-block diagnostics. (a) Code-capacity BP–OSD on the truncated single-chart incidence family (superseded). (b) The exact-DEM circuit-level gauging primitive under SI1000-like noise. Error bars are pointwise Wilson 95% intervals. Open downward triangles denote the upper end of the Wilson interval for a zero-failure observation. These are finite-model controls.

joints are introduced simultaneously, and the code verifies $H_X^{\text{gauge}}(H_Z^{\text{gauge}})^T = 0$ rather than deleting residual anticommuting rows. The exit implements the inverse boundary relation. Detector records span both transition edges and the repeated deformed rounds in one Stim circuit.

For each $L = 3, 5, 7, 9, 11$, the final archived campaign tests $G = d, 2d, 3d$ (Fig. S14 and Table S16). The dataset contains 255 parameter points and 4.534×10^6 shots. All 15 $p = 0$ controls have zero failures, every transition-identity check passes, and every headline DEM is graph-like. At $p = 0.0035$ the end-to-end failure probabilities

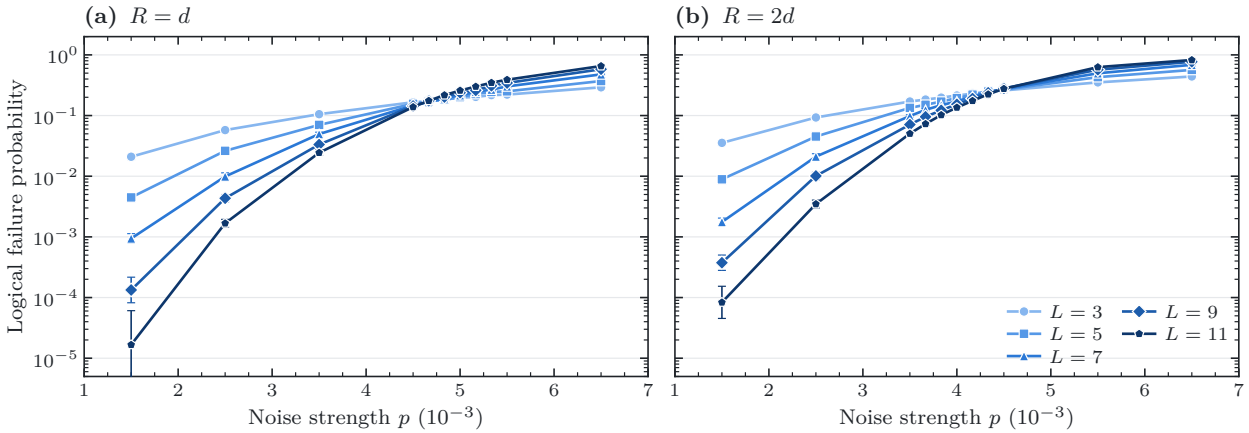


FIG. S13. Stage-composed SI1000-like stress test on planar surface codes (hypergraph products of repetition codes). Five sizes $L = 3, 5, 7, 9, 11$ retain low-p suppression for both $R = d$ and $R = 2d$; the smallest-versus-largest crossings are approximately 4.64×10^{-3} and 4.43×10^{-3}.

are

G	$L = 3$	5	7	9	11
d	.1111	.0795	.0609	.0438	.0324
$2d$	.1308	.0924	.0729	.0511	.0399
$3d$	.1326	.1125	.0811	.0576	.0428

and the consecutive-size Wilson intervals are separated for these rows. Five-size suppression remains monotone through at least $p = 0.004$ for all three durations. Consecutive-size Wilson intervals are separated at every tested p from 0.0025 to 0.00375 for all three durations, and also at $p = 0.0015$ for $G = d$ and $3d$; for $G = 2d$ at $p = 0.0015$ the $L = 9$ and $L = 11$ intervals overlap.

Sampling independent Jeffreys posteriors for the $L = 3$ and $L = 11$ binomial rates and linearly interpolating the first crossing gives medians 4.471×10^{-3}, 4.431×10^{-3}, and 4.328×10^{-3} for $G = d, 2d, 3d$. The corresponding conditional 95% posterior crossing bands are $[4.426, 4.522]$, $[4.392, 4.466]$, and $[4.265, 4.393] \times 10^{-3}$. Their full envelope is therefore $[4.265, 4.522] \times 10^{-3}$, but this envelope describes Monte Carlo/interpolation uncertainty only; it does not include finite-size-scaling or microscopic-model systematics. Tripling the noisy gauged interval shifts the median crossing by about 3.2%.

E. Finite-size scaling diagnostics

A two-size crossing can be accidental, so the final campaign also records all-size monotonicity, Wilson-separated consecutive-size intervals, and the slope of $\log p_L$ versus $\log L$. For each duration the slope is strongly negative at low p and changes sign in the same 4.3–4.5 $\times 10^{-3}$ region as the endpoint crossing. This all-size trend is independent of the posterior calculation, which uses only the smallest and largest sizes.

At $p = 0.0015$ the $G = d$ circuit decreases from $p_L = 2.425 \times 10^{-2}$ at $L = 3$ to 3.75×10^{-5} at $L = 11$, a factor of about 6.5×10^2. This checks that the suppression seen at $L = 3, 5, 7$ continues when $L = 9, 11$ are added.

Point-level CSV output includes shots, failures, Wilson bounds, exact-DEM and transition-check flags, detector/error counts, code sizes, round counts, and stopping conditions. The archived final dataset and deterministic crossing analysis are in `numerical/continuous_transition/threshold_estimate/`.

F. Negative controls, factories, and interpretation

The truncated single-chart incidence family (superseded; Sec. SXX F) does not show a clean $L = 3 > 4 > 5 > 6$ SI1000-like circuit-level crossing under the tested schedule. Earlier WB circuit runs also required approximate DEM decomposition. For that reason the headline circuit results use planar surface codes, whose detector error models are exactly graphlike; they do not test the layer-code geometry.

The source package retains the Steane and punctured Reed–Muller factory maps, finite SI1000-like probes with correct noiseless behavior, and three-dimensional routing coordinates. These check the displayed gadget maps; they are not a multilevel factory-threshold calculation. Coherent-overrotation and leakage toy models are also exploratory. The local-stochastic Pauli theorem is not extended to general CPTP or leakage noise, and no efficient-decoder threshold is inferred from MWPM or BP–OSD performance.

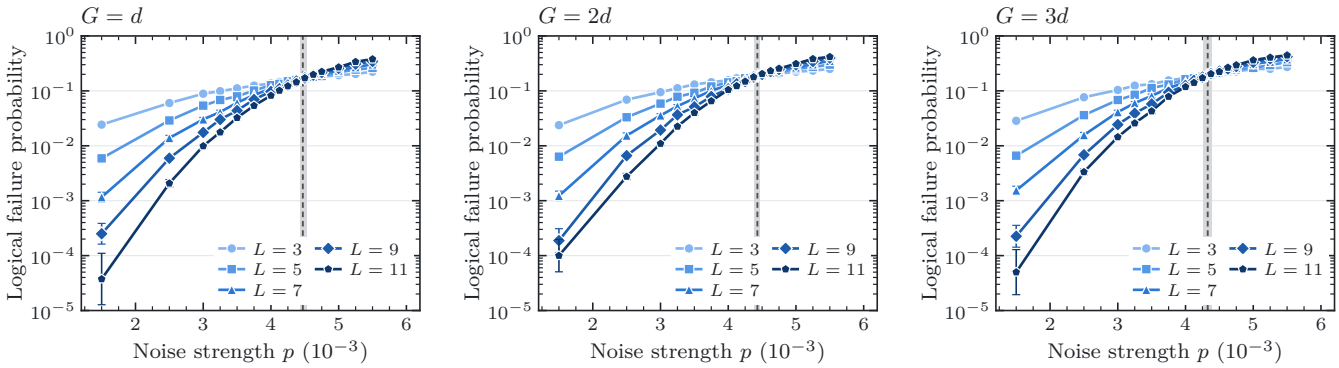


FIG. S14. Continuous algebraic base→gauged→base code-switch circuit on the planar surface code of distance L under the SI1000-like parameterization for $G = d, 2d, 3d$ (left to right). The same encoded block is used throughout. Pointwise bars are Wilson 95% intervals; the vertical markers and gray bands show the smallest-versus-largest Jeffreys posterior crossing medians and conditional 95% intervals. The transition rules implement the stabilizer/check relations used by the model rather than a gate-by-gate rendering of every Williamson–Baspin defect cell.

TABLE S16. Continuous-circuit finite-size diagnostics. The posterior bands are conditional on the smallest-versus-largest interpolation and are not systematic uncertainty bands for an asymptotic threshold.

G	monotone through	Wilson-separated at	Jeffreys crossing median	conditional 95% band
d	0.00425	0.0015–0.00375	4.471×10^{-3}	$[4.426, 4.522] \times 10^{-3}$
$2d$	0.004125	0.0025–0.00375	4.431×10^{-3}	$[4.392, 4.466] \times 10^{-3}$
$3d$	0.004125	0.0015–0.00375	4.328×10^{-3}	$[4.265, 4.393] \times 10^{-3}$

G. Files and interpretation boundary

The principal files are `numerical/continuous_transition/final_transition_heavy.py`, the archived final crossing campaign in `numerical/continuous_transition/threshold_estimate/` (including `all_points.csv` and `fit_threshold_estimate.py`), `numerical/stage_composed/paper_stim_validation.py`, `numerical/stage_composed/all_points.csv`, and `numerical/stage_composed/HEAVY_VERIFICATION_REPORT.md`, together with the core verification scripts listed in `run_verification.sh`. The final continuous crossing campaign contains 4.534×10^6 shots and the stage-composed campaign 3.955×10^6 shots.

The arbitrary-size slab theorem for the explicit realization, Williamson–Yoder spacetime cleaning, the global restriction/composition proof, the existence of a positive committed-MW threshold, and the asymptotic resource bounds are analytic results. The continuous model is an algebraic surface-code code-switch schedule with noisy detector histories and persistent encoded data; it does not reconstruct each published Williamson–Baspin defect cell as a gate-by-gate Stim circuit. Accordingly, the observed 4.3–4.5×10^{-3} structure is only a finite-size diagnostic of that model.

SXIV. COMPLETE TWO-BLOCK MEASUREMENT ON EXPLICIT LAYER-CODE BLOCKS

The surface-code campaigns of Sec. SXIII prepare Z-basis data and check one final logical-Z observable, so they do not test the reported measurement bit, both logical sectors, spectator logicals, or the action on an unknown, entangled input. The script `two_block_measurement.py` tests all four on a complete circuit for one joint logical measurement between two explicit layer-code blocks.

a. Circuit. Each block is the explicit realization of the $[[4, 2, 2]]$ input at spacing $s = 2$ ($n = 224$, $K = 2$, $d_X = 5$, $d_Z = 4$; Table S2). Logical qubit 1 of each block is measured jointly, $P = \overline{X}_1^{(a)} \overline{X}_1^{(b)}$, by Williamson–Yoder gauging: an ancilla qubit on every edge of a planar grid G attached to the single-layer slab representatives of Theorem SI.3 (12 port qubits per block), Gauss-law checks on the grid vertices, unit-square flux checks, and the original Z checks deformed along shortest grid paths. Logical qubit 2 of each block is a spectator. Every check is measured with its own ancilla and CNOTs, X checks before Z checks in each round, for $2s = 4$ rounds before, during, and after the deformation. The edges are prepared in $|0\rangle$ and read in Z, and the gauging byproduct fixed by the edge readouts is applied to the Pauli frame. The noise is uniform circuit noise of strength p: two-qubit depolarizing after every CNOT, one-qubit depolarizing on every data and edge qubit in every round, and flips

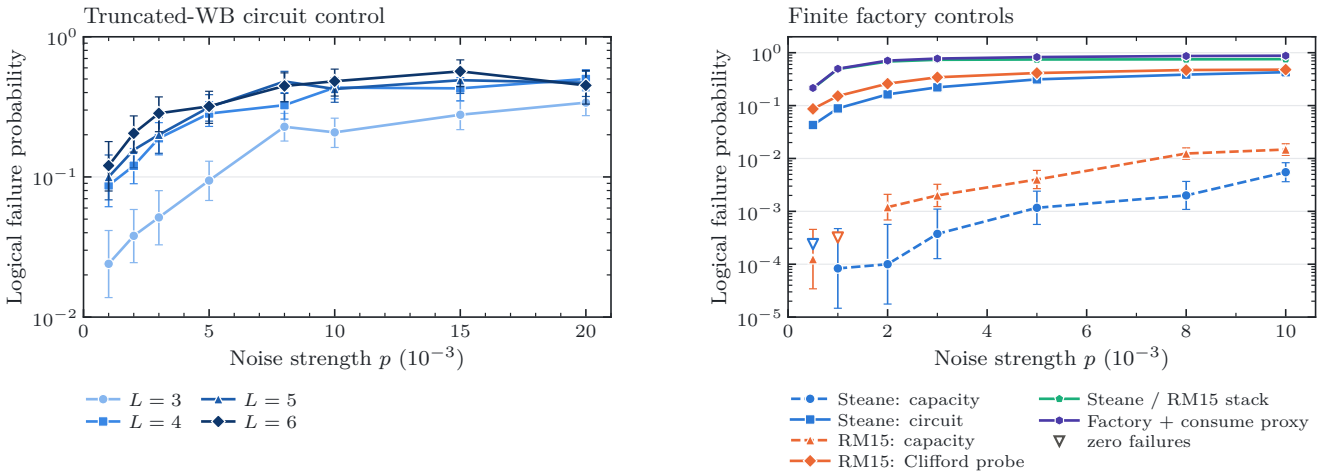


FIG. S15. Diagnostic controls. Left: the truncated single-chart (superseded) SI1000-like schedule, which lacks the clean all-size ordering used for the headline result. Right: finite Steane and Reed–Muller capacity, Clifford-circuit, and consumption-proxy controls with the proper noiseless convention. Open downward triangles denote the upper end of a Wilson 95% interval at zero observed failures.

after every reset and before every measurement, including the final data readout.

b. Input and observables. Each of the four logical qubits starts maximally entangled with a noiseless reference qubit; the entangled input is prepared by noiseless multi-qubit measurements whose outcomes enter the observables. After the protocol, the output Choi state is fixed by eight stabilizer generators, which are tested in two experiments. An X-basis readout tests the reported bit, $m + x_{R_1^a} + x_{R_1^b}$, and the four X-sector frames $\overline{X}_j^{(\cdot)} + x_{R_j^{(\cdot)}}$; a Z-basis readout tests $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)} + z_{R_1^a} + z_{R_1^b}$ and the two spectator Z frames. Without noise all detectors and all eight observables are deterministic and the protocol has zero failures, which certifies that the circuit implements the projective measurement of P with the correct reported outcome and acts as the identity on both spectators, for an arbitrary entangled input.

c. Interfaces. We compare grids that join the two port regions through a bridge of $w = 1, 2, 3$ parallel rows (the ports of each block have their own planar grid), a single planar grid spanning both blocks, and the same grid with two ancillas on every edge. The cut between the two blocks' ports has $1, 2, 3, 5$ and 10 edges, respectively.

d. Short undetectable fault patterns. For every observable we compute the fault distance of the complete circuit, the minimum number of fault mechanisms that flip it without firing a detector, with the CP-SAT formulation of `min_weight_logical.py` (exactly where the solver proves optimality, otherwise as a bracket) and the CSS sector reduction described there (Table S17). The deformed-code distances are computed the same way at the code level. Cut obstruction: the deformed code's X distance equals the bridge width, and in the circuit a single fault on the one-edge bridge flips $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ undetectably. This is the Proposition SIII.2 logical $X(\delta T)$, realized as an X fault on the bridge edge that changes the byproduct frame by $\overline{X}_1^{(a)}$.

e. Results. The reported bit m has fault distance 4 for every interface. This equals the number of deformed rounds, which bounds a timelike string of flipped Gauss-law records. The block distance $d_Z = 4$ does not limit it: the weight-4 Z logicals flip only the spectator, and the lightest Z-type operator of one block that anticommutes with $\overline{X}_1$ has weight 10 (exact; `reported_bit_spacelike_check.py`). With $R = 5$ rounds the fault distance of m is exactly 5 through the three bridges and the spanning grid (Sec. SXIV A). The spectators are unaffected by the interface: their X frames have fault distance $4 = d_Z$ for every interface, and their Z frames are at least 4 (at least 5 for four of the five interfaces). The measured pair's Z sector is where the interface matters. With a bridge of w rows, the fault distance of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ is exactly w for $w = 1, 2, 3$: a single fault on a one-edge bridge flips it undetectably, although both blocks have distance at least four. The planar grid spanning both blocks restores it to exactly 5, the blocks' X distance; with two ancillas per edge the solver proves at least 4 and finds a pattern of weight 5. The deformed-code distance $\widetilde{d}_X$ shows the same pattern at the code level (w for the bridges, 5 for both grids). The sampled failure rates of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ follow the same ordering: 71, 68 and 5 failures in 1000 shots for $w = 1, 2, 3$ (Wilson 95% intervals $[5.7, 8.9]\%$, $[5.4, 8.5]\%$ and $[0.2, 1.2]\%$), none for the grid (upper end 0.38%), and 2 for the doubled grid ($[0.05, 0.73]\%$). The one- and two-row bridges fail at similar rates because an even width detects a single bridge fault but cannot correct it (Sec. SXIV A). The failure rate of the reported bit, 1.8–6.1%, does not improve with

the interface, consistent with its interface-independent fault distance; it is highest for the doubled grid, whose Gauss-law checks have twice the weight. The spectator frames fail in at most 3 of 1000 shots in the X sector and never in the Z sector. Every run uses a recorded integer seed, listed in `two_block_mc.csv`. These blocks are small ($d \leq 5$), and the extraction measures each check with its own ancilla rather than through the routed schedule of Proposition SV.8; the experiment tests the complete logical action and the interface mechanism. Section SXIV A repeats it at four block sizes, and Sec. SXIV B with the routed schedule on three input codes.

A. Scaling with block size

A second campaign repeats the measurement at four block sizes. The circuits come from `two_block_measurement.py`, generalized to iceberg inputs $[[k+2, k, 2]]$ and free spacing; for the parameters of Table S17 the generator produces the same circuits, character for character. Each block is the explicit realization of the $[[4, 2, 2]]$ input at spacing $s = 2, 3, 4, 5$ ($n = 224$, 504, 896, 1400; $d_X = 2s + 1$ and $d_Z = 2s$, Table S2), or of the $[[6, 4, 2]]$ input ($K = 4$) at $s = 2, 3, 4$. Every phase has $R = 2s + 1$ rounds, at least both block distances. The interfaces are the one-row bridge at every size, bridges of $w = 2, 3$ rows at $s = 3, 4$, the spanning grid with one or two ancillas per edge, and, as a baseline, the same blocks idling for the same $3R$ rounds without gauging. The noise model, input and observables are those above, at $p \in \{0.5, 1, 2, 3, 4, 6\} \times 10^{-3}$.

a. Decoding and sampling. At these sizes the full-model decoder of Table S17 takes seconds per shot, so each experiment is decoded on the detectors of its own CSS sector (the X checks and Gauss laws for the X experiment, the Z and flux checks for the Z experiment) with the same BP and LSD settings; mechanisms with the same restricted syndrome and observable effect are merged. Every fault is still sampled; the decoder only ignores the correlation that a Y fault induces between the sectors. A paired check on identical shots at $s = 2$ (`decoder_check.py`, 2000 shots per point) compares it with the full-model decoder. Through the spanning grid at $p = 3 \times 10^{-3}$ the full-model decoder fails in 79 shots of the X experiment against 162 for the per-sector decoder (exact McNemar test on the discordant shots, p-value 7×10^{-10}) and in 20 against 31 shots of the Z experiment (p-value 0.11); through the one-row bridge at $p = 2 \times 10^{-3}$ it fails in 212 against 224 shots (p-value 0.07). The per-sector decoder is therefore pessimistic for the spanning grid, by up to a factor of two at this point, and nearly matches the full-model decoder for the bridge, whose failures are undetectable faults; the contrast between the interfaces reported below is, if anything, understated. The per-sector decoder is 34 to 62 times faster. Shots are drawn in batches of fixed size with recorded seeds. Each point ran until 200 failed shots, a shot cap, or the end of a 36-hour budget on eight cores, with unfinished points sharing the remaining time: 3.0×10^6 shots and 237 core-hours in total. At $s \geq 4$ and $p \geq 3 \times 10^{-3}$ the decoder exceeded the 16 GB memory in several runs, so ten of the 206 points have no data and several have few shots; these points lie near or above the crossing and are not used below. The runner, the circuits and every batch with its seed are in `scaling_campaign/` and `numerical/scaling_campaign/`; `make_scaling_figure.py` generates Table S18, Fig. S16 and every number quoted here from the batch file, the paired-check file and the distance files.

b. Thin bridge versus spanning grid. Through the one-row bridge the failure rate of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ rises with the block size: from 3.9% at $s = 2$ to 8.8% at $s = 5$ at $p = 10^{-3}$, and from 2.3% to 4.8% at $p = 5 \times 10^{-4}$. A fault on the bridge edge is never detected, and the edge is exposed in every one of the R deformed rounds, which grow with the blocks. Through the spanning grid the same observable falls, from 8.2×10^{-4} at $s = 2$ to 1.1×10^{-4} at $s = 5$ (one failure in 9300 shots) at $p = 10^{-3}$, and at every size it agrees within the Wilson intervals with the idle blocks. At $s = 5$ the two interfaces differ by more than two orders of magnitude (Table S18, Fig. S16a,b). This is Proposition SIII.2 at growing size: the block distance grows, the cut does not.

c. Fault distances at two sizes. The fault distances of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ and of the reported bit in Table S17 were recomputed for the circuits of this section ($R = 2s + 1$ rounds per phase) at $s = 2$ and $s = 3$, for the three bridges and the spanning grid with one ancilla per edge, with the same CP-SAT formulation and 30 minutes per observable (`run_distances.py`, `numerical/scaling_campaign/fault_distance.csv`). Through a bridge of $w = 1, 2, 3$ rows the fault distance of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ is exactly w at both sizes, although the blocks' X distance grows from 5 to 7. Through the spanning grid it is exactly $5 = d_X$ at $s = 2$; at $s = 3$ the solver finds an undetectable pattern of weight $7 = d_X$ and proves a lower bound of 4. The reported bit has fault distance exactly $5 = R$ for all four interfaces at $s = 2$, and between 5 and $7 = R$ at $s = 3$. This is consistent with a limit set by the number of rounds, as at $R = 4$: the lightest spacelike Z operator that flips it has weight 10 at $s = 2$ and 15 at $s = 3$.

d. Suppression with size. For the spanning grid every observable improves with size below the crossing. A linear fit of $\ln(\text{failure})$ in s, weighted by the failure counts, gives a suppression factor per unit of spacing $\Lambda = 4.8$ [3.6, 6.3] for the X experiment (reported bit and all X frames) and $\Lambda = 2.6$ [1.7, 3.8] for the Z experiment at $p = 10^{-3}$, decreasing to 1.11 [1.01, 1.21] and 0.98 [0.86, 1.12] at $p = 6 \times 10^{-3}$ (Fig. S16c; brackets are 95% intervals). With this decoder the finite-size crossing of the complete joint measurement therefore lies near $p \approx 6 \times 10^{-3}$. The reported bit alone fails in 101 of 58000 shots at $s = 2$ and in 3 of 22200 at $s = 4$ ($p = 10^{-3}$).

TABLE S17. Complete two-block measurement on explicit $[[4,2,2]]$ layer-code blocks ($s=2$, four rounds per phase). "Cut" is the number of grid edges separating the two blocks' ports; $\widetilde{d}_X$, $\widetilde{d}_Z$ are the distances of the deformed code (exact; *exact because the solver found 4 and Proposition SIII.1 gives $\widetilde{d}_Z \geq d_Z = 4$). The fault distances are for the complete circuit and each observable; an entry a–b means the solver proved a lower bound a and found an undetectable fault pattern of weight b within its time limit. Failure rates are BP+LSD decoded at $p = 0.002$ with 1000 shots per basis (four runs of 250 with recorded seeds); a rate of 0.0% has Wilson 95% upper end 0.38%. "Spect." is the spectator $\overline{X}_2^{(a)}$ and $\overline{Z}_2^{(a)}$ frames.

					circuit fault distance					failure rate		
Interface	edges	cut	$\widetilde{d}_X$	$\widetilde{d}_Z$	m	$\overline{X}_1^{(a)}$	$\overline{X}_2^{(a)}$	$\overline{Z}_1\overline{Z}_1$	$\overline{Z}_2^{(a)}$	m	$\overline{Z}_1\overline{Z}_1$	spect. X/Z
bridge, $w=1$	156	1	1	4*	4	4–10	4	1	4–12	2.7%	7.1%	0.1%/0.0%
bridge, $w=2$	163	2	2	4*	4	4–12	4	2	5–12	1.8%	6.8%	0.2%/0.0%
bridge, $w=3$	170	3	3	4*	4	4–10	4	3	5–13	3.2%	0.5%	0.2%/0.0%
grid, multiplicity 1	184	5	5	4*	4	4–10	4	5	5–12	3.1%	0.0%	0.3%/0.0%
grid, multiplicity 2	368	10	5	4*	4	4–10	4	4–5	5–12	6.1%	0.2%	0.1%/0.0%

e. Bridge width. At $s=3$ and 4 and $p = 2\times10^{-3}$ the two-row bridge fails in 10.8% and 16.0% of shots, hardly better than the one-row bridge (13.8%, 17.0%), while the three-row bridge fails in 2.6% and 2.4%. This is the pattern of an even distance: the two bridge edges that cross the cut bound the same flux square, so an X fault on either has the same syndrome, and the two corrections differ by the cut logical $X(\delta T)$. A single bridge fault is detected but not corrected; three rows correct it.

f. More logical qubits and doubled edges. The $[[6,4,2]]$ blocks show the same ordering: at $p = 10^{-3}$, $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ fails in 4.9%, 5.2% and 8.7% of shots through the bridge and at rates 1.2×10^{-3}, 1.8×10^{-4} and 1.3×10^{-4} through the grid at $s=2,3,4$, and the six spectator X frames have 240, 42 and 2 failures in 26000, 20400 and 17100 shots. At $p = 10^{-3}$ two ancillas per edge leave the rates of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ within the intervals of one ancilla per edge, but make the reported bit worse at the smallest size (5.6×10^{-3} against 1.7×10^{-3} at $s=2$, $p=10^{-3}$), since each Gauss law has twice the weight; the difference closes with size (1.7×10^{-4} against 1.4×10^{-4} at $s=4$).

g. Scope. These are finite sizes of a family that grows the spacing of $d_{\text{in}} = 2$ inputs, with one ancilla per check rather than the routed schedule of Proposition SV.8 (Sec. SXIV B uses it), decoded by a particular heuristic decoder. They show the cut obstruction and the protection of the spanning grid at growing size, and a suppression factor above one for the complete joint measurement below $p \approx 6\times10^{-3}$; they do not estimate the threshold of Theorem SV.31, which concerns the committed minimum-weight decoder.

B. Routed nearest-neighbor extraction and growing inputs

The circuits of Secs. SXIV and SXIV A measure every check with its own ancilla, and they grow only the spacing of inputs with $d^{\text{in}} = 2$. A third campaign replaces that extraction by the routed schedule of Proposition SV.8 and grows the input code at fixed spacing $s=2$. It reports noiseless certification and exact fault distances; it reports no failure rates, for the reasons given at the end of this subsection.

a. Circuit. Every check has one routing ancilla, prepared and measured in the basis of the check. The ancilla moves by nearest-neighbor SWAPs along a depth-first traversal of the lexicographically first spanning tree of a reserved routing graph, applies the data–ancilla CNOT on its first visit to the contact site of each support qubit, and retraces its path before readout. Extraction words whose sites overlap are separated by a greedy coloring and run in sequential color classes. The routing sites are an explicit three-dimensional embedding of the layer-code edges, with an additional sheet for the gauging edge qubits. The noise is that of Sec. SXIV with the routing included: two-qubit depolarizing noise of strength p after every CNOT and every SWAP, flips of probability p at every preparation and measurement, and one-qubit depolarizing noise of strength p on every data qubit, and on every routing qubit of the active color class, in every time step in which it waits. Each phase (before, during and after the deformation) has R rounds. The interfaces are the one-row bridge and the spanning grid with one ancilla per edge, and the control is the same blocks idling for $3R$ rounds.

b. Inputs. The inputs are the $[[4,2,2]]$ code of Sec. SXIV; a six-qubit band code with $H_X = H_Z$ formed by the checks 111100 and 001111, a $[[6,2,2]]$ code with $\eta = 2$; and the Steane code $[[7,1,3]]$. At $s=2$ their realizations have $n = 224$, 720 and 1792 qubits and $L_x = L_z = 4$, 6 and 8. Their distances are $(d_X, d_Z) = (5,4)$ (Table S2); $d_X \geq 7$ and $d_Z = 6$, where the bounds are those of Theorem SI.6 and an operator of weight six attains the second (below); and $d_X \geq 18$, $d_Z \geq 16$ by the same theorem. The routed circuits have 2352 to 3144, 7200 to 9842 and 14118 to 16293 qubits from idle memory to the spanning grid, of which 2507, 7448 and 12085 are routing sites in the grid circuit. The coloring uses 7, 10 to 11, and 11 to 13 classes for the X and Z checks of the blocks, and 8 to 11, 20 to 31, and 12 to 19 for the Gauss-law and flux checks; the median ancilla tour has 24, 26 and 24 shuttle

TABLE S18. Failures of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ (failed shots/shots) in the scaling campaign: two $[[4,2,2]]$-input blocks at spacing s, block distances $(d_X, d_Z) = (2s+1, 2s)$, $R = 2s+1$ rounds per phase, uniform circuit noise p, per-sector BP+LSD. "Idle" is the same blocks for the same time without gauging. An entry $0/n$ has Wilson 95% upper end about $3.8/n$; "–" was not run.

		spacing s (d_X, d_Z)			
p	interface	2 (5, 4)	3 (7, 6)	4 (9, 8)	5 (11, 10)
0.5×10^{-3}	thin bridge, $w = 1$	46/2000	279/9600	145/3600	72/1500
	spanning grid	24/196000	1/72600	0/23400	0/9450
	idle	27/264000	0/78000	0/23100	0/9000
1×10^{-3}	thin bridge, $w = 1$	78/2000	30/600	342/4500	252/2850
	spanning grid	28/34000	13/69000	3/22800	1/9300
	idle	31/44000	10/66600	0/23400	0/9150
2×10^{-3}	thin bridge, $w = 1$	176/2000	83/600	51/300	24/150
	spanning grid	212/38000	26/14400	19/13500	1/1200
	idle	64/20000	23/19200	18/19800	6/9000
3×10^{-3}	thin bridge, $w = 1$	315/2000	135/600	56/300	36/150
	spanning grid	32/2000	49/4200	26/5100	0/150
	idle	178/16000	34/5400	22/5700	22/4800
2×10^{-3}	thin bridge, $w = 2$	–	65/600	48/300	–
2×10^{-3}	thin bridge, $w = 3$	–	46/1800	22/900	–

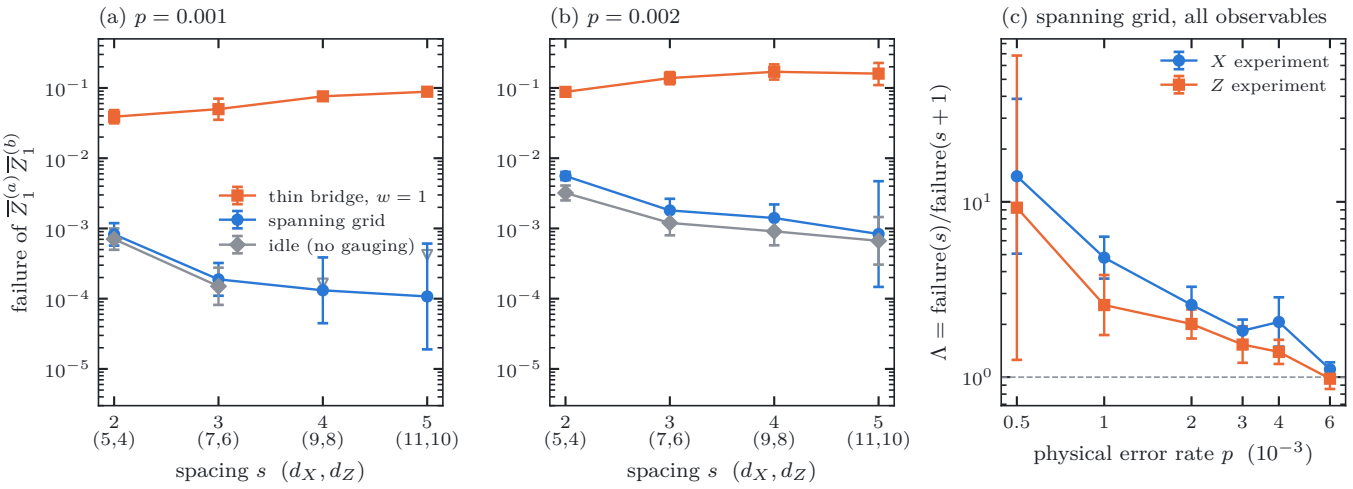


FIG. S16. Joint measurement at growing block size. (a,b) Failure of the measured pair's Z sector through the one-row bridge, the spanning grid, and for the same blocks idling without gauging, at $p = 10^{-3}$ and 2×10^{-3}; bars are Wilson 95% intervals and open triangles the upper end at zero failures. The bridge gets worse as the blocks grow, the grid gets better and tracks the idle blocks. (c) Suppression factor per unit of spacing for the spanning grid (all observables of each experiment), weighted fit over the available sizes with 95% intervals; it reaches one near $p \approx 6 \times 10^{-3}$.

steps (the longest 143, 189 and 123). The bridge crosses between the two blocks in one edge for every input, the grid in 5, 17 and 9 edges.

c. Certification. Without noise every detector is silent and every observable takes its correct value in all 40 circuits run, 28 measurements and 12 idle controls (three inputs, both bases, and the round numbers of Table S19): the reported bit, the X and Z frames of the measured pair, and those of the spectators. The routed circuit therefore implements the projective measurement with the correct outcome and leaves the spectators untouched, as the colocated circuit does.

d. Fault distances. The fault distances use the CP-SAT formulation and sector reduction of Sec. SXIV, with a time limit per observable between a few minutes and a few hours; for the band grid and idle circuits at $R = 7$ the solver instead decides whether an undetectable pattern of weight at most six exists, for up to 8.7 hours (Table S19). Four results hold.

(i) The cut obstruction survives routing and growing

inputs. Through the one-row bridge the fault distance of $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ is exactly one for all three inputs and every R tested, while the blocks' X distance grows from 5 to at least 7 and at least 18.

(ii) On the $[[4,2,2]]$ blocks the routed grid keeps the full distance. Its fault distance for $\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$ is exactly 5 at $R = 2$, 3 and 5, and the same blocks left idle also give exactly 5, as do the colocated circuits (Table S17). The grid therefore removes nothing from the distance that storage alone provides. For the band and Steane blocks the grid distance is not resolved: the lower bounds are 4 to 5, and the lightest patterns found have weight 17 and 9, the number of grid edges crossing between the blocks. For the Steane input that number is 9, so by Proposition SIII.2 the deformed code has X distance at most 9, below the block's $d_X \geq 18$. This is the case in which Corollary SIII.4 needs more than one ancilla per edge; a run with two (crossing number 18, the lower bound on d_X) was prepared but not carried out.

(iii) The reported bit is limited by the number of rounds or by the lightest Z-type operator that anticommutes with the measured logical, whichever is smaller. On the $[[4,2,2]]$ blocks its fault distance is exactly R for $R = 2$, 3 and 5 through both interfaces; the lightest such operator has weight 10 (Sec. SXIV). On the band blocks it is exactly R for $R = 2$ and 4. At $R = 7$ the solver finds undetectable patterns of weight 6 through both interfaces, which flip $\overline{X}_1^{(a)}$ together with the bit, and a pattern of weight 6 for the $\overline{X}_1^{(a)}$ frame of the idle blocks. A spacelike operator of the block accounts for this weight: the lightest Z-type operator of the band block that anticommutes with $\overline{X}_1$ has weight exactly 6 (`band_spacelike_check.py`), which is the block's Z distance, so no extraction schedule can give the bit more than $\min\{R, 6\}$, and $R = 7$, the lower bound on d_X, cannot give it 7. This is consistent with the bound $\delta_g^{\mathrm{rel}} \geq \min\{\beta_0 d, R_g\}$ of Eq. (A15d), in which d is the least distance of the blocks involved. The solver proves only 4 (bridge) and 2 (grid, idle), so whether routing adds lighter patterns is open. The spacelike limit depends on the logical class: for the class of the input word 110000 the lightest anticommuting Z-type operator has weight 16 to 24 (CP-SAT bracket), against 6 for $\overline{X}_1$ and for the third class.

(iv) The frames of the blocks do not depend on the interface. On the $[[4,2,2]]$ blocks at $R = 5$ the spectator X frames have fault distance exactly $4 = d_Z$ for the bridge, the grid and idle memory, and 4 is also the weight of the lightest Z-type operator anticommuting with $\overline{X}_2$ (Sec. SXIV). For all three, the measured X frames have patterns of weight 10, the spacelike value, with lower bounds 5 to 6, and the spectator Z frames have patterns of weight 12 with lower bound 6. On the band blocks the spectator frames are not resolved (lower bounds 3 to 5).

e. Sampled failure rates and the decoder. At $R \leq 2$ and $p = 5\times10^{-4}$ to 2×10^{-3}, with the per-sector BP+LSD decoder of Sec. SXIV A, the target observable fails in between a quarter and two thirds of the shots for every input, interface and noise strength, idle memory included (for example 41 of 80 shots for the idle $[[4,2,2]]$ blocks at $p = 5 \times 10^{-4}$), so these runs do not compare interfaces. Two features of these runs limit what they can show. The routed schedule is long: the longest ancilla tour on the $[[4,2,2]]$ blocks has 143 shuttle steps, so a round lasts at least that many time steps, in each of which almost every data qubit receives depolarizing noise of strength p; at $p = 5 \times 10^{-4}$ this alone gives each data qubit an error probability of about 0.07 per round. The decoder, moreover, is not minimum weight. An exhaustive test of every single mechanism on the $[[4,2,2]]$ blocks at $R = 3$ finds that it leaves 56 of 81,332 uncorrected in the Z sector of idle memory, 102 of 85,260 in its X sector, and 2943 of 93,385 for the reported bit through the bridge, although a minimum-weight decoder corrects every single mechanism at these fault distances and none of them is undetectable. The detector error models are not graphlike, so matching does not apply. For the routed circuits we therefore use exact fault distances only.

f. Scope. The inputs grow in length but stay far from the balanced regime ($n^{\mathrm{in}} \leq 7$, $K \leq 2$), the grid has one ancilla per edge, and the grid distance is resolved only for the $[[4,2,2]]$ blocks; the band grid, the band idle Z frame and a long-schedule Steane run remain open. The campaign tests the routed construction of Proposition SV.8 at finite size and does not estimate a threshold. The directory `numerical/routed_campaign/` holds every table, the noiseless certification, the solver logs and the lightest undetectable pattern found for every observable with an upper bound; `routed_campaign_check.py` checks the certification rows and that every archived pattern is undetectable and flips its observable.

SXV. HARDWARE DEMONSTRATIONS

a. The idea in plain terms. The distance results of Sec. SIII rest on three mechanisms that already appear in very small circuits. A joint logical measurement through a narrow interface can be undone by a few faults on that interface (Proposition SIII.2); parallel ancillas on an interface edge widen every wall that crosses it (Corollary SIII.4); and a wrong outcome in one round of the measurement is caught by the next round (Proposition SV.17). This section tests each mechanism on superconducting processors with blocks of three qubits. It does not test layer codes, the grid expansion, or the architecture, which need far larger devices.

A. Devices, placement, and checks before submission

a. Devices. IQM Emerald (54 qubits on a square lattice, native CZ) was used through the Open Quantum platform [13] on its public plan, with qubit rewiring disabled. The IBM Heron processors `ibm_fez` and `ibm_kingston`

TABLE S19. Fault distances of the routed two-block circuits (Proposition SV.8; spacing $s = 2$, one ancilla per grid edge, R rounds per phase). An entry a–b means the solver proved a lower bound a and found an undetectable pattern of weight b; $\geq a$ means a lower bound without a pattern; "–" was not computed. "Cut" is the number of edges crossing between the blocks (bridge/grid). The block distances of the band and Steane inputs are the bounds of Theorem SI.6, except $d_Z = 6$ for the band (attained, see text). The lightest Z-type operator anticommuting with $\overline{X}_1$ has weight 10 for $[[4, 2, 2]]$ and 6 for the band block, and with $\overline{X}_2$ weight 4 for $[[4, 2, 2]]$.

				$\overline{Z}_1^{(a)}\overline{Z}_1^{(b)}$			reported bit m		$\overline{X}_1^{(a)}$			spectator $\overline{X}_2^{(a)}$		
Input	(d_X, d_Z)	cut	R	bridge	grid	idle	bridge	grid	bridge	grid	idle	bridge	grid	idle
$[[4, 2, 2]]$	$(5, 4)$	1/5	2	1	5	–	2	2	–	–	–	–	–	–
			3	1	5	5	3	3	–	–	6–10	–	–	–
			5	1	5	5	5	5	5–10	5–10	6–10	4	4	4
band $[[6, 2, 2]]$	$(\geq 7, 6)$	1/17	2	1	5–17	–	2	2	–	–	–	–	–	–
			4	1	5–17	–	4	4	–	–	–	–	–	–
			7	1	≥ 4	≥ 4	4–6	2–6	4–6	4–6	2–6	5–24	≥ 3	–
Steane $[[7, 1, 3]]$	$(\geq 18, \geq 16)$	1/9	1	1	4–9	–	1	1	–	–	–	–	–	–

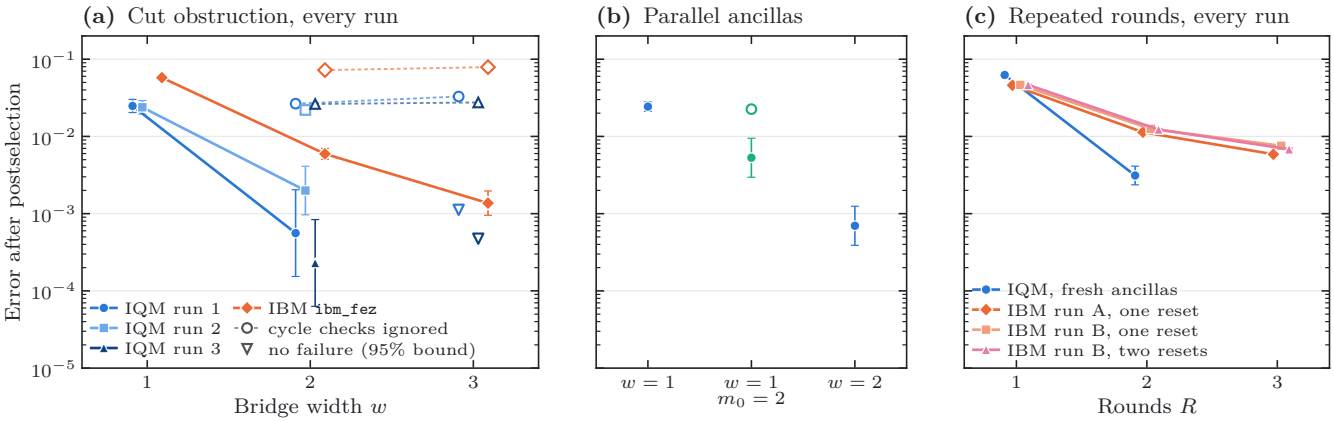


FIG. S17. Every hardware run separately (the main-text hardware figure pools the IQM runs); error after postselection on every detector, with Wilson 95% intervals (open triangles: Wilson upper ends of zero counts). (a) Two three-qubit blocks joined by a bridge of w edges; open symbols: the same samples with the bridge cycle checks ignored. (b) One bridge edge with two parallel ancillas ($m_0 = 2$, aqua) against one bridge qubit and two bridge edges, on IQM, each width pooled over every IQM run that has it; open symbol: the bundle cycle ignored. (c) Reported-bit error against the number of rounds; IQM uses fresh ancillas in every round, IBM measures and resets them, once or twice. Data in Tables S20 and S21.

(156 qubits on a heavy-hex lattice, native CZ) were used through the IBM Quantum Platform with the Qiskit Runtime sampler; the runs took 54 s of QPU time on `ibm_fez` and 485 s and 407 s on `ibm_kingston`. Two attempts on a Rigetti processor are reported under excluded runs below.

b. Placement. In every circuit except the `ibm_fez` cut-obstruction circuits, each two-qubit gate acts on a coupler, so these circuits contain no SWAP gate. The cut-obstruction graph does not embed in the heavy-hex lattice, so the `ibm_fez` circuits were routed by the Qiskit transpiler, and their checks below were repeated on the routed circuits. On IQM the circuits are written in OpenQASM 3 on physical qubits. The platform's validation does not check connectivity, and its qubit numbering (counted from zero or from one) is not documented, so every IQM placement was chosen to be SWAP-free under both readings. On IBM the circuits are compiled with a fixed initial layout. For the repeated-rounds runs the layout is the best pair of disjoint 13-qubit paths under the live calibration (CZ errors of the path couplers, readout errors of the measured ancillas, and dephasing of the other qubits during a measure-and-reset window), and idle windows carry an XX dynamical-decoupling sequence. Figure S18 shows every fixed placement and the couplers that its circuit uses.

c. Checks before submission. Each circuit as submitted was checked to (i) on IQM, parse with Qiskit's OpenQASM 3 importer and equal the reference circuit as a Clifford operation; (ii) pass every detector with the correct observable in a noiseless stabilizer simulation; (iii) have the stated exact circuit fault distance, computed by the CP-SAT solver of `min_weight_logical.py` on the Stim detector error model of the circuit with every operation

TABLE S20. Cut obstruction and edge multiplicity on hardware. Two three-qubit blocks joined by a bridge of w edges ($w = 0$: no gauging); $m_0 = 2$ puts two parallel ancillas on a single bridge edge. Errors are postselected on every detector, in percent, with Wilson 95% intervals; a zero count shows the upper end. Control: the same shots postselected on the block checks only, ignoring the cycle checks that the extra bridge edges create. Model: uniform circuit noise (IQM: CZ error 0.5%, readout 2%/4%) or typical published medians (IBM), fixed before the run. Generated by `hardware/hardware_summary.py`.

Device, run	Bridge	Shots	Kept	Wrong	Error (%)	Control (%)	Model (%)
IQM Emerald, run 1	$w = 1$	5,000	3,911	97	2.5 [2.0, 3.0]		4.2
	$w = 2$	5,000	3,571	2	0.056 [0.015, 0.20]	2.7	0.22
	$w = 3$	5,000	3,424	0	0 (≤ 0.11)	3.3	0.022
IQM Emerald, run 2	$w = 1$	5,000	3,820	91	2.4 [1.9, 2.9]		4.2
	$w = 2$	5,000	3,513	7	0.20 [0.097, 0.41]	2.2	0.19
IQM Emerald, run 3	$w = 2$	12,000	8,700	2	0.023 [0.0063, 0.084]	2.6	0.22
	$w = 3$	12,000	8,105	0	0 (≤ 0.047)	2.8	0.022
IBM `ibm_fez`	$w = 0$	50,000	47,715	1	0.0021 [0.00037, 0.012]		0.0023
	$w = 1$	50,000	27,931	1,611	5.8 [5.5, 6.0]		3.3
	$w = 2$	50,000	24,220	144	0.59 [0.51, 0.70]	7.2	0.16
	$w = 3$	50,000	21,181	29	0.14 [0.095, 0.20]	7.9	0.012
IQM Emerald	$w = 1$, $m_0 = 2$	3,000	2,075	11	0.53 [0.30, 0.95]	2.3	

TABLE S21. Repeated rounds on hardware. Phase-flip blocks ($d_Z = 3$) joined by one gauging edge; the two vertex stars and the four block checks are measured in R rounds and the reported bit is the round-one star product. IQM: fresh ancillas in every round, all measured at the end. IBM (`ibm_kingston`): ancillas measured and reset between rounds, once or twice; runs A and B on different days. Errors postselected on every detector, in percent, pooled over two copies. Control: the round-to-round star comparisons ignored. Persistence: a check ancilla reads 1 in rounds 1 and 2 while the ancillas sharing its data qubits show no change; in brackets, the rate that independent errors would give. Model: uniform noise (IQM, CZ 0.5%) or the device calibration (IBM). Samples: shots times the two copies. Generated by `hardware/hardware_summary.py`.

Device	Rounds	Samples	Kept	Wrong	Error (%)	Control (%)	Persistence (%)	Model (%)
IQM Emerald	$R = 1$	10,000	8,979	561	6.2 [5.8, 6.8]			6.9
	$R = 2$	20,000	15,703	49	0.31 [0.24, 0.41]	7.7		0.34
IBM, run A	$R = 1$	400,000	349,885	16,071	4.6 [4.5, 4.7]			3.7
	$R = 2$, one reset	1,100,000	765,042	8,677	1.1 [1.1, 1.2]	4.4	0.969 (0.041)	0.18
	$R = 3$, one reset	2,100,000	1,212,715	7,127	0.59 [0.57, 0.60]	4.3	0.972 (0.041)	0.0086
IBM, run B	$R = 1$	200,000	177,413	8,250	4.7 [4.6, 4.7]			2.3
	$R = 2$, one reset	600,000	419,450	5,258	1.3 [1.2, 1.3]	4.4	1.15 (0.051)	0.062
	$R = 2$, two resets	600,000	405,271	4,978	1.2 [1.2, 1.3]	4.4	1.11 (0.052)	0.061
	$R = 3$, one reset	800,000	459,123	3,469	0.76 [0.73, 0.78]	4.3	1.15 (0.051)	0.00081
	$R = 3$, two resets	800,000	431,173	2,938	0.68 [0.66, 0.71]	4.3	1.11 (0.050)	0.0022

noisy (and, in the scheduled IBM repeated-rounds circuits, every idle window); and (iv) for the IBM repeated-rounds circuits, use exactly the planned physical qubits and pass the platform's instruction-set and timing validation. For the run repeated on the same IQM qubits, the platform's recorded input hashes confirm byte-identical circuit files.

d. Analysis. Every result is the error of the observable after postselection on all detectors, with Wilson 95% intervals; comparisons use one-sided Fisher exact tests (Table S22). For the later runs, marked P in that table, the test and its level 0.05 were fixed in the analysis script before the jobs were submitted. For the first runs the analysis code was fixed before the data, but the tests were chosen afterwards. The bit order of each output is checked against the data wherever possible: on IQM the intended order gives detector rates near the noise model and every other candidate order gives rates near one half. The IBM repeated-rounds circuits are symmetric under reversal of the bit string, so there the documented order is used; the `ibm_fez` run confirmed it, since the reversed reading puts every detector at one half.

B. Cut obstruction

a. Circuit. Each block is a three-qubit code with Z checks Z_0Z_1 and Z_1Z_2, $\overline{X} = X_0X_1X_2$ (a weight-three

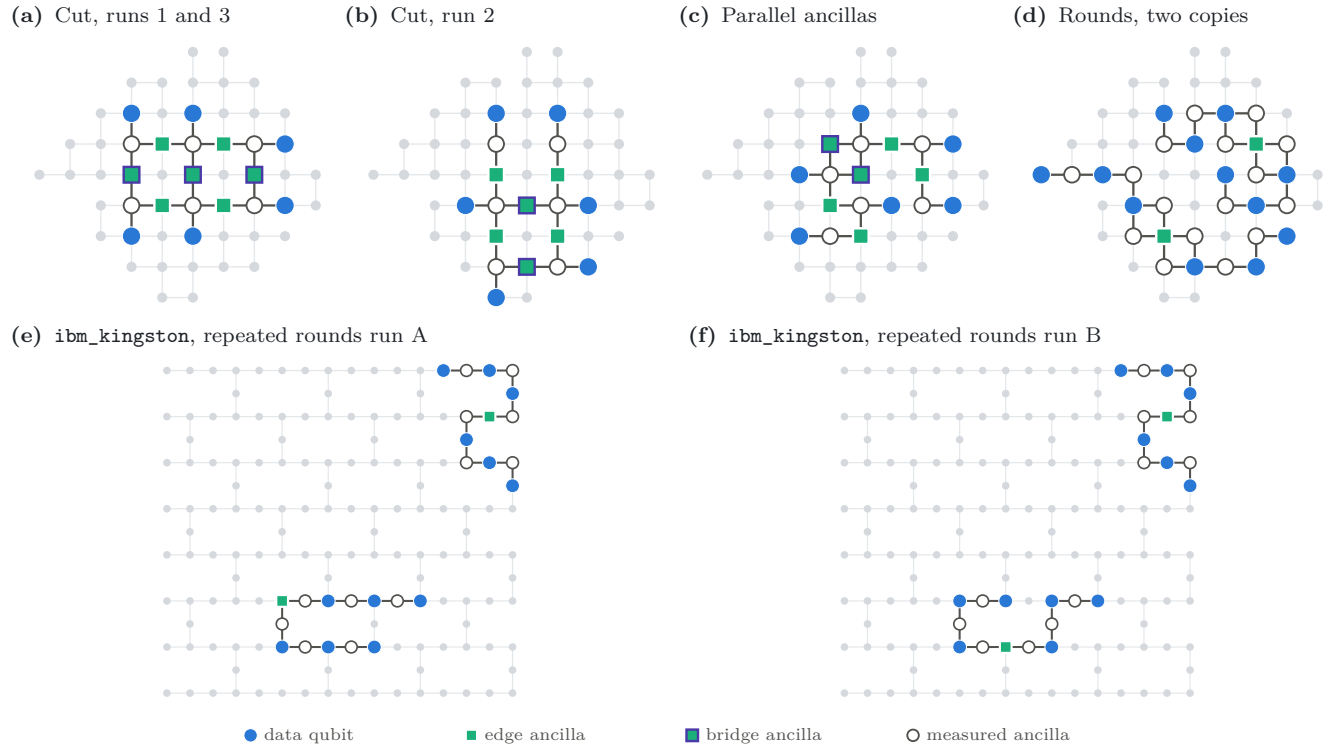


FIG. S18. Placements on the chips. Grey: unused qubits and couplers; dark lines: the couplers that the submitted circuit uses. (a)–(d) IQM Emerald, 54 qubits, with the qubit numbers of the Open Quantum coupling map read from zero (every placement is also SWAP-free when they are read from one). (a) Cut obstruction, width three, runs 1 and 3 (widths one and two use a subset). (b) Run 2, width two (width one uses a subset). (c) Two parallel ancillas on one bridge edge. (d) Repeated rounds, two copies of the circuit, with the star and check ancillas of both rounds. (e), (f) `ibm_kingston`, heavy-hex lattice of 156 qubits, in the repeated-rounds runs A and B: each run uses two 13-qubit paths, one for each copy of the circuit. The `ibm_fez` cut-obstruction circuits were routed by the transpiler and have no fixed placement. The chip coordinates are derived from the coupling maps.

membrane, the stand-in for a slab port) and $\overline{Z} = Z_0$. The gauging graph has one vertex per data qubit, the edges (a_0, a_1), (a_1, a_2), (b_0, b_1), (b_1, b_2), and the bridge edges (a_i, b_i) for $i < w$; an ancilla on every edge starts in $|0\rangle$. Each vertex star is measured once by its own ancilla, which is read only at the end, so the circuit needs no mid-circuit measurement; at the end every data and edge qubit is read in Z. The blocks start in $|000\rangle$, so the deformed $\overline{Z}_a\overline{Z}_b = Z_{a_0}Z_{b_0}Z_{(a_0,b_0)}$ must read $+1$. The detectors are the deformed Z checks $Z_xZ_yZ_{(x,y)}$ and, for $w \geq 2$, the cycle checks of the bridge squares. By Proposition SIII.2, X on the w bridge edges is equivalent to $\overline{X}_a$, so the deformed X distance is w, and the exact circuit fault distance is also w for $w = 1, 2, 3$, both for the reference circuits and for the compiled IBM circuits. On IQM the circuits use 16, 18 and 20 CZ gates; on the heavy-hex lattice of `ibm_fez` the compiled circuits need 50, 55 and 60 CZ gates.

b. Results. Table S20 and Fig. S17(a) give the data. IQM runs 1 and 3 use one set of qubits; run 2 uses a set that shares 8 of its 19 qubits and none of its bridge qubits; run 3 repeats run 1 with byte-identical files. On both devices the error drops sharply from $w = 1$ to $w = 2$ ($p \leq 9.5 \times 10^{-19}$ in each IQM run with both widths, $p = 1.7 \times 10^{-278}$ on IBM), and on IBM again from $w = 2$ to $w = 3$ ($p = 1.1 \times 10^{-16}$). On IQM width three had no failure in 11,529 kept samples, but width two was already so rare there (0.070%, pooled) that the test fixed before run 3 was not met ($p = 0.071$). The decoded error without postselection, a secondary measure chosen afterwards, does fall from width two to width three, from 3.0% to 1.7% in run 3 ($p \leq 6.4 \times 10^{-12}$ in runs 1 and 3). The control column shows that the cycle checks cause the gain: postselecting the same samples on the block checks alone returns widths two and three to the width-one level. The IBM errors exceed the model, which assumed typical published median error rates and no idle or crosstalk errors.

C. Edge multiplicity

a. Circuit. The width-one bridge edge (a_0, b_0) carries two parallel ancillas, as in Corollary SIII.4 with $m = 2$ on that edge. The bundle cycle $Z_eZ_{e'}$ of the two parallel edges is a new detector, so a single X fault on one of them is detected and the deformed X distance is two. On the square lattice the vertex ancillas of a_0 and b_0 share both

TABLE S22. Significance of the hardware comparisons: one-sided Fisher exact tests that the second group fails less often than the first. P: test and level 0.05 fixed in the analysis script before the data were taken. Generated by `hardware/hardware_summary.py`.

Comparison	p	P
IQM run 1: $w = 2$ below $w = 1$	3.1×10^{-25}	
IQM run 2: $w = 2$ below $w = 1$	9.5×10^{-19}	
IQM runs 1+3: $w = 3$ below $w = 2$	0.071	P
IBM fez: $w = 2$ below $w = 1$	1.7×10^{-278}	
IBM fez: $w = 3$ below $w = 2$	1.1×10^{-16}	
IQM run 1, decoded: $w = 3$ below $w = 2$	1.4×10^{-12}	
IQM run 3, decoded: $w = 3$ below $w = 2$	6.4×10^{-12}	
IQM: two parallel bridge qubits below one bridge qubit (runs 1+2)	5.1×10^{-10}	
IQM: two separate bridge edges below two parallel bridge qubits (runs 1+2)	0.0017	
IQM rounds: $R = 2$ below $R = 1$	2.4×10^{-188}	P
IBM rounds A: $R = 2$ below $R = 1$	$< 10^{-300}$	P
IBM rounds A: $R = 3$ below $R = 2$	$< 10^{-300}$	P
IBM rounds B (single reset): $R = 2$ below $R = 1$	$< 10^{-300}$	P
IBM rounds B (single reset): $R = 3$ below $R = 2$	9.3×10^{-123}	P
IBM rounds B (double reset): $R = 2$ below $R = 1$	$< 10^{-300}$	P
IBM rounds B (double reset): $R = 3$ below $R = 2$	2.5×10^{-148}	P
IBM rounds B: $R = 2$ error lower with two resets	0.15	P
IBM rounds B: $R = 3$ error lower with two resets	1.8×10^{-5}	P
IBM rounds B: persistence lower with two resets	8.8×10^{-6}	P

parallel edge qubits as neighbors, which fits the circuit on IQM without SWAP gates.

b. Gate order. In the first version of the circuit each of the two vertex ancillas applied its CNOTs to the two parallel edges one after the other. The exact circuit fault distance of that circuit is one: an X fault on the vertex ancilla just before these two CNOTs spreads to both parallel edges, which flips the observable and leaves the bundle cycle silent. Ordering each star as first parallel edge, data qubit, second parallel edge, block edge makes every such spread detectable, and the exact circuit fault distance becomes two. This is an instance of the constant c_1 of Sec. SV, the number of phenomenological mechanisms that one circuit fault can produce; with parallel ancillas the extraction order decides whether that constant costs distance.

c. Results. With all checks the error is 0.53% (Table S20), against 2.4% for one bridge qubit in runs 1 and 2 ($p = 5.1 \times 10^{-10}$); ignoring the bundle cycle gives 2.3%. Two parallel ancillas protect less than two separate bridge edges ($p = 0.0017$). Both need two faults, but the parallel version has more two-fault paths, as the noise model also predicts.

D. Repeated rounds

a. Circuit. The reported bit is the product of the vertex-star outcomes. To test its protection we use two three-qubit phase-flip blocks (X checks X_0X_1, X_1X_2; $\overline{X} = X_0$; $\overline{Z} = Z_0Z_1Z_2$, so $d_Z = 3$) in $|{+}{+}{+}\rangle$, so that $\overline{X}_a\overline{X}_b = +1$, joined by one edge qubit e in $|0\rangle$. The stars $A_{a_0} = X_{a_0}X_e$ and $A_{b_0} = X_{b_0}X_e$ are random individually, and their product $X_{a_0}X_{b_0}$, the reported bit, must be $+1$. In each of R rounds the two stars and the four block checks are measured with ancillas. The detectors are every block-check outcome and, from the second round on, every star outcome against the previous round; the observable is the round-one product. A phase error on a port qubit flips one star in every round but also a block check, so the bit is limited by R and not by the toy code: the exact circuit fault distance is $\min\{R, 3\}$, that is 1, 2 and 3 for $R = 1, 2, 3$ and still 3 for $R = 4$ (computed on the compiled IBM circuits), as in Eq. (A15s).

b. IQM: fresh ancillas. The IQM platform rejects any operation on a qubit after it has been measured, so each round uses its own ancillas, all read at the end. The edge qubit has four couplers, which allows two rounds. The error falls from 6.2% to 0.31% (Table S21; $p = 2.4 \times 10^{-188}$, fixed in advance), close to the model, and ignoring the round-to-round comparison returns it to 7.7%.

c. IBM: measure and reset. On `ibm_kingston` the ancillas are measured and reset between rounds. Both steps, from $R = 1$ to 2 and from 2 to 3, are clear in two runs on different days (every step $p \leq 9.3 \times 10^{-123}$, fixed in advance), and ignoring the round-to-round comparison returns the error to 4.3 to 4.4%. The gain is, however, far below the calibrated model: 1.1% and 0.59% at $R = 2, 3$ in run A, against 0.18% and 0.0086%.

d. Errors that survive the reset. The excess comes from ancilla errors that persist across rounds. A check ancilla that reads 1 in rounds 1 and 2 while the ancillas sharing its data qubits show no change cannot be explained by a data error. Independent readout and gate errors would make this happen in about 0.051% of samples, but it happens in 1.15% (run B, one reset; Table S21). On a star ancilla the same event repeats a wrong outcome in every round, which no round-to-round comparison can see. Run B tests whether failed resets cause it by applying two resets between rounds, interleaved in time with the single-reset circuits on the same qubits. The second reset lowers the persistence only from 1.15% to 1.11% (a 3% reduction, $p = 8.8 \times 10^{-6}$) and the error at $R = 3$ from 0.76% to 0.68% ($p = 1.8 \times 10^{-5}$); at $R = 2$ the change is not significant ($p = 0.15$). Most of the persistence is therefore something that a reset cannot undo, most

likely leakage out of the computational subspace, which a measure-and-flip reset leaves in place. The IQM runs, whose ancillas are never reused, agree with their model. A persistent ancilla error is one physical event that produces faults in every round, while the local-stochastic model of Sec. SV bounds the probability of faults in R rounds by p^R; leakage reduction is the hardware measure that makes such events rare enough for the model to apply.

e. No $R = 4$ on hardware. For $R = 4$ the exact circuit fault distance stays at $d_Z = 3$, but the calibrated model predicts an error far below what the available QPU time could resolve, so the cap $\min\{R, d_Z\}$ is shown only by the exact distance computation.

E. Excluded runs

Two earlier attempts used a 108-qubit Rigetti processor through Open Quantum. In the first the compiler chose the placement; in the second the cut-obstruction circuits were placed without SWAP gates and qubit rewiring was disabled. Both gave detector violation rates between about 0.2 and 0.5, most of them near one half, and in the second one physical qubit read 1 in 30,000 of 30,000 shots although the platform listed it as operational. The platform provides no calibration data, so faulty qubits could not be avoided; both attempts are excluded from the analysis, and their outputs and diagnosis are archived. A probe of mid-circuit measurement on IQM failed at execution, because an operation followed a measurement, and returned no data.

F. Files

All hardware material is in the folder `hardware`. `cut_obstruction_hw.py` defines the cut-obstruction circuits, detectors, observable and lookup decoder, and submits and analyzes the `ibm_fez` run; `circuit_distance_check.py` computes their exact circuit fault distances. `embed_iqm.py`, `export_iqm.py` and `final_check_iqm.py` place, write and check the IQM circuits, and `analyze_iqm.py`, `cycle_check_control.py` and `analyze_pooled_w23.py` analyze them. `multiplicity_hw.py`, `export_multiplicity.py` and `analyze_multiplicity.py` do the same for the parallel-ancilla circuit; `rounds_hw.py`, `export_rounds.py` and `analyze_rounds.py` for the IQM rounds; and `ibm_rounds_hw.py`, `ibm_rounds_plan.py`, `ibm_rounds_submit.py`, `ibm_rounds_analyze.py`, `ibm_rounds_persistence.py` and the three `ibm3_reset` scripts for the IBM rounds. The submitted circuits (`qasm3_*.qasm`, `*.qpy`), the platform outputs (`oq_*output*.json`, `hw_counts.json`, `ibm2_rounds_bits.npz`, `ibm3_reset_bits.npz`), the job identifiers, the IBM calibration snapshots and the Open Quantum backend descriptions are included. The submission helpers in `hardware/submission` read access keys from files that are not part of the package. `hardware_summary.py` recomputes every hardware number of the main text and of this section and writes Tables S20, S21 and S22 and `hardware_macros.tex`; `make_hardware_figure.py` draws the main-text hardware figure and Figs. S17 and S18.

SXVI. SPACE–TIME TRADEOFF AND A CONCRETE RESOURCE ESTIMATE

a. The depth bound. Theorem SV.31 pairs the space bound $O(Wf + (B_f(A)+1)f^{3/2})$, which is $O(Wf + f^{3/2})$ for a bounded-aspect box (Corollary SV.30), with the depth bound $O(Tf^{3/2}\Psi)$, $\Psi = O(f^{\log_2 15})$, i.e. $O(Tf^{5.41})$ rounds. This depth is the price of the minimal-space schedule: every data cell has one depth-first factory, a distilled resource costs $O(\Psi)$ serialized gadget slots of $O(f)$ rounds, and a logical layer waits for the $O(K)$ resources of its cell. A logical layer of lattice surgery, by contrast, takes $O(f)$ rounds. The two schedules are the ends of one tradeoff. With P factories per data cell ($1 \le P \le K$), the space is $O(WfP + f^{3/2})$ and the depth $O(TKf\Psi/P)$, so the qubit-time $O(WTf^{5/2}\Psi)$ does not depend on P; the serial schedule minimizes space, not qubit-time. The exponent $\log_2 15$ comes from the quadratic recursion $q_{r+1} \le Cq_r^2$ proved for always-output distillation under local-stochastic noise. The standard cubic suppression of 15-to-1 distillation would give $\Psi = O(f^{\log_3 15}) = O(f^{2.47})$, but it is not proved here for jointly local-stochastic inputs.

b. A concrete estimate. The theorem fixes exponents, not constants. The script `resource_estimate.py` evaluates the accounting of Theorem SV.31 with every constant stated as a named, adjustable assumption (listed with its value in `resource_estimate.md`). Every schedule receives an explicit integer allocation of blocks, or surface-code tiles, by role: data, working slots, factories, and couriers. Each schedule's distance is chosen from its complete runtime, including the time spent waiting for resources: f is the least odd distance with $L_{\text{live}} R_{\text{total}}\, p_L(f) \le \varepsilon/2$, where L_{live} counts every live logical qubit, R_{total} is the runtime in rounds, and $p_L(f) = 0.1\,(p/p_{\text{th}})^{(f+1)/2}$ with $p_{\text{th}} = 10^{-2}$ for both architectures. The final resources are budgeted separately, $WT\, q_{\text{final}} \le \varepsilon/2$. The target is $W = 100$ logical sites and $T = 10^6$ logical layers with one distilled $|A\rangle$ per site and layer, at total failure $\varepsilon = 10^{-2}$.

Three distillation models are kept separate. (i) *Proved*: the always-output bound (A8), $q_{r+1} = C_{\text{dist}} q_r^2$, with $C_{\text{dist}} = \binom{30}{2} = 435$ on layer-code blocks (fifteen principal and fifteen S-correction slots per run), and, as an assumed analog, $\binom{15}{2} = 105$ for a surface-code unit, whose rotations need no $|Y\rangle$ correction (Lemmas SV.4 and SV.5 are proved for the factory (A4a), not for that unit). (ii) *Exact*: the exact error map of the ideal always-output factories for independent Z-type candidate errors, from an enumeration of all 2^{15} patterns. Every pair of errors is malignant (the syndrome of a pair is that of a third coordinate, and the three form a weight-three logical),

so $q_{\rm out} = 105q_c^2 - 1330q_c^3 + O(q_c^4)$; on layer-code blocks the per-coordinate error $q_c = q_A + q_Y/2 - q_A q_Y$ includes the S correction, used with probability one half. (iii) *Postselected*: standard 15-to-1 and 7-to-1 distillation that discards nontrivial syndromes ($q_{\rm out} = 35q^3 + O(q^4)$, with retries). The theorem does not cover this model. A 15-to-1 run on layer-code blocks takes 30 gadget slots of $3f$ rounds and consumes fifteen S-correction resources assigned in the fixed schedule and produced just in time, themselves distilled by 7-to-1 runs of 14 slots; a raw resource costs two slots of injection and courier transfer.

The layer-code block is the explicit realization at $s = 2$ of a balanced input with rate 0.1, relative distance 0.1, check weight 8, and supports spanning the input. Its size n follows from the Euler count of Proposition SI.2 and $K = \lfloor 0.1N \rfloor$. Its distance is bracketed by Theorem SI.6 between $L_z\lceil d^{\rm in}/4\rceil$ and $L_z d^{\rm in} + s\, m_X n_{\rm in}$. The estimate uses the ratio $d = L_z d^{\rm in}/2$ of the exact small instances and reports two further scenarios in `resource_estimate.md`: the proved lower bound, and the data-string part $L_z d^{\rm in}$ of a slab (not a proved upper bound). A data block holds $K - 1$ data sites and one working slot (one site and a separate working block when $K = 1$). The theorem's serial schedule gives each data block a depth-first factory of $15\ell+7(\ell-1)+2$ logical slots and two live courier blocks; a shared serial schedule uses one such factory for all data blocks, as a surface-code layout with one factory unit does; and a throughput-limited schedule runs factories in parallel so that each logical layer takes two gadget slots. Every block carries one extraction ancilla per qubit and a gauging grid of multiplicity two, and an extraction round has 28 CNOT layers. For the surface code we use $2d^2$ qubits per tile, two tiles per logical site, a logical layer of $2d$ rounds, 15-to-1 units of 11 tiles for $11d$ rounds, and 8 layers per round. One layer is taken to last 50 ns.

c. What the estimate shows. Three conclusions hold for all three distance scenarios in `resource_estimate.md`. First, distillation. At $p = 10^{-3}$ with injected error $q_0 = 10^{-2}$, neither always-output model converges on layer-code blocks: the proved recursion needs $q_0 < 1/435 \approx 2.3 \times 10^{-3}$, and the exact map of the ideal factory, with same-level S corrections, needs $q_0 < 4.7\times10^{-3}$ (1.1×10^{-2} without corrections). The postselected factory converges up to $q_0 \approx 8\times10^{-2}$, so the comparison at this noise point uses it for both architectures. At $p = 10^{-4}$ and $q_0 = 10^{-3}$ all three models converge, in five, four and two levels on layer-code blocks. Second, the block. At the distances needed here ($f \approx$ 15–37) a block of this family holds one to three logical qubits (8×10^3 to 3×10^5 qubits per block across the scenarios), so it stores one or two data sites. With the same accounting as the tables (extraction ancillas and gauging grid for the layer code, a routing tile for the surface code), a layer-code block stores a logical more cheaply than two surface-code tiles from $d = 1103$, and for every larger tabulated distance from $d = 1169$; the corresponding values are 609 for the data-string scenario and 2000 (stably from 2309) for the proved lower bound, where a block holds 15 (8, 28) logical qubits. Comparing a bare block with extraction ancillas against a single tile gives the same distances. Third, space and time. At these distances the layer code loses to the surface-code baseline in peak space, in wall time, and in qubit-time; the qubit-time gap is 4.7 to 6.2 orders of magnitude for the rows of Table S23 (three to six across all models and scenarios), because each factory run occupies whole blocks for 30 gadget slots and each 15-to-1 run also consumes fifteen distilled $|Y\rangle$ corrections.

Four changes would move these numbers most: inputs whose blocks carry many logical qubits already at moderate distance; factories that do not tie up a whole block per gadget, or distillation delegated to a different code with a protected local transfer; an always-output protocol with cubic suppression under local-stochastic noise; and a lower raw injection error. The asymptotic advantage of the architecture is real, but the present constants place it far from the practical regime. (The superseded estimate of an earlier version, and the audit that reproduces its errors, are recorded in Sec. SXX.)

SXVII. SYMPLECTIC-OVERLAP EXTENSIONS AND BENCHMARKS

The following consequences of the overlap-corrected cleaning theorem are independent of the fault-tolerance proof. Besides pairwise overlap benchmarks, they expose a two-polarization mobility tradeoff linking rate, distance, disjointness, and shallow logical gates.

A. Hierarchy and arbitrary stabilizer codes

Using the CSS notation of the detailed overlap section, one may refine the invariant to

$$\chi_r^{XZ}(Q) = \min_{A,B:\, r(A,B)\ge r} |A \cap B|, \qquad 1 \le r \le K. \quad \text{(S12)}$$

The rank-factorization argument in the detailed overlap theorem implies immediately

$$\chi_r^{XZ}(Q) \ge r. \quad \text{(S13)}$$

This hierarchy records how much physical intersection is required to carry r independent dual logical commutators, rather than how much support is required for an r-dimensional subcode. The detailed port-packing theorem is the corresponding many-port statement: summing the overlap inequality over regions of bounded physical multiplicity gives $\sum_i g_X(A_i) \le c|B|$ whenever B supports all pure-Z logical classes. The preceding analysis also gives a direct injectivity proof of the complete-B case.

There is also a general stabilizer version with a sharp factor of two. Let $\mathcal{L} = S^\perp/S$ be the $2K$-dimensional logical symplectic space of an $[[n, K]]$ stabilizer code and choose complementary logical Lagrangians $\mathcal{L} = U \oplus W$,

TABLE S23. Resource estimate for $W = 100$, $T = 10^6$, $\varepsilon = 10^{-2}$, with explicit integer allocations and distances chosen from each schedule's complete runtime (`resource_estimate.py`). Upper block: $p = 10^{-3}$, injected error $q_0 = 10^{-2}$, where only the postselected factory (not covered by the theorem) converges on layer-code blocks. Lower block: $p = 10^{-4}$, $q_0 = 10^{-3}$, with the always-output bound (A8) (an assumed analog for the surface-code unit). "Serial" gives each data block its own factory, as in Theorem SV.31; "shared" uses one factory for all data blocks; "throughput" adds factories until a logical layer takes two gadget slots (layer code) or $2d$ rounds (surface code). All constants are stated assumptions; the table is not a hardware prediction.

Architecture	Schedule	levels	f	block n	K	peak qubits	wall time	qubit-seconds
$p = 10^{-3}$, $q_0 = 10^{-2}$, postselected distillation								
Layer code	serial	3	35	7.6×10^4	2	1.0×10^9	61 d	5.5×10^{15}
Layer code	shared	3	37	9.8×10^4	2	5.2×10^7	18 yr	2.9×10^{16}
Layer code	throughput	3	33	5.8×10^4	2	1.8×10^{12}	282 s	4.9×10^{14}
Surface code	serial	2	31	1922	1	4.1×10^5	2.9 d	1.0×10^{11}
Surface code	throughput	2	29	1682	1	1.9×10^8	23 s	4.4×10^9
$p = 10^{-4}$, $q_0 = 10^{-3}$, always-output bound (A8)								
Layer code	serial	5	19	4.1×10^4	1	1.8×10^9	18 yr	1.0×10^{18}
Layer code	shared	5	21	4.6×10^4	2	2.8×10^7	2×10^3 yr	1.8×10^{18}
Layer code	throughput	5	19	4.1×10^4	1	4.0×10^{14}	741 s	3.0×10^{17}
Surface code	serial	4	17	578	1	1.2×10^5	313 d	3.3×10^{12}
Surface code	throughput	4	17	578	1	1.3×10^{10}	14 s	1.7×10^{11}

$\dim U = \dim W = K$. Suppose region A supports all classes in U and region B supports all classes in W. If $C = A \cap B$ has m qubits, choose linear supported representative maps $\sigma_U : U \to S^\perp \cap V_A$ and $\sigma_W : W \to S^\perp \cap V_B$. All cross commutation occurs in C, so

$$\omega_{\mathcal{L}}(u, w) = \omega_C(\sigma_U(u)|_C, \sigma_W(w)|_C)\,. \tag{S14}$$

The perfect rank-K logical pairing therefore factors through $V_C \cong \mathbb{F}_2^{2m}$, which gives

$$K \le 2m, \qquad |A \cap B| \ge \lceil K/2 \rceil. \tag{S15}$$

The factor two is sharp. For the $[[3, 2, 1]]$ code $S = \langle X_1X_2X_3 \rangle$, take $A = \{1, 2\}$ and $B = \{2, 3\}$ and the two commuting logical Lagrangians

$$U = \mathrm{span}\{X_1X_2,\, Z_1Z_2\},$$
$$W = \mathrm{span}\{Z_2Z_3,\, X_2X_3\}.$$

Their cross-pairing matrix is I_2, while $|A \cap B| = 1$ and $K = 2$. The stronger CSS bound arises because a pure-X restriction on m qubits lives in an m-dimensional binary space rather than the full $2m$-dimensional local Pauli symplectic space.

B. Symplectic rederivation of the BPT tradeoff

The geometric part of the Bravyi–Poulin–Terhal (BPT) argument partitions a D-dimensional local code into

$$\Lambda = R_1 \sqcup R_2 \sqcup C, \tag{S16}$$

where R_1 and R_2 are correctable and, for a block scale R chosen with $R^{D-1} = \Omega(d)$, the separator obeys

$$|C| = O(n/R^2) = O\Big(n/d^{2/(D-1)}\Big) \tag{S17}$$

[14]. For a CSS stabilizer code, correctability of R_1 lets every pure-X logical be cleaned into $A = R_1^c$, and correctability of R_2 lets every pure-Z logical be cleaned into $B = R_2^c$; this is the standard cleaning mechanism [2]. Since

$$A \cap B = (R_1 \cup R_2)^c = C, \tag{S18}$$

the detailed universal bound $\chi_{XZ} \ge K$ gives the sandwich

$$K \le \chi_{XZ}(Q) \le O\Big(n/d^{2/(D-1)}\Big)\,, \tag{S19}$$

and hence

$$K d^{2/(D-1)} = O(n). \tag{S20}$$

Thus the standard BPT geometry can be combined with the overlap rank theorem to give a purely symplectic rederivation of the stabilizer/CSS scaling law. The geometric construction is BPT's; the new point here is the interpretation of its separator as a carrier of the logical X/Z commutation rank. This argument is inherently Pauli/stabilizer and does not extend the original BPT result for more general local commuting-projector codes.

For the present three-dimensional layer codes, $n = \Theta(N^3)$, $d = \Theta(N^2)$, and $K = \Theta(N)$, so both ends of (S19) scale as $\Theta(N)$. Their native transverse slabs therefore attain the algebraic lower scale and the three-dimensional BPT upper scale simultaneously:

$$\chi_{XZ} = \Theta(K) = \Theta(n/d). \tag{S21}$$

C. Price as the union analog

Pastawski and Preskill define the price $p(Q)$ as the minimum number of physical qubits in a region supporting

the complete logical algebra and prove the strong quantum Singleton bound

$$p(Q) - K \geq d - 1 \tag{S22}$$

for subspace codes [15]. For the same feasible CSS pairs used in the definition of χ_{XZ},

$$p(Q) = \min_{\substack{A,B: \\ g_X(A)=K,\, g_Z(B)=K}} |A \cup B|. \tag{S23}$$

Indeed, $A \cup B$ supports the product of any pure-X and pure-Z logical representatives and hence the full logical Pauli algebra, while an optimal price region R gives the feasible choice $A = B = R$.

Consequently, for every feasible pair,

$$\begin{aligned} |A \cup B| &\geq d + K - 1, \\ |A \cap B| &\geq K, \end{aligned}$$

and therefore

$$|A| + |B| \geq d + 2K - 1. \tag{S24}$$

For the native layer-code slabs, bounded density gives $|A \cup B| = O(N^2) = O(d)$ while the detailed layer-code overlap corollary gives $|A \cap B| = \Theta(K)$. Equation (S22) then supplies the matching union lower bound, so the same transverse pair is simultaneously efficient in the two complementary senses

$$(|A \cup B|, |A \cap B|) = (\Theta(d), \Theta(K)). \tag{S25}$$

This yields the dimensional hierarchy

$$\begin{aligned} \Theta(n) &: \text{ bulk volume}, \\ \Theta(d) &: \text{ complete-logical interface}, \\ \Theta(K) &: \text{ symplectic intersection}. \end{aligned} \tag{S26}$$

D. Bipolar mobility, price, and the crossbar corner

Let $A_1, \ldots, A_r$ be pairwise-disjoint complete pure-X ports and $B_1, \ldots, B_s$ pairwise-disjoint complete pure-Z ports of an $[[n, K]]$ CSS code. The detailed symplectic-crossbar theorem gives

$$rsK \leq n. \tag{S27}$$

The price relation also constrains the same two families. Applying (S24) to every pair (A_i, B_j) and summing gives

$$\begin{aligned} rs(d + 2K - 1) &\leq \sum_{i,j} (|A_i| + |B_j|) \\ &= s \sum_i |A_i| + r \sum_j |B_j| \\ &\leq (r + s)n, \end{aligned}$$

so

$$rs(d + 2K - 1) \leq (r + s)n. \tag{S28}$$

For balanced bipolar mobility $r = s = m$, Eqs. (S27)–(S28) become

$$n \geq \max\left\{ m^2 K,\ \frac{m}{2}(d + 2K - 1) \right\}. \tag{S29}$$

In particular, if $R = K/n$ then $Rm^2 \leq 1$. If $m = \Omega(K)$, any such CSS family requires $n = \Omega(K^3)$. The balanced layer-code family has $m = \Theta(K)$ and $n = \Theta(K^3)$, so it saturates this physical-qubit lower bound up to constants.

The scaling can be summarized by exponents. Suppose $K = n^{\alpha + o(1)}$, $d = n^{\beta + o(1)}$, and $m = n^{\mu + o(1)}$. The crossbar bound, the balanced price bound, and the three-dimensional BPT bound respectively imply

$$2\mu + \alpha \leq 1, \qquad \mu + \max\{\alpha, \beta\} \leq 1, \qquad \alpha + \beta \leq 1. \tag{S30}$$

If all three inequalities are saturated and $\mu > 0$, the only possible triple is

$$(\alpha, \beta, \mu) = \left(\frac{1}{3}, \frac{2}{3}, \frac{1}{3} \right). \tag{S31}$$

Indeed, the first and third equalities give $\mu = \beta/2$. If $\beta \geq \alpha$, saturation of the middle inequality gives $3\beta/2 = 1$ and hence (S31). If $\alpha \geq \beta$, the same equations force $\beta = \mu = 0$, excluded by $\mu > 0$. Thus the layer-code exponents form the unique nontrivial simultaneous-saturation corner of these three constraints.

There is a parallel gate-theoretic consequence. Put $m = \min\{r, s\}$. For any nontrivial logical Pauli coset $G = [X(u)Z(v)]$, choose the ith pure-X and pure-Z representatives and multiply them. Pairwise disjointness within each polarization makes these m representatives 2-disjoint, exactly as in the detailed layer-code argument. Hence

$$\Delta(Q) \geq m/2. \tag{S32}$$

If $d_\uparrow/d_\downarrow$ remains bounded while $m \to \infty$, the Jochym-O'Connor–Kubica–Yoder hierarchy criterion therefore excludes bounded-arity fixed-depth non-Clifford code-preserving unitary gates for all sufficiently large codes. Bipolar mobility is thus simultaneously a resource for local logical access and a source of static shallow-unitary rigidity.

E. Related boundary localization in tile codes

A recent independent result for two-dimensional tile codes proves, under its stated topological-order assumptions, a canonical symplectic basis with X- and Z-logical operators supported in transverse boundary strips and logical pairs labeled by physical qubits in their $D \times D$ corner overlap [16]. For a fixed tile rule this is the two-dimensional analog of a bulk $\to$ boundary-interface $\to$ transverse-overlap picture. We do not infer the layer-code theorem from the tile-code construction (or vice versa); the comparison only indicates that localized transverse logical polarizations occur in more than one geometrically local CSS family.

SXVIII. UNIFORM-GAP MOBILITY AND DISJOINTNESS CHECKS

Two facts remove any dependence of the shallow-unitary corollary on a privileged slab.

A. Why the prefix construction is gap-uniform

In the explicit realization of Sec. SI A the slab operators of Theorem SI.3 exist at every height z_0 and every column x_0: the gluing forms of item (1) of that theorem do not depend on the coordinate, because every junction and wall rule adds qubits that the slab operators do not use. In particular no admissibility condition is needed, and the prefix equations

$$p_i = \sum_{h \le i} x_h, \qquad p_{i-1} + p_i + x_i = 0 \tag{S33}$$

are the same at every height. The certificate of Sec. SII checks this at every height and column of every tested code. Hence

$$r_X = L_z = \Theta(N), \qquad r_Z = L_x + 1 = \Theta(N) \tag{S34}$$

disjoint complete ports exist for balanced inputs (Corollary SI.4).

B. Mixed logical Paulis and two-disjointness

Let $G = [X(u)Z(v)]$ be any nontrivial logical Pauli coset and set $r = \min(r_X, r_Z)$. Choose $X_i(u)$ and $Z_i(v)$ in the ith slabs and set $P_i = X_i(u)Z_i(v)$. Since the X slabs are mutually disjoint, a physical qubit occurs in at most one X_i; the same is true of the Z_i. Therefore

$$\#\{i : q \in \operatorname{supp} P_i\} \le 2 \tag{S35}$$

for every physical qubit q. These representatives are distinct. If both polarizations are nontrivial and $P_i = P_j$ up to phase, uniqueness of the binary Pauli decomposition would make the pure-X operator X_iX_j equal a pure-Z operator Z_iZ_j, forcing both to be identity; pairwise disjoint nonempty same-polarization supports exclude this for $i \neq j$. The pure cases are immediate. Thus every G has a 2-disjoint set of size r, and the single-qubit-partition disjointness obeys

$$\Delta \ge r/2 = \Omega(N). \tag{S36}$$

Lemma 2 of Jochym-O'Connor, Kubica, and Yoder gives $\Delta \le n_{\mathrm{phys}}/d_\uparrow$ [3]. The fixed transverse slab pair represents every mixed logical Pauli in $O(N^2)$ qubits, whereas the ordinary layer-code distance is $\Theta(N^2)$; hence

$$d_\downarrow = \Theta(N^2), \qquad d_\uparrow = \Theta(N^2), \qquad \Delta = \Theta(N). \tag{S37}$$

C. Exact scope of the shallow-unitary corollary

For a q-local depth-h physical unitary circuit, Theorem 9 of Ref. [3] shows that the induced logical unitary belongs to level M of the Clifford hierarchy if

$$d_\uparrow q^{(2^M-1)h} < d_\downarrow \Delta^{M-1}. \tag{S38}$$

At $M = 2$ and fixed q, h, Eq. (S37) makes the left-to-right ratio $O(N^{-1})$. Thus sufficiently large blocks are Clifford-only under such static bounded-arity constant-depth unitary circuits, including transversal circuits. This is stronger for this family than the generic three-dimensional Bravyi–König geometric-locality restriction, which allows the third level of the Clifford hierarchy [4].

The scope distinction is essential. The theorem does not prohibit the architecture in the companion manuscript because that construction uses protected logical Pauli measurements, encoded resource states, classical feed-forward, ancilla systems, and time-dependent code deformation. These operations are not a single bounded-depth unitary circuit preserving one fixed layer-code codespace.

SXIX. RELATED ARCHITECTURES AND IMPLEMENTATION DIRECTIONS

The construction complements gauging [5], parallel and parsimonious surgery [17, 18], homological measurement [19], adapters and extractors [20, 21], lifted surgery [22], and recent universal good-qLDPC architectures [23, 24]. Recent layer-code work also studies partial self-correction, random layer codes, weight reduction, and non-CSS embeddings [25–28]. Constant-overhead fault tolerance with LDPC codes [29–31] and magic-state injection into LDPC codes [32, 33] use nonlocal connectivity. Geometrically local approaches include surface-code lattice surgery [34, 35], three-dimensional surface codes with transversal gates [36], local simulation of LDPC codes [37], local fault-tolerance constructions [38], wire codes [39], and planar logical measurements [40]. The distinction relevant here is one sentence: universal fault-tolerant processing is carried out on a native three-dimensional Euclidean nearest-neighbor layer-code lattice (the explicit realization of the Williamson–Baspin defect network), with no nonlocal auxiliary connectivity. Table S24 gives a compact comparison.

A. Algorithmic and noise-model outlook

a. Practical two-pass decoder. For algorithmic and numerical exploration, one may decode a layer-code block by (i) running a local decoder independently on each surface-code sheet, (ii) resolving the residual syndrome along the $O(1)$-width junction corridors by matching or belief propagation, and (iii) applying OSD only on the $O(1)$-width slab regions needed to settle the logical frame.

TABLE S24. Compact positioning against selected related constructions. The table describes each cited construction itself, not every capability that could be obtained by combining separate schemes.

Work	Geometry	Logical-operation mechanism	End-to-end circuit FT?	Principal resource scaling / emphasis
Williamson–Yoder [5]	General code plus auxiliary gauging graph	Gauging an arbitrary logical Pauli	Spacetime fault-distance theorem for the gauging primitive	Ancilla overhead near-linear in logical weight up to polylogarithmic factors
Parallel / parsimonious surgery [17, 18]	General sparse stabilizer connectivity	Parallel logical Pauli measurements via surgery	Fault-distance-preserving measurement framework	Parallel measurements; parsimonious ancilla systems of $O(w \log w)$ qubits for weight-w logical Paulis
Homological measurement [19]	General CSS-code geometry	Edge-expanded / homological measurement	Distance-preserving logical-measurement framework	Ancilla count linear in logical weight for the edge-expanded protocol
Extractors [21]	Fixed sparse qLDPC connectivity	Extractor systems / Pauli-based computation	Universal PBC architecture, with high-fidelity non-Clifford supply as an interface	$\widetilde{O}(n)$ extractor size; logical cycles scale with syndrome rounds
Good-qLDPC architectures [23, 24]	General good-qLDPC or SHYPS connectivity	Parallel surgery/state preparation or transversal/fold-transversal Clifford processing	Universal or universal-completable FT architectures	Constant/low qubit overhead and reduced logical time in their respective settings
Lattice surgery [34, 35]	2D nearest-neighbor surface-code patches	Merging and splitting patch boundaries	Standard surface-code fault tolerance	$\Theta(d^2)$ physical qubits per logical qubit
This work	3D Euclidean nearest-neighbor layer-code lattice	Constant-width slabs + thick local 3D grid gauging + encoded courier	Scheduled-circuit distance and resource theorem; full distance kept in every measurement	$\Theta(f)$ qubits per logical, $O(Wf + (B_f(A)+1)f^{3/2})$ space; $O(Tf^{3/2}\Psi)$ depth with $\Psi = O(f^{\log_2 15})$ (see Sec. SXVI for constants)

A positive-threshold proof for this decoder would require a separate cluster-expansion analysis of the residual failures, and a near-linear runtime claim would require an implementation and complexity analysis. Neither is claimed here. The theorem of Sec. SV continues to use committed minimum-weight decoding.

b. Next numerical and algorithmic steps. Four directions follow from the present results. (i) *A growing family of the explicit realization.* The layer-code tests here vary the spacing of small inputs up to $s = 5$ (Sec. SXIV A) and, with the routed schedule, the input up to seven qubits (Sec. SXIV B); neither probes $K = \Theta(N)$ and $d = \Theta(N^2)$. The next campaign should grow the input into that regime, carry the multiplicity of Corollary SIII.4 where the grid's crossing number falls below the block distance, and shorten the routed schedule, whose long idle periods leave the present sampled rates uninformative, so that failure rates can be measured with a decoder close to minimum weight. (ii) *A decoder for deformation histories.* The layer-code decoders of Ref. [26] are a natural starting point, but changing checks, inherited frames and records at cuts, and joint measurements require new work. (iii) *Parallel access.* How many logical measurements can share the slab interfaces while keeping bounded congestion and full distance, compared with parallel code surgery [17] under the additional restriction of three-dimensional Euclidean locality? (iv) *Resource supply.* Specialized factories that deliver resources to layer-code data blocks through a protected local transfer would remove the dominant cost identified in Sec. SXVI.

c. Constructivity of the component codes. Leverrier–Zémor's random-method result is sufficient for the existential input family used in Lemma SV.28. A deterministic version would require component codes satisfying the needed distance and robustness conditions, or a derandomization that controls the global robustness predicate. Limited-independence or conditional-expectation methods become applicable only after the relevant bad events and independence requirements are identified. Random layer codes built from random CSS inputs [26], suggested by D. J. Williamson (private communication), are a different route; the slab construction applies to the explicit realization of any CSS input, but the resource accounting would have to be redone with their parameter scaling.

d. Noise beyond local-stochastic Paulis. Randomized compiling and Pauli/Clifford twirling can tailor suitable Markovian noise toward stochastic Pauli descriptions under additional assumptions [41, 42], while leakage-reduction gadgets can return population to the computational subspace at additional local circuit cost [43]. Extending the theorem requires a separate fault-path analysis for coherent temporal correlations, gate dependence, leakage propagation, and the interaction between leakage-reduction units and the detector schedule. The exploratory probes of Sec. SXIII F are diagnostics only.

SXX. ARCHIVE: REVISION RECORD AND SUPERSEDED MATERIAL

This section records what changed in successive revisions and keeps the certificates and arguments that those revisions superseded, so that earlier versions remain traceable. Nothing in this section is used by the results of Secs. SI–SXIX.

A. Revision of 14 October 2026

Added the hardware demonstrations of Sec. SXV, a main-text section and figure that summarize them, a summary in the abstract and the introduction, the corresponding acknowledgments, and the Open Quantum citation [13]. The main-text figure shows what was run next to what was measured; Fig. S17 shows every run and Fig. S18 every placement. In the main text, the surface-code threshold figure moved to this supplement (Fig. S14, same data), and a figure of the two-block measurement on layer-code blocks took its place. The runs found two practical points, discussed in Sec. SXV and in the main-text Discussion: the CNOT order inside a vertex star decides whether parallel ancillas keep their distance at circuit level, and ancilla errors that survive a reset limit repeated rounds on hardware. No theoretical result changed.

B. Revision of 9 October 2026

An audit of the 8 October revision accepted the invariants (I1)–(I3) for the abstract schedule but noted that no executable trace showed them for births, cell reuse and adaptive branches, and that the hypothesis $R_{\mathrm{pad}}(B) > \tau$ of Lemma SV.19 needed an explicit parameter choice. The choices $d_0 > \tau$ and $R > \tau$ are now stated with the courier estimate and Eq. (A19), and `schedule_verifier.py` exports a complete schedule and checks it, with nine negative controls (paragraph "An executable schedule certificate"). The audit also found that the coordinates of the routing sketch `factory_3d_routing_coords.csv` overlap; that file is schematic and is now marked as such.

C. Revision of 8 October 2026

a. Rectangle distance without cleaning. The relative rectangle bound (Lemma SV.18) previously rested on the cleaning interface of Proposition SV.15, whose weight step WY6 was argued by a timeline-parity sweep across entry, exit and idle boundaries. A reader asked for that step to be checked at every boundary. It is now replaced by the direct slice argument of Proposition SV.17, which needs no sweep: the X-type faults leave one deformed logical at the right cut, the Z-type faults leave a logical at the right cut or, in a round without a record fault, an operator anticommuting with the measured logical, and in each case every qubit of that operator carries a fault. The bound is slightly sharper, $\min\{\widetilde{d}_X, d_Z, R\}$, and `rectangle_slice_check.py` shows by exact minimization that it is attained (Table S8). Lemmas SV.21 and SV.23 now use the same argument with effects recorded modulo the resource subgroup.

b. Separation of resource neighborhoods. Lemma SV.25 previously derived the linking length $\rho \geq \kappa R$ from a bounded-degree coloring of conflicting event cores, which did not explicitly exclude paths through intermediate cells. The schedule invariants (I1)–(I3) are now stated, and Lemma SV.19 shows that any path between the regions of two primitives stays on one block for that block's whole padding; the separation follows with $\kappa = 1/\tau$ for every schedule satisfying the invariants. The same lemma closes a gap in the proof of Theorem SV.20, which inferred from the temporal span of a short component alone that it meets at most one primitive; that inference did not cover two primitives running at the same time on disjoint blocks.

D. Revision of 26 September 2026

a. Boundary composition. The earlier worked gluing example wrote the T0 row of each primitive as $f_t + \mu_t + s_t^{\mathrm{in}} + s_t^{\mathrm{out}} = 0$, with s the physical syndrome at the cut. That lets a record error change the quantum boundary class: with $f_1 = f_2 = 0$, $\mu_1 = \mu_2 = 1$ and $s_{\mathrm{in}} = s_{\mathrm{out}} = 0$ the equations are satisfied by $s_{\mathrm{mid}} = 1$, although no data error occurred, whereas the physical detector history of two flipped records between clean records is $(1, 0, 1)$; summing the two rows loses both endpoints. The script that accompanied the example verified the elimination of the equations it was given, not that they describe the circuit. The boundary interface now carries the last-round records ρ separately from the quantum class (s, λ) (Eqs. (A15-rec) and (A15a)). The worked example, Proposition SV.10, Lemmas SV.11, SV.13 and SV.14 (which now asks for a round free of the component before the right cut), Eq. (A15-rel), the WY5 interface, and the proof of Theorem SV.20 were revised accordingly, and `boundary_composition_check.py` compares the direct and composed maps on explicit blocks through idle $\to$ gauging $\to$ idle $\to$ gauging $\to$ idle.

b. Resource noise. The earlier "noisy A-twirl closure" lemma assumed that every incoming $|A\rangle$ candidate is diagonal in $\{|A\rangle, Z|A\rangle\}$ and used a twirl $T_A = e^{-i\pi/4} S \cdot X$, implemented with its own $|Y\rangle$ resource, to keep it so. A Pauli X on a raw candidate gives $X|A\rangle$, which has coherence $1/2$ in that basis. If the twirl's $|Y\rangle$ is also bad, the nontrivial branch implements ZT_A, and $X|A\rangle$ is an eigenvector of ZT_A, so $\frac{1}{2}(\rho + ZT_A \rho T_A^\dagger Z) = \rho$ for $\rho = X|A\rangle\langle A|X$: the faulty twirl removes no coherence. This joint event is allowed by local-stochastic Pauli noise. The twirl is removed. Defective slots may now hold arbitrary, possibly entangled states (Definition SV.2); Lemmas SV.3, SV.4 and SV.5 prove the recursion (A8) with $C_{\mathrm{dist}} = \max\{\binom{15+c_Y}{2}, \binom{7}{2}\}$ in place of the earlier $4C_{\mathrm{dist}}$, and `resource_defect_check.py` checks the lemmas and the counterexample.

c. Distance. Theorem SI.6 previously stated $d_X \geq (L_x + 1) d_X^{\mathrm{in}} / w_X$ and $d_Z \geq L_z d_Z^{\mathrm{in}} / w_Z$. Exchanging an occupied backbone prefix for its complementary suffix, and rounding the number of occupied pieces up in each plane before summing, gives Eq. (P7). For the $[[4, 2, 2]]$ input at $s = 5$ this closes the CP-SAT brackets 10–11 and

9–10 to $d_X = 11$, $d_Z = 10$.

d. Resource estimate. The earlier estimate contained three accounting errors, all reproduced by the audit section of `resource_estimate.py`. (A) Its "quadratic" model $q \mapsto 35q^2$ was labeled as the proved always-output bound; the proved bound is $C_{\mathrm{dist}}q^2$, and the exact map of the ideal always-output factory starts at $105q^2$ because every pair of Z errors is malignant. (B) The serial surface-code distance $f = 23$ was chosen from a $2f$-round logical layer, but the reported serial runtime of 4.048×10^{11} rounds gives $WR_{\mathrm{total}}\, 0.1(p/p_{\mathrm{th}})^{(f+1)/2} = 4.048$ against a budget of 0.005; $f = 31$ is needed under the same model. (C) The serial layer-code space used $\lceil 16/K \rceil$ blocks per factory run for the timing but never allocated them; at $K \approx 2.4$ half a data block cannot hold a 16-slot factory. The superseded table quoted 2.4×10^7 qubits and 18.5 h for the serial layer-code schedule, 2.2×10^5 qubits and 45 h for a serial surface-code schedule, 1.0×10^8 qubits for 18 s for a throughput-limited one, and storage crossovers $d = 546$, 1083 and 4337. Table S23 replaces it.

e. Reproducibility and presentation. The Monte Carlo seeds of `two_block_measurement.py` were derived from Python's `hash()` of strings, which is salted per process, so the archived failure rates could not be regenerated exactly. Seeds are now recorded integers (a base seed plus a CRC32 offset for each interface and basis), and the sampling of Table S17 was rerun with more shots. The references to Williamson and Yoder now use the theorem numbering of their published version. A junction and wall figure was added (Fig. S1), and the revision notes that were spread through the text were collected here, leaving short pointers in the body. An independent read of the revised text then led to four further repairs: a decomposition of the residual into components, evaluated at clean idle cuts, shows that the large-cluster complement G is contained in the event G_{res}, also at resource boundaries, and the injection events E_ι are defined; Lemma SV.18 now states its zero-syndrome condition at the right cut and covers idle rectangles of any length and rectangles extended by idle pads; Lemma SV.14 now treats births, merges, splits, discards and the edge-readout terms of the gauging exit explicitly; and the storage crossover uses the same qubit accounting as the space tables.

E. Earlier revision: from the single-chart cells to the explicit realization

a. What changed. An earlier version derived the slab theorem from a transcription of the six WB line-junction cells into a single unit-cubic chart (Table S3), audited by a finite-patch certificate. An independent review found that the certificate did not discriminate: it omitted every ordinary check touching a designated port, and its "other-boundary" test counted occupied frontier edges instead of evaluating their syndrome, so the $\hat{z}$ cells accepted words outside $\ker H_{\mathrm{end}}$ and $\ker H_{\mathrm{mid}}$. Glued into complete codes, the single-chart cells enforce the all-equal relation $\{000, 111\}$ at every middle junction, not $\ker H_{\mathrm{mid}}$, and the resulting codes do not encode the input logical qubits (Table S5, last two columns). The slab theorem was therefore restated and proved for the explicit realization of Sec. SI A. The same revision restored the syndrome terms $C_g s$ and $G_g s$ of (A15b)–(A15c), which an earlier version had omitted, and retired the truncated layer-code family built from the single-chart cells, which shows no clean all-size crossing; it is kept only as the negative control of Sec. SXIII F.

TABLE S25. Archived cell solver run on every boundary word. Starred words lie outside $\ker H_{\mathrm{end}}$ or $\ker H_{\mathrm{mid}}$. The last column counts checks violated by the archived witnesses of the nonzero kernel words once the boundary-adjacent checks are included.

Cell	Words accepted by the archived solver	Violated checks
first/last $\hat{z}$	$00, 01^*, 10^*, 11$	6
middle $\hat{z}$	all eight, incl. $001^*, 010^*, 100^*, 111^*$	4–6
first/last $\hat{x}$	$00, 11$	2
middle $\hat{x}$	$000, 011, 101, 110$	2

b. The archived cell certificate. Feeding every boundary word, not only the kernel words, to the archived solver reproduces the review's finding (Table S25): the $\hat{z}$ cells accept the odd endpoint words 01, 10 and all eight middle words. Evaluating each archived witness on every check of its patch, including the omitted boundary-adjacent checks, shows nonzero syndrome for every nonzero kernel word (two to six violated checks). The archived certificate therefore established neither the boundary relations nor their cancellation under gluing. It dropped every ordinary check touching a designated port, so an X on a single port edge had zero "retained" syndrome, and its frontier test counted occupied edges rather than syndrome. (This is part A of `slab_gluing_certificate.py`.)

F. Archived finite-cell certificate

The previous version used `wb_exact_cell_patch_audit.py` as its proof-critical cell certificate. As explained in the preceding subsection and quantified in Table S25, that certificate omitted the checks touching the designated ports and did not test boundary syndrome, so it cannot certify $\ker H_{\mathrm{end}}$ or $\ker H_{\mathrm{mid}}$, and the single-chart cells it audited do not realize $\ker H_{\mathrm{mid}}$ when glued. The script and its outputs are retained unchanged for provenance and are superseded by the complete-code certificate of Sec. SII. The same applies to the bullet/plus audits, the radius-2/3 exterior-syndrome stress test, and the truncated incidence-based instance of `wb_layer_code_instance.py`, whose construction discarded noncommuting candidate checks and condensed spurious boundary modes until $k = 1$.

a. Finite WB-incidence-derived test instance (superseded). Script `wb_layer_code_instance.py` constructs

a truncated CSS instance from the single-chart junction incidences for the 3-qubit Z-repetition input ($L = 3$): $n = 66$, $k = 1$. Its construction discards noncommuting candidate checks and condenses spurious boundary modes until $k = 1$, so it is not evidence for the slab theorem; the explicit realization replaces it.

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